Introduction to

Artificial Intelligence

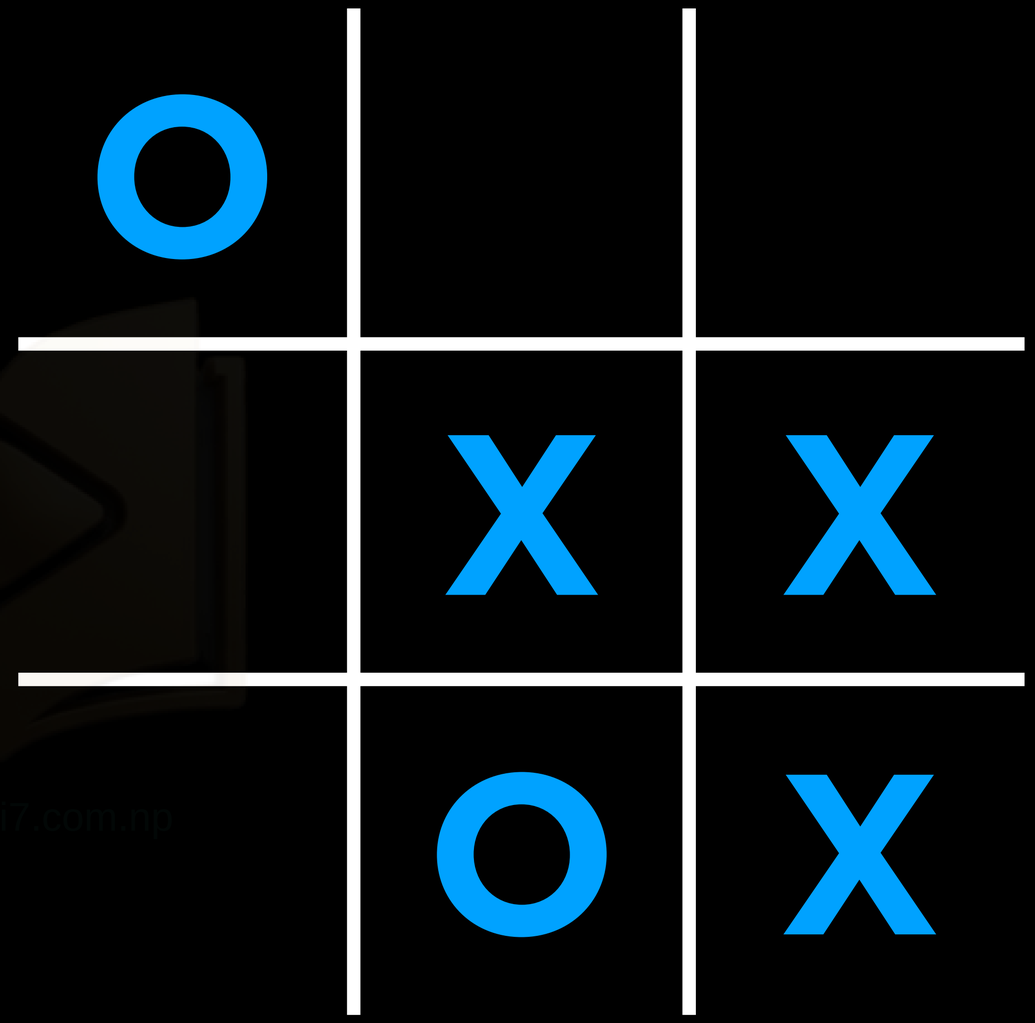
with Python

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Artificial Intelligence

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Search



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Knowledge

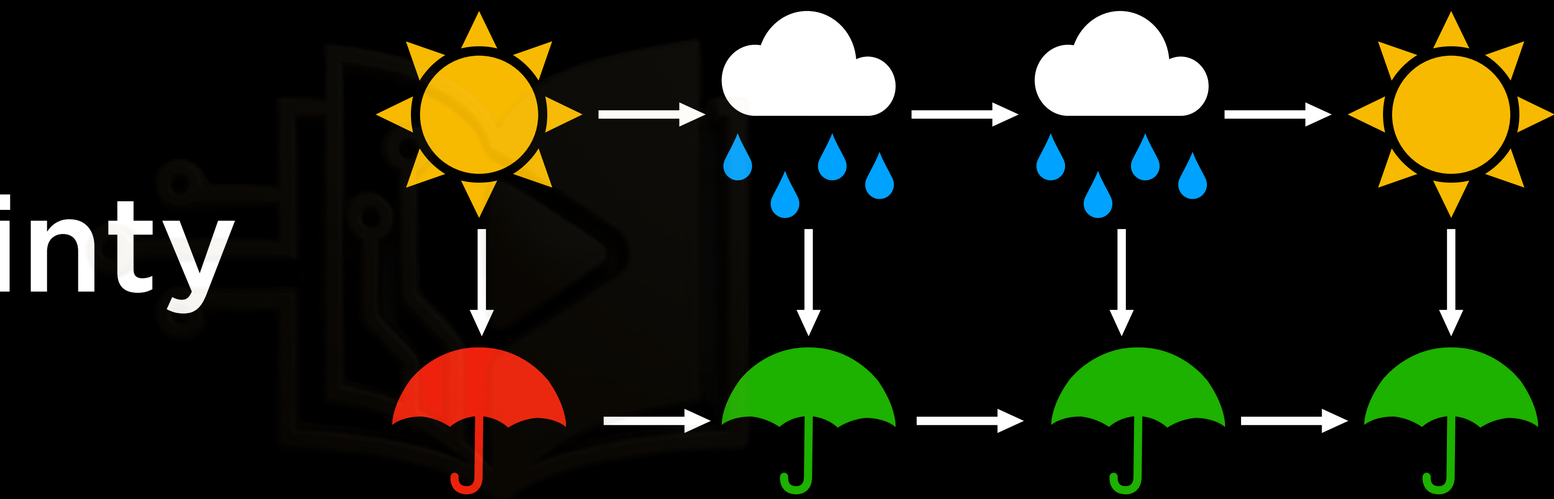
$$P \to Q$$

$$P$$

$$Q$$

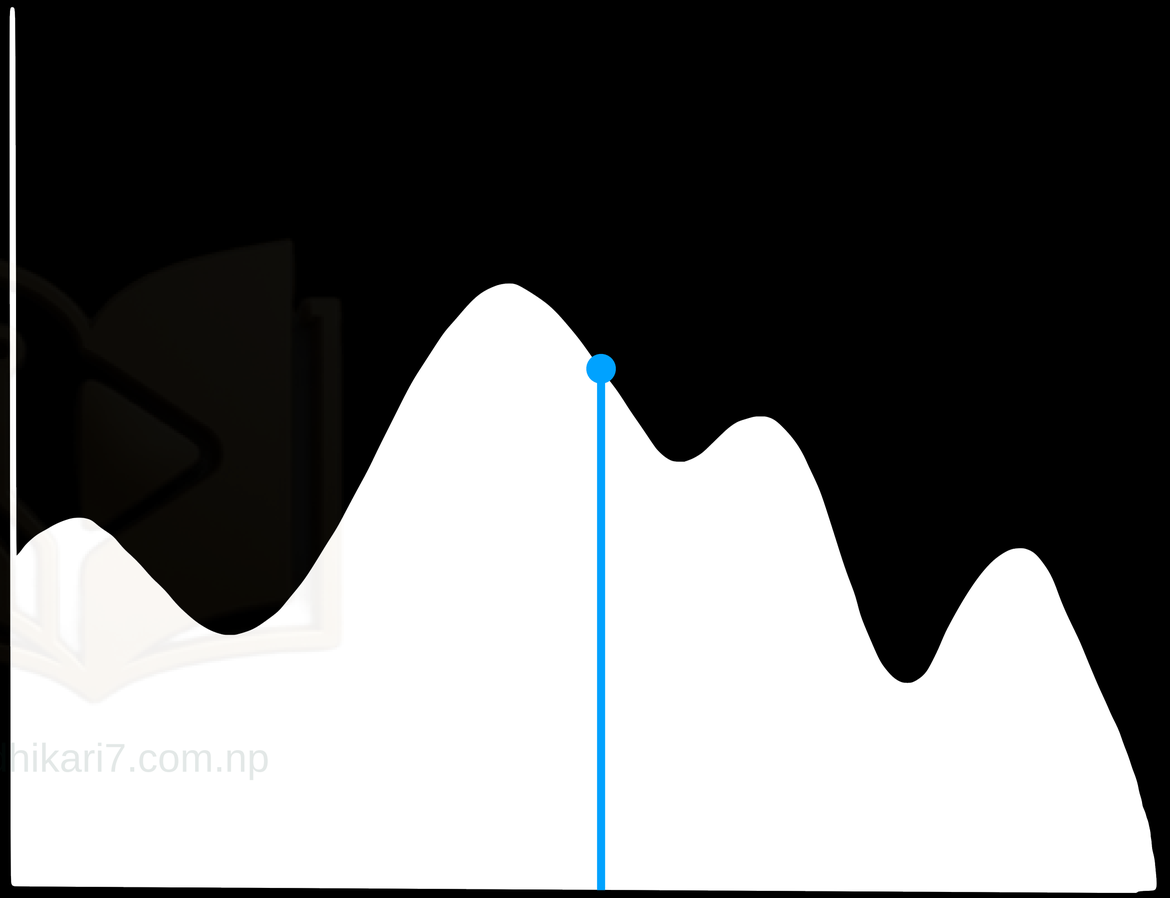
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Uncertainty



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Optimization



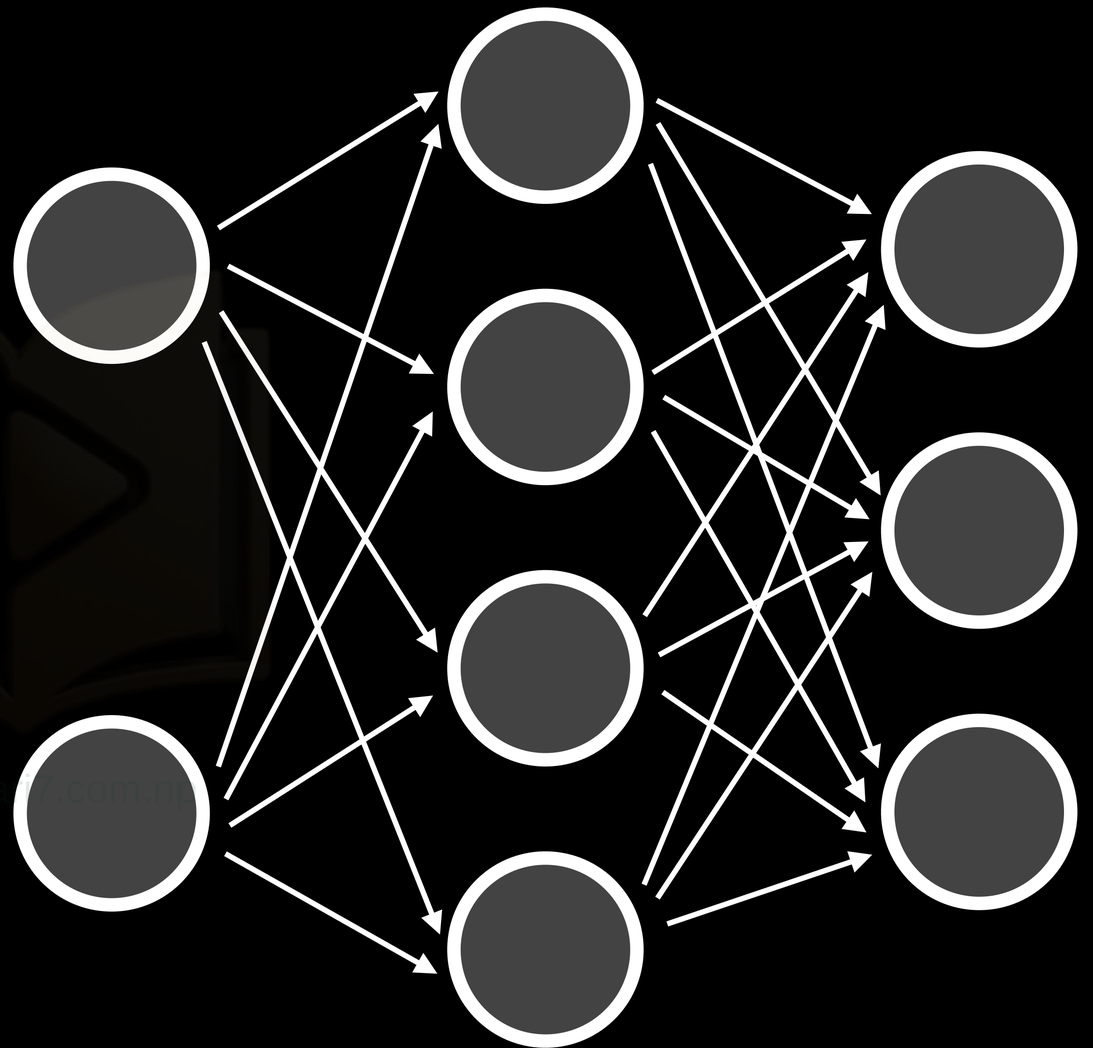
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Learning



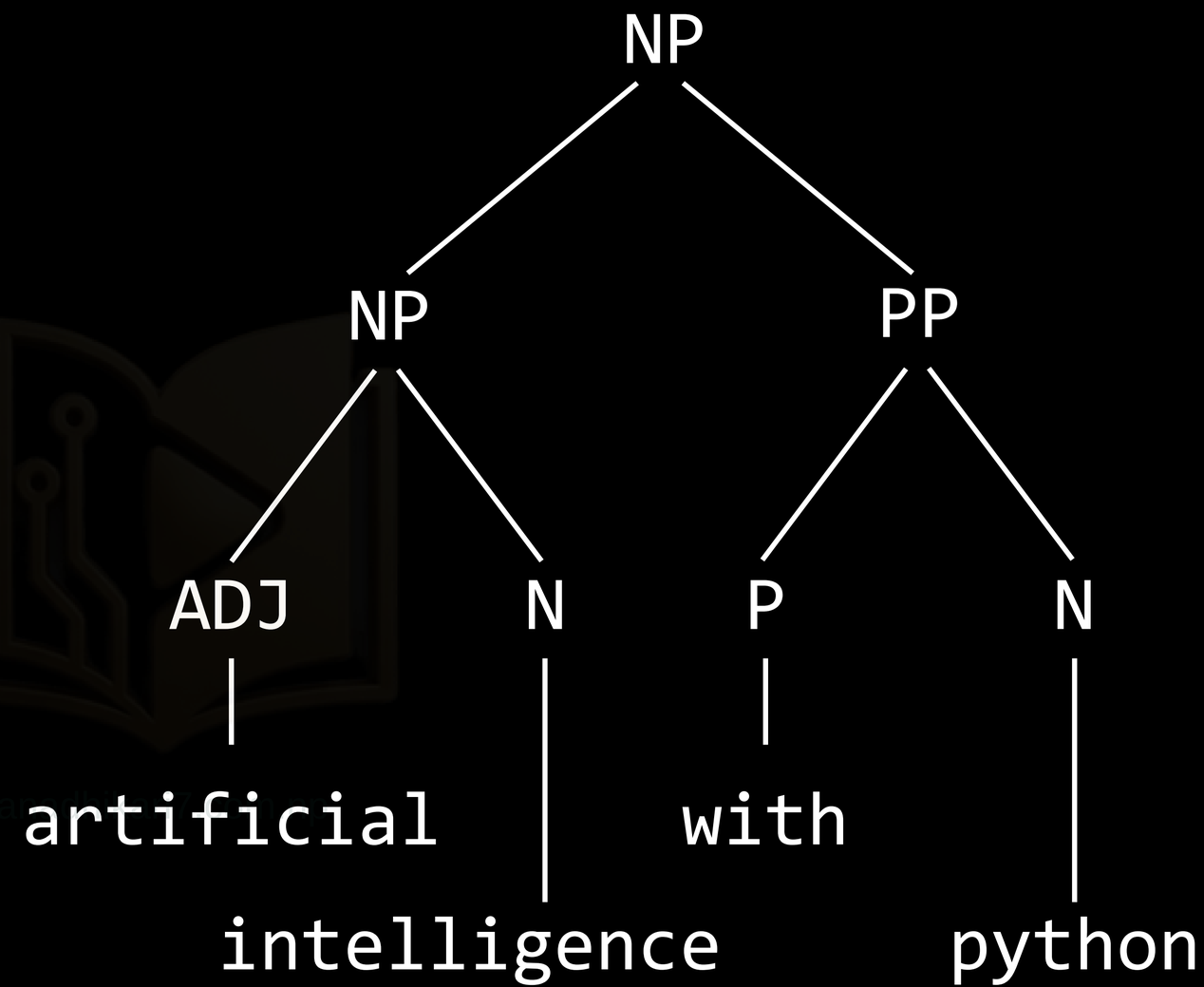
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Neural Networks



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Language



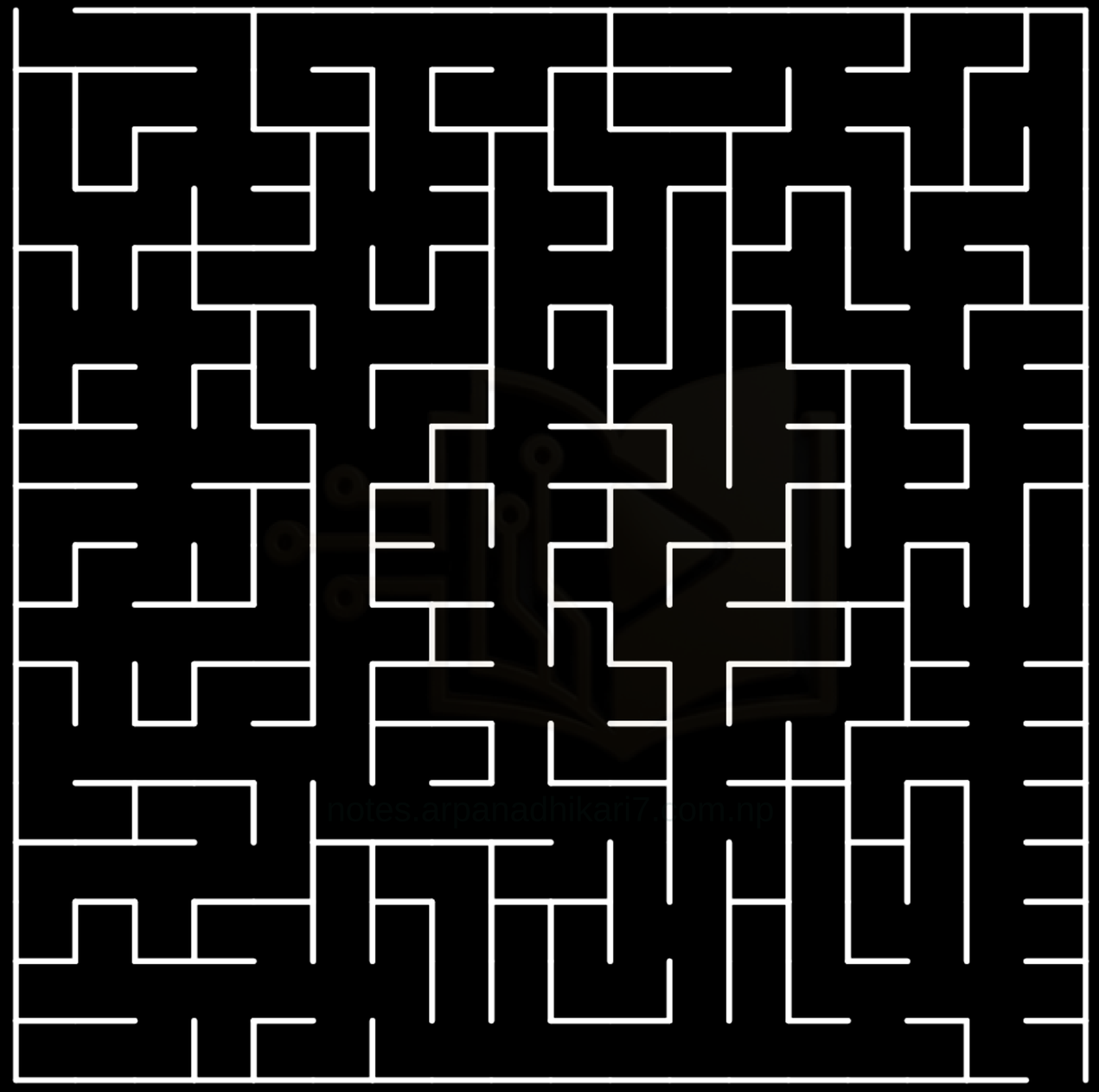
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Search

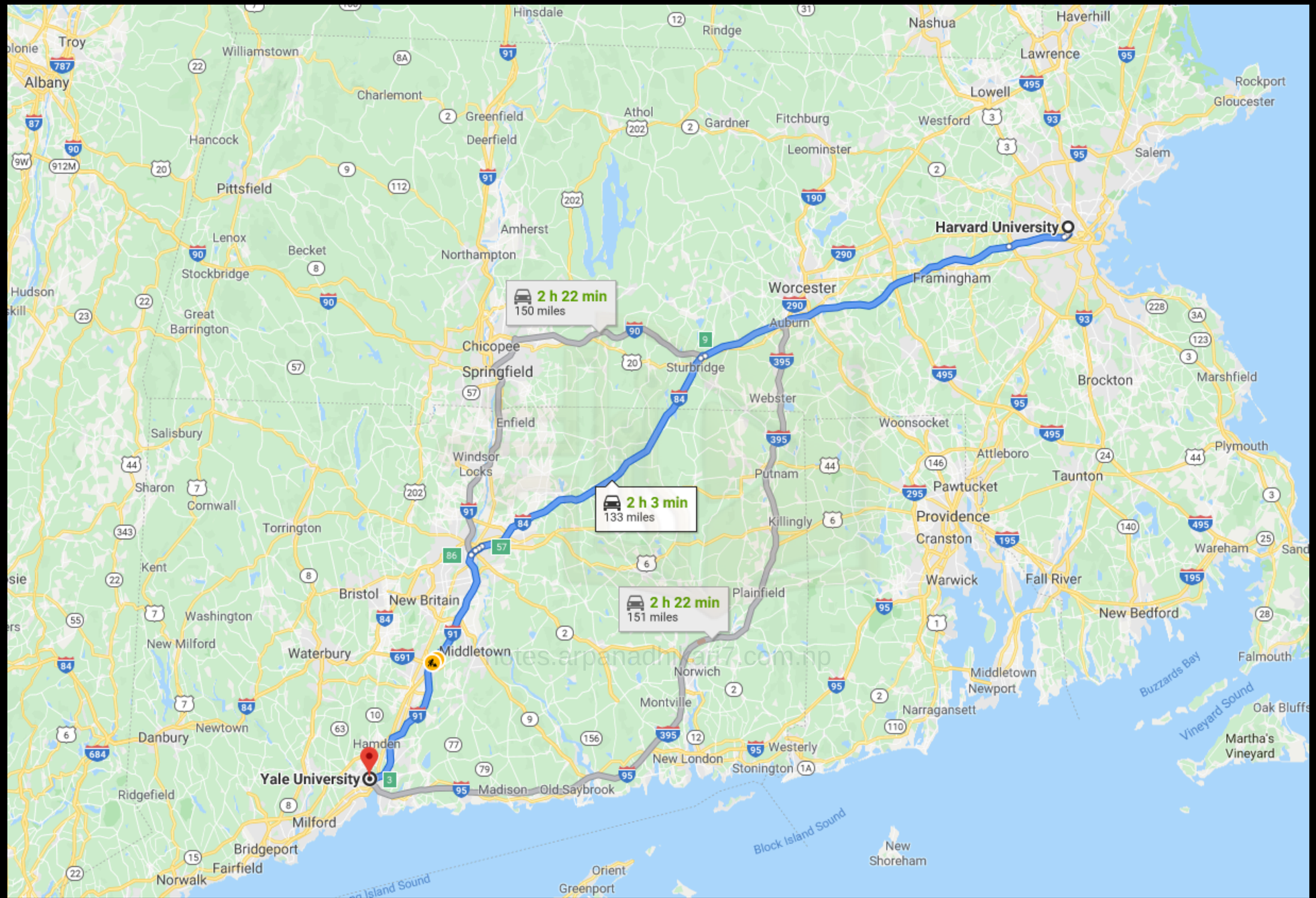
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1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

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Search Problems

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agent

entity that perceives its environment and acts upon that environment

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state

a configuration of the agent and its environment

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2	4	5	7
8	3	1	11
14	6		10
9	13	15	12

12	9	4	2
8	7	3	14
	1	6	11
5	13	10	15

15	4	10	3
13	1	11	12
9	5	14	7
6	8		2

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initial state

the state in which the agent begins

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initial state

2	4	5	7
8	3	1	11
14	6		10
9	13	15	12

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actions

choices that can be made in a state

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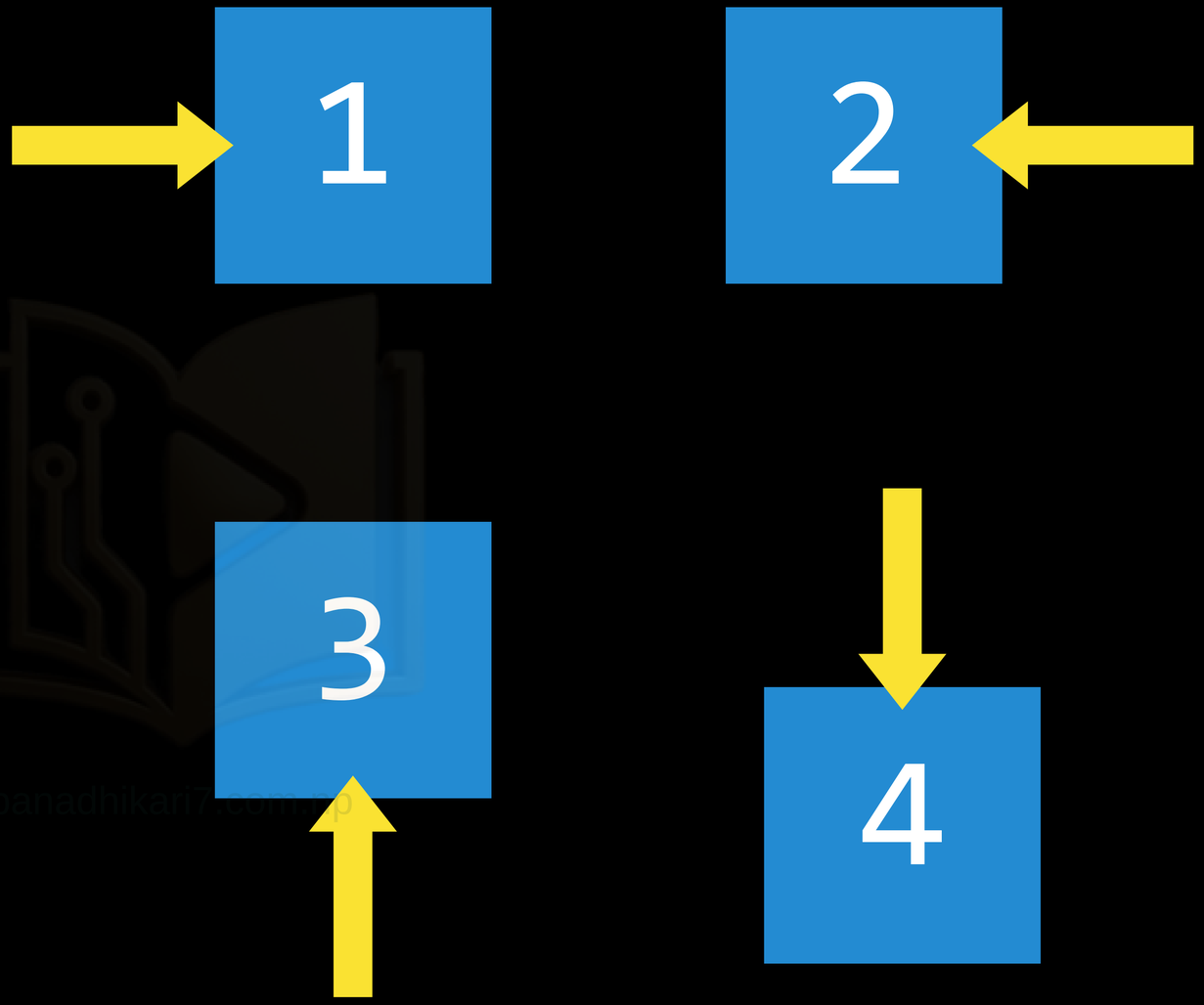
actions

$\textsc{Actions}(s)$ returns the set of actions that can be executed in state s

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actions



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transition model

a description of what state results from performing any applicable action in any state

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transition model

$\textsc{Result}(s, a)$ returns the state resulting from performing action a in state s

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RESULT(

2	4	5	7
8	3	1	11
14	6	10	12
9	13	15	

, ) =

2	4	5	7
8	3	1	11
14	6	10	12
9	13		15

RESULT(

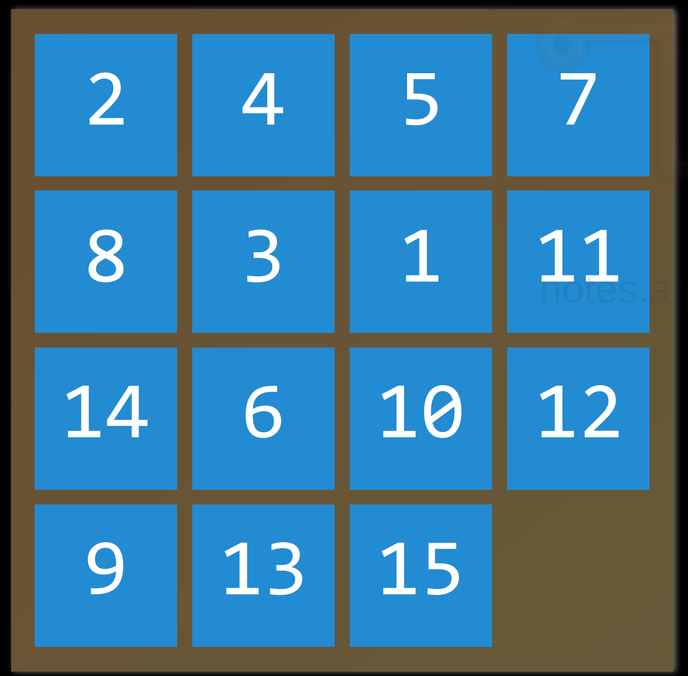
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8	3	1	11
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9	13	15	

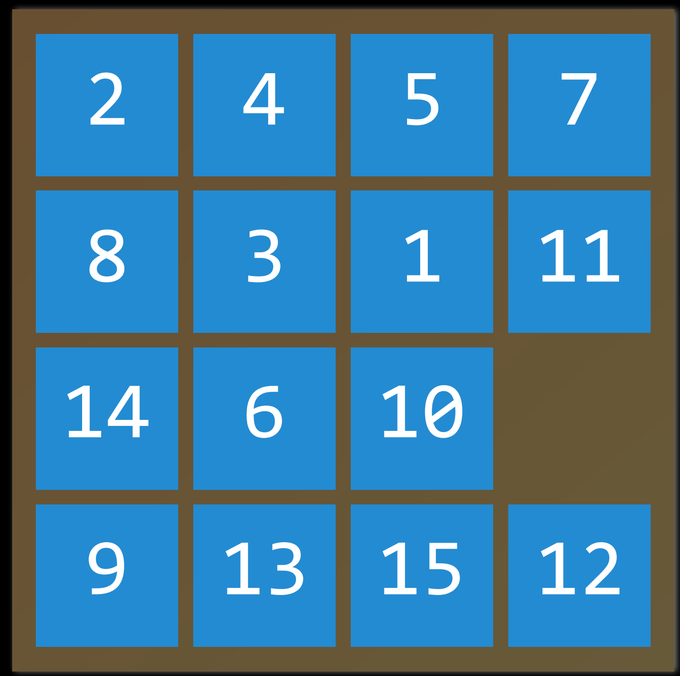
, ) =

2	4	5	7
8	3	1	11
14	6	10	
9	13	15	12

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transition model

RESULT(

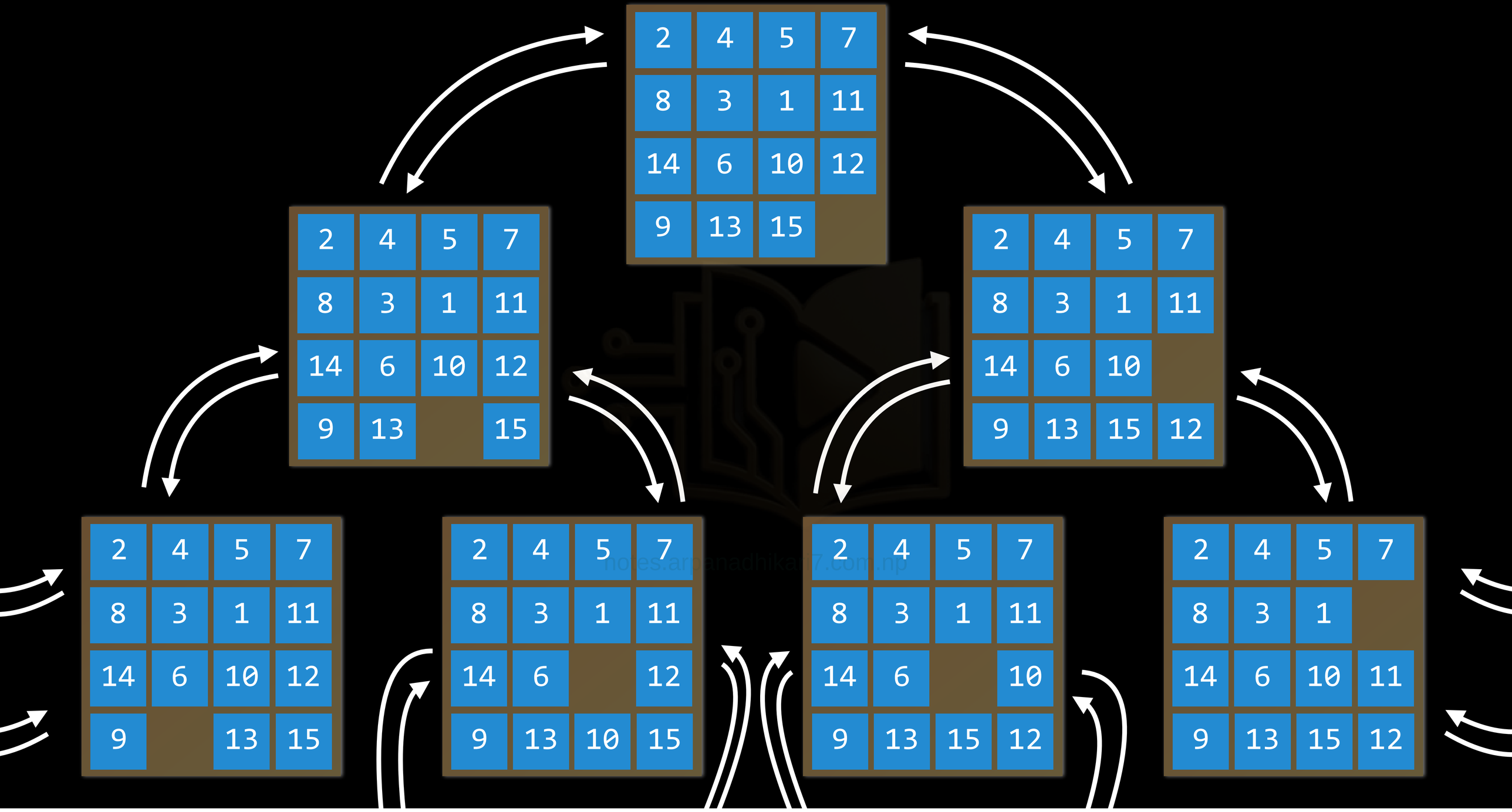
, ) =


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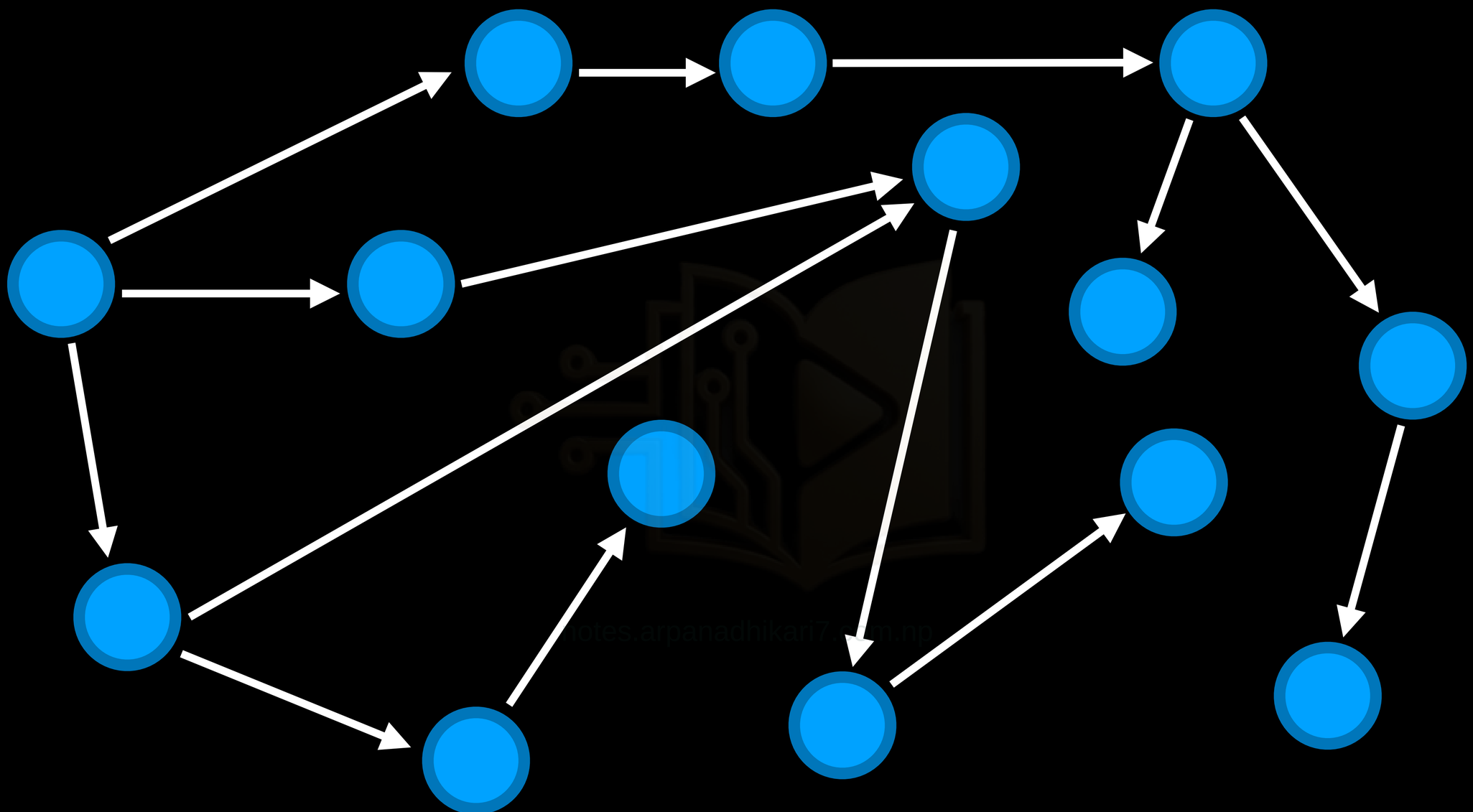
state space

the set of all states reachable from the initial state by any sequence of actions

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goal test

way to determine whether a given state is a goal state

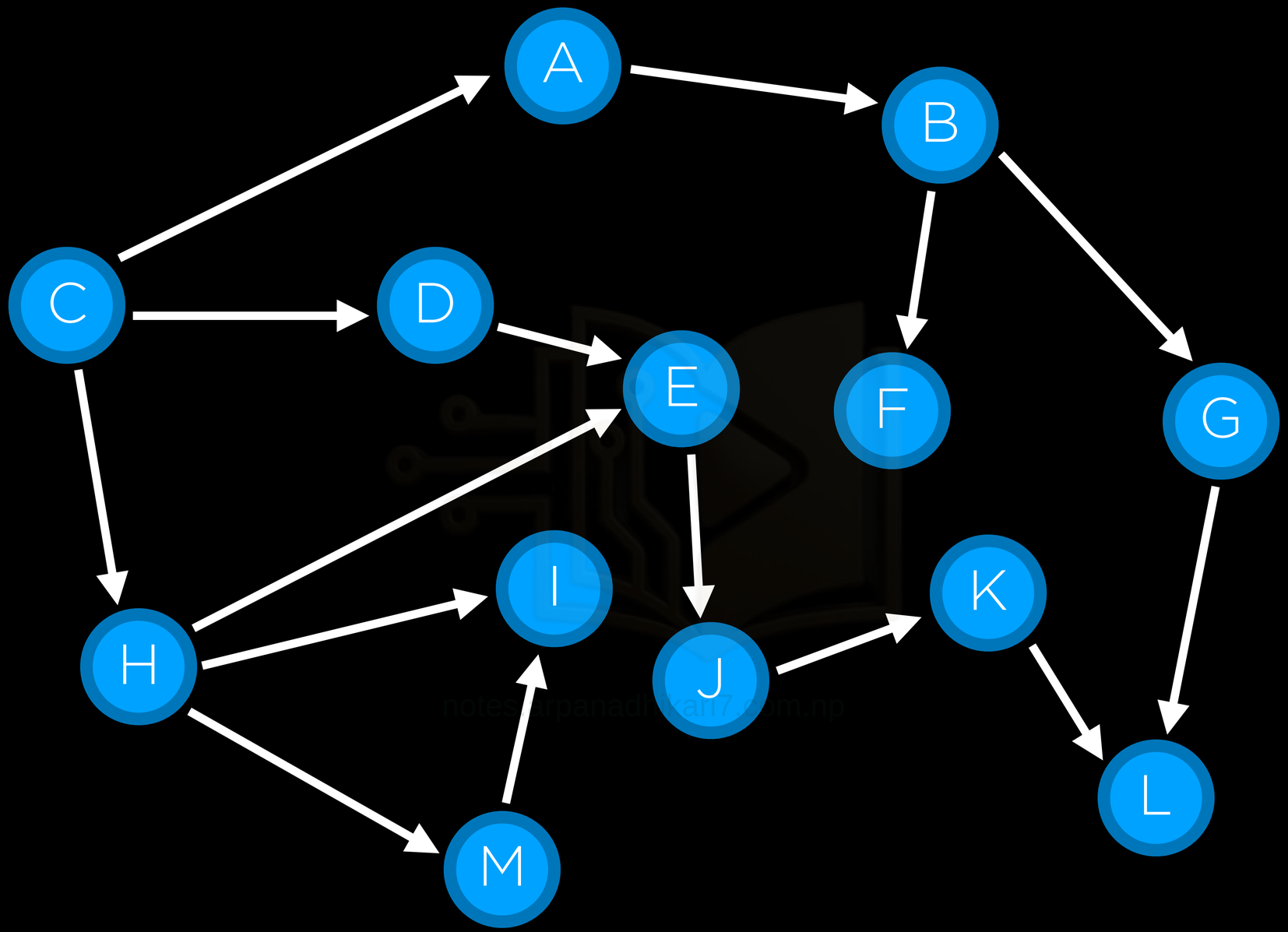
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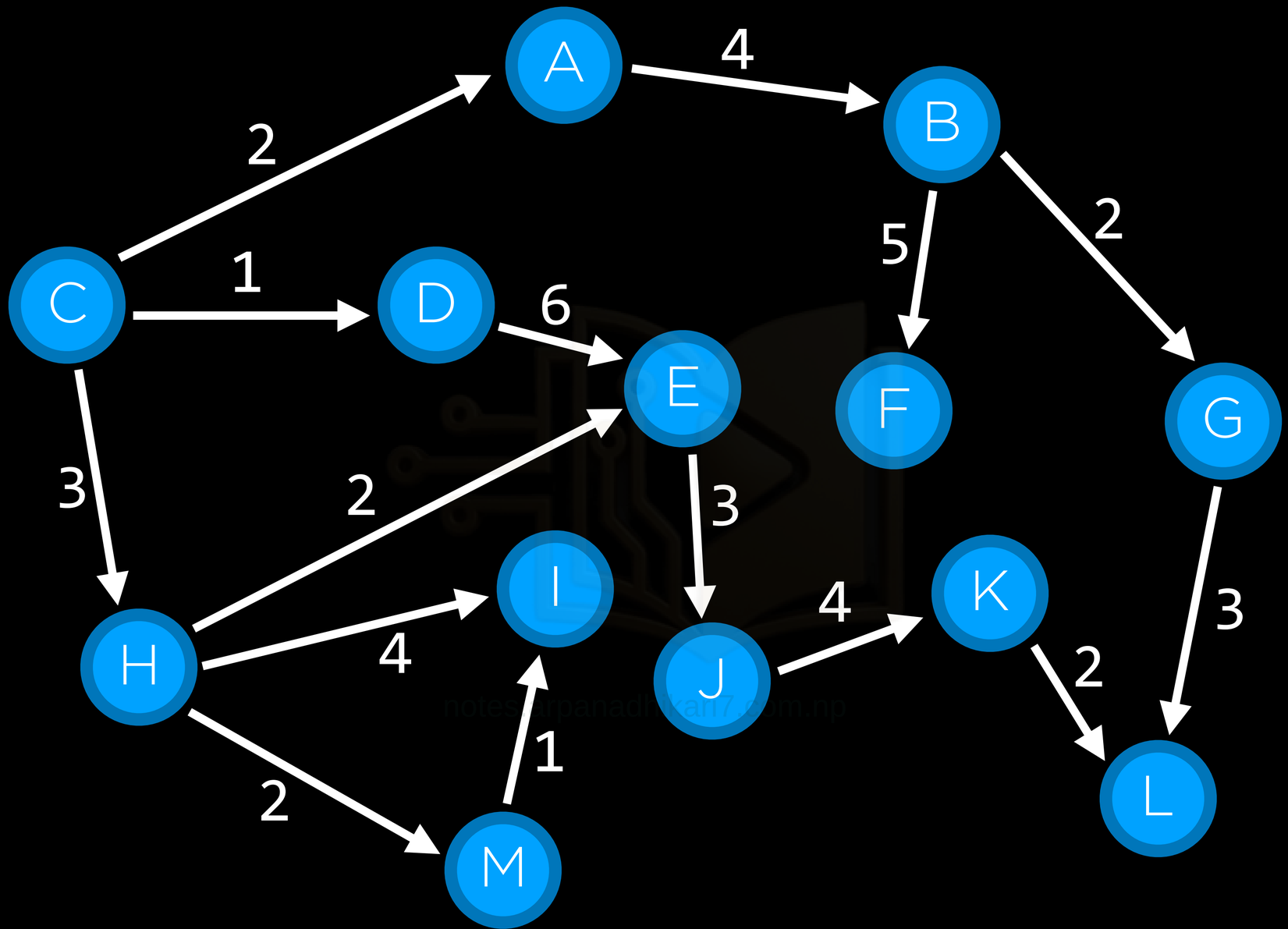
path cost

numerical cost associated with a given path

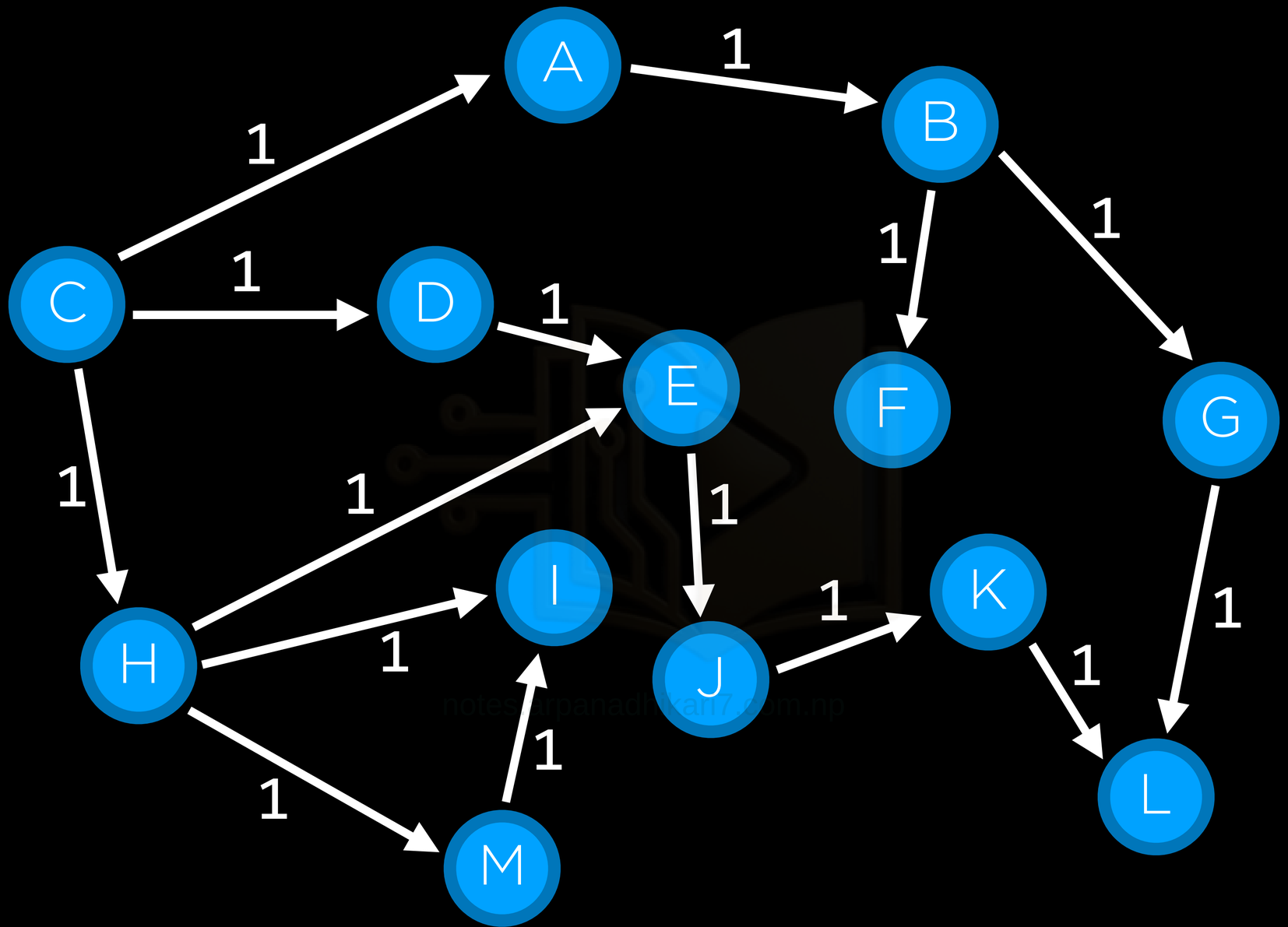
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Search Problems

- initial state
- actions
- transition model
- goal test
- path cost function

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solution

a sequence of actions that leads from the initial state to a goal state

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optimal solution

a solution that has the lowest path cost among all solutions

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node

a data structure that keeps track of

- a **state**
- a **parent** (node that generated this node)
- an **action** (action applied to parent to get node)
- a **path cost** (from initial state to node)

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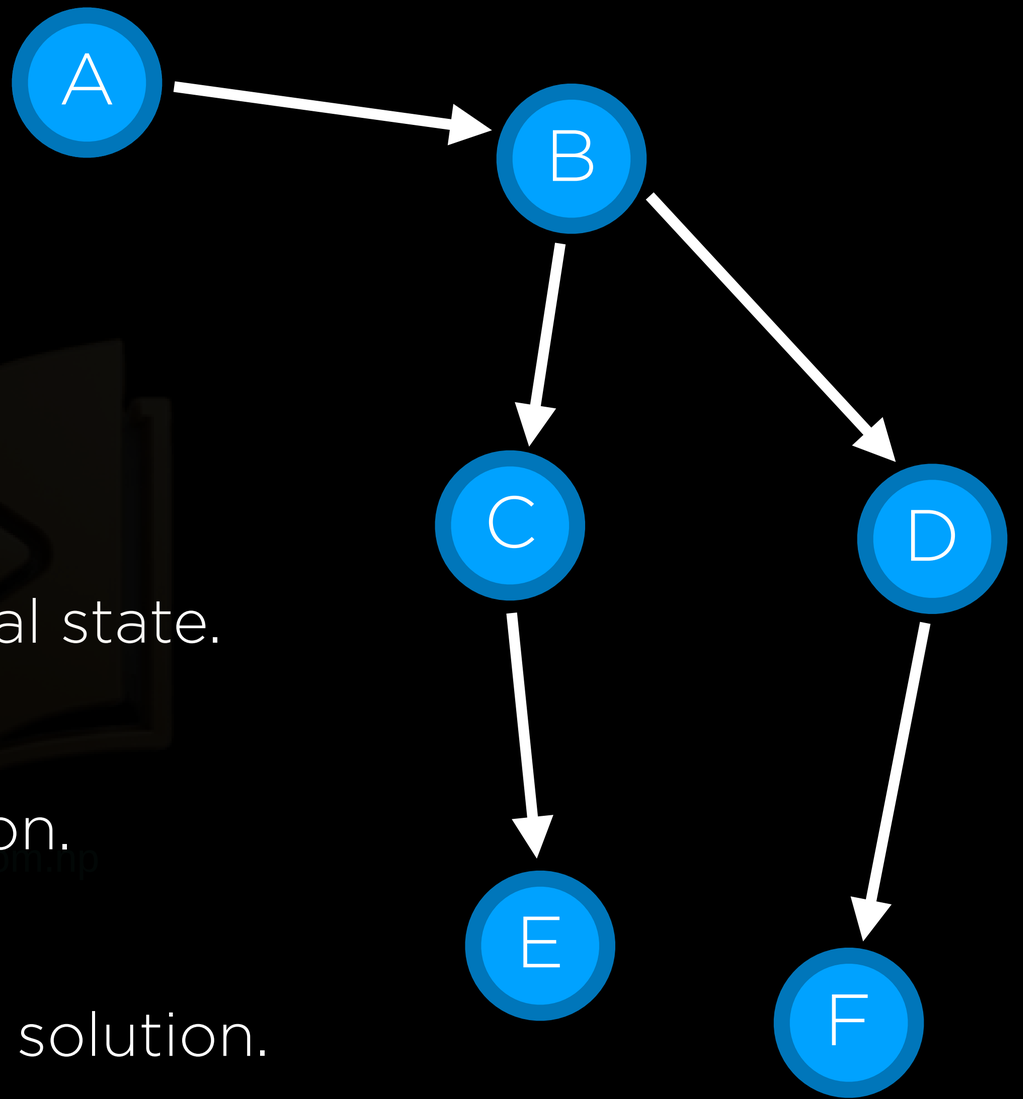
Approach

- Start with a **frontier** that contains the initial state.
- Repeat:
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  - Remove a node from the frontier.
  - If node contains goal state, return the solution.
  - **Expand** node, add resulting nodes to the frontier.

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Find a path from A to E.

Frontier



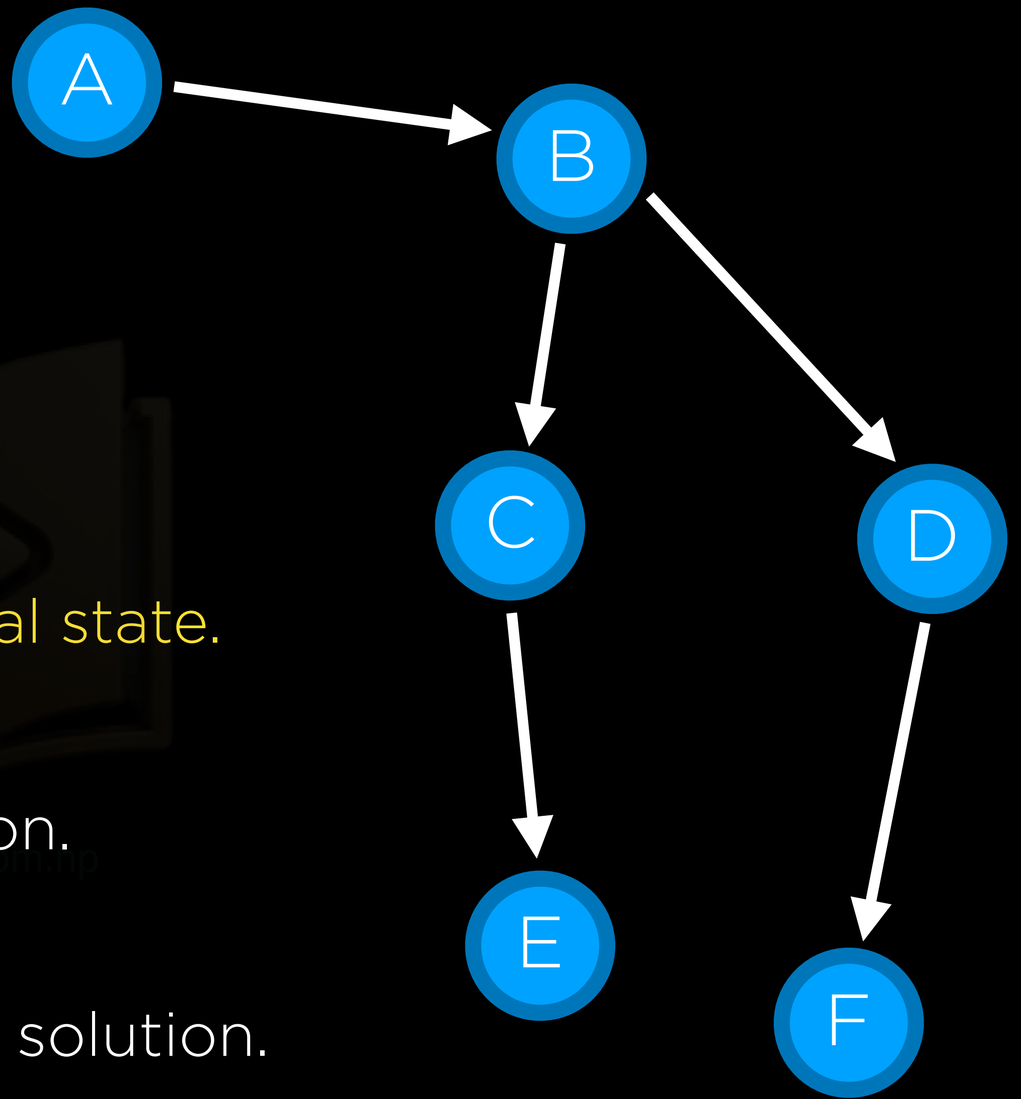
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Find a path from A to E.

Frontier

A

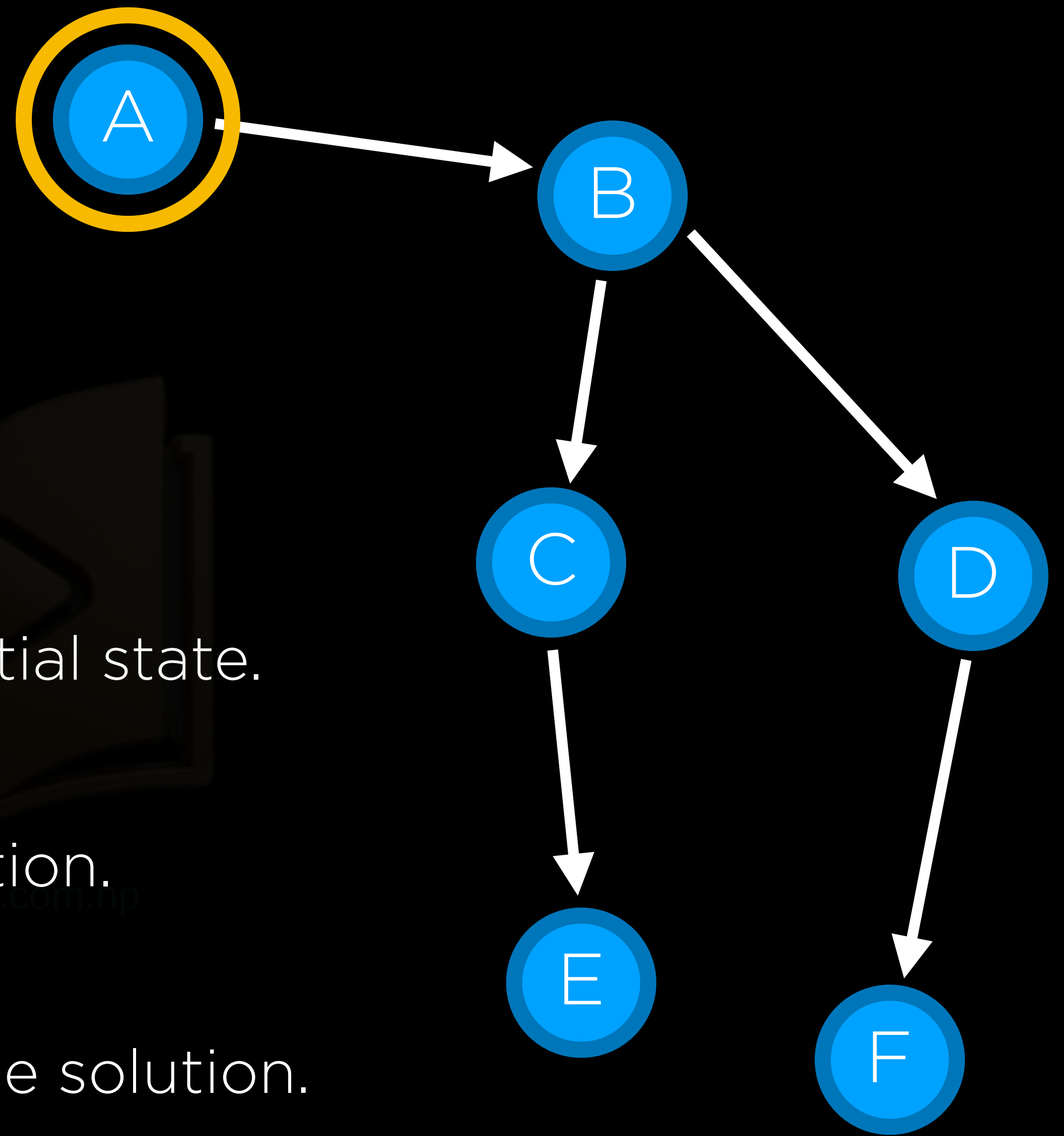


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Find a path from A to E.

Frontier



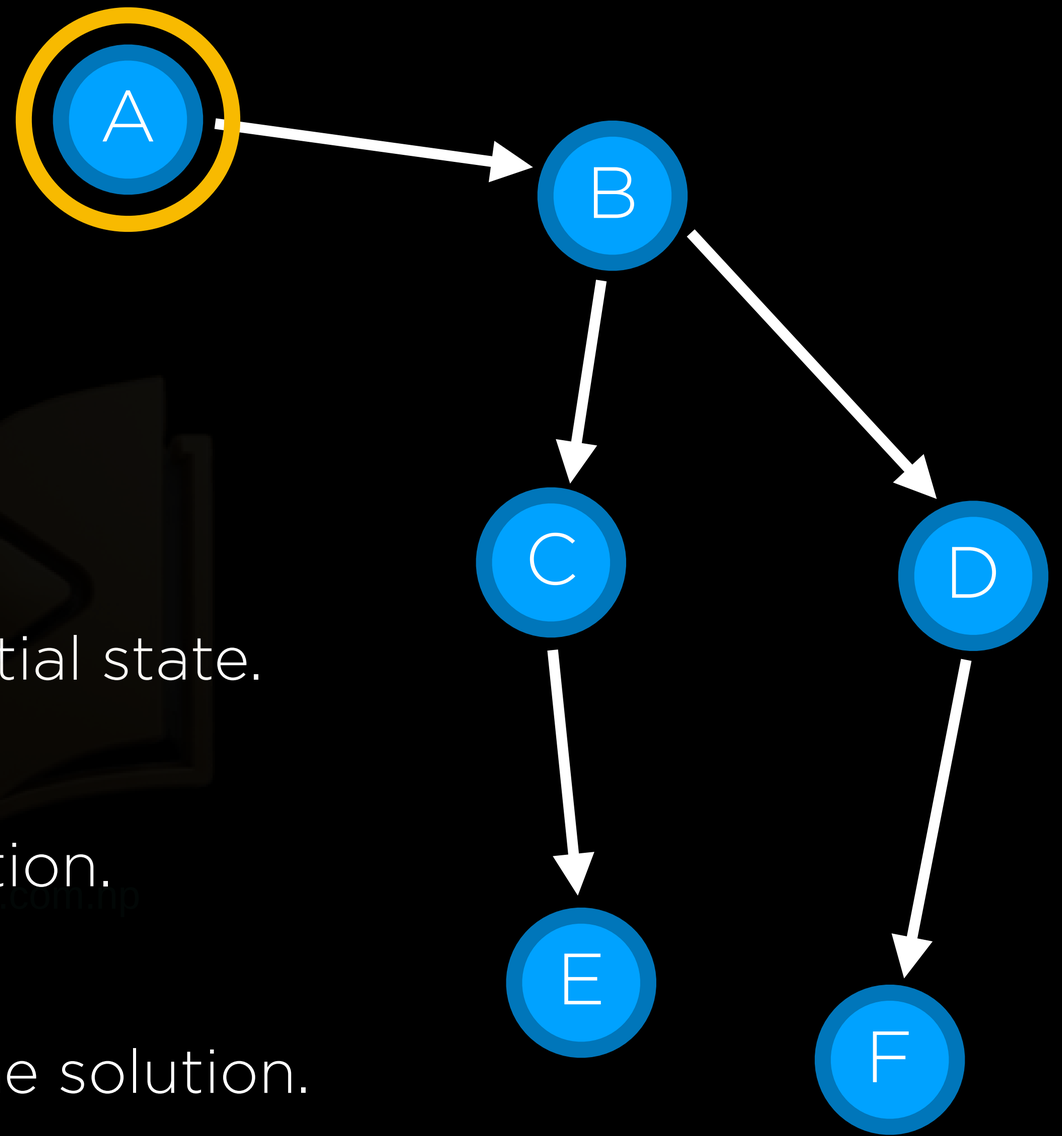
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Find a path from A to E.

Frontier

B

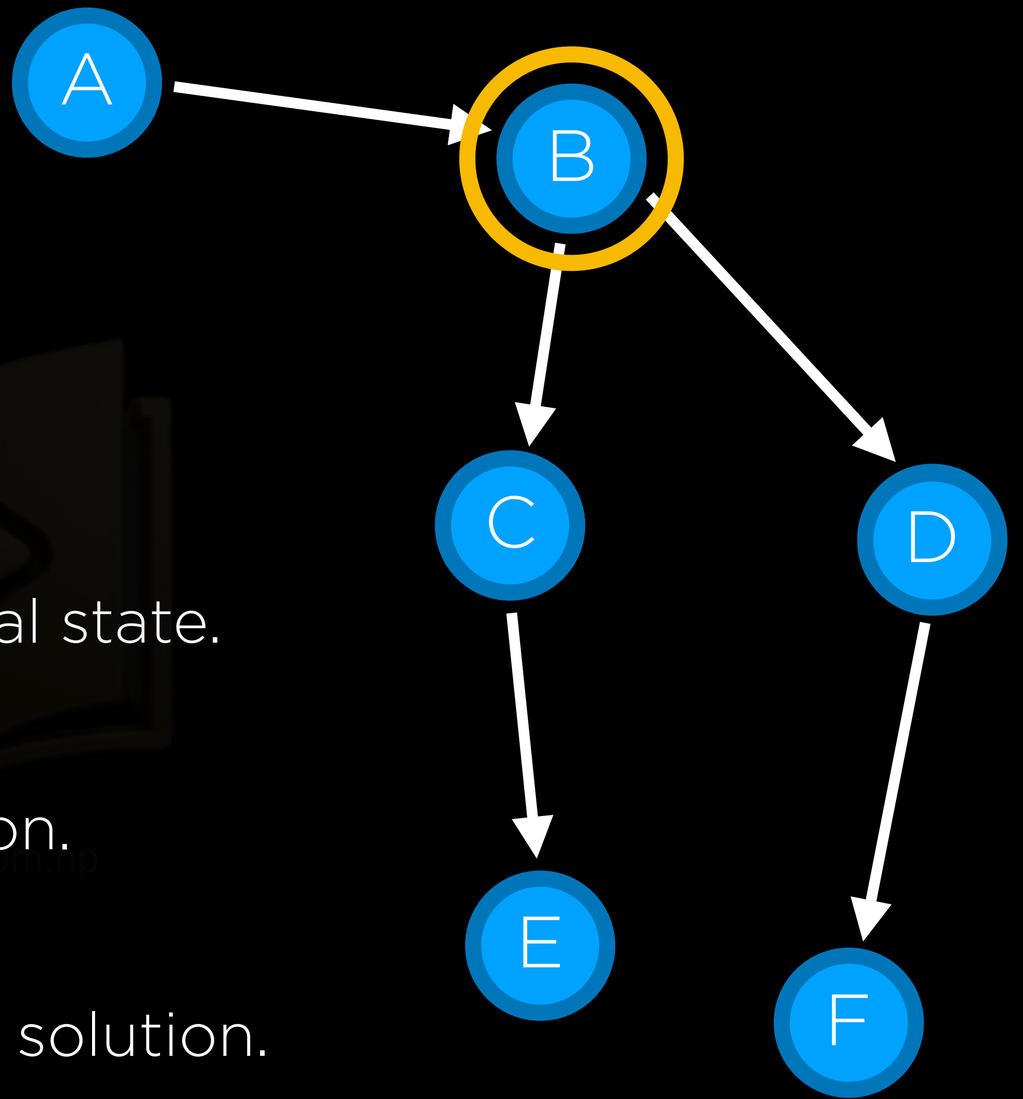


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Find a path from A to E.

Frontier



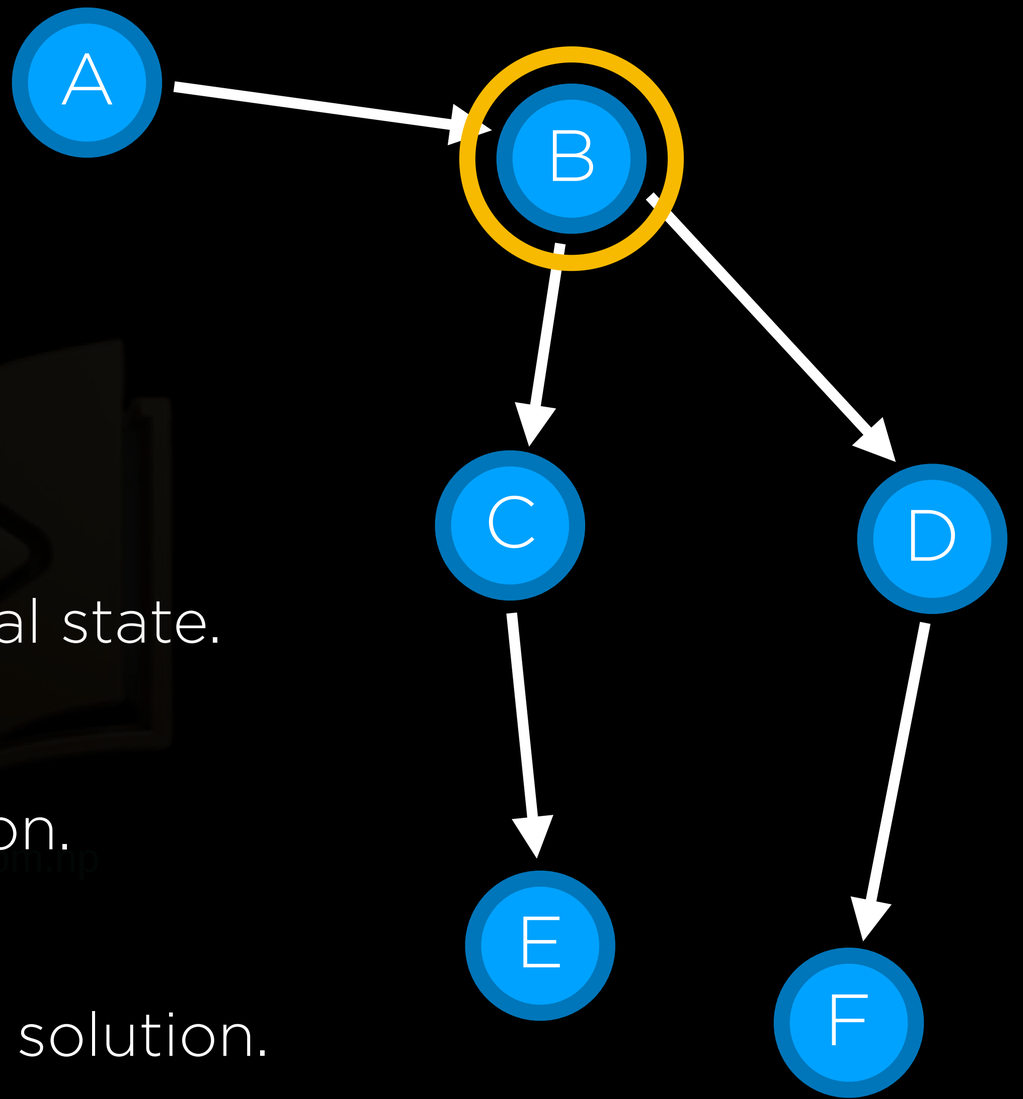
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Find a path from A to E.

Frontier

C D



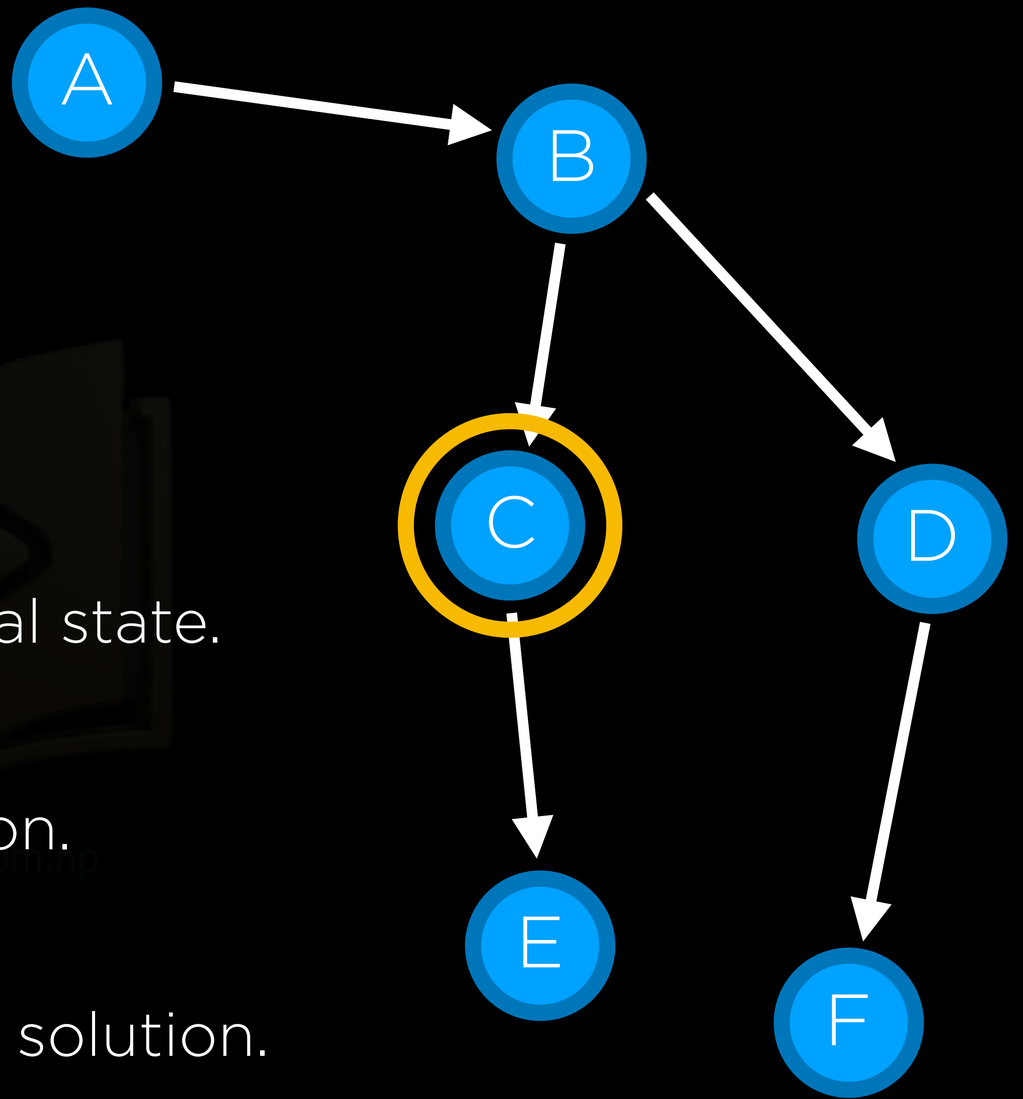
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Find a path from A to E.

Frontier

D



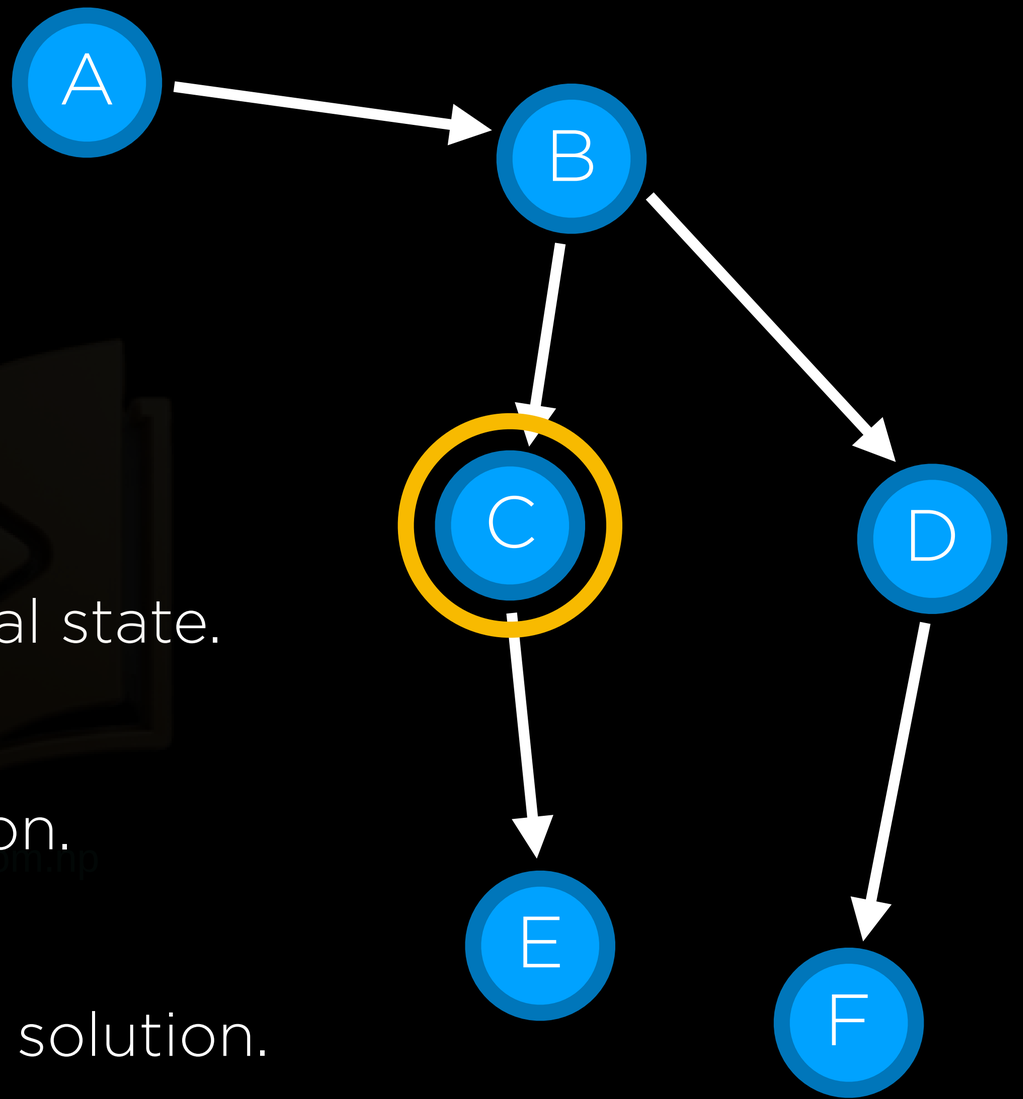
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Find a path from A to E.

Frontier

E D



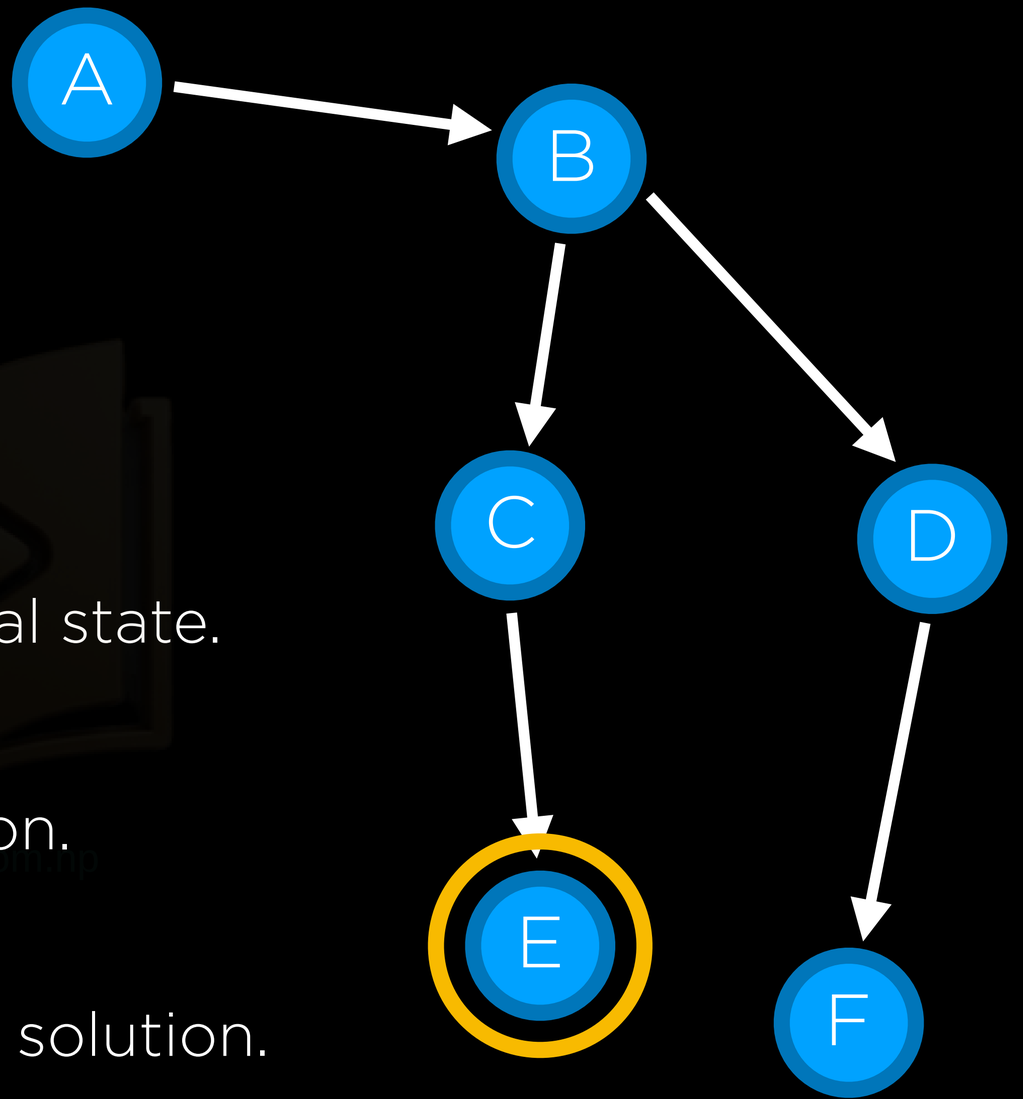
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Find a path from A to E.

Frontier

D



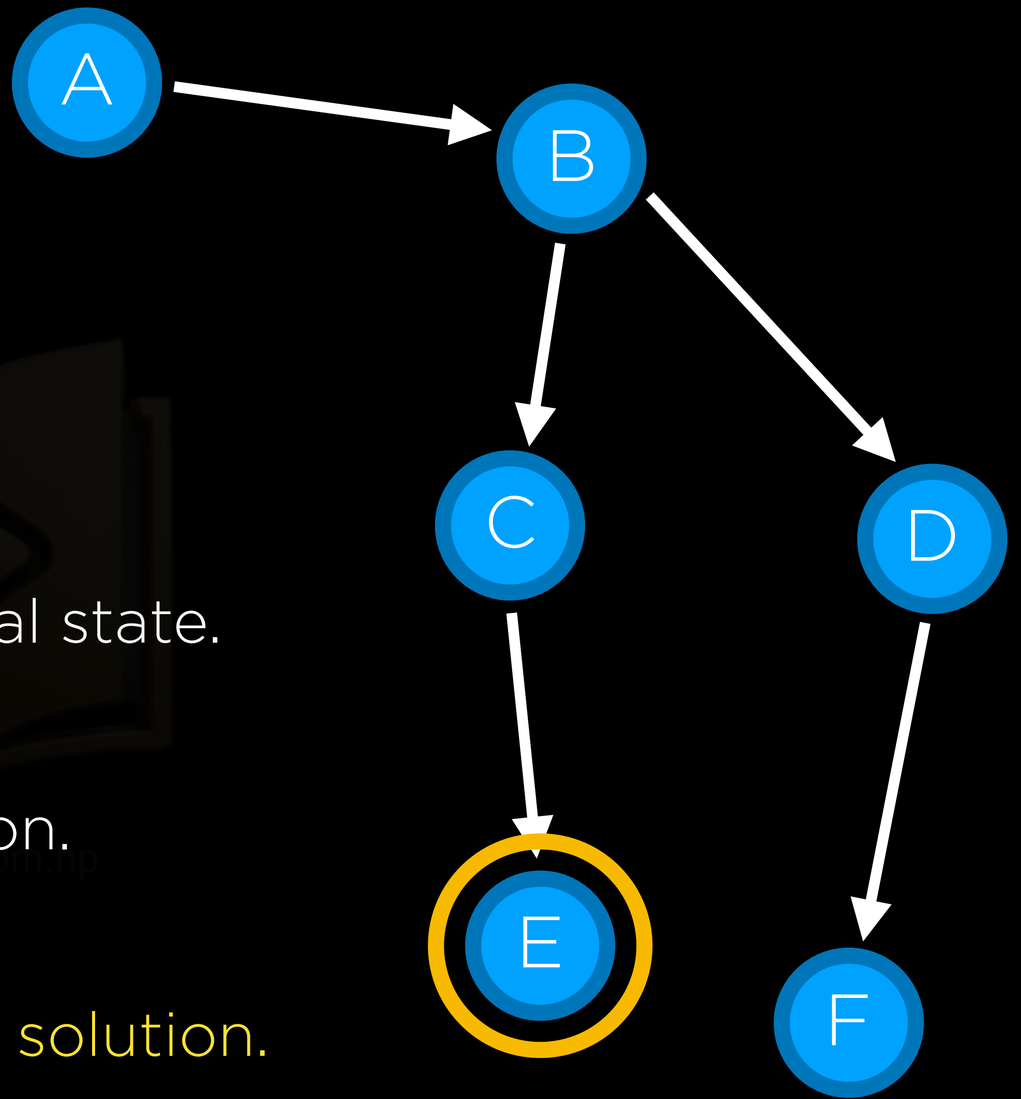
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Find a path from A to E.

Frontier

D



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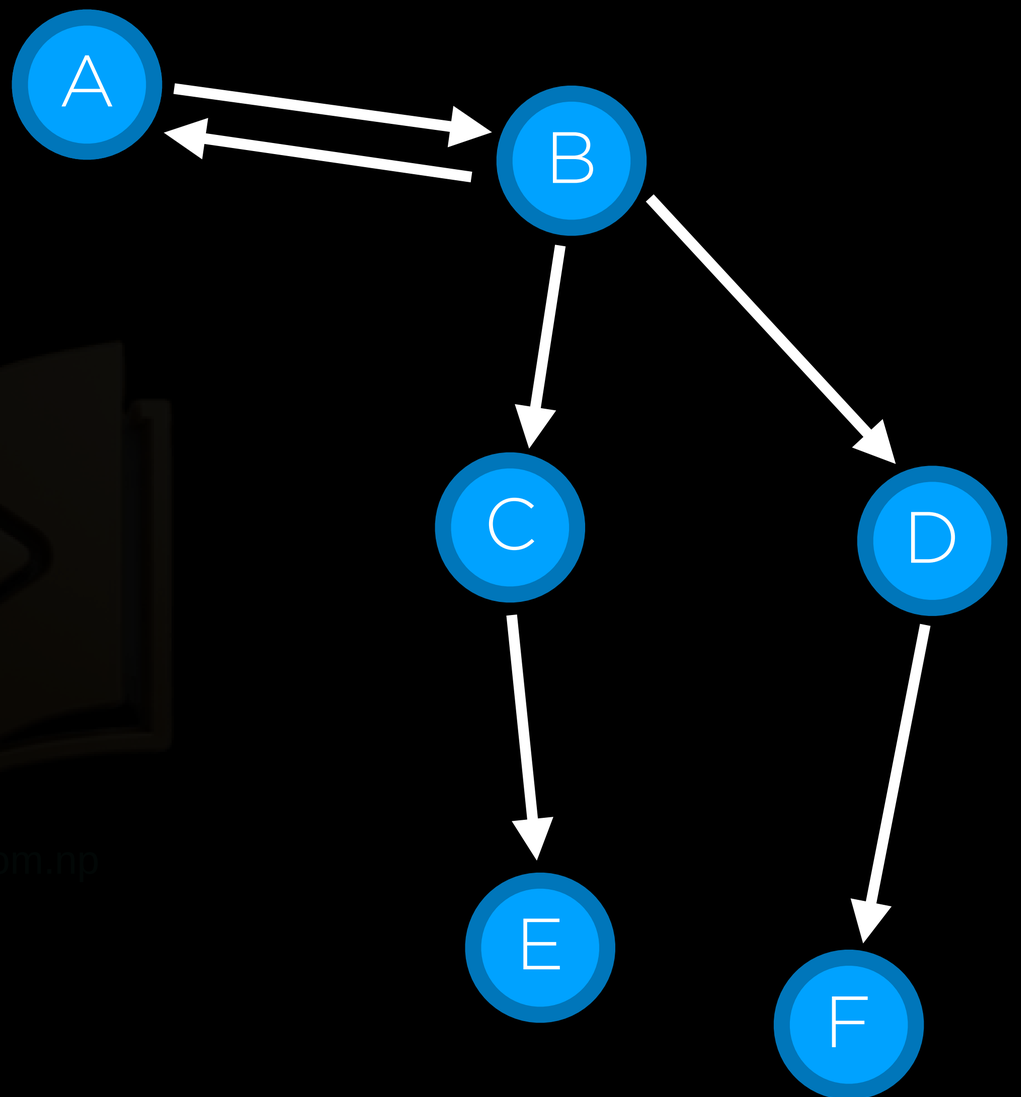
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What could go wrong?

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Find a path from A to E.

Frontier

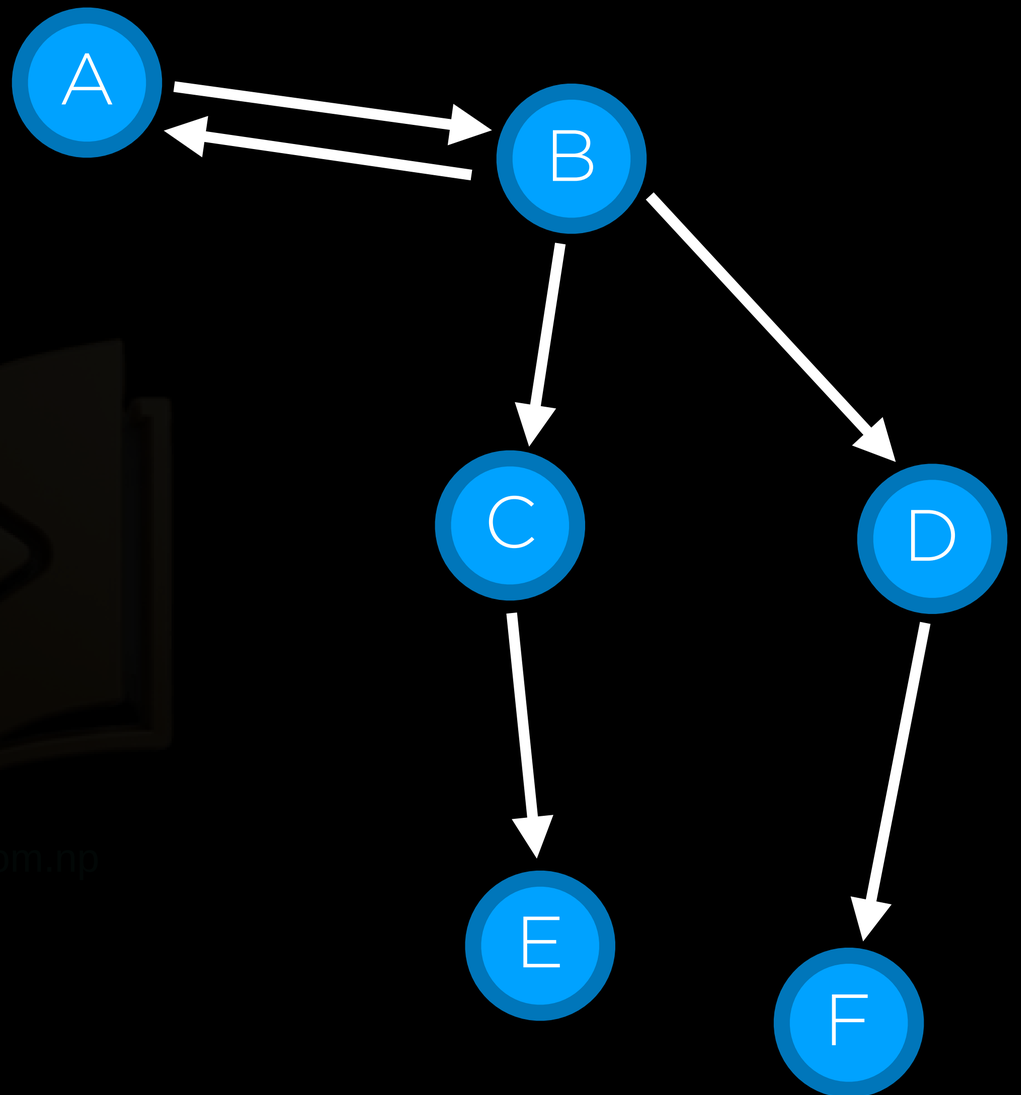


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Find a path from A to E.

Frontier

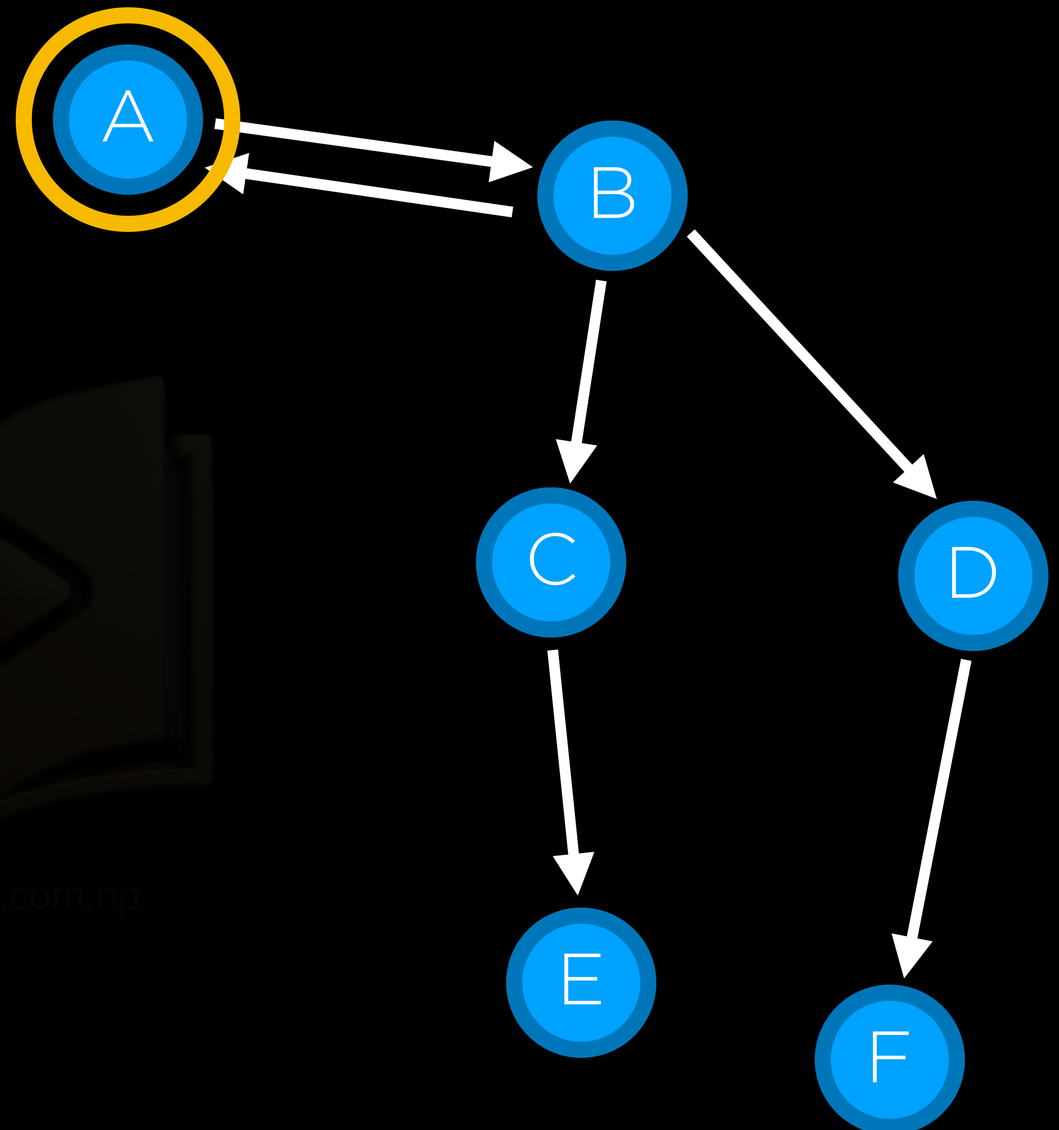
A



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Find a path from A to E.

Frontier

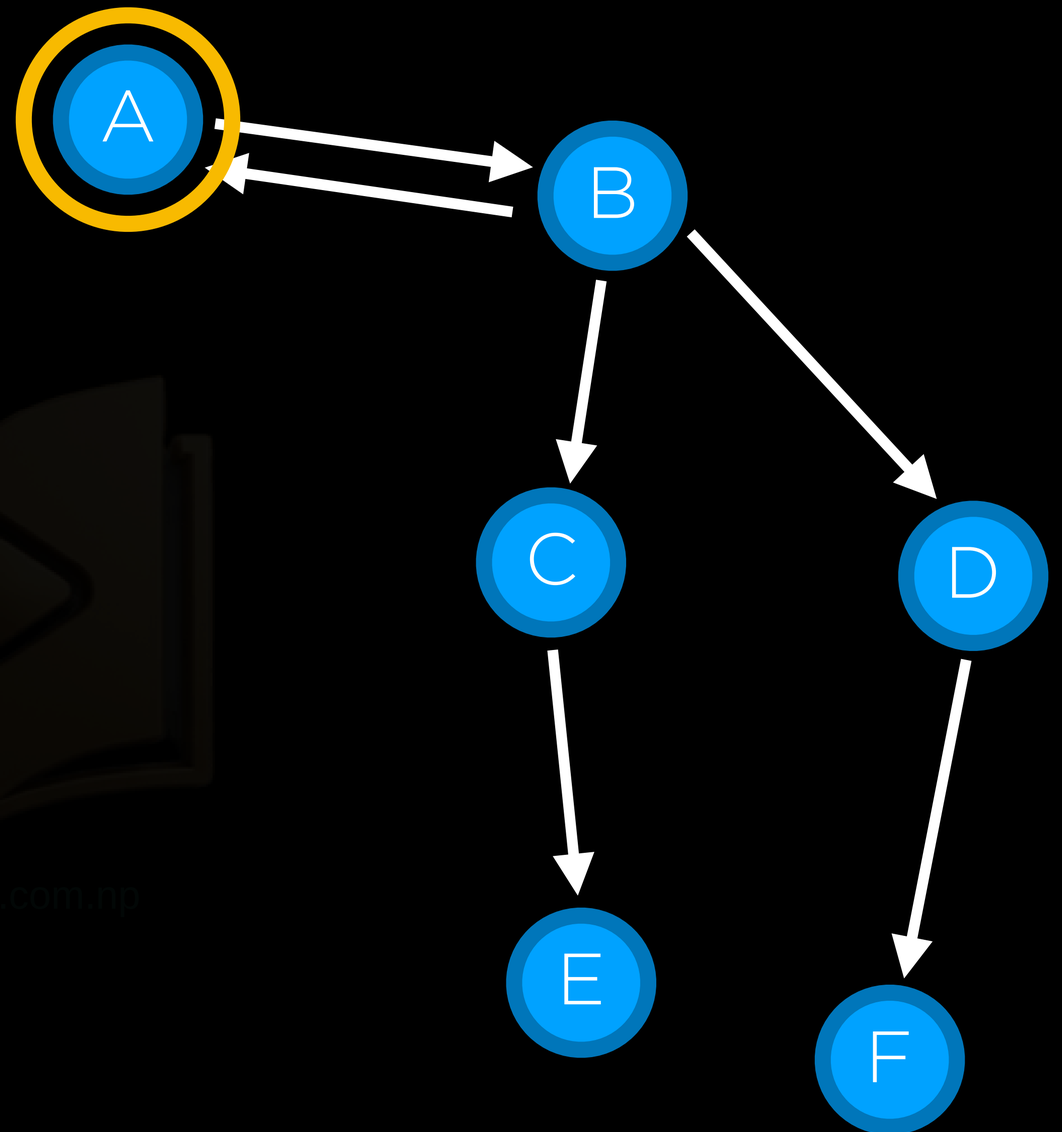


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Find a path from A to E.

Frontier

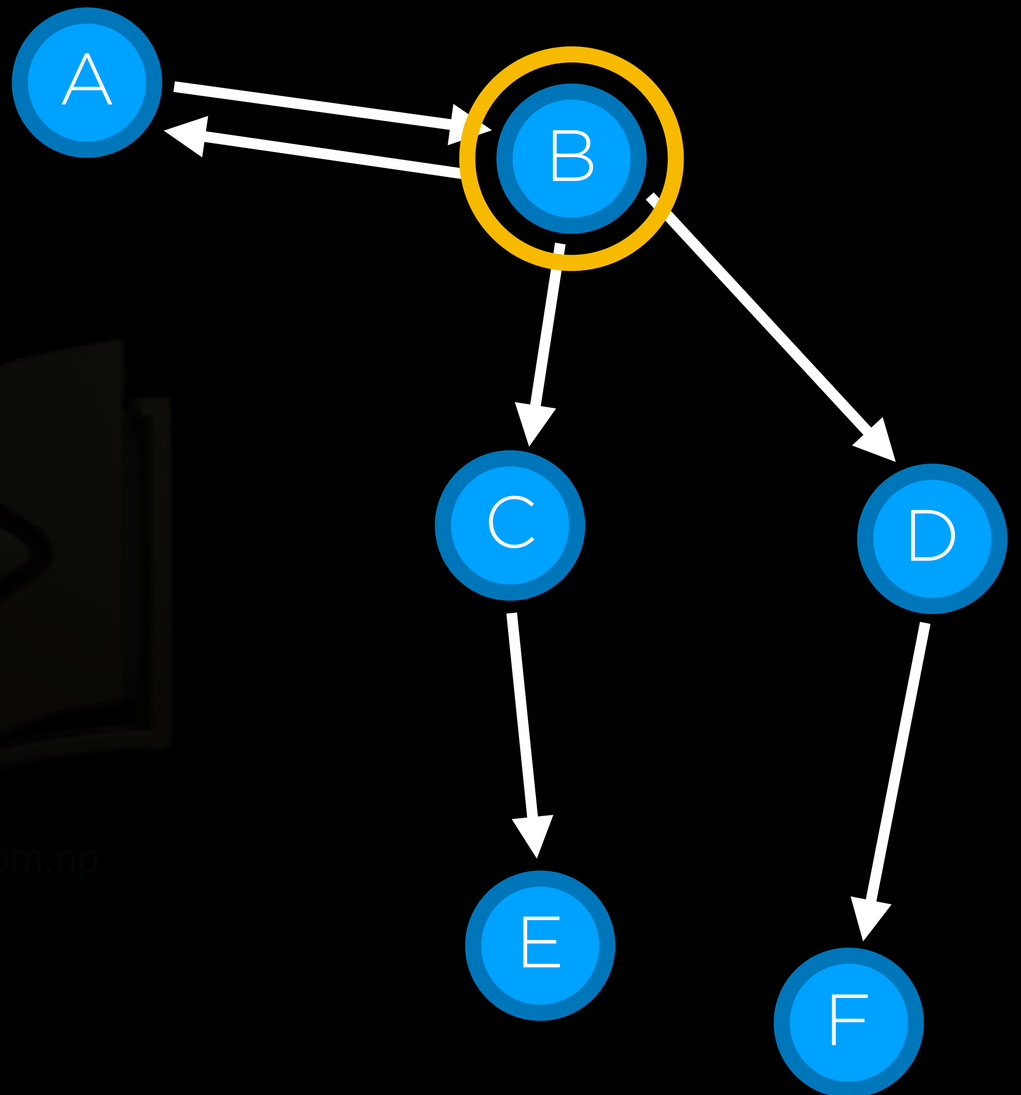
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Find a path from A to E.

Frontier

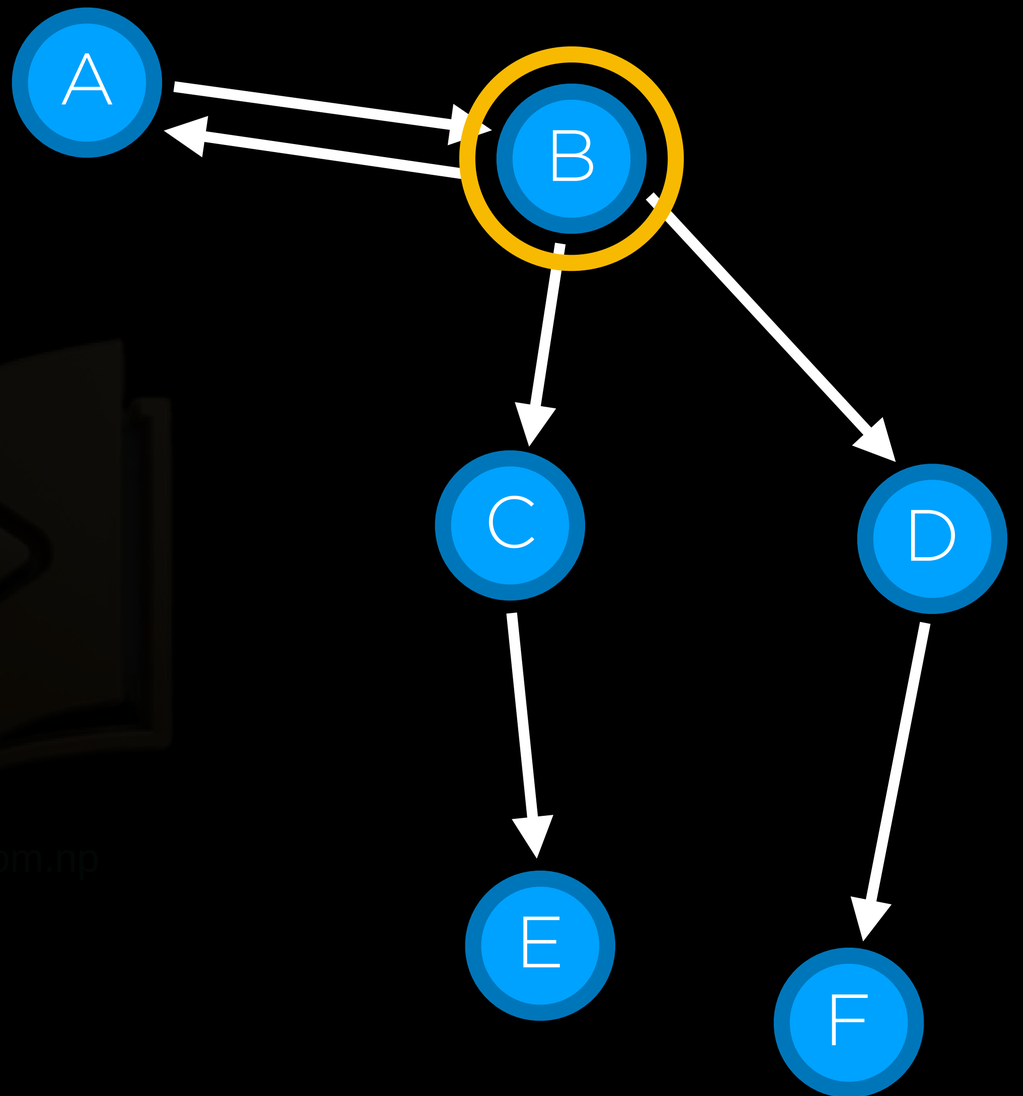


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Find a path from A to E.

Frontier

A C D

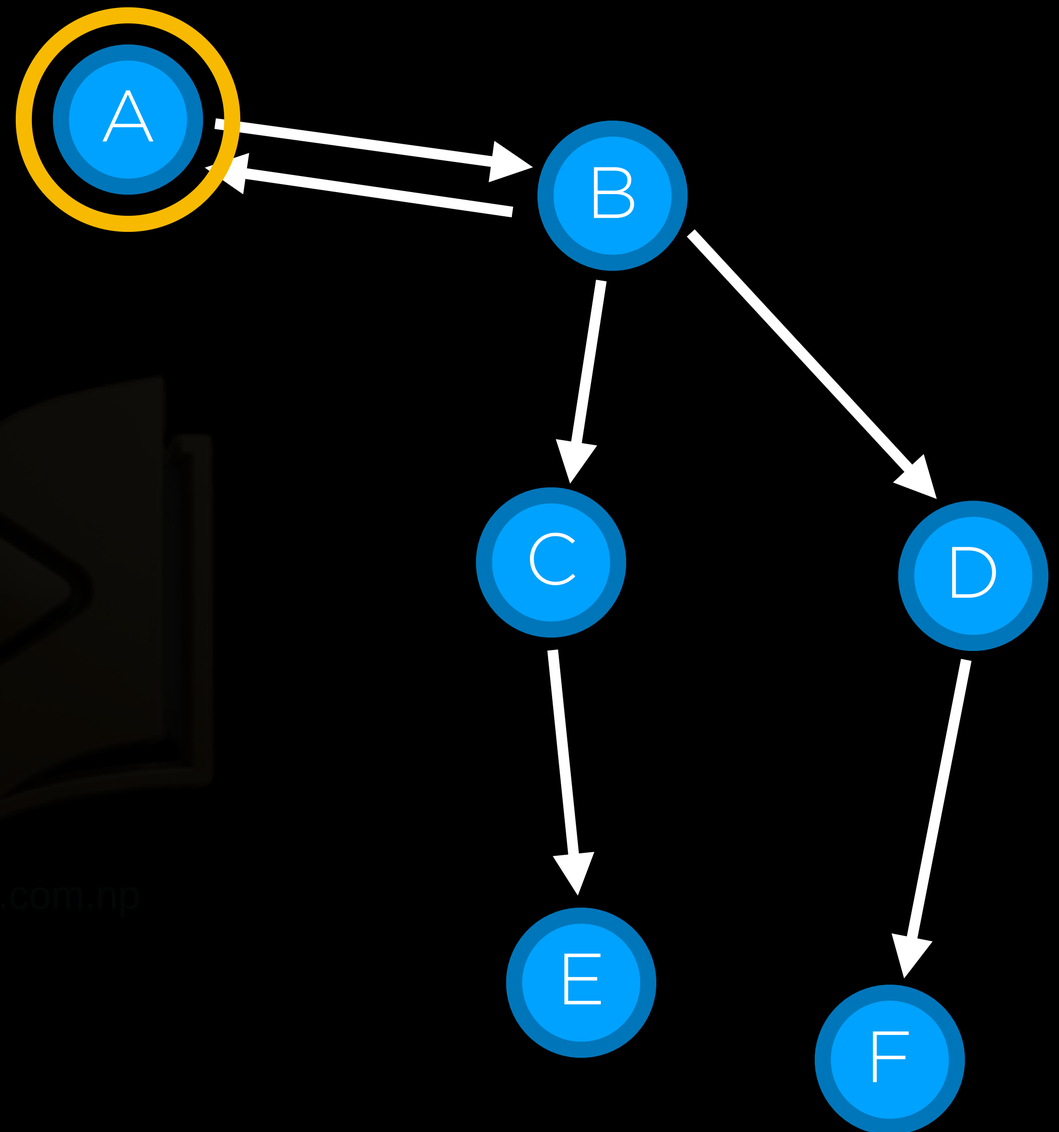


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Find a path from A to E.

Frontier

C D



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Revised Approach

- Start with a **frontier** that contains the initial state.
- Start with an empty **explored set**.
- Repeat:
    - If the frontier is empty, then no solution.
    - Remove a node from the frontier.
    - If node contains goal state, return the solution.
    - Add the node to the explored set.
    - **Expand** node, add resulting nodes to the frontier if they aren't already in the frontier or the explored set.

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Revised Approach

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stack

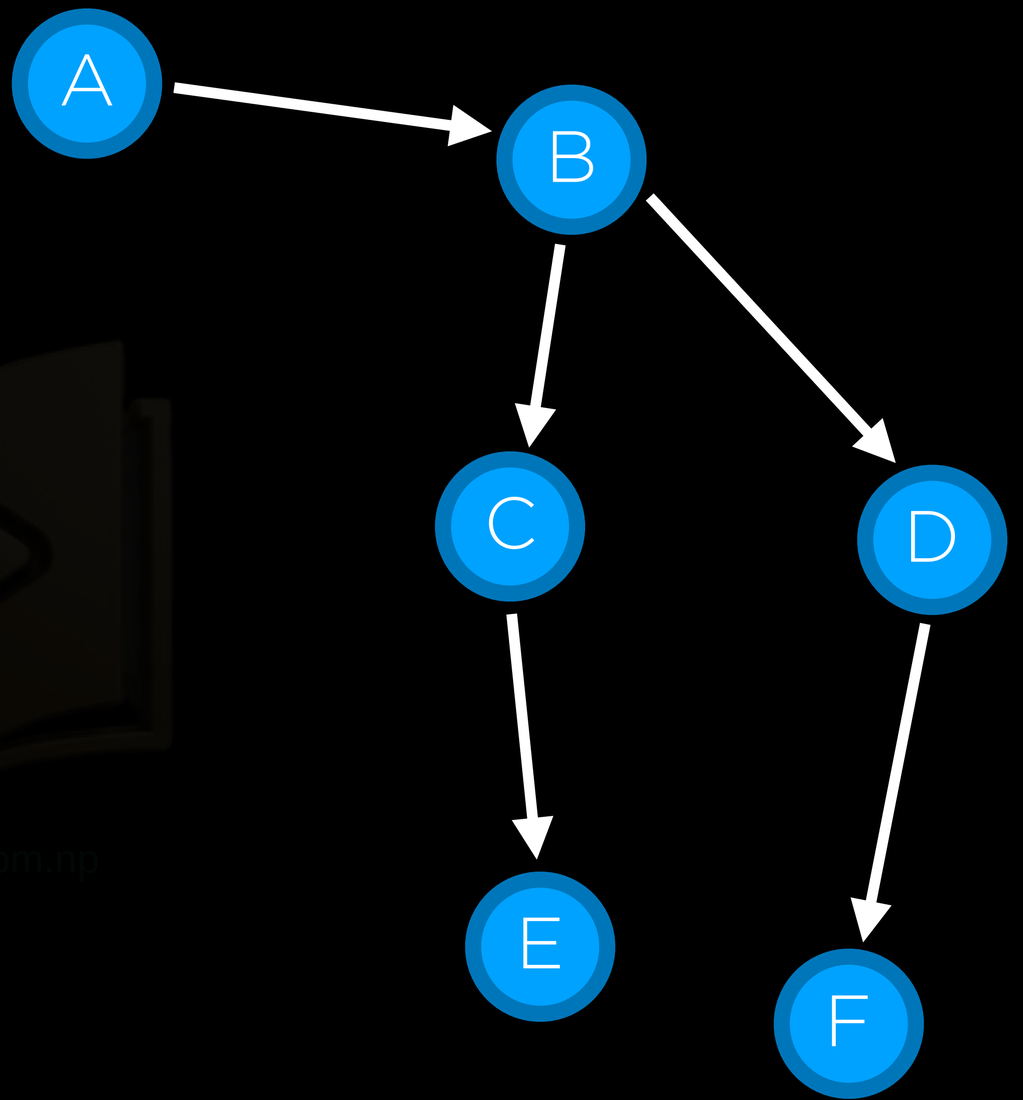
last-in first-out data type

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Find a path from A to E.

Frontier

Explored Set



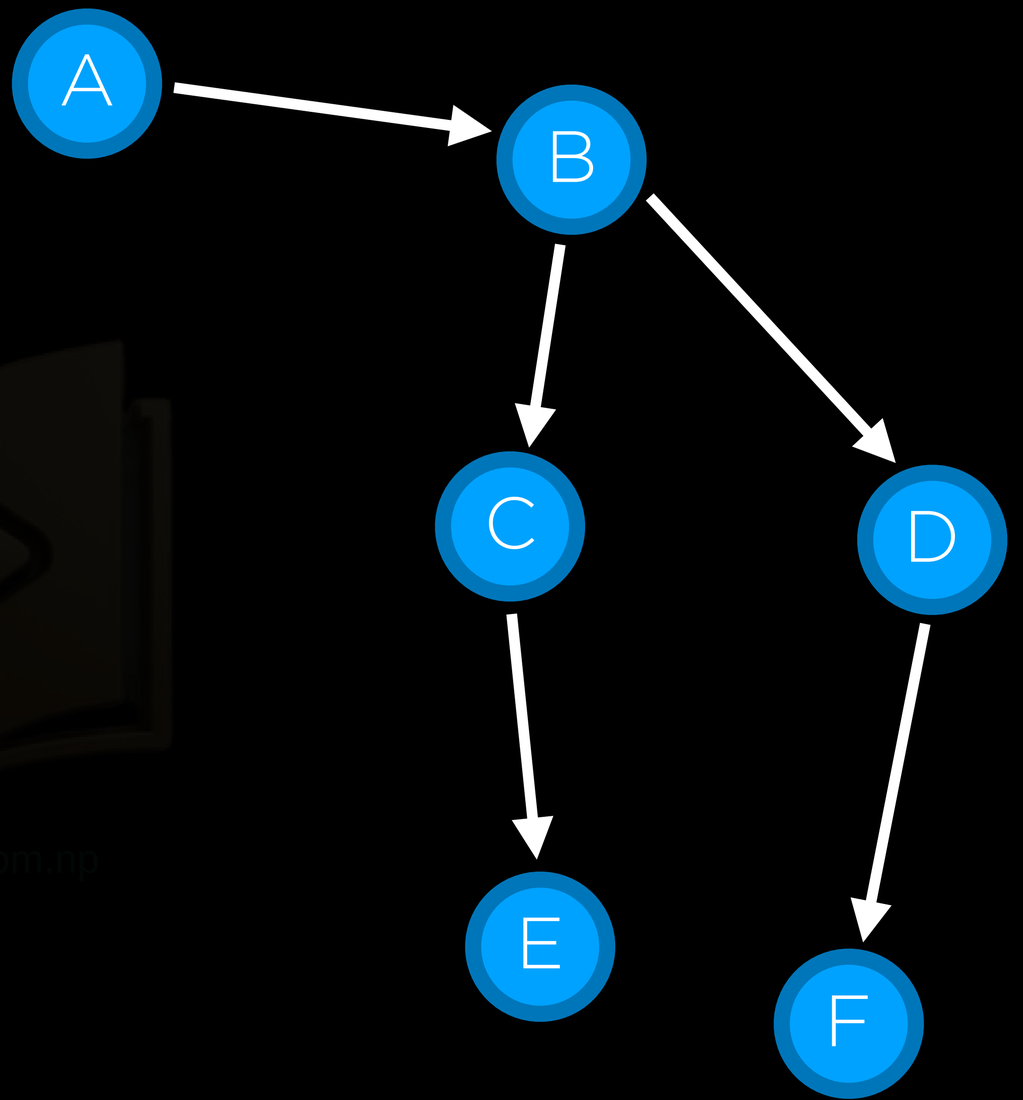
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Find a path from A to E.

Frontier

A

Explored Set



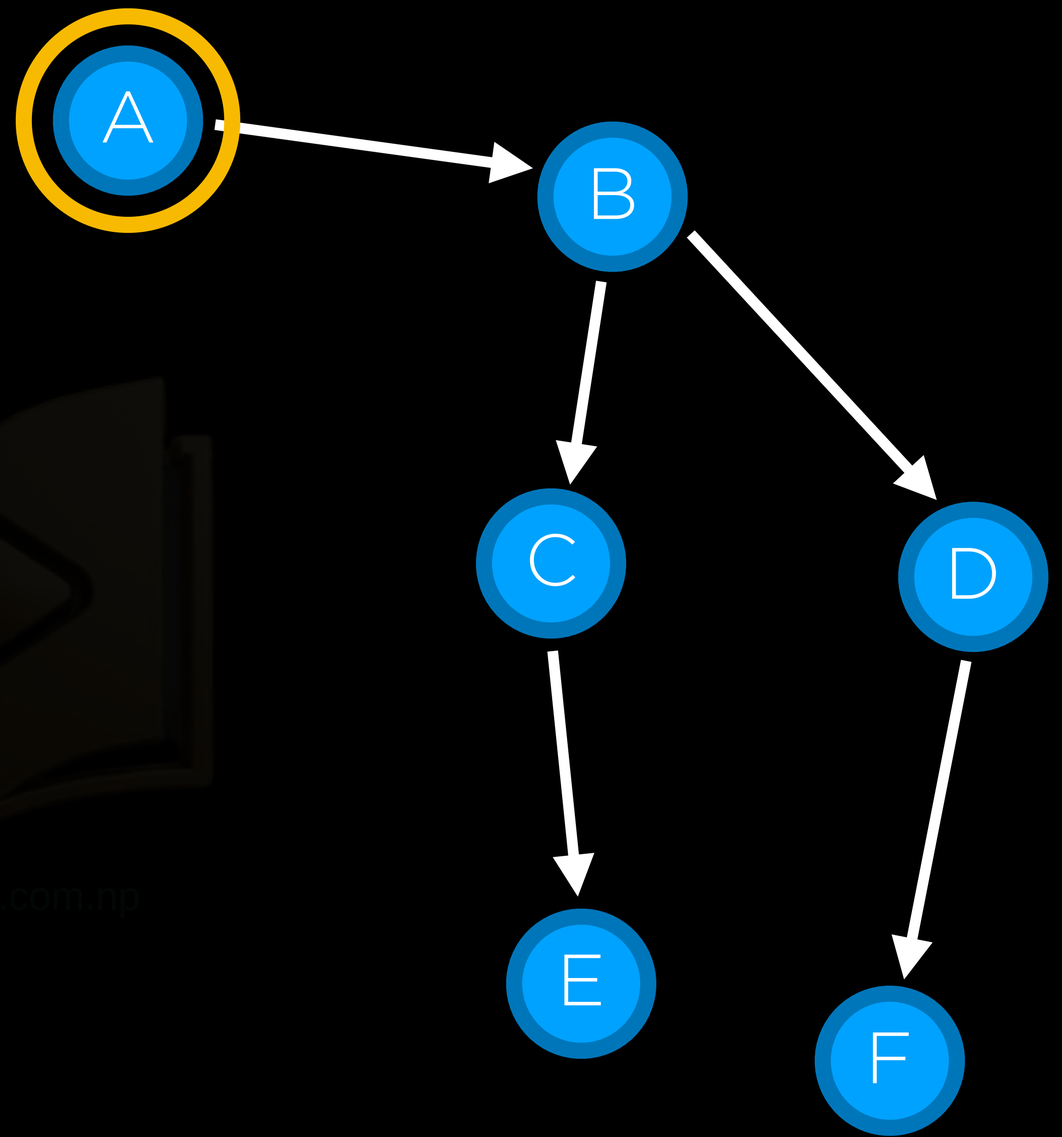
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Find a path from A to E.

Frontier

Explored Set

A



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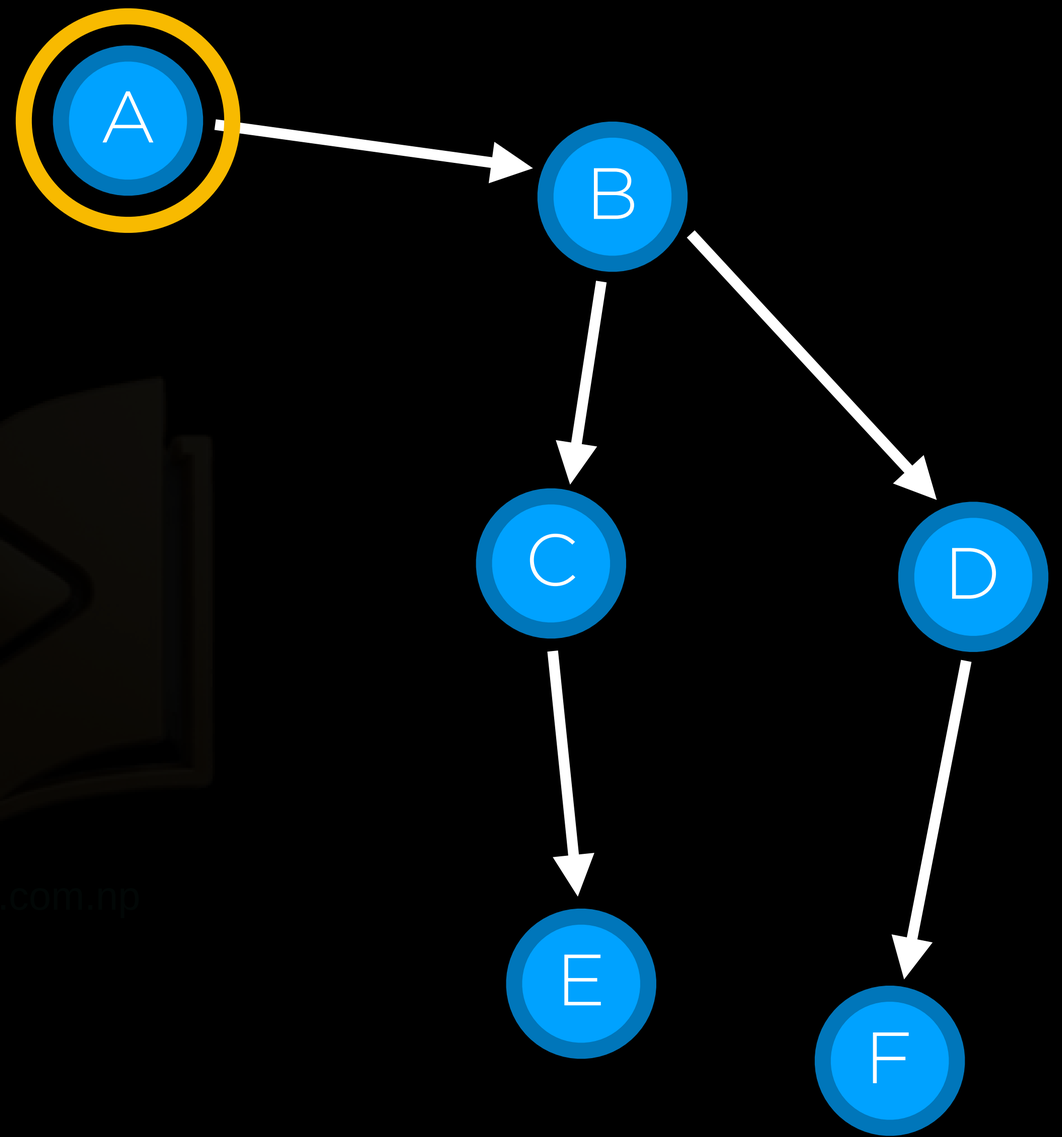
Find a path from A to E.

Frontier

B

Explored Set

A



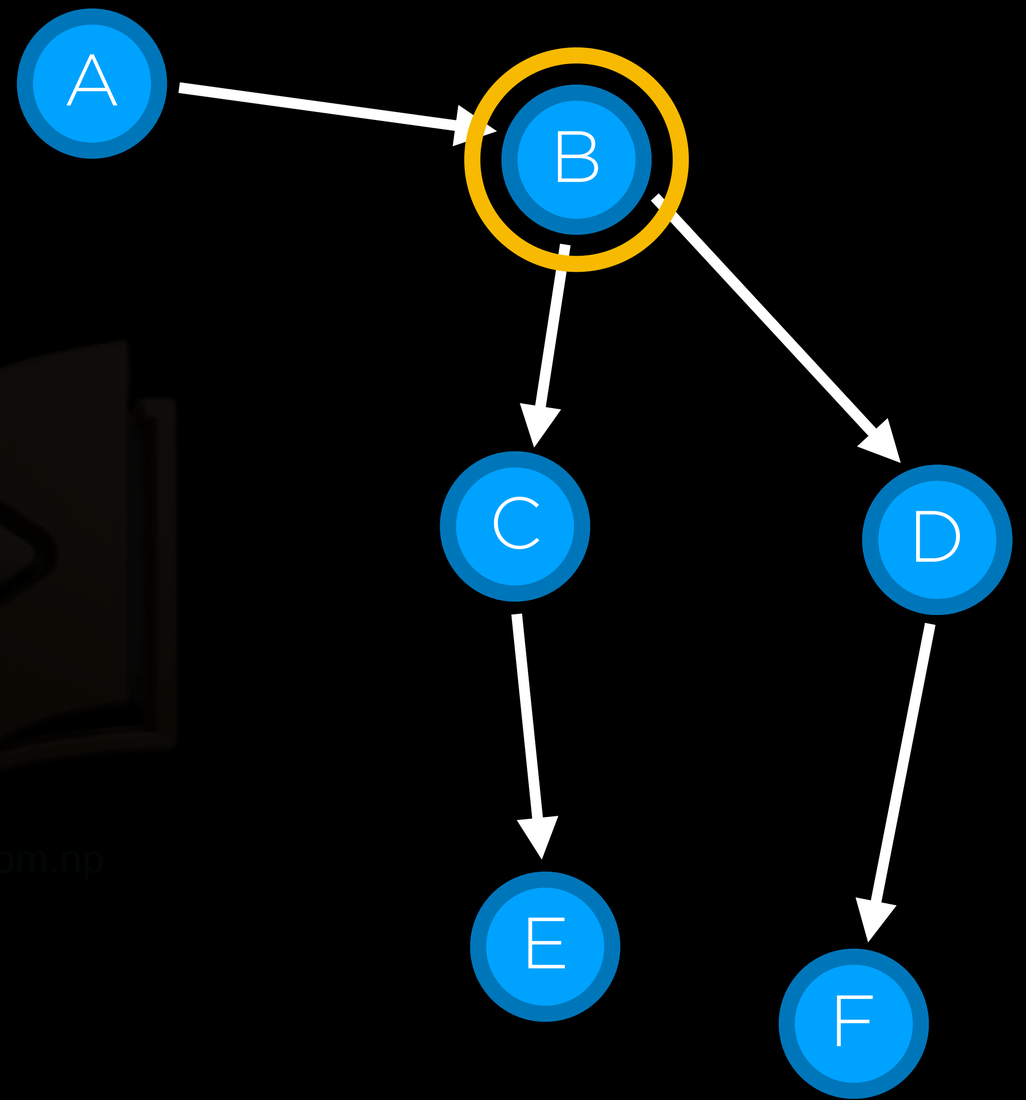
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Find a path from A to E.

Frontier

Explored Set

A B



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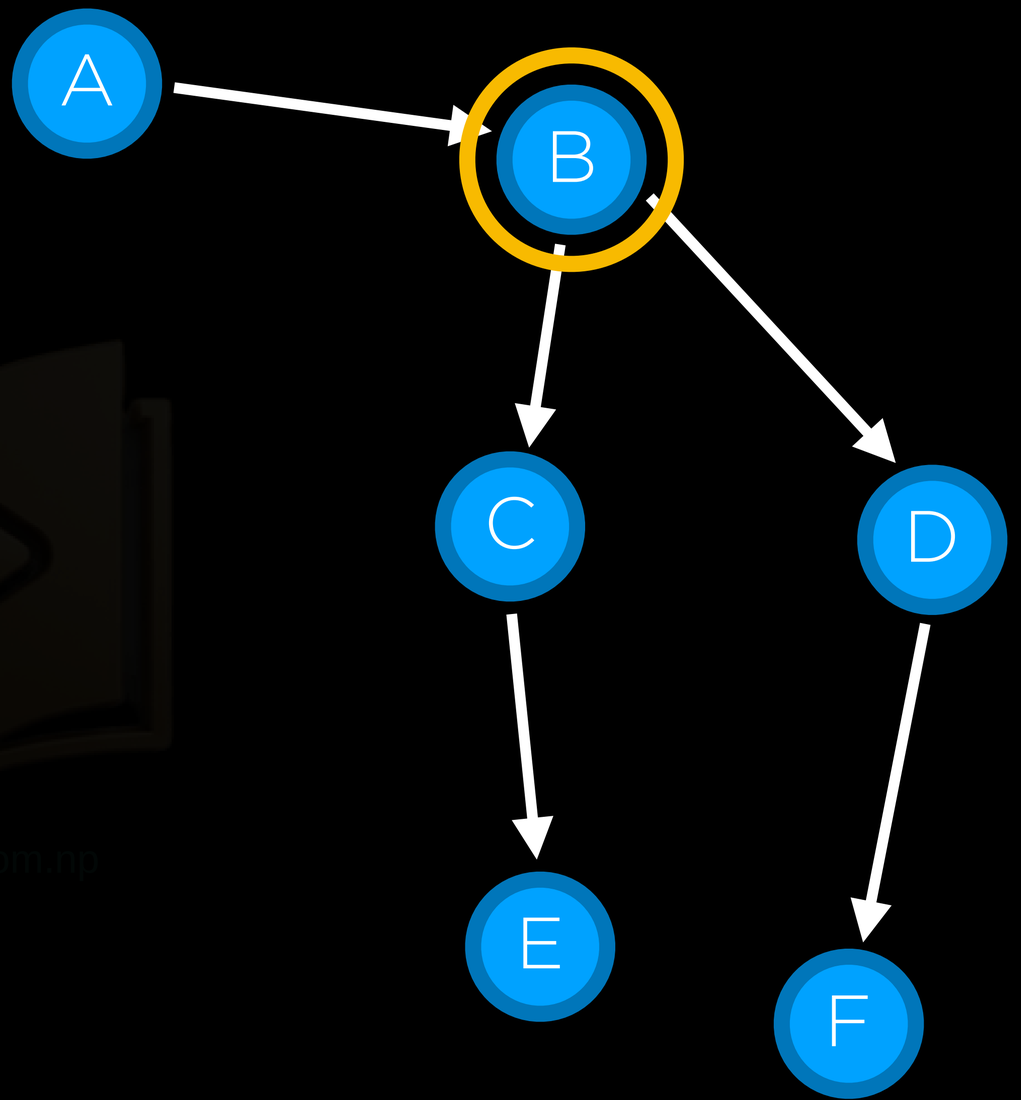
Find a path from A to E.

Frontier

C D

Explored Set

A B



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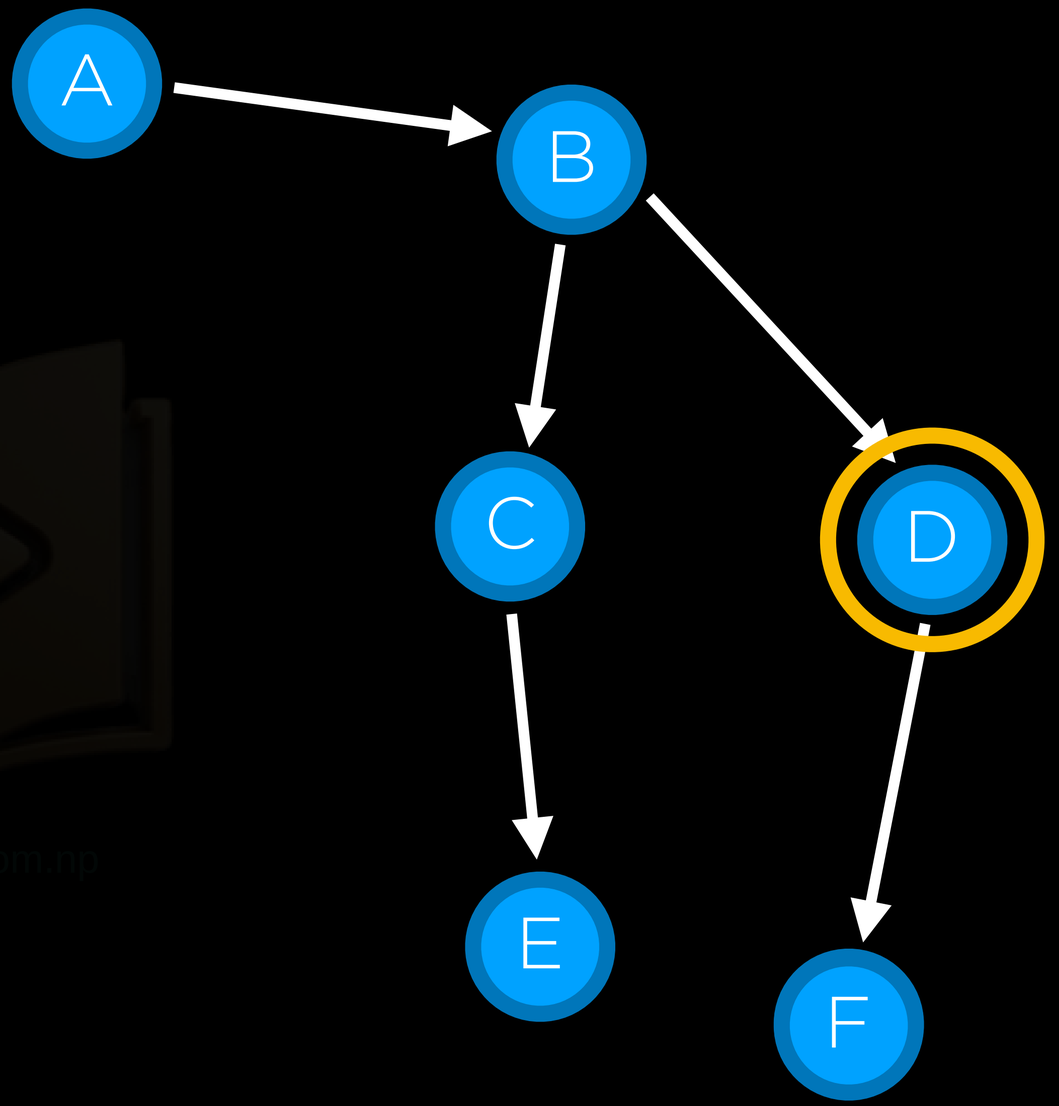
Find a path from A to E.

Frontier

C

Explored Set

A B D



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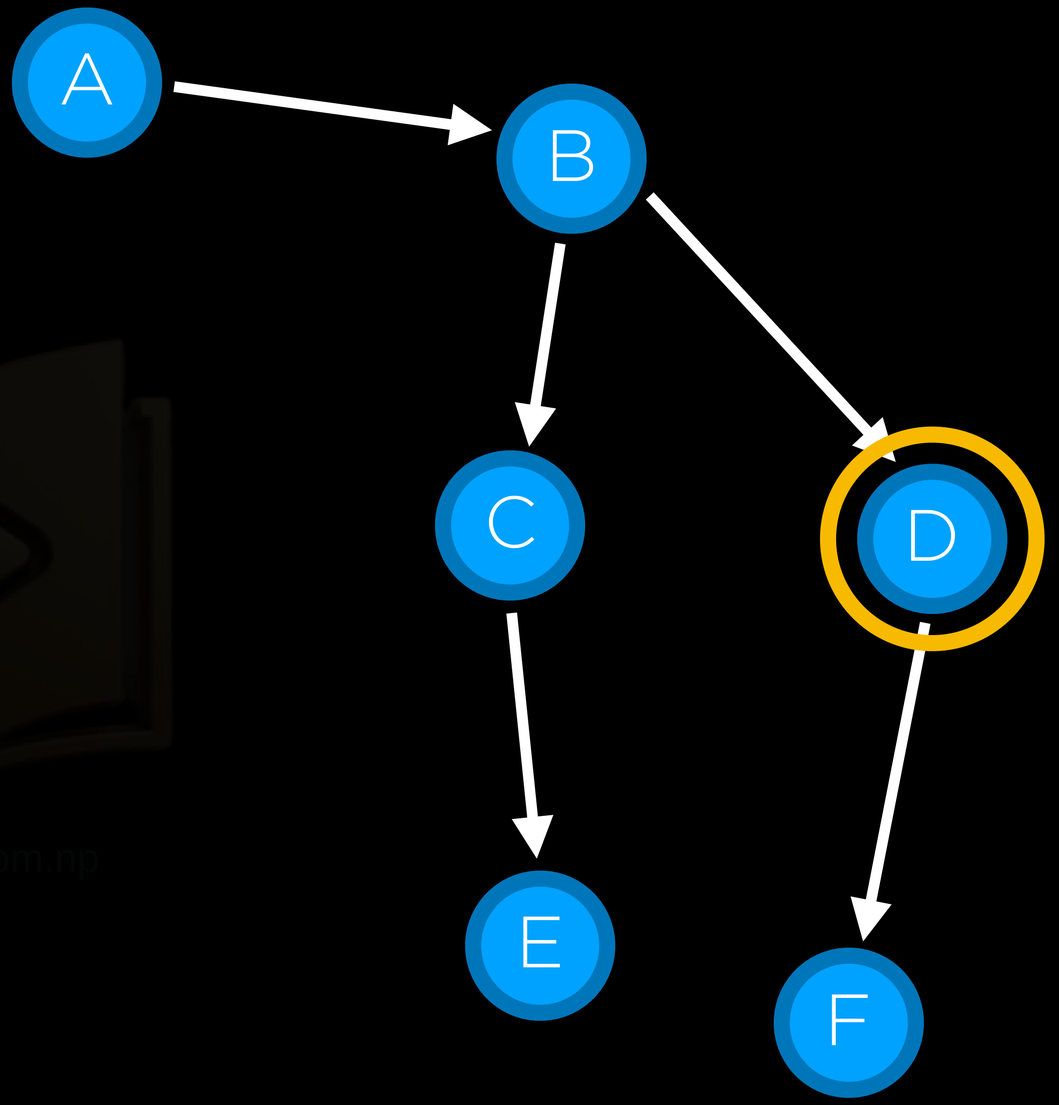
Find a path from A to E.

Frontier

C F

Explored Set

A B D



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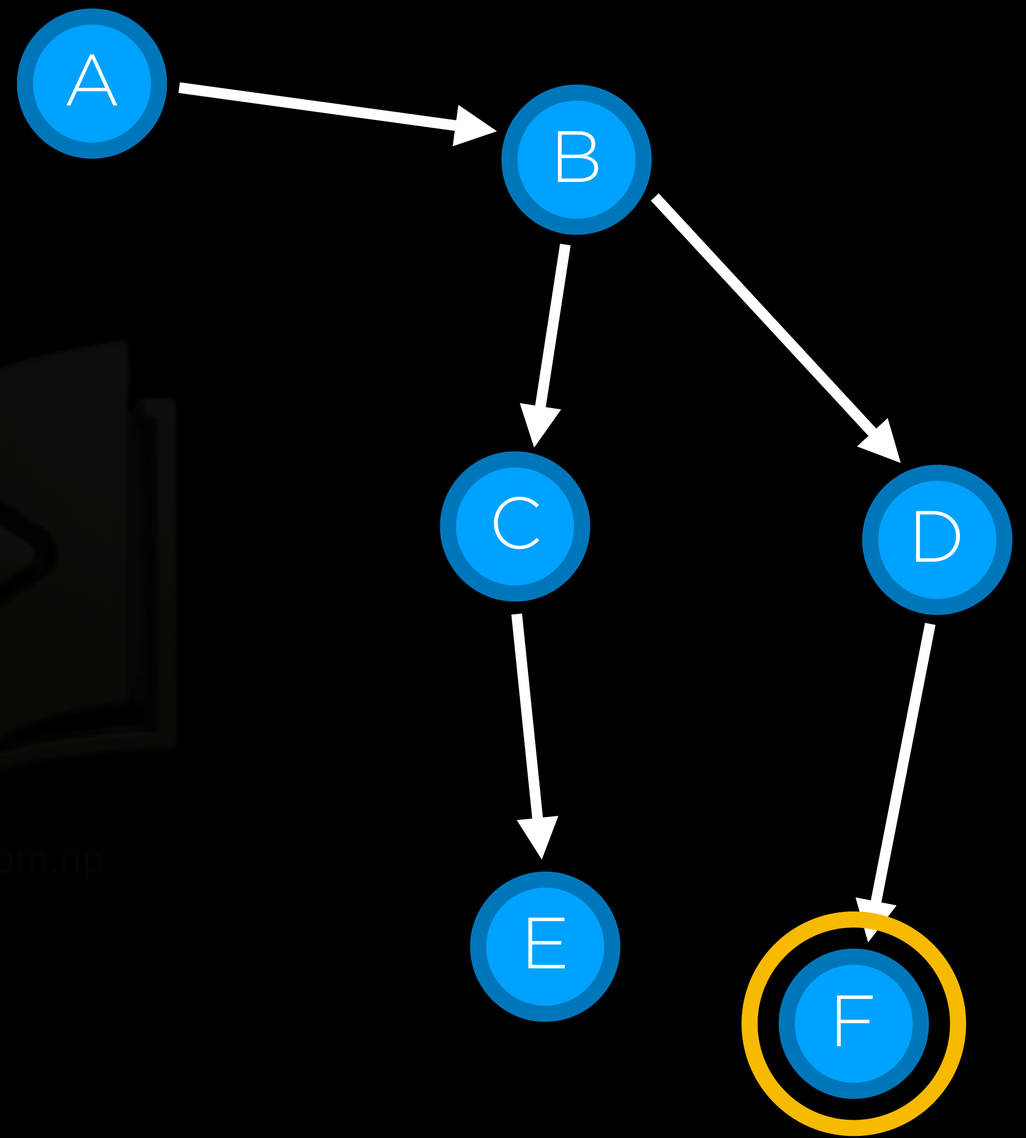
Find a path from A to E.

Frontier

C

Explored Set

A B D F



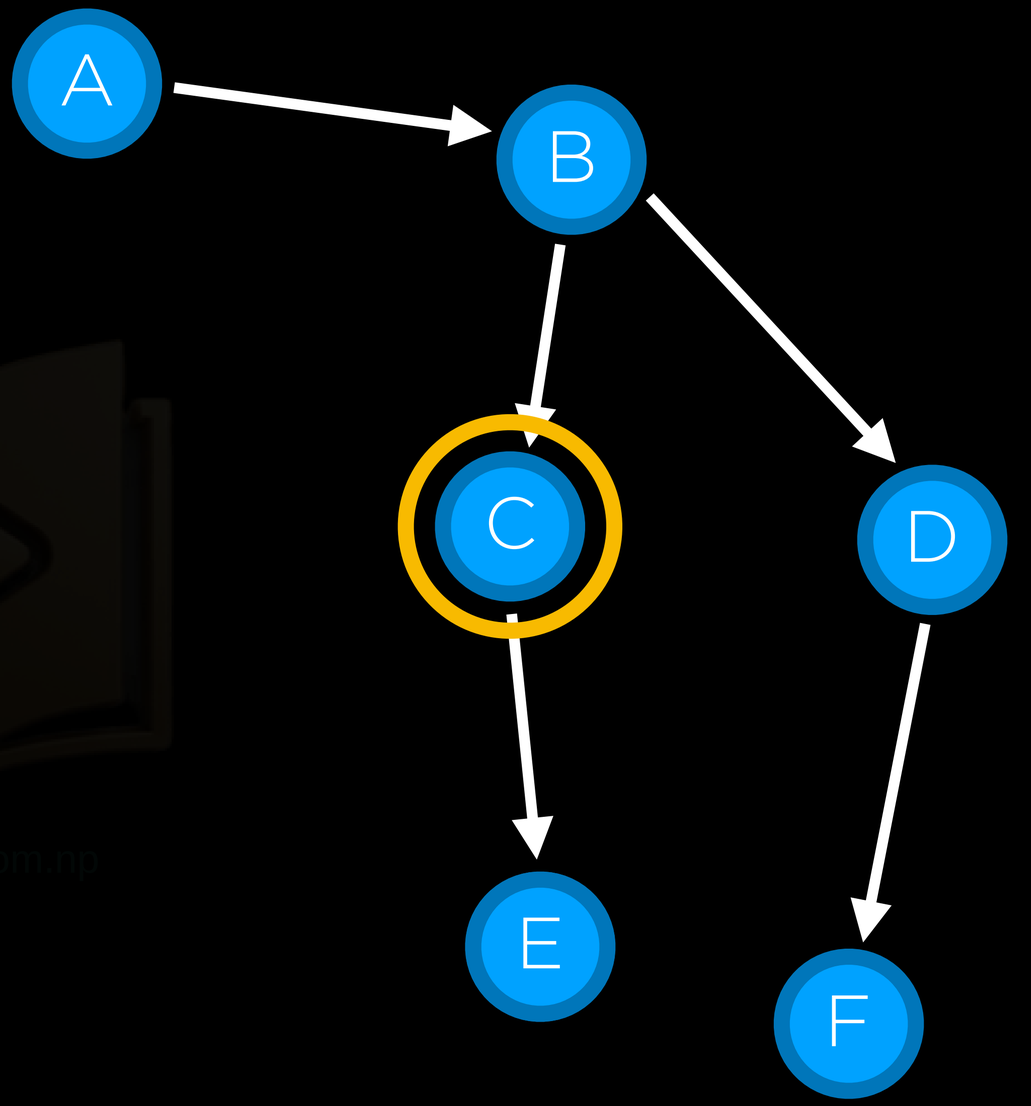
Available on notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

Explored Set

A B D F C



Available on notes.arpanadhikari7.com.np

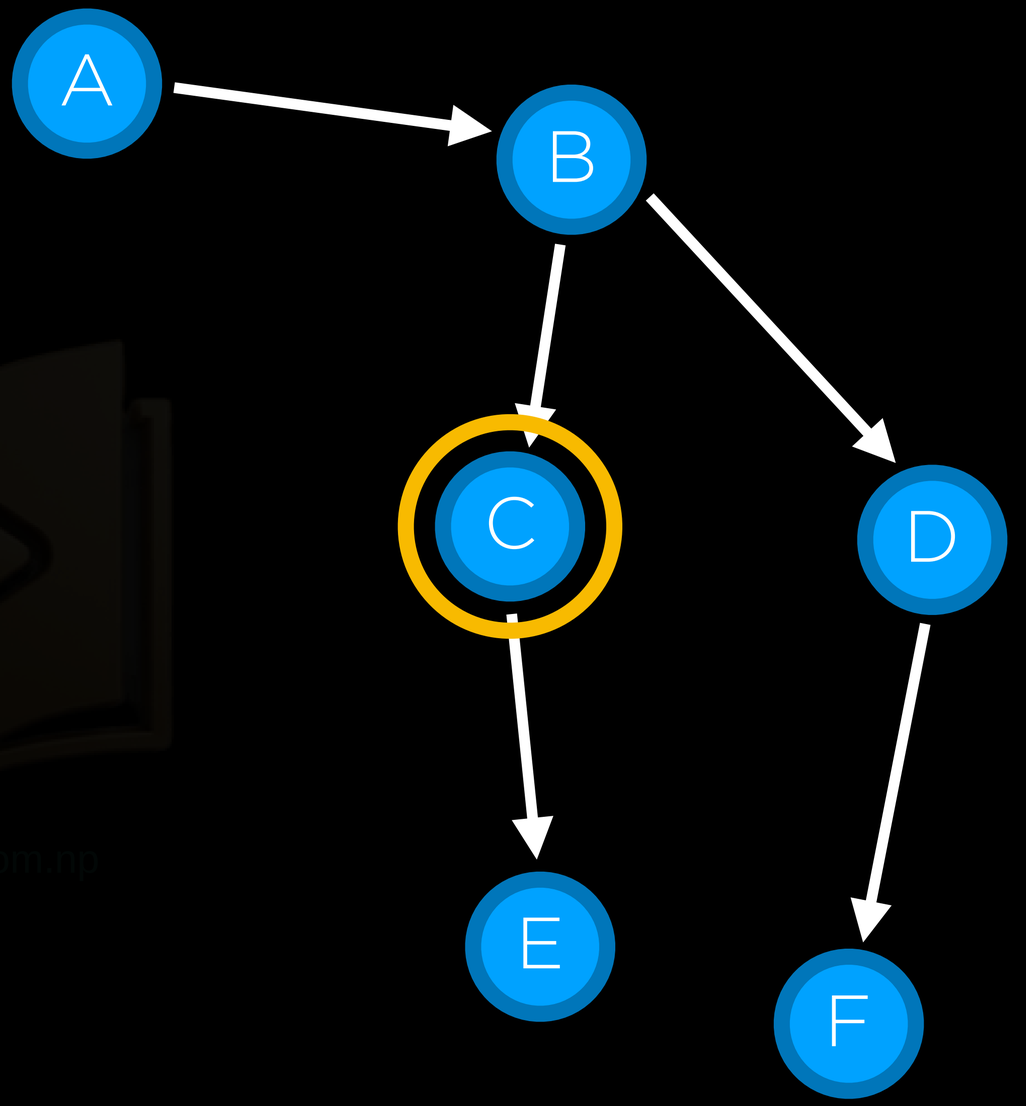
Find a path from A to E.

Frontier

E

Explored Set

A B D F C



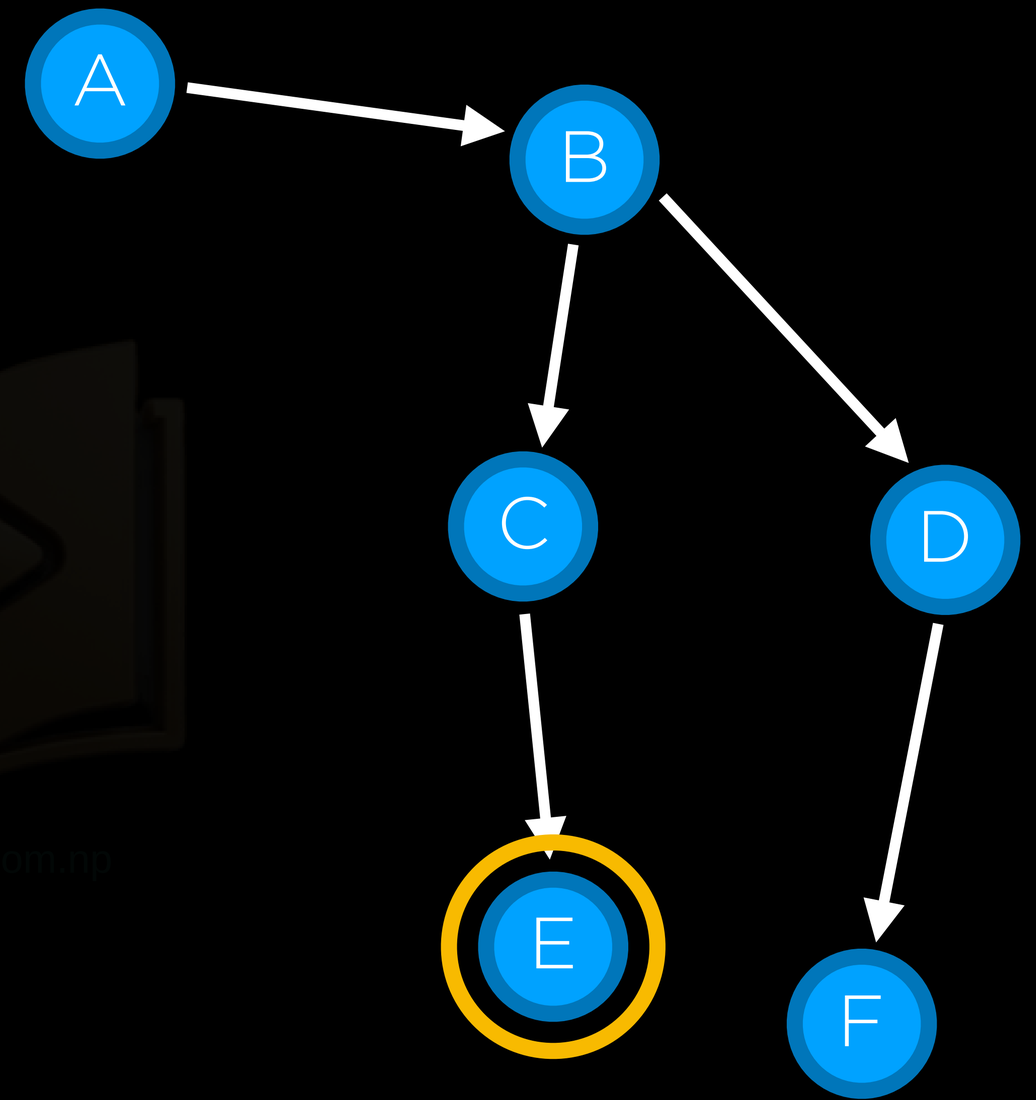
Available on notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

Explored Set

A B D F C



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Depth-First Search

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depth-first search

search algorithm that always expands the deepest node in the frontier

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Breadth-First Search

notes.arpanadhikari7.com.np

breadth-first search

search algorithm that always expands the shallowest node in the frontier

notes.arpanadhikari7.com.np

queue

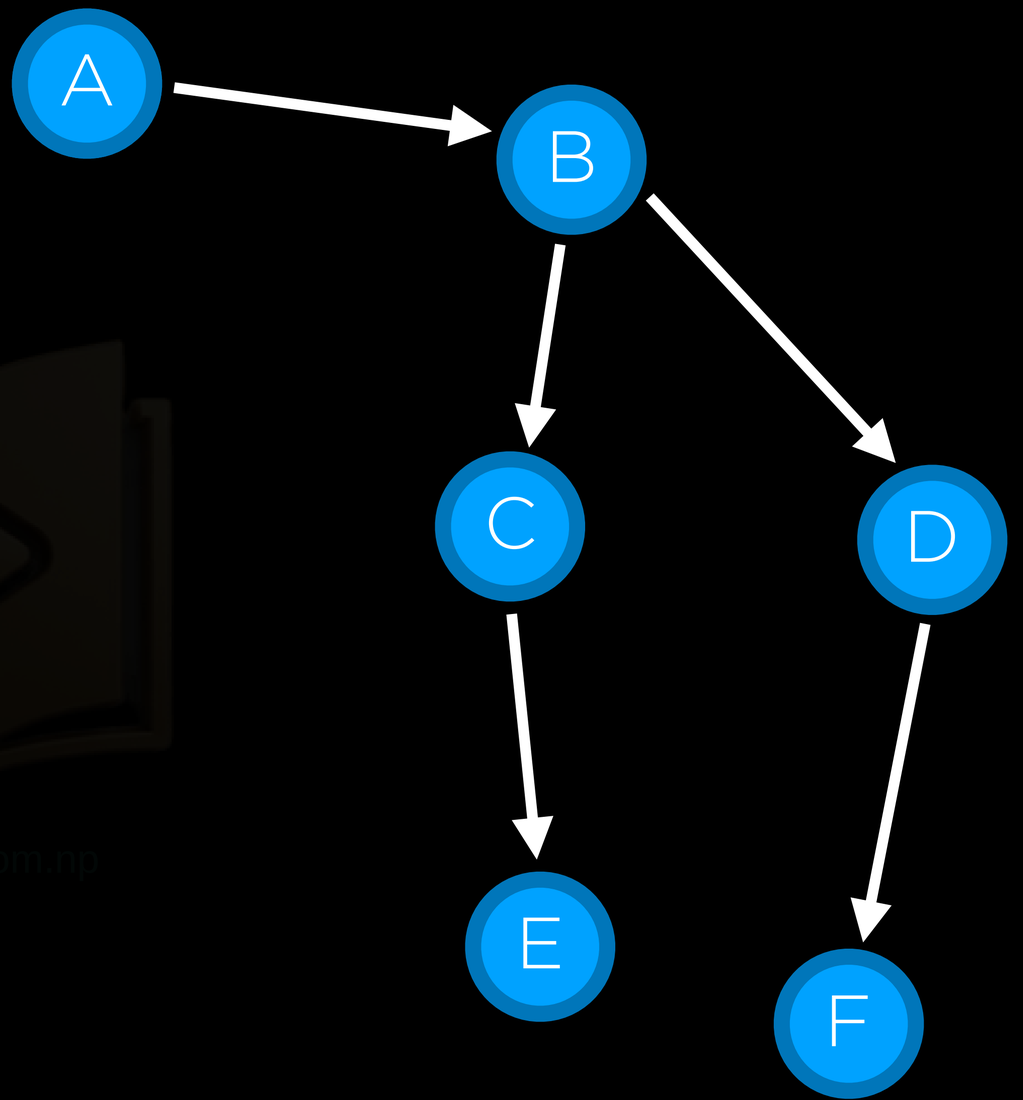
first-in first-out data type

notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

Explored Set



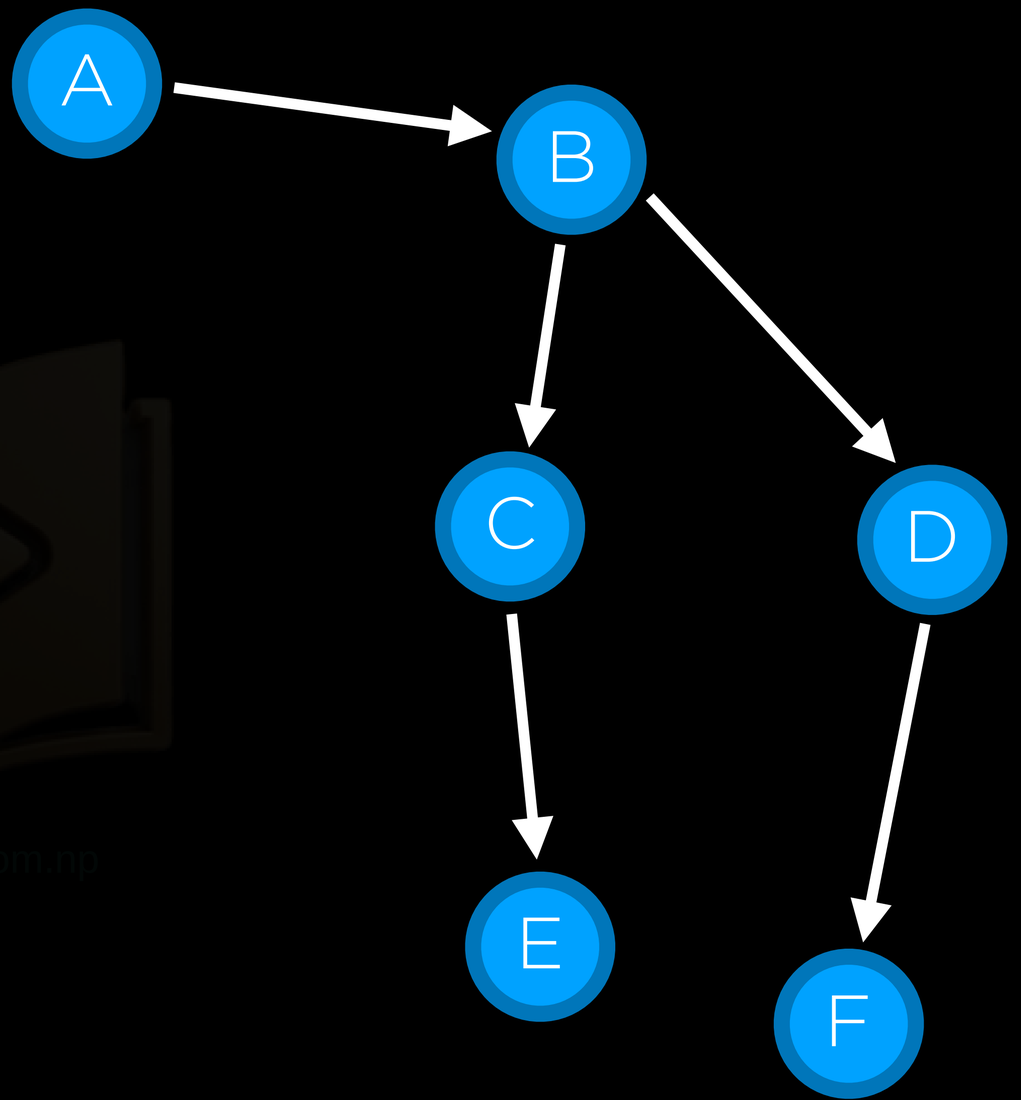
Available on notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

A

Explored Set



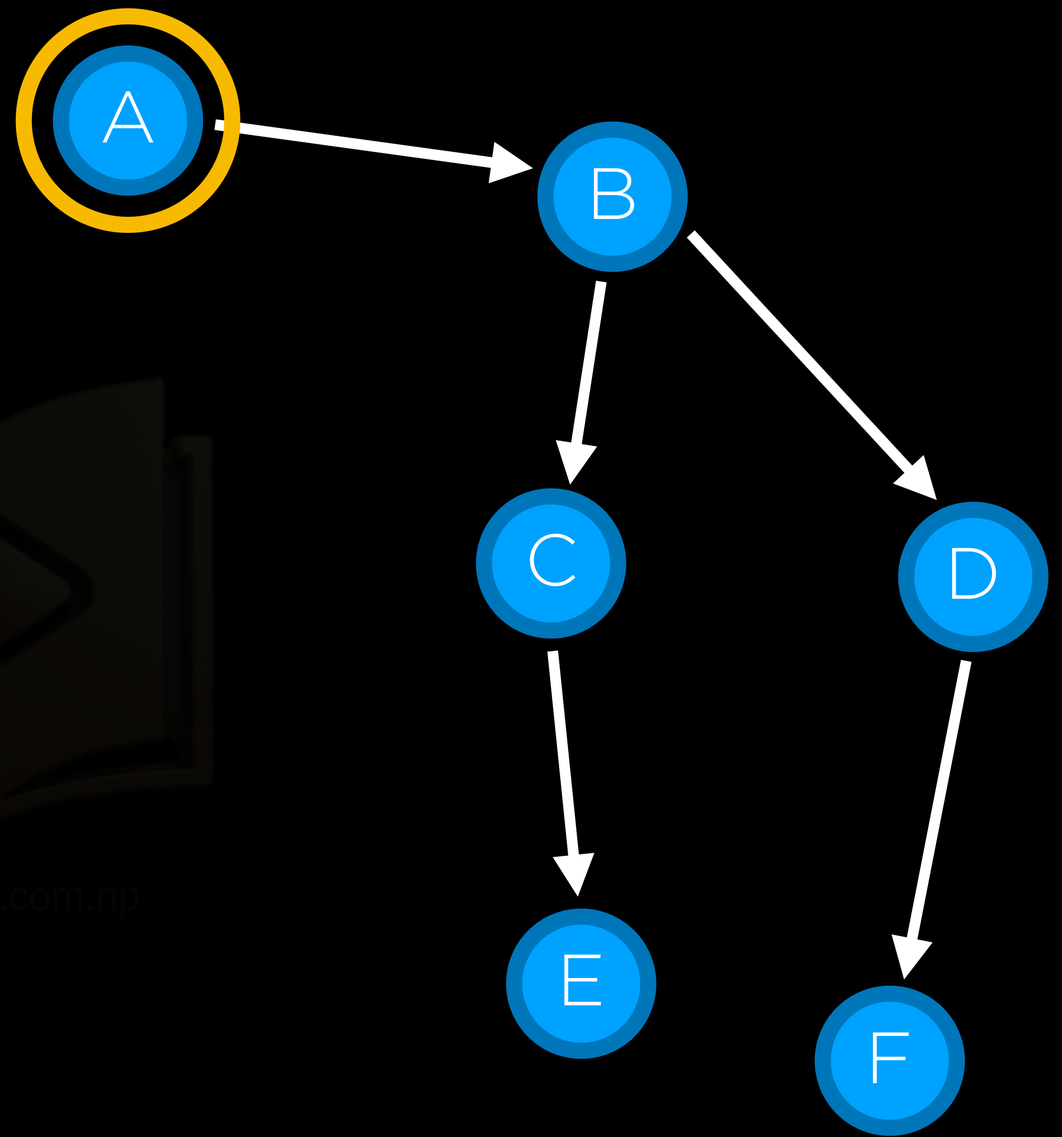
Available on notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

Explored Set

A



Available on notes.arpanadhikari7.com.np

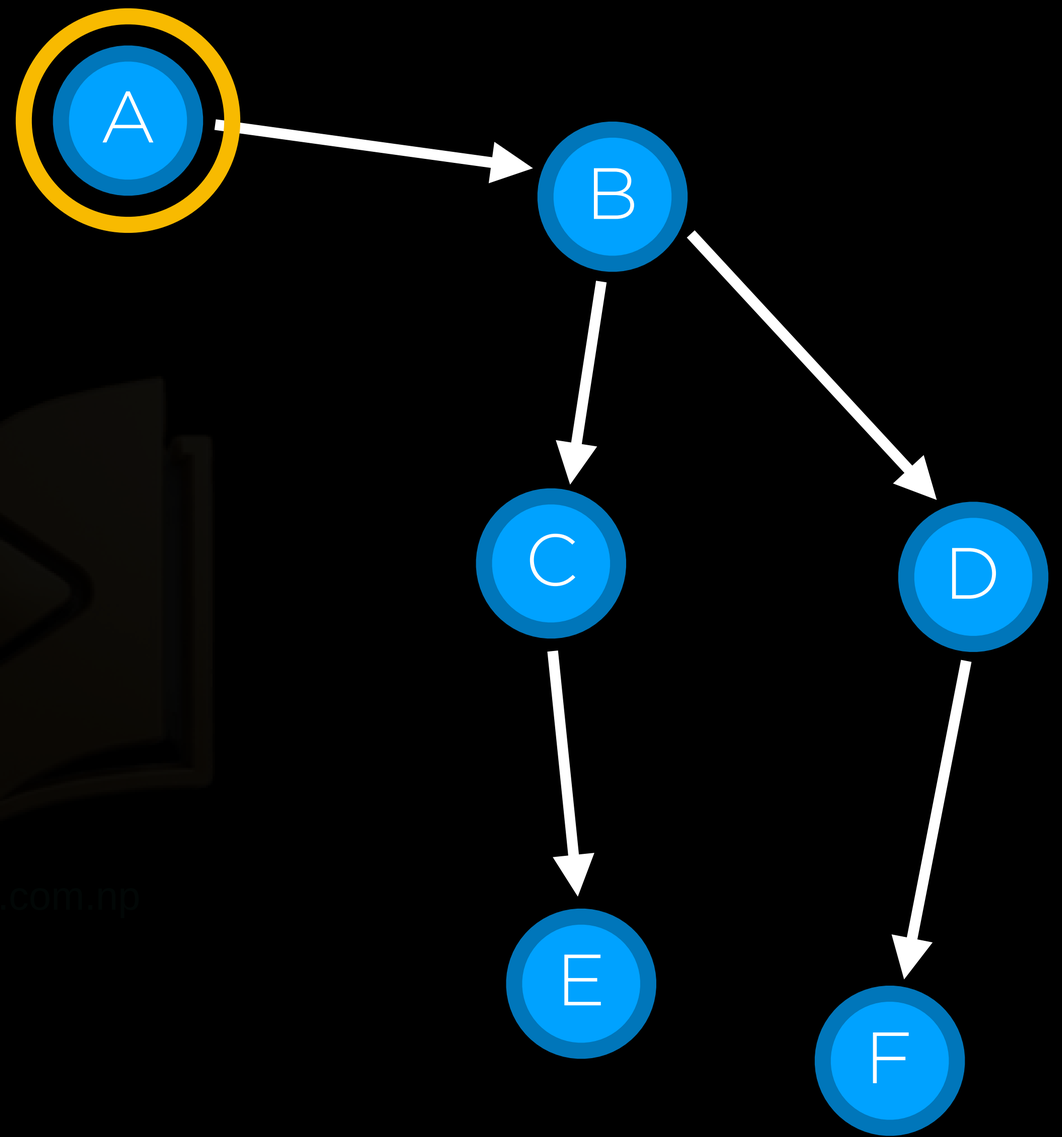
Find a path from A to E.

Frontier

B

Explored Set

A



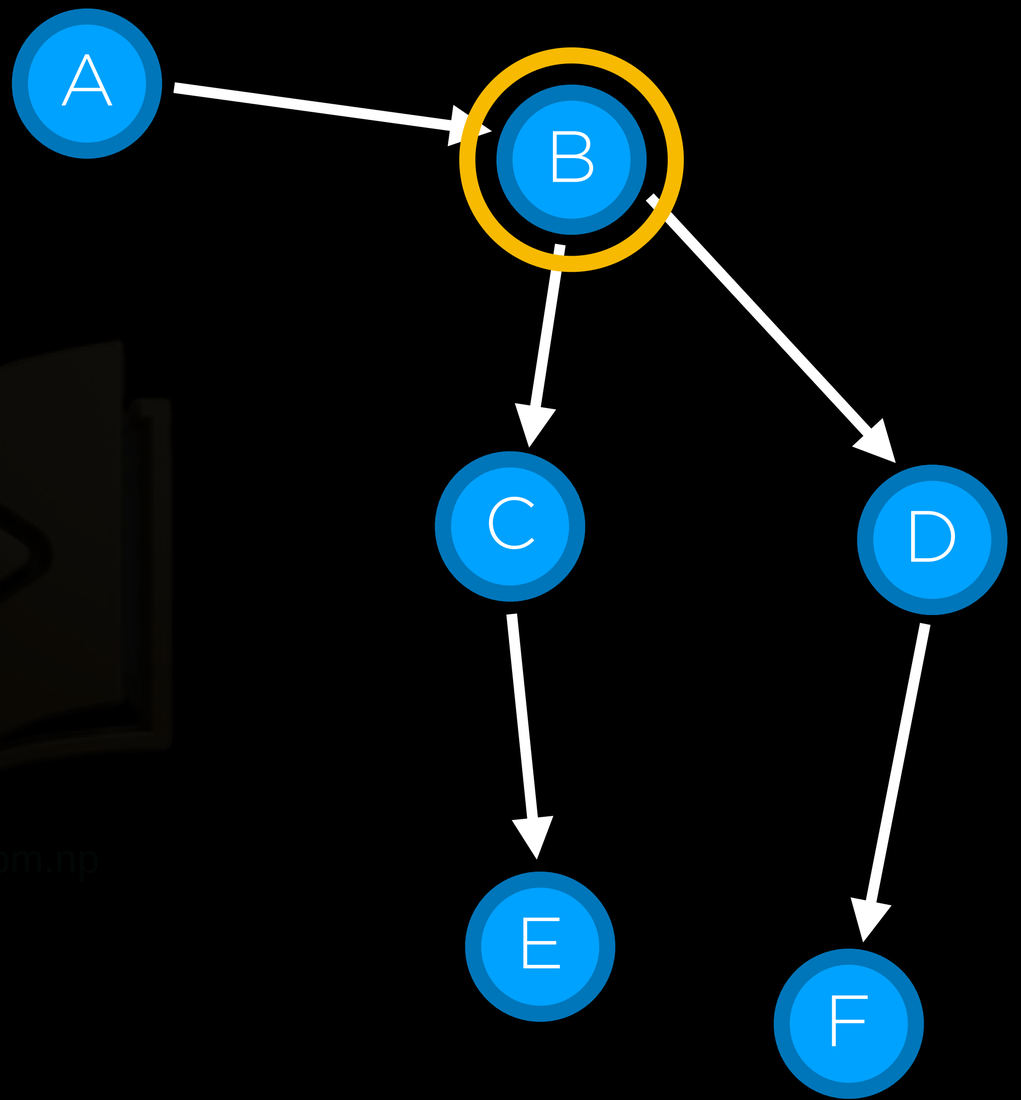
Available on notes.arpanadhikari7.com.np

Find a path from A to E.

Frontier

Explored Set

A B



Available on notes.arpanadhikari7.com.np

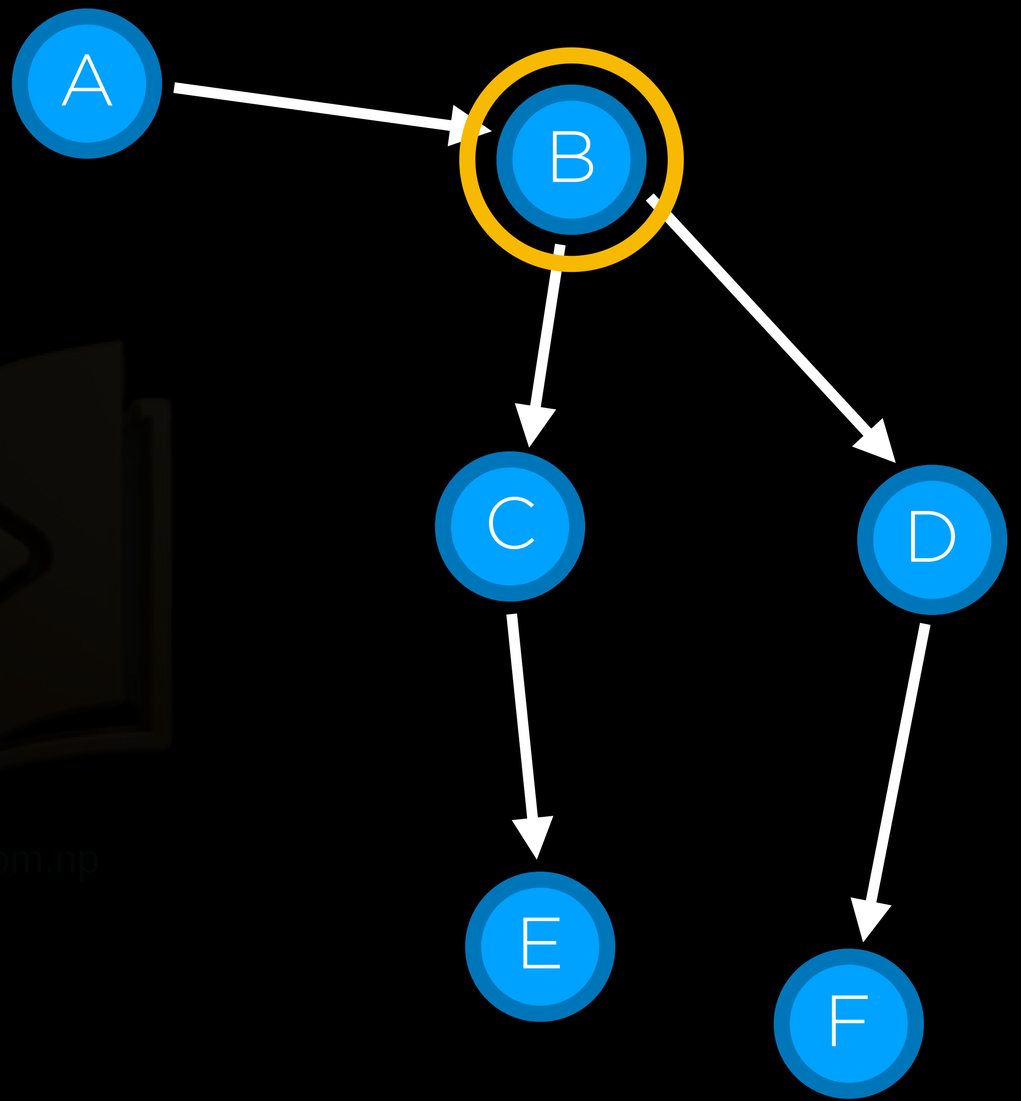
Find a path from A to E.

Frontier

C D

Explored Set

A B



Available on notes.arpanadhikari7.com.np

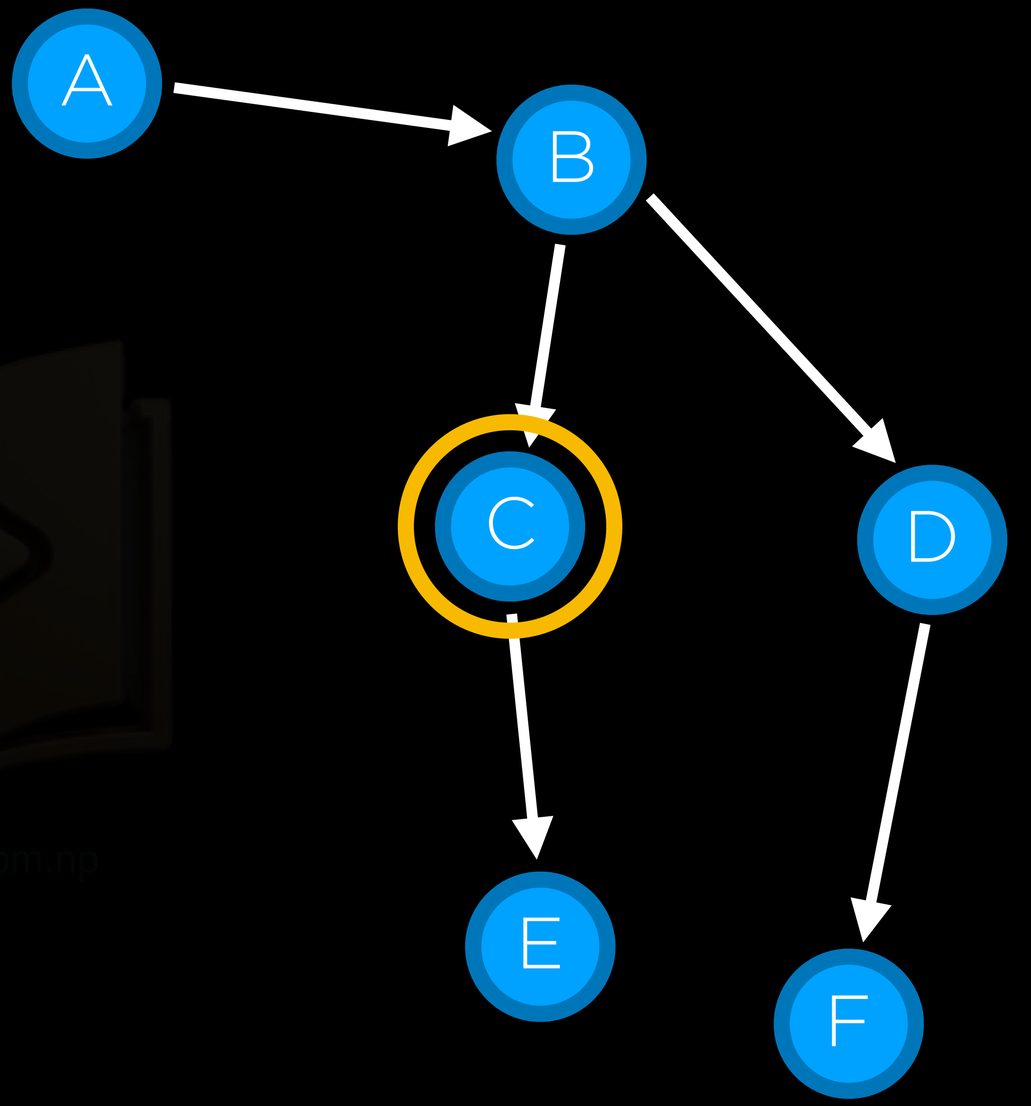
Find a path from A to E.

Frontier

D

Explored Set

A B C



Available on notes.arpanadhikari7.com.np

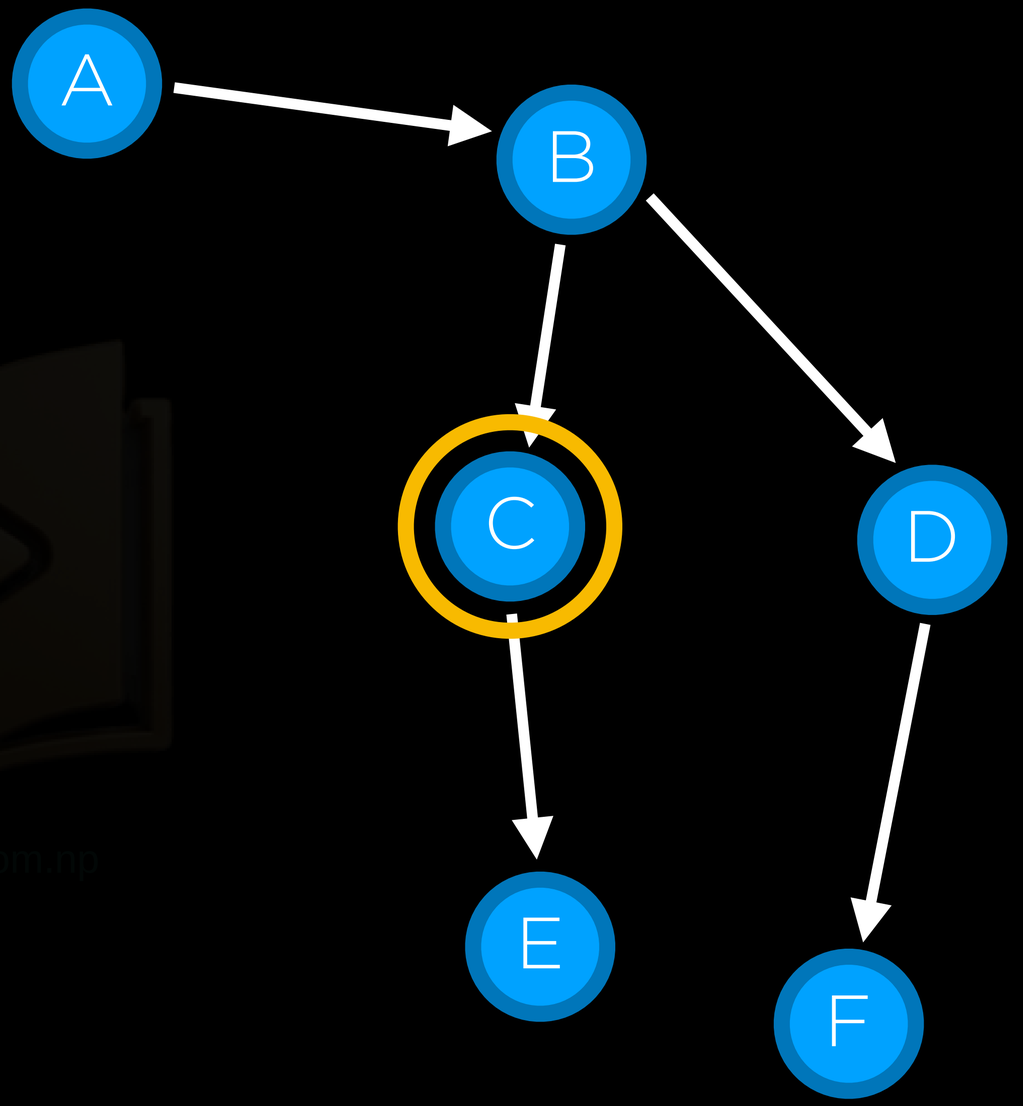
Find a path from A to E.

Frontier

D E

Explored Set

A B C



Available on notes.arpanadhikari7.com.np

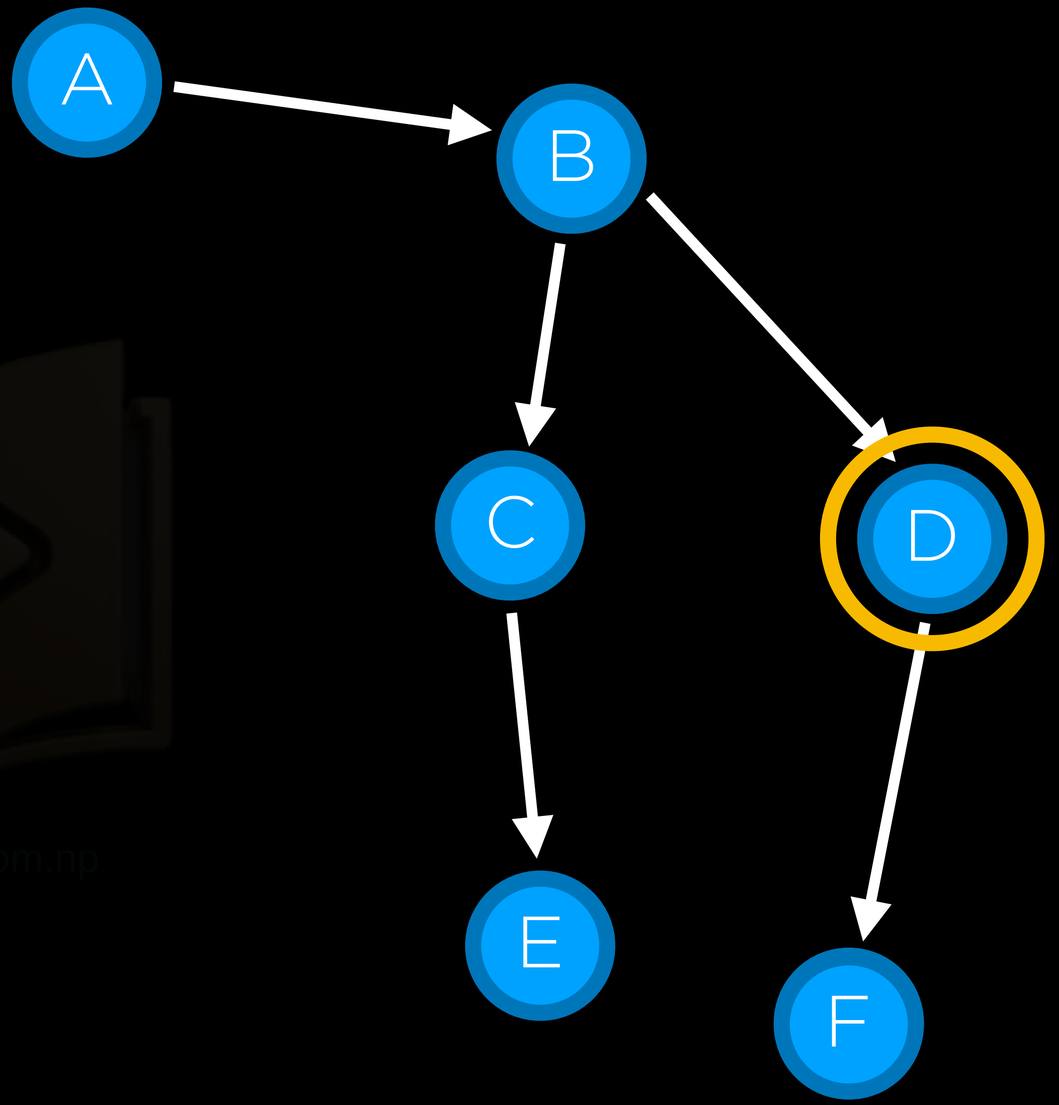
Find a path from A to E.

Frontier

E

Explored Set

A B C D



Available on notes.arpanadhikari7.com.np

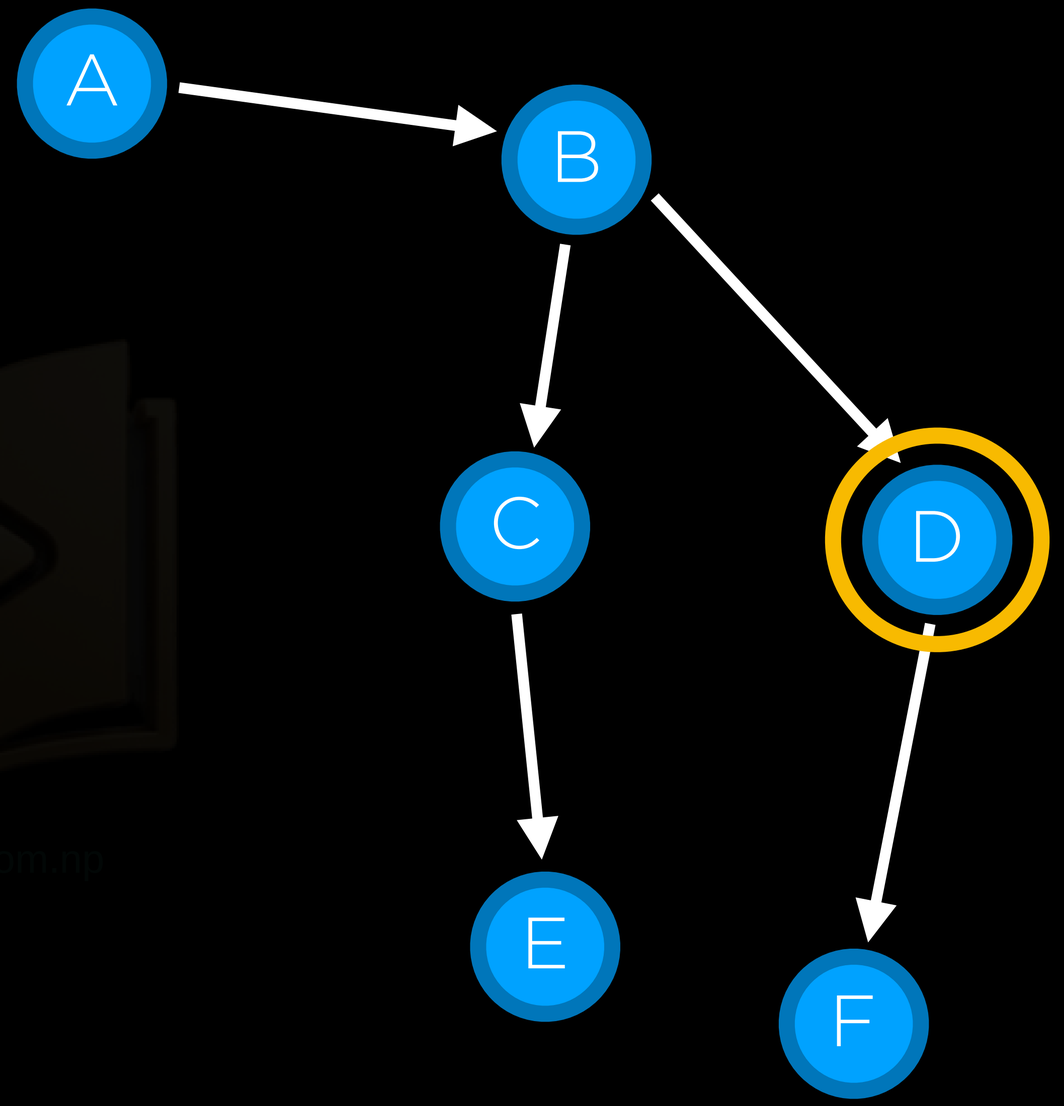
Find a path from A to E.

Frontier

E F

Explored Set

A B C D



Available on notes.arpanadhikari7.com.np

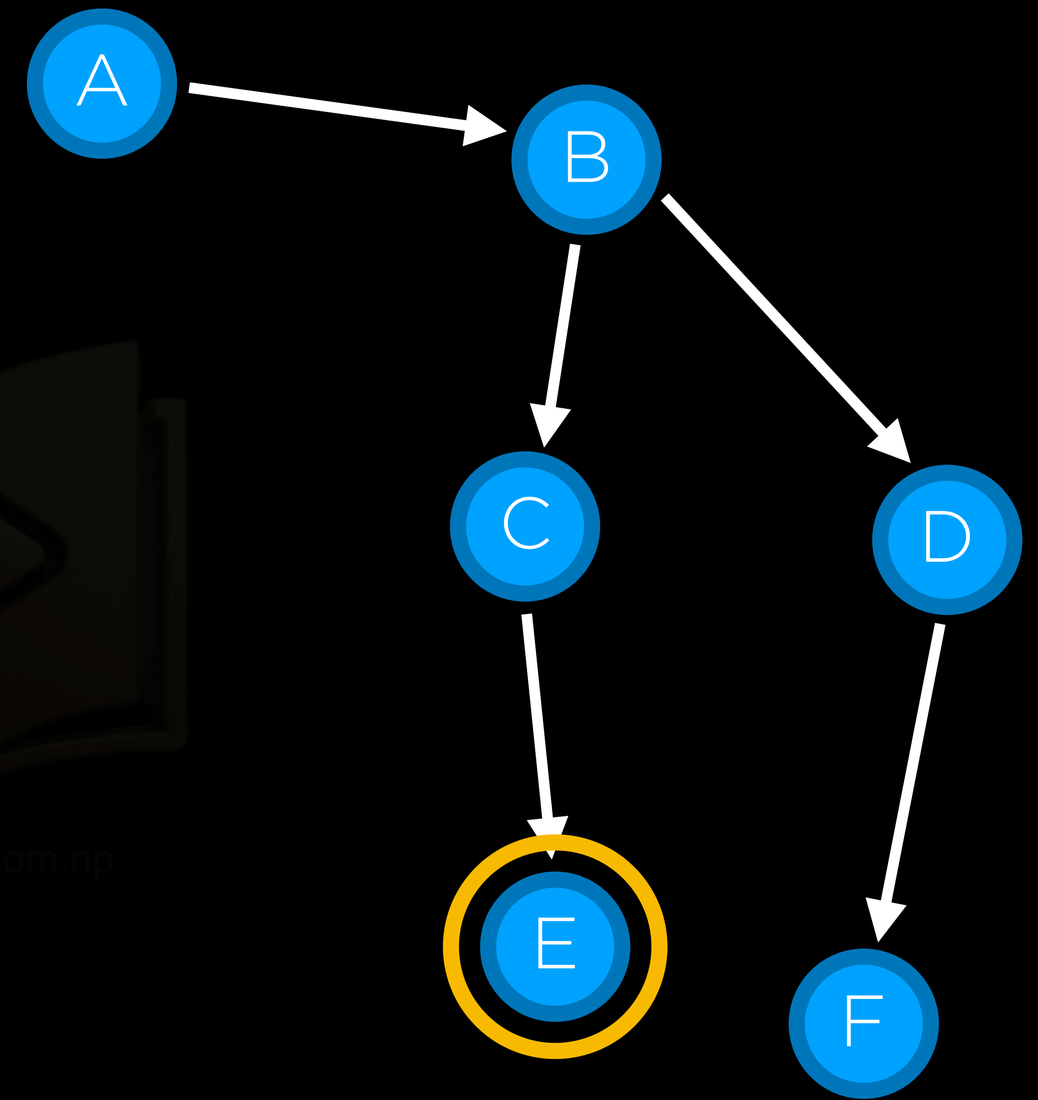
Find a path from A to E.

Frontier

F

Explored Set

A B C D



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Depth-First Search



Available on notes.arpanadhikari7.com.np

Depth-First Search



Available on notes.arpanadhikari7.com.np

Depth-First Search



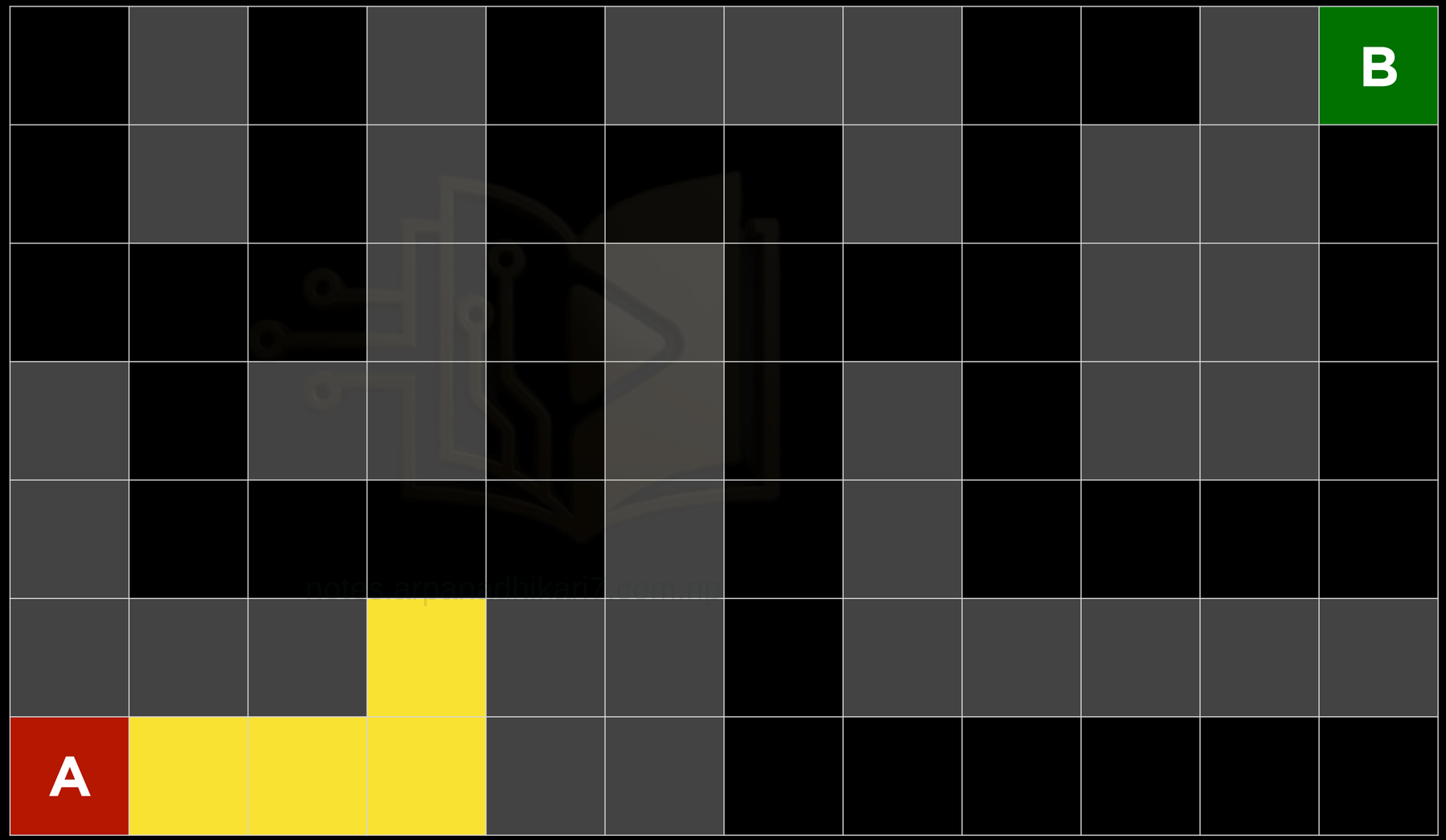
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Depth-First Search



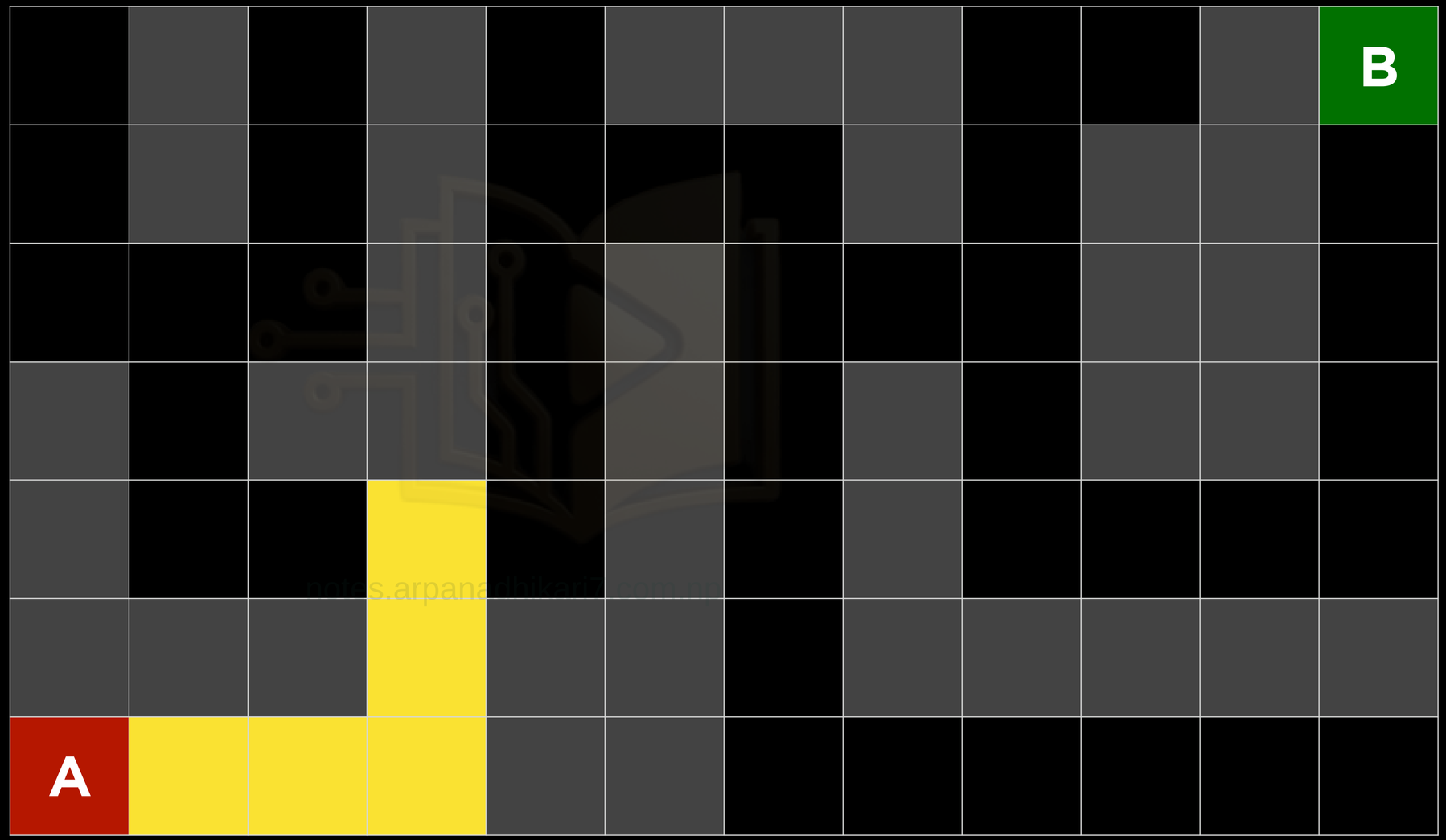
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Depth-First Search



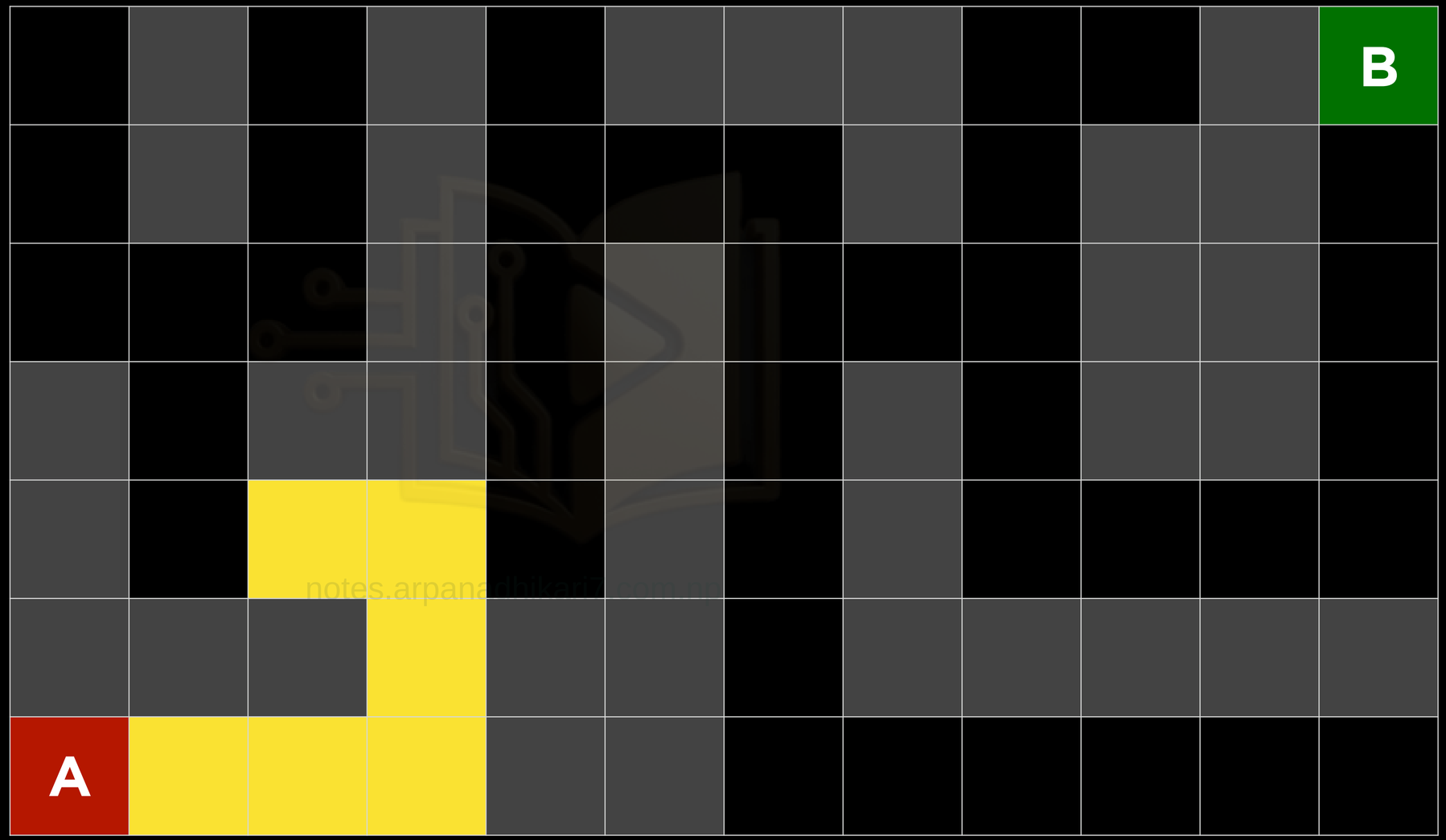
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Depth-First Search



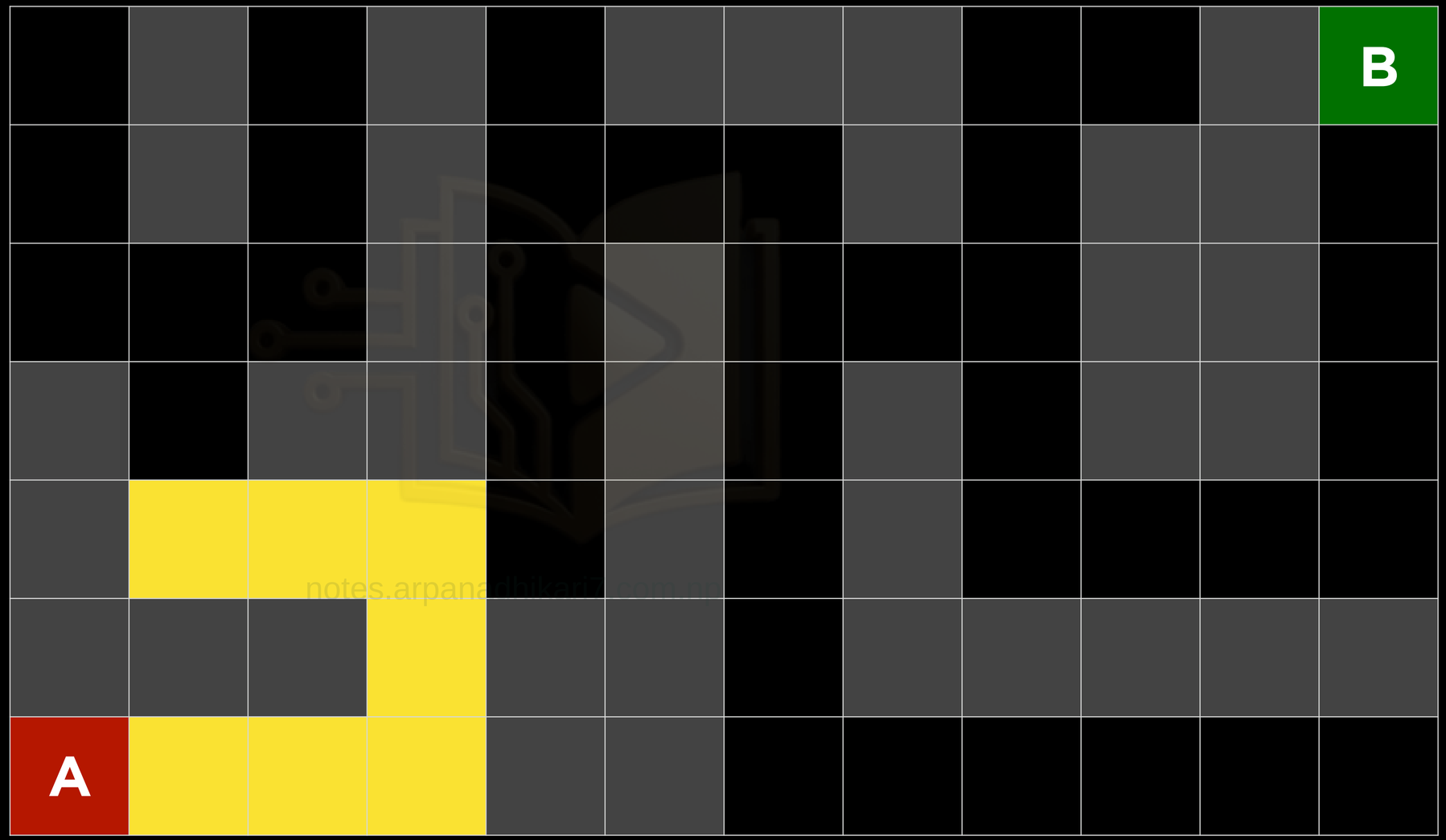
notes.arpanadhikari7.com.np

Depth-First Search



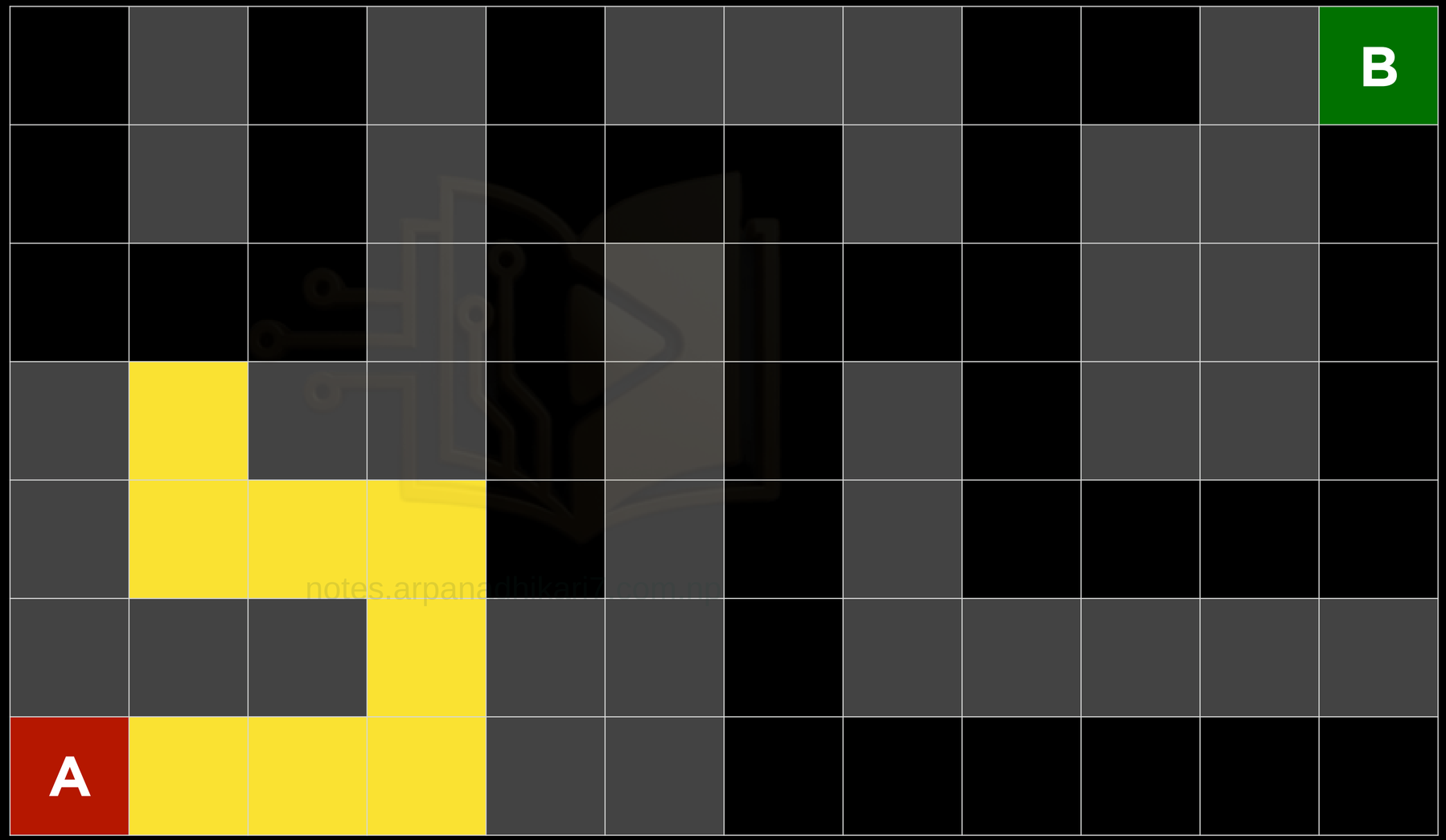
Available on notes.arpanadhikari7.com.np

Depth-First Search



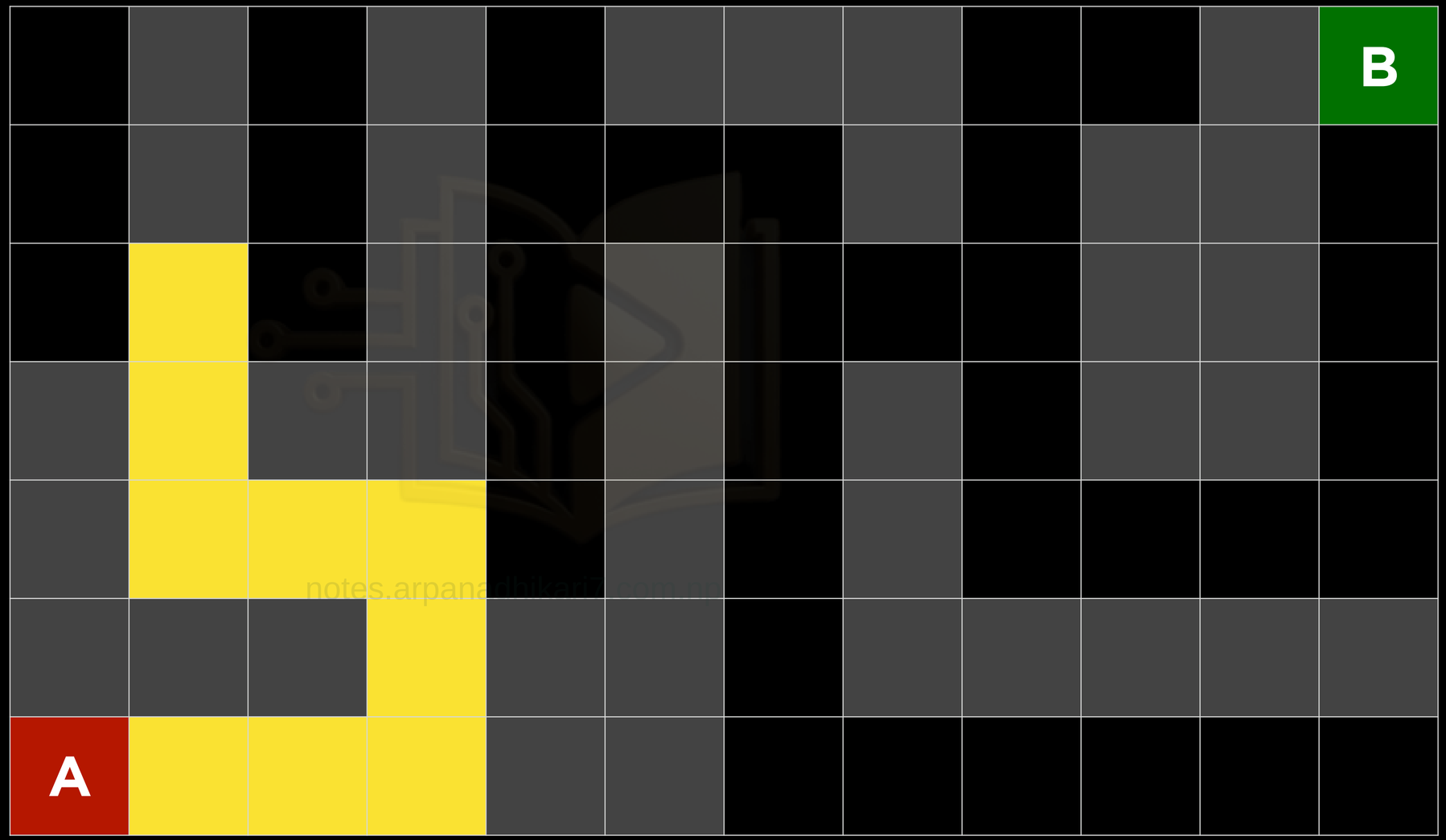
notes.arpanadhikari7.com.np

Depth-First Search



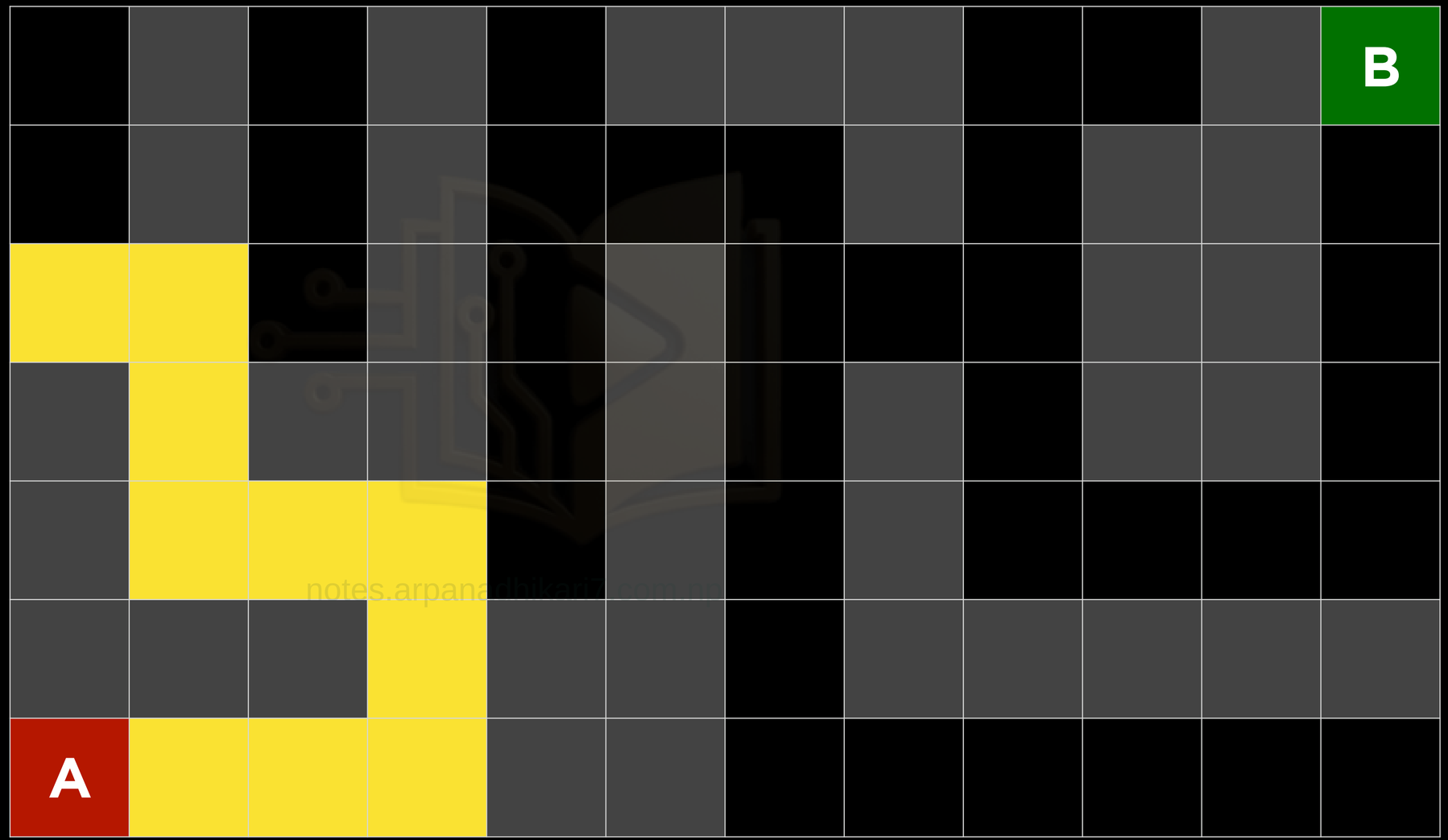
notes.arpanadhikari7.com.np

Depth-First Search



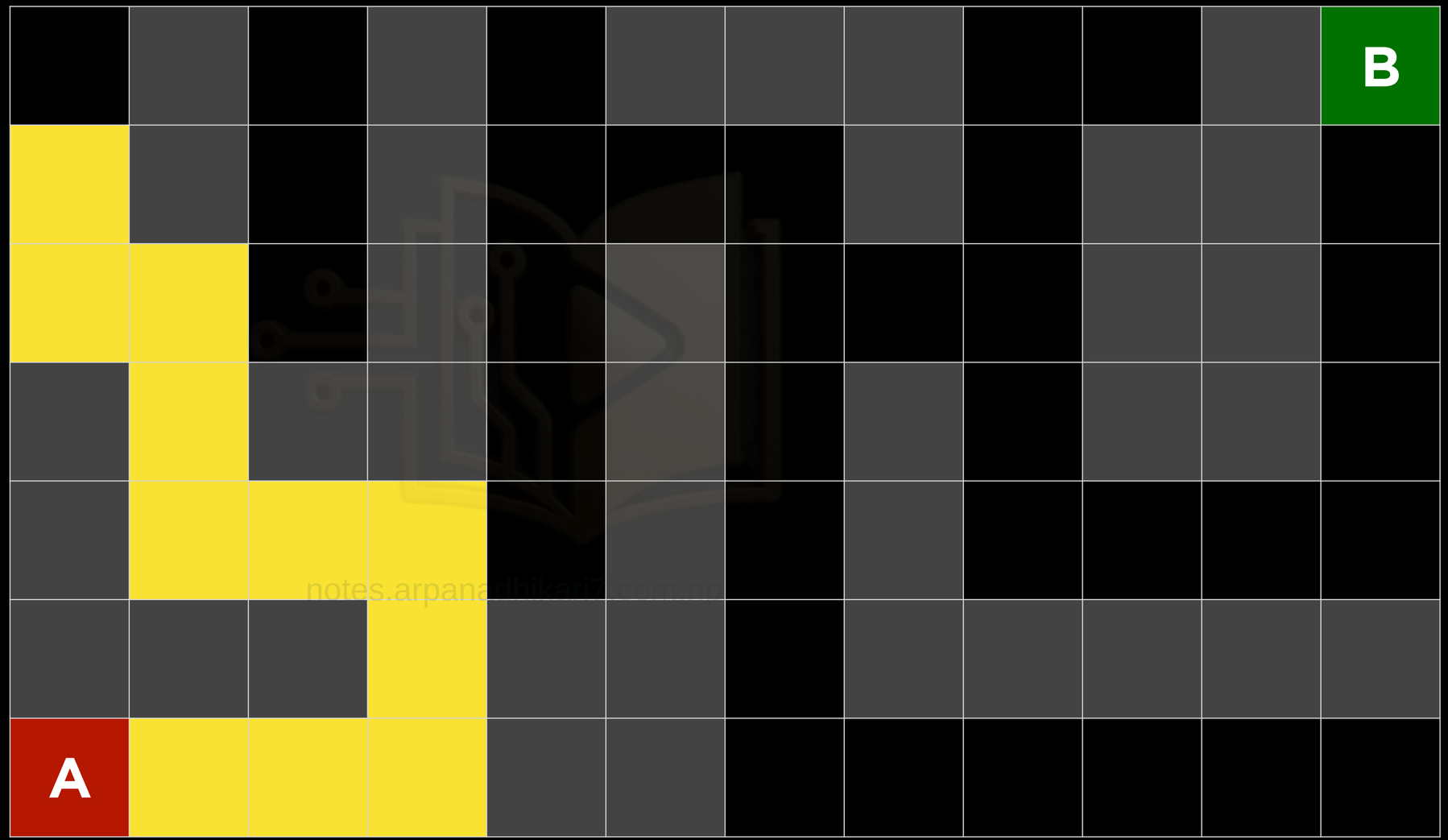
notes.arpanadhikari7.com.np

Depth-First Search



notes.arpanadhikari7.com.np

Depth-First Search



notes.arpanadhikari7.com.np

Depth-First Search



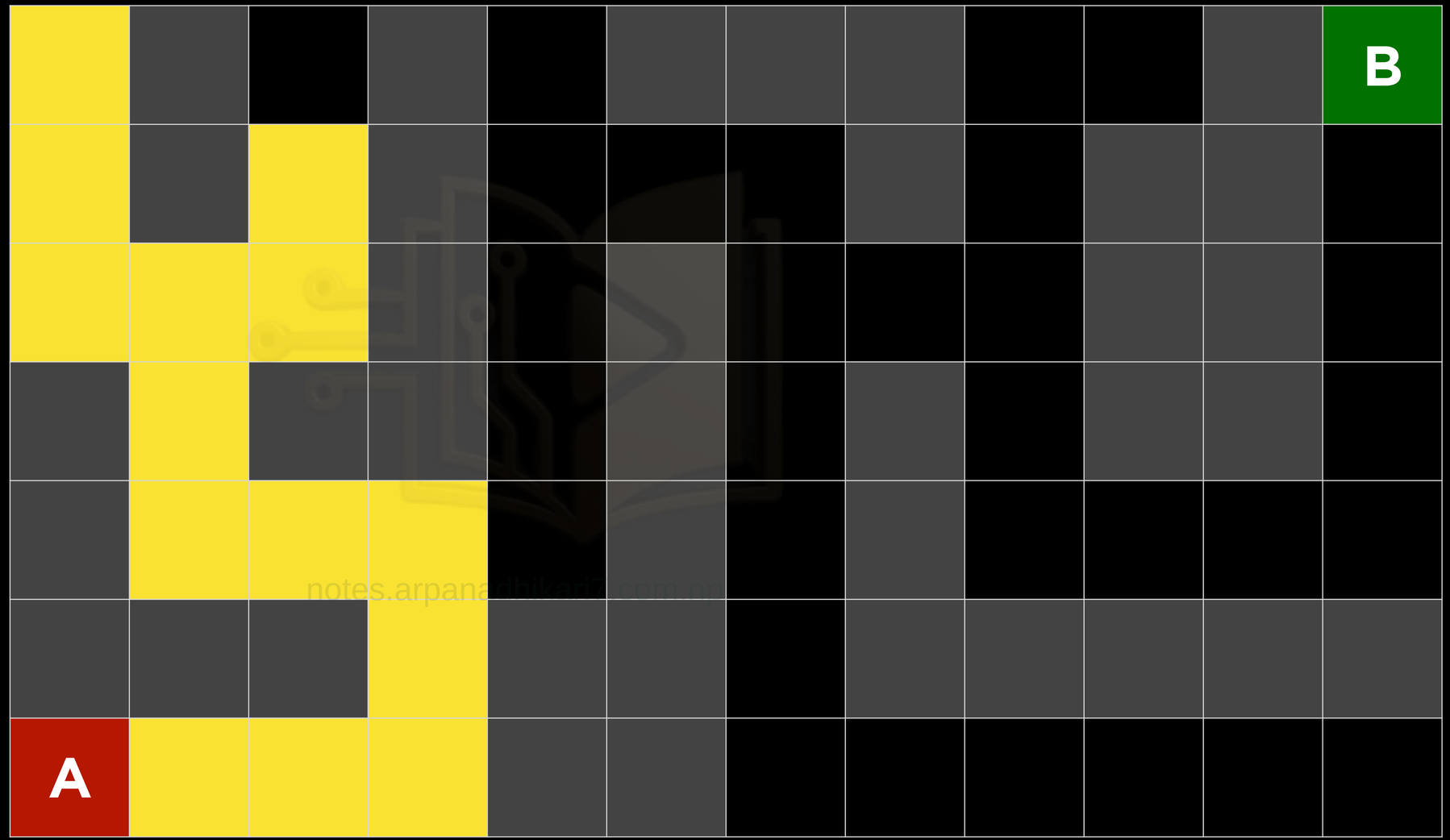
notes.arpanadhikari7.com.np

Depth-First Search



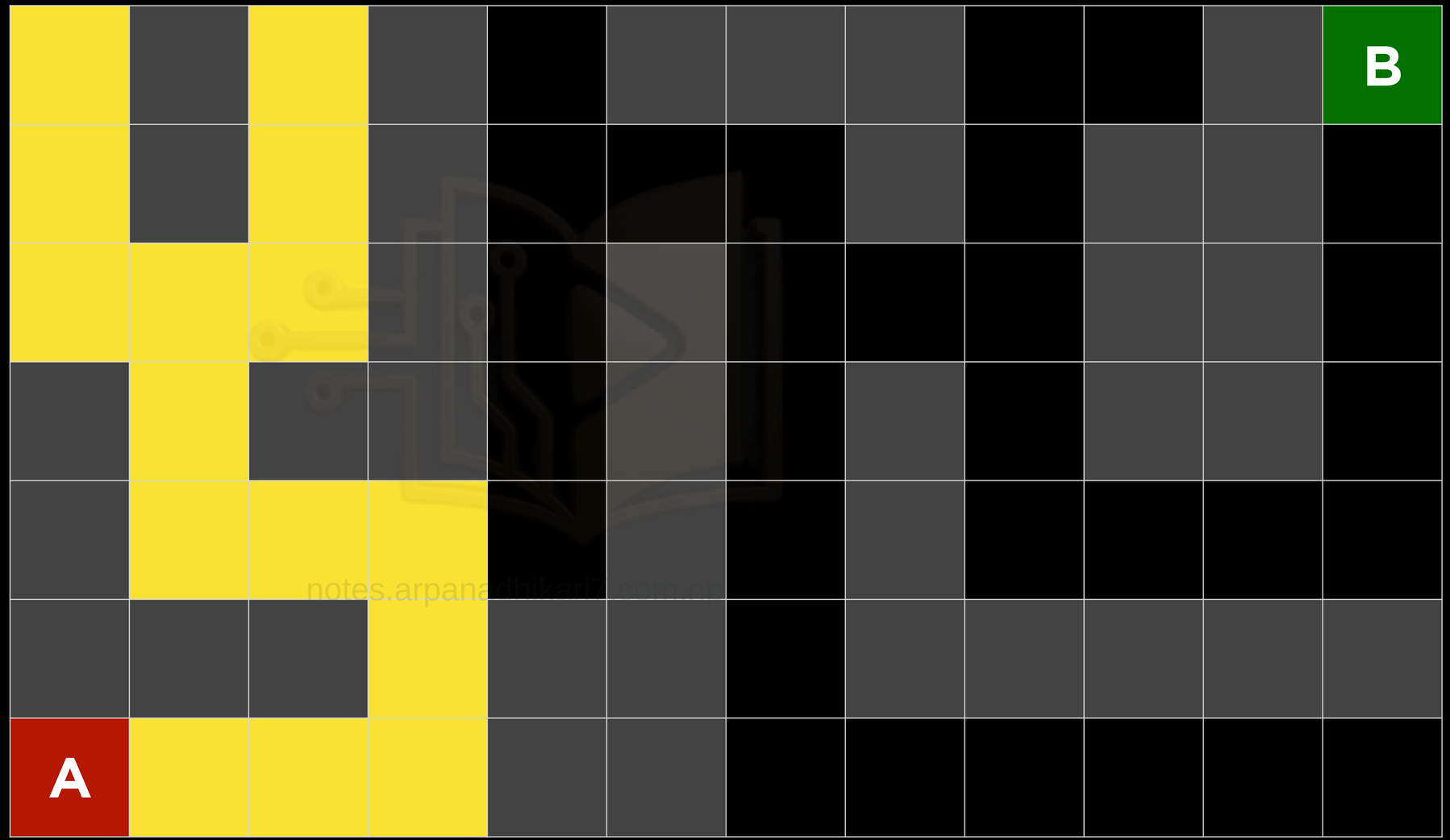
notes.arpanadhikari7.com.np

Depth-First Search



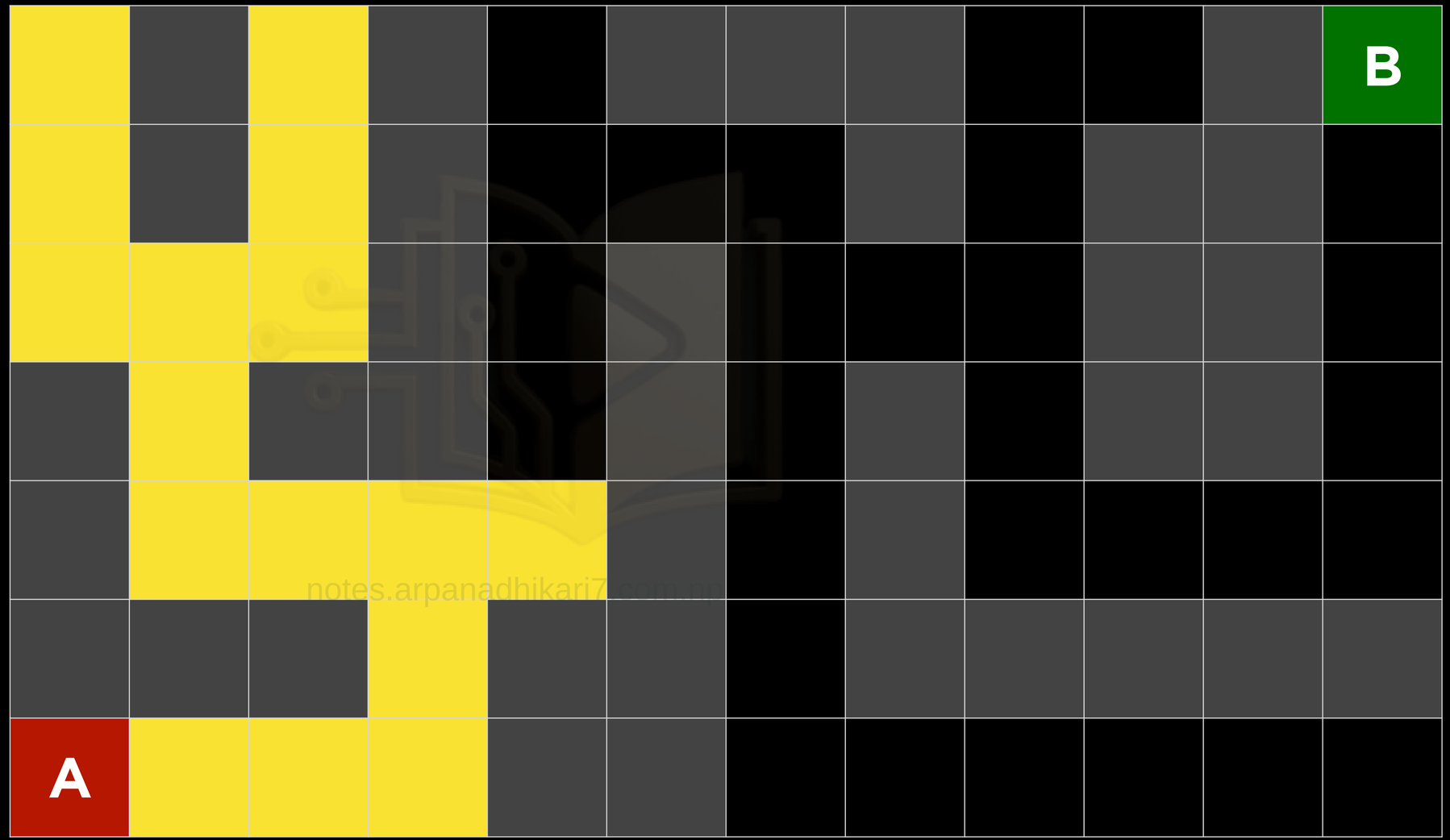
notes.arpanadhikari7.com.np

Depth-First Search



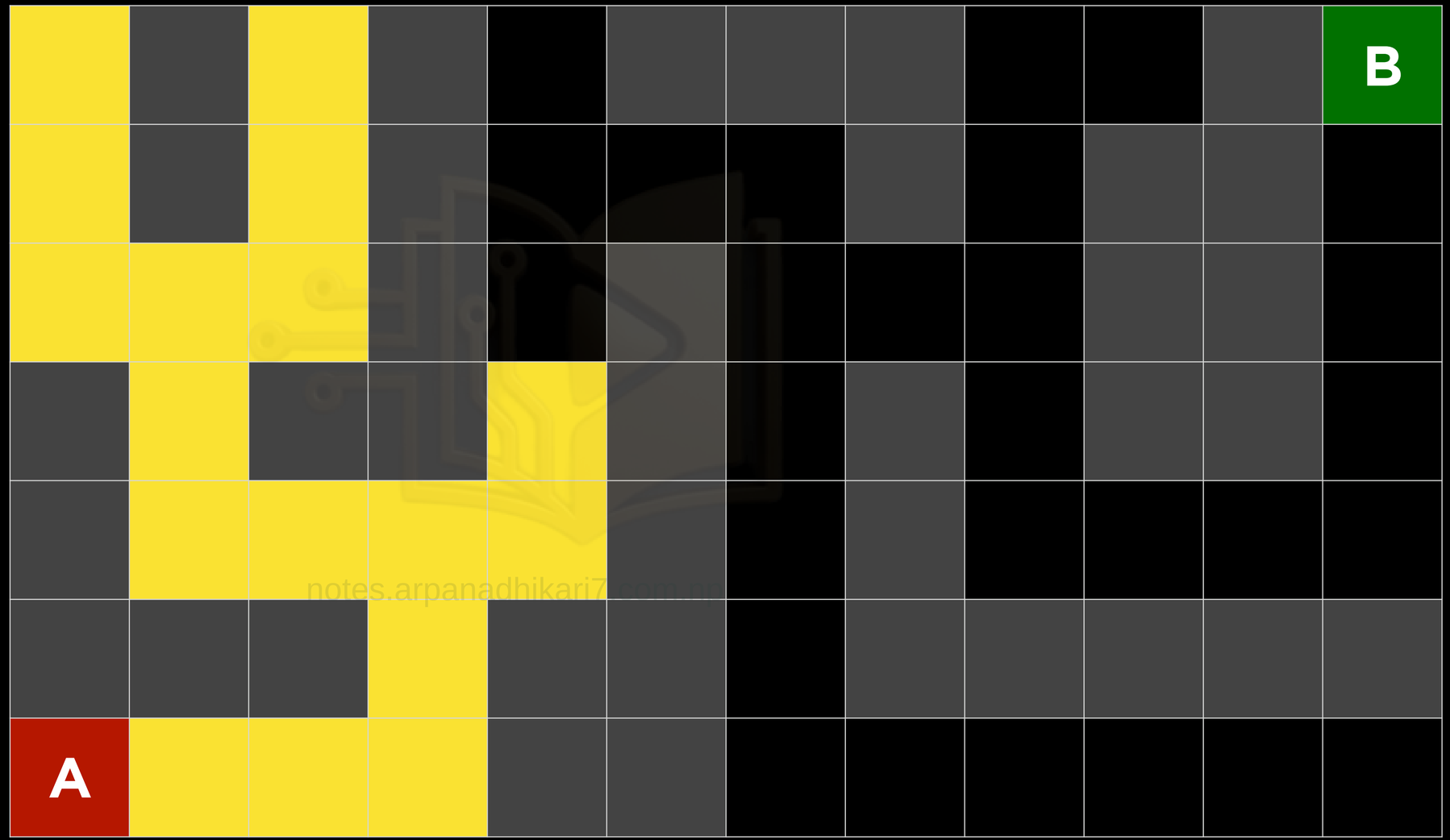
notes.arpanadhikari7.com.np

Depth-First Search



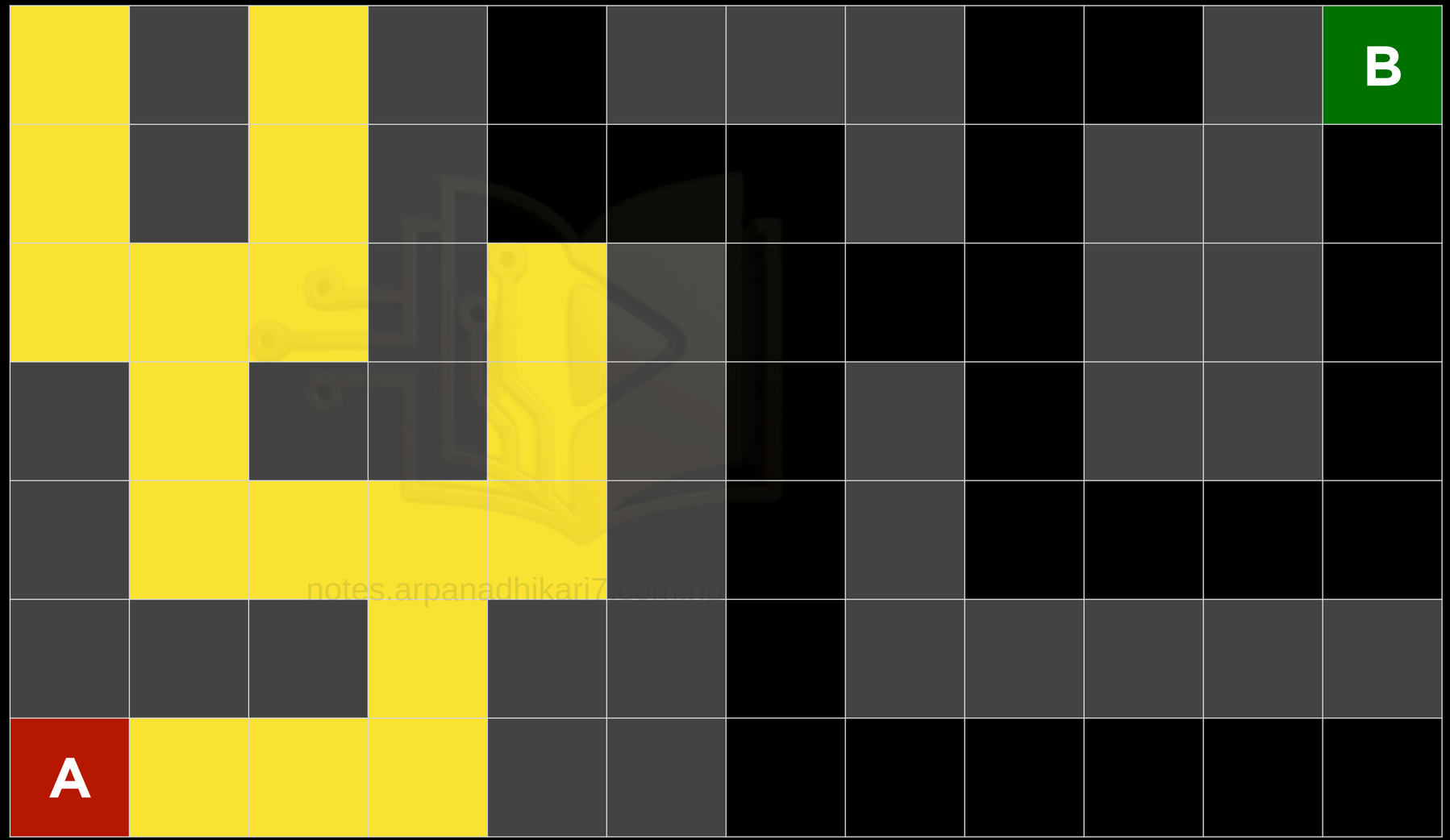
notes.arpanadhikari7.com.np

Depth-First Search



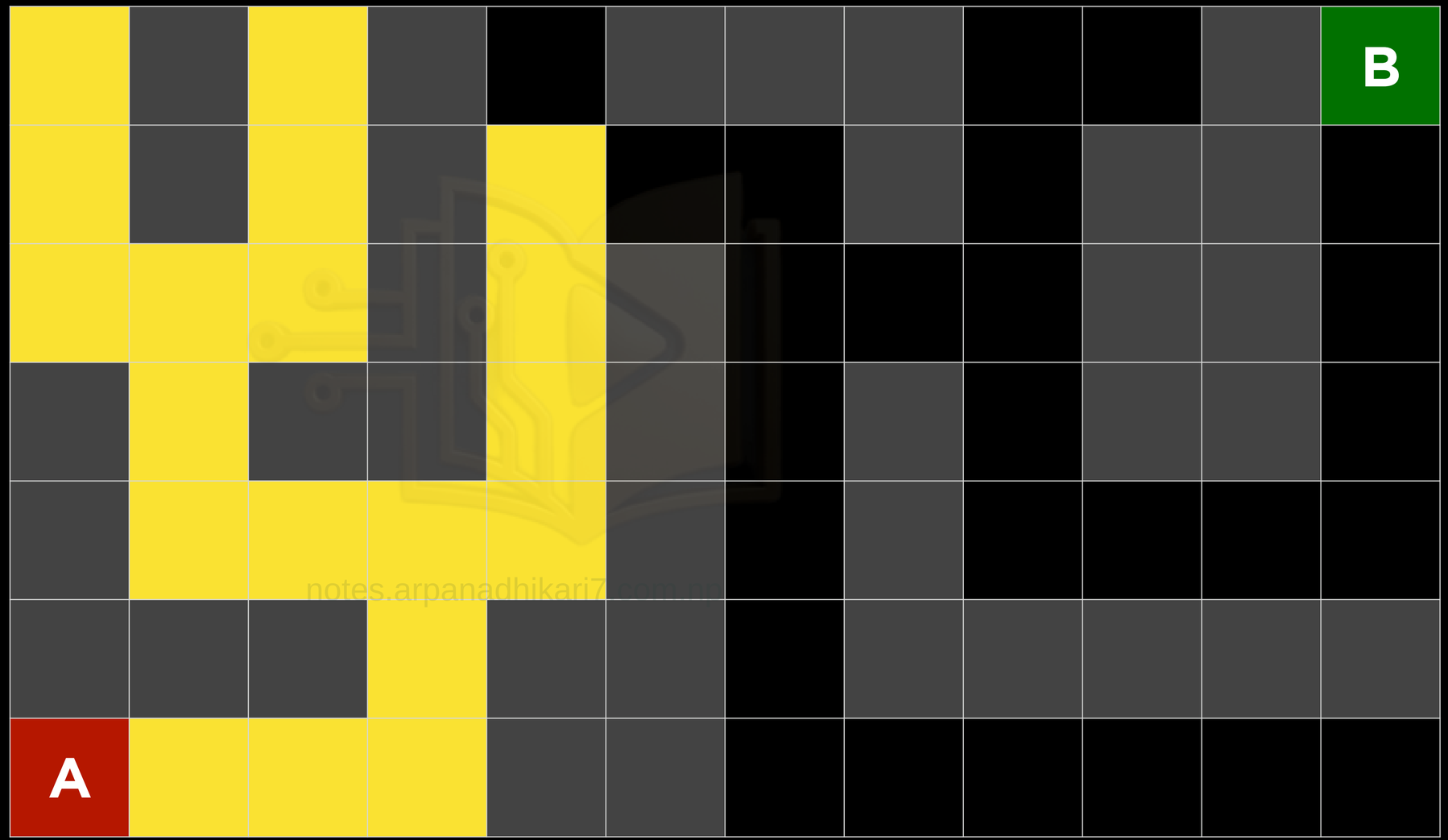
notes.arpanadhikari7.com.np

Depth-First Search



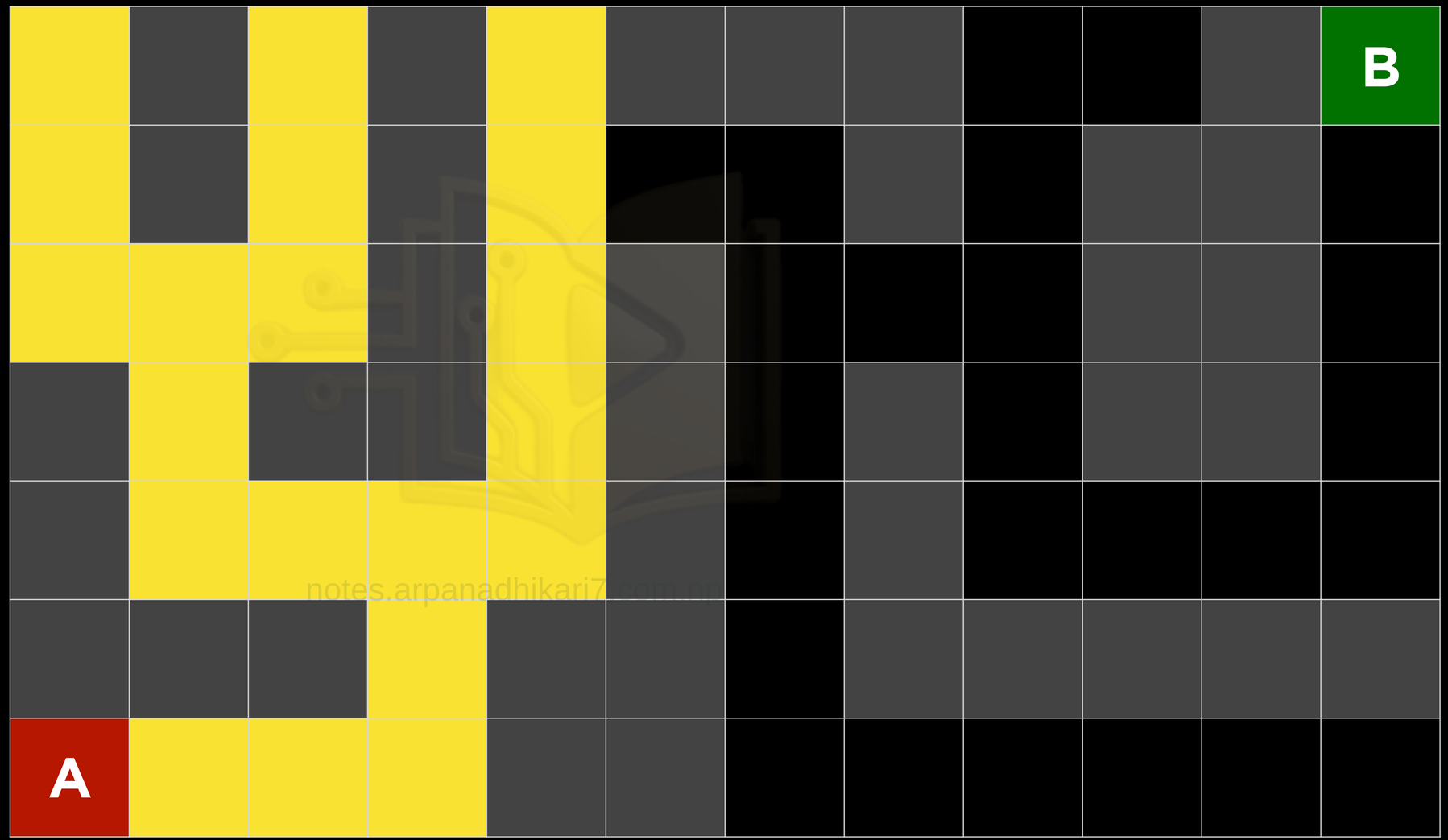
notes.arpanadhikari7.com.np

Depth-First Search



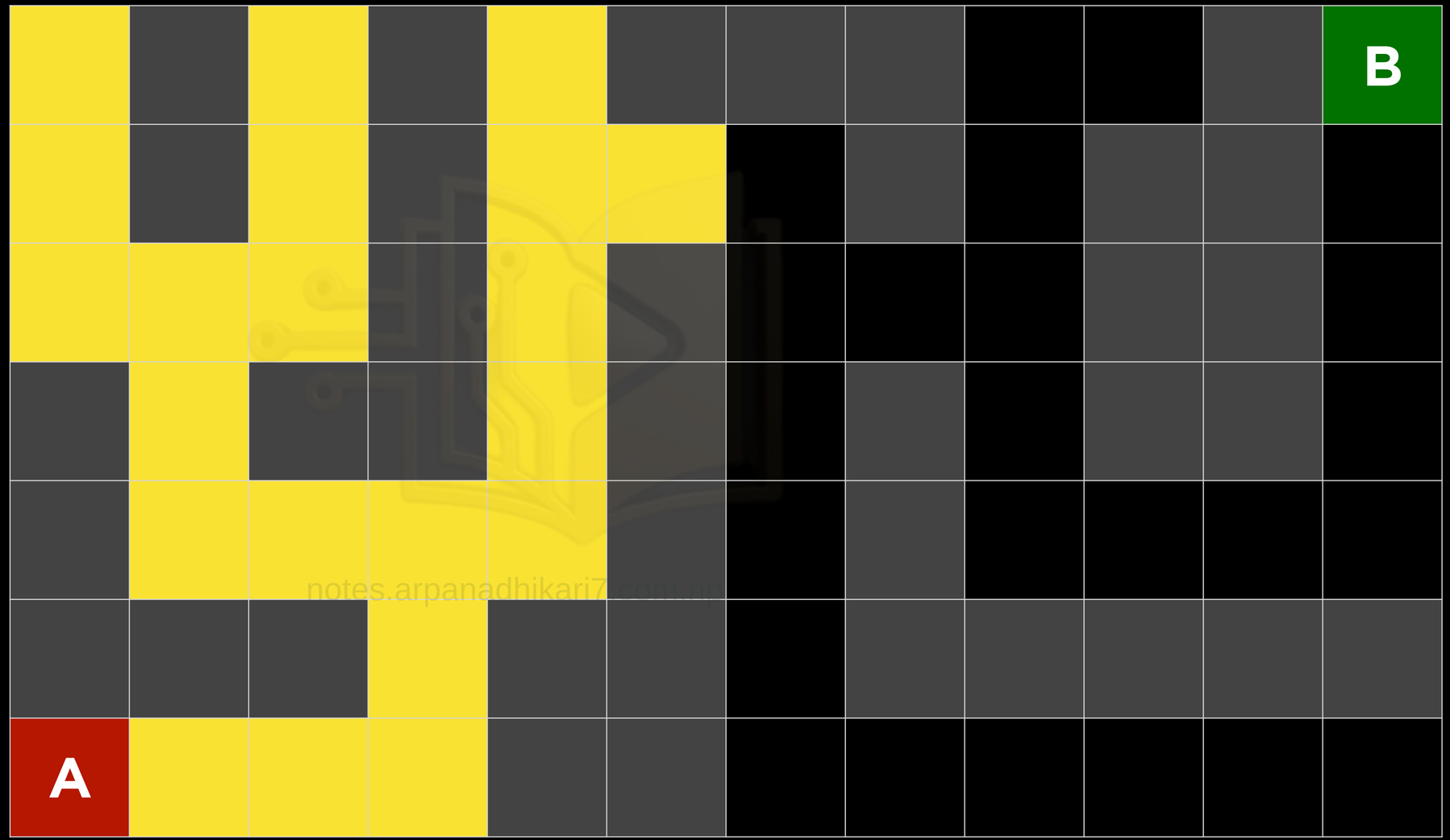
notes.arpanadhikari7.com.np

Depth-First Search



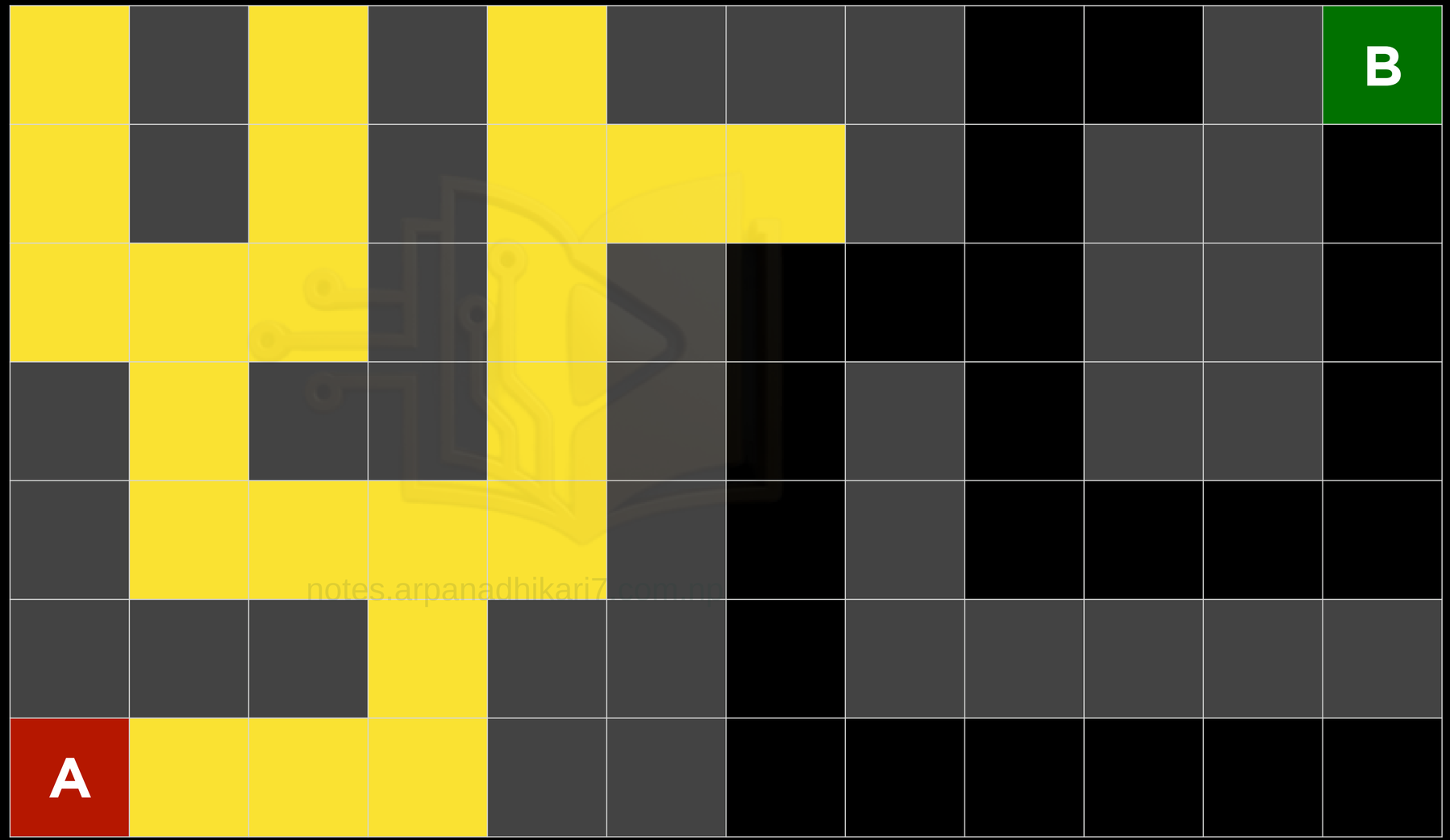
notes.arpanadhikari7.com.np

Depth-First Search



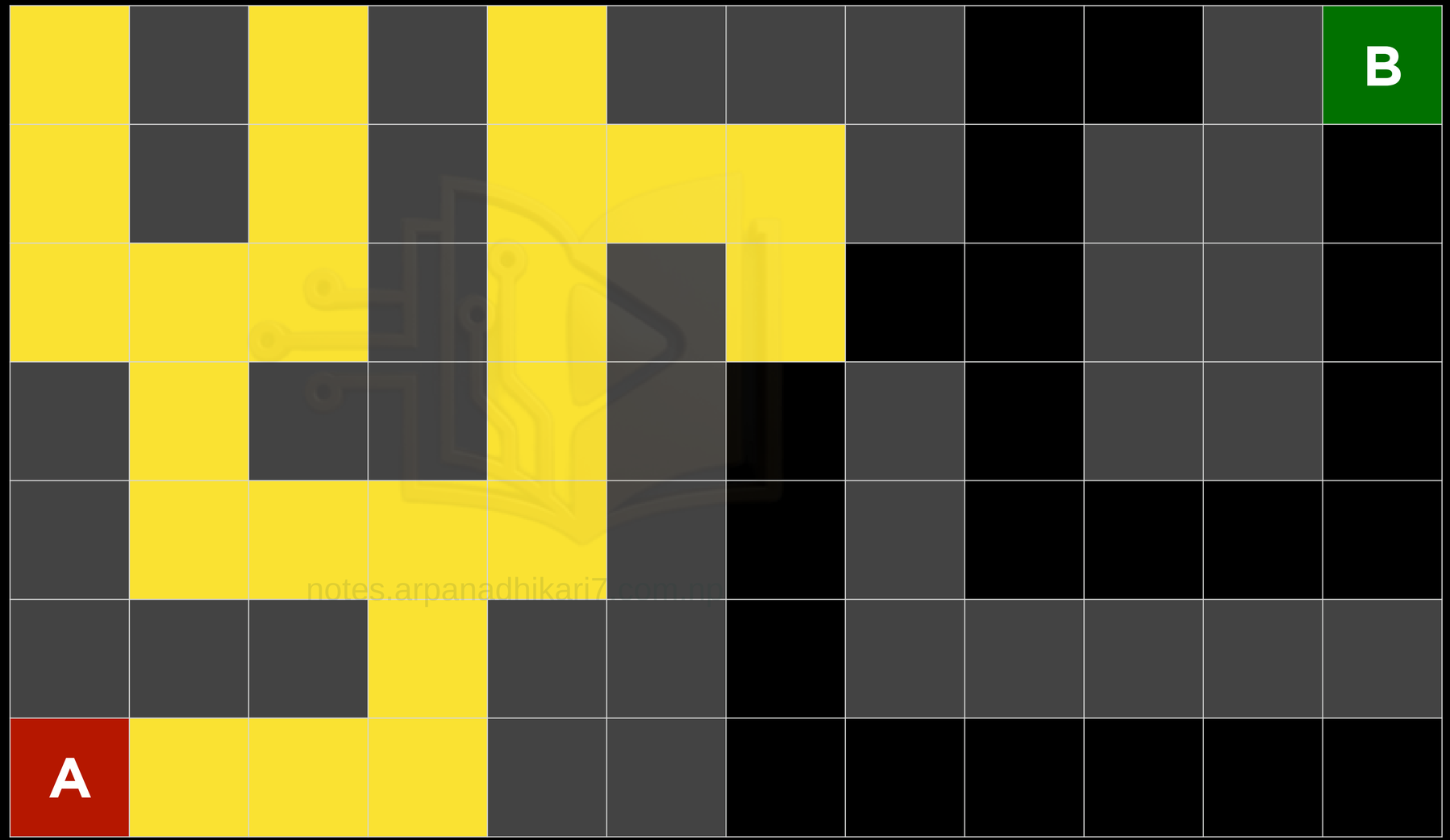
notes.arpanadhikari7.com.np

Depth-First Search



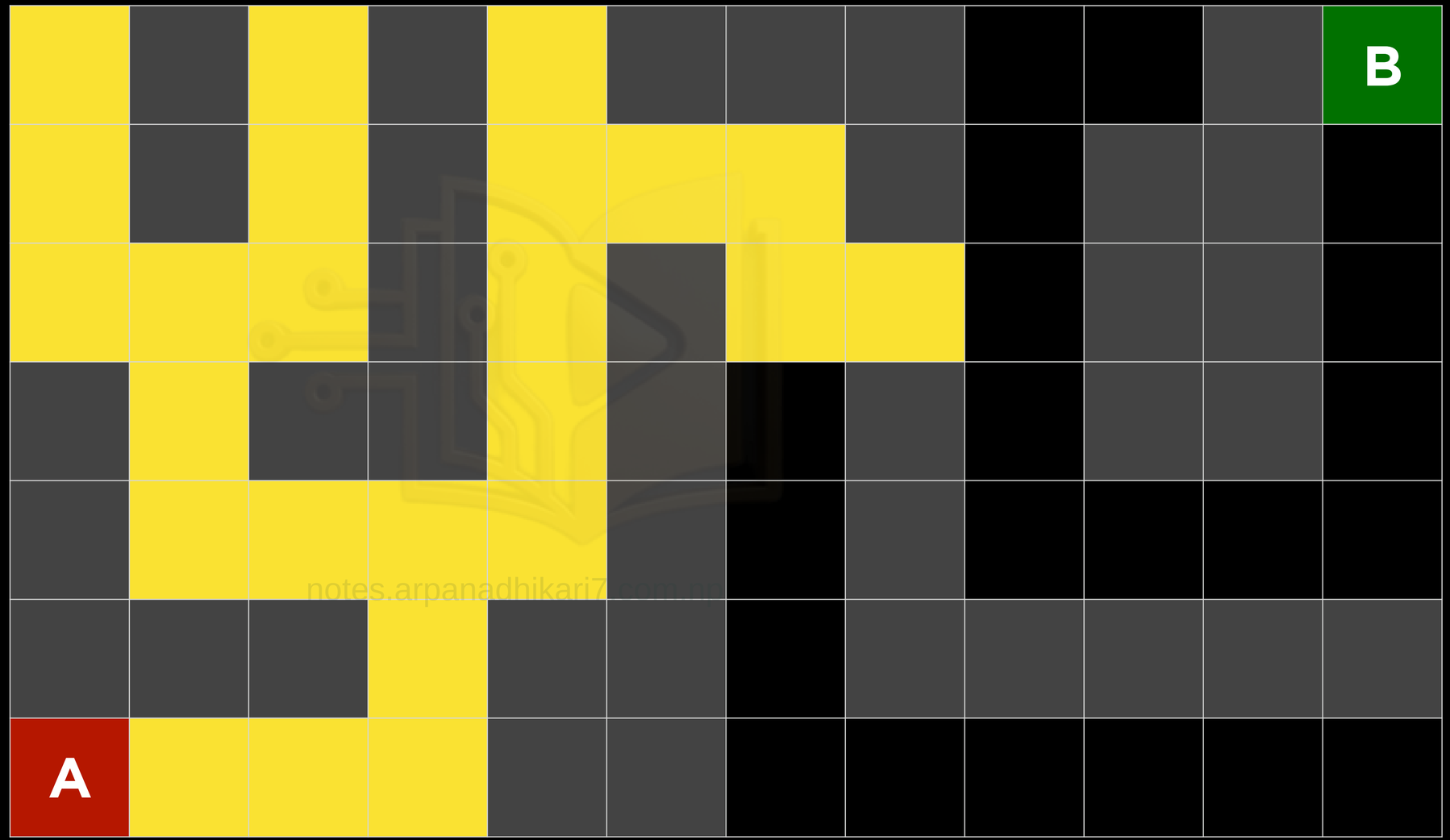
notes.arpanadhikari7.com.np

Depth-First Search



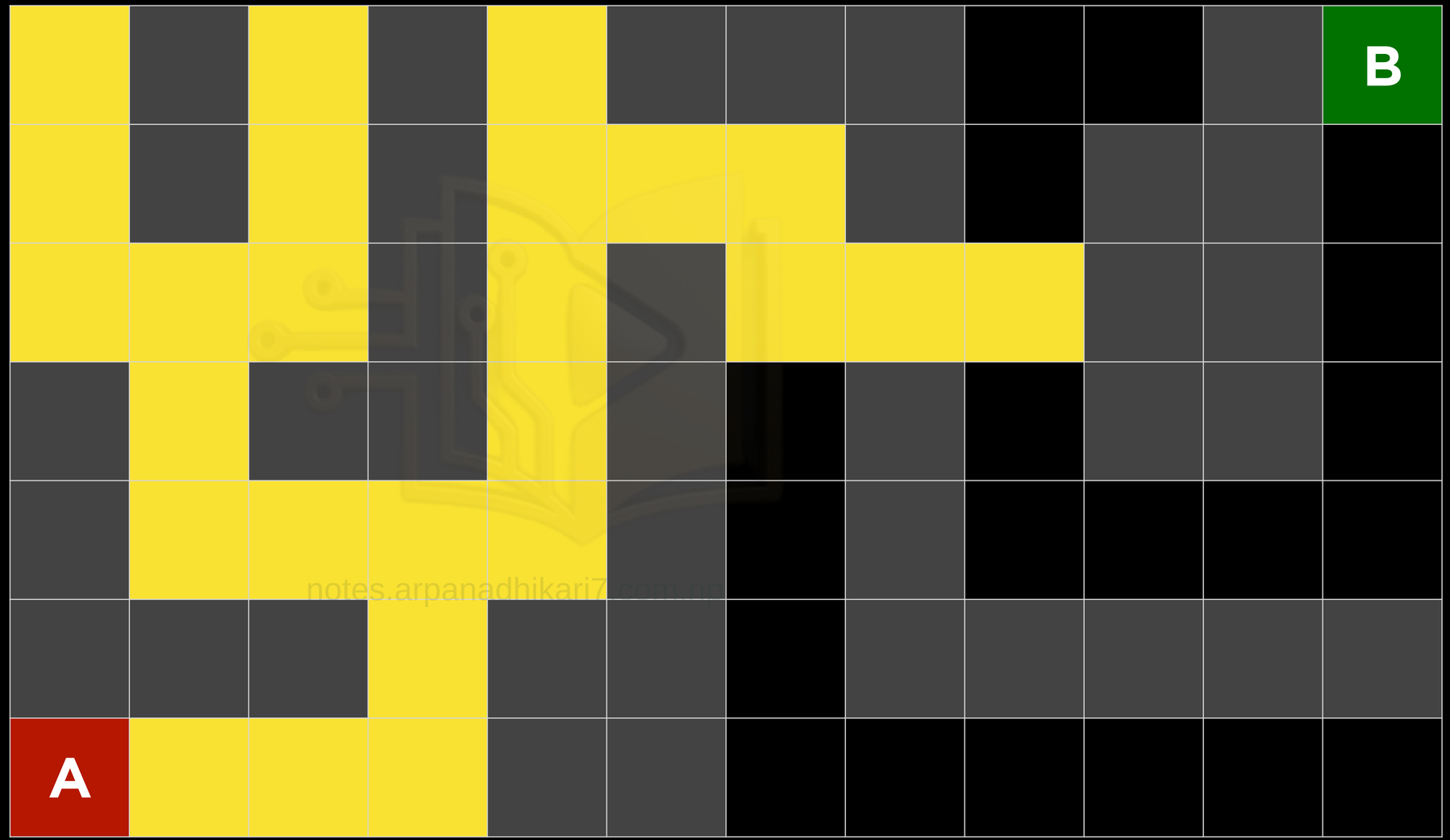
notes.arpanadhikari7.com.np

Depth-First Search



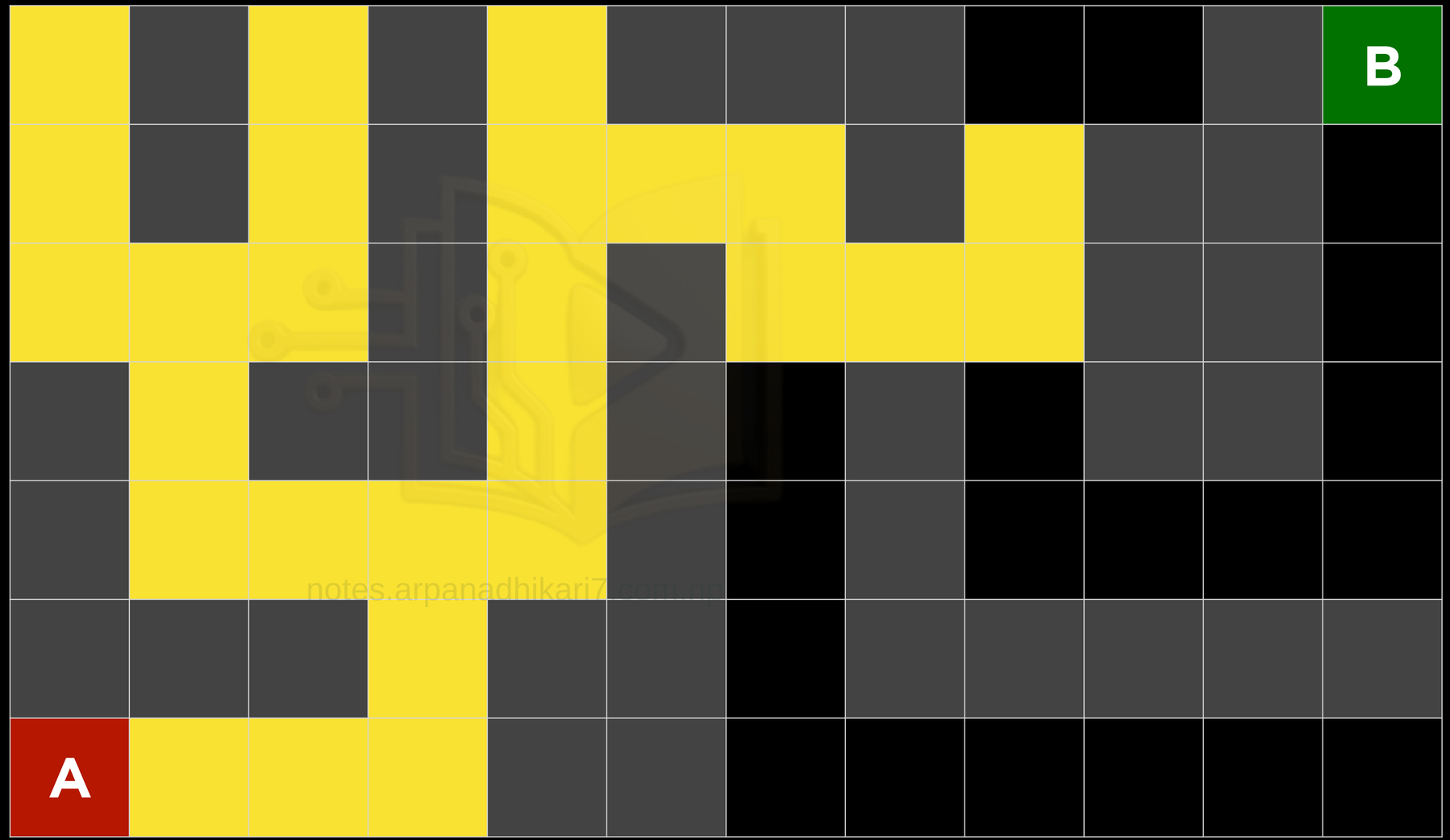
notes.arpanadhikari7.com.np

Depth-First Search



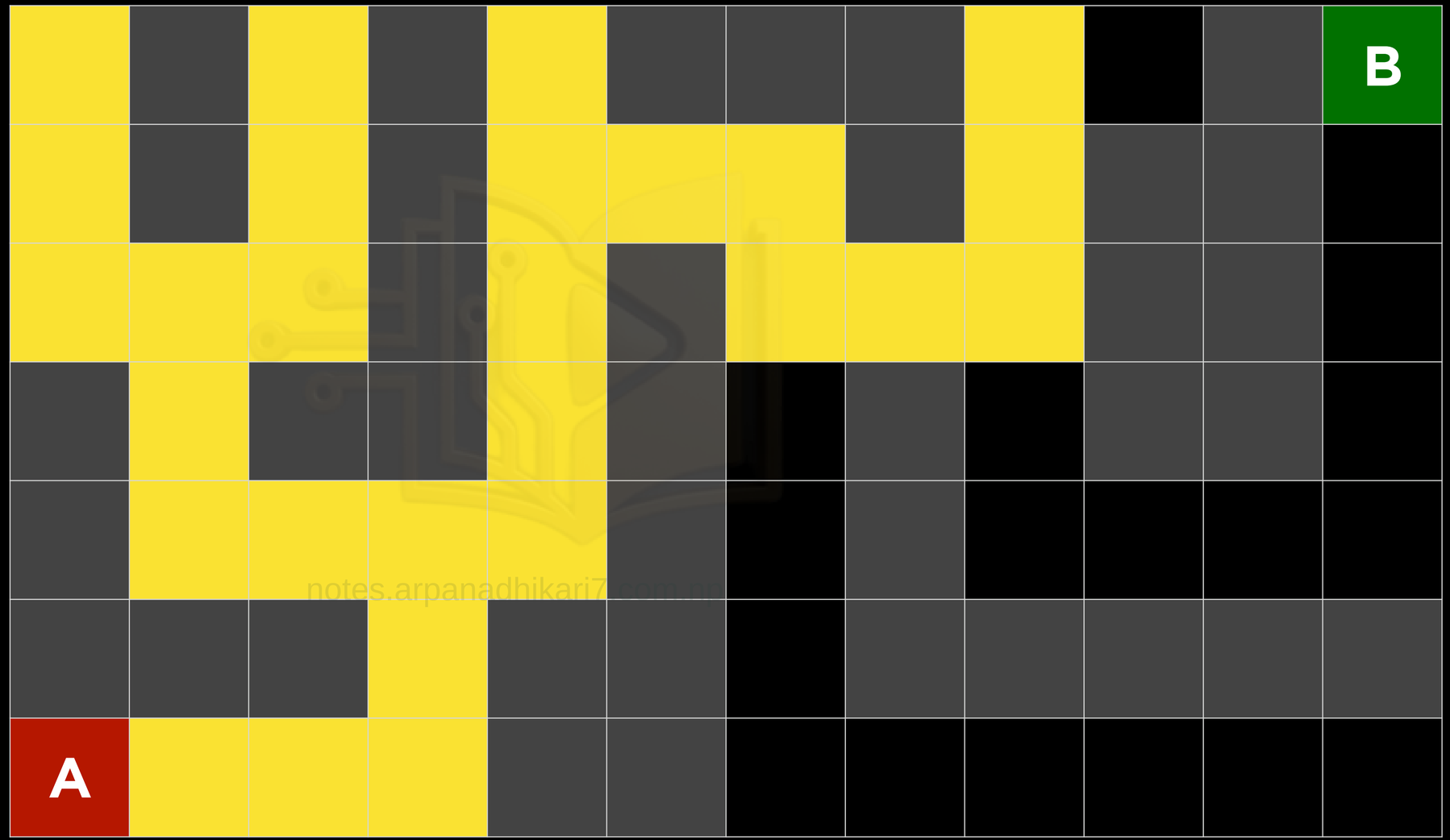
notes.arpanadhikari7.com.np

Depth-First Search



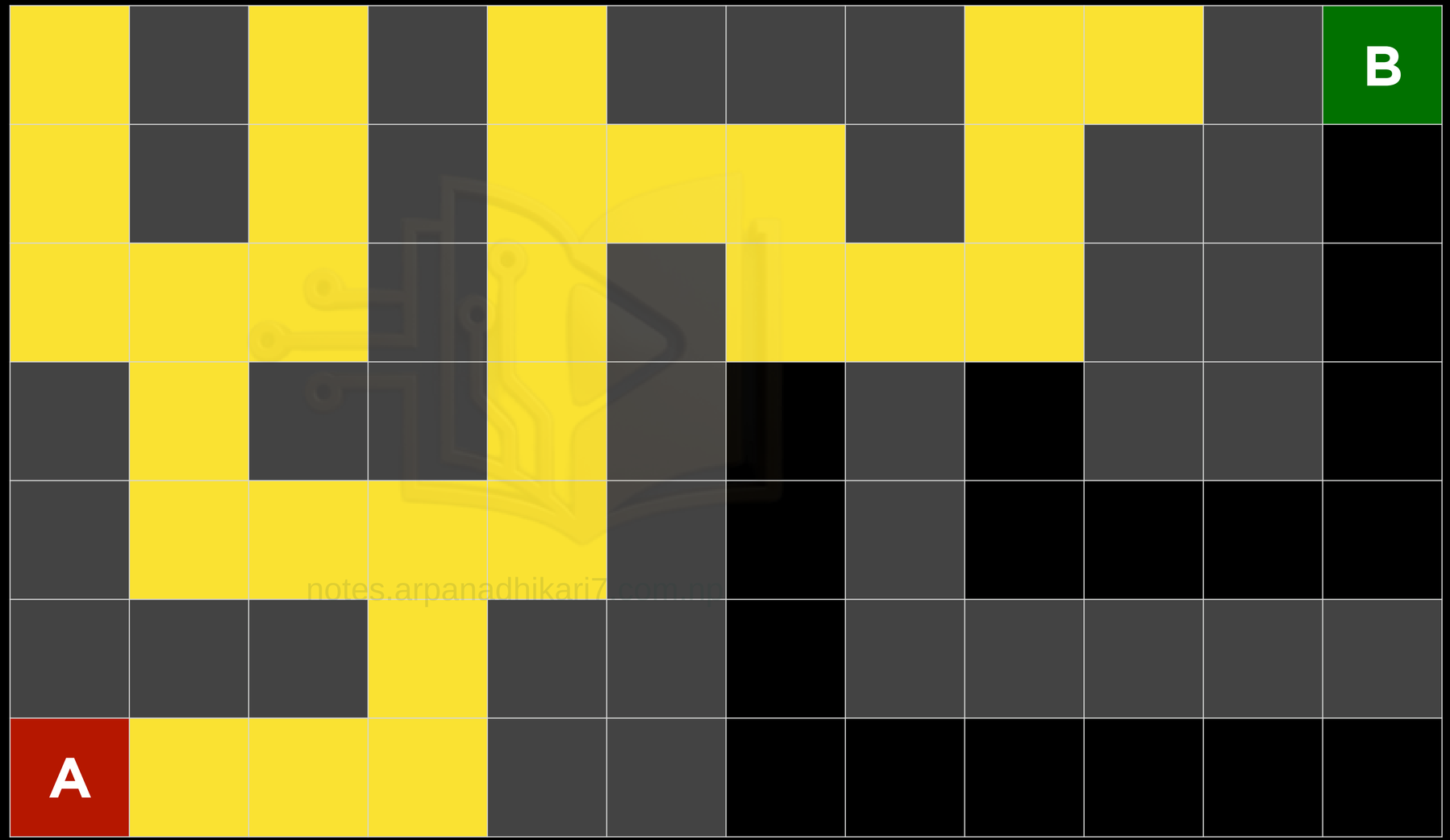
notes.arpanadhikari7.com.np

Depth-First Search



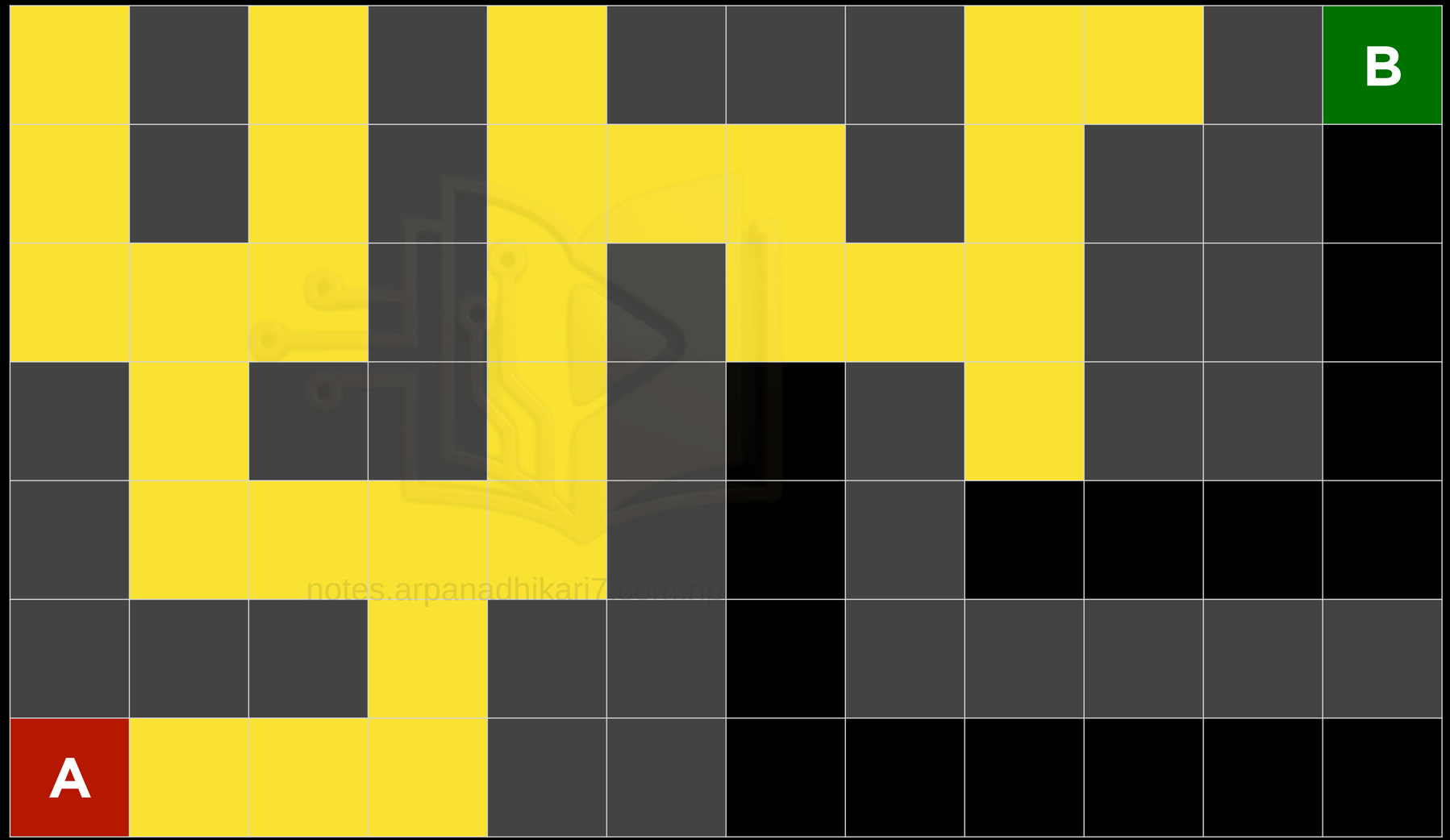
notes.arpanadhikari7.com.np

Depth-First Search



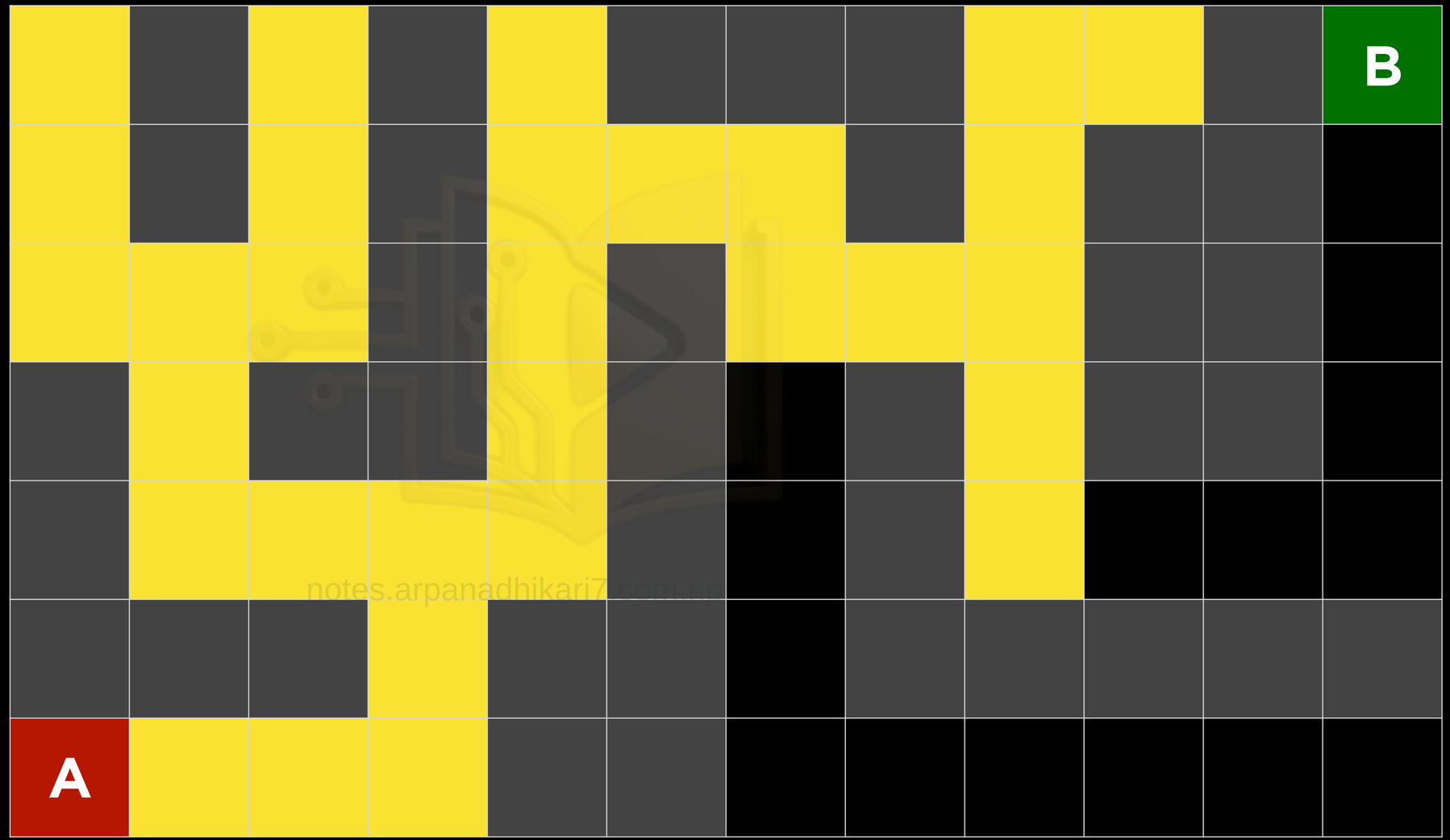
notes.arpanadhikari7.com.np

Depth-First Search



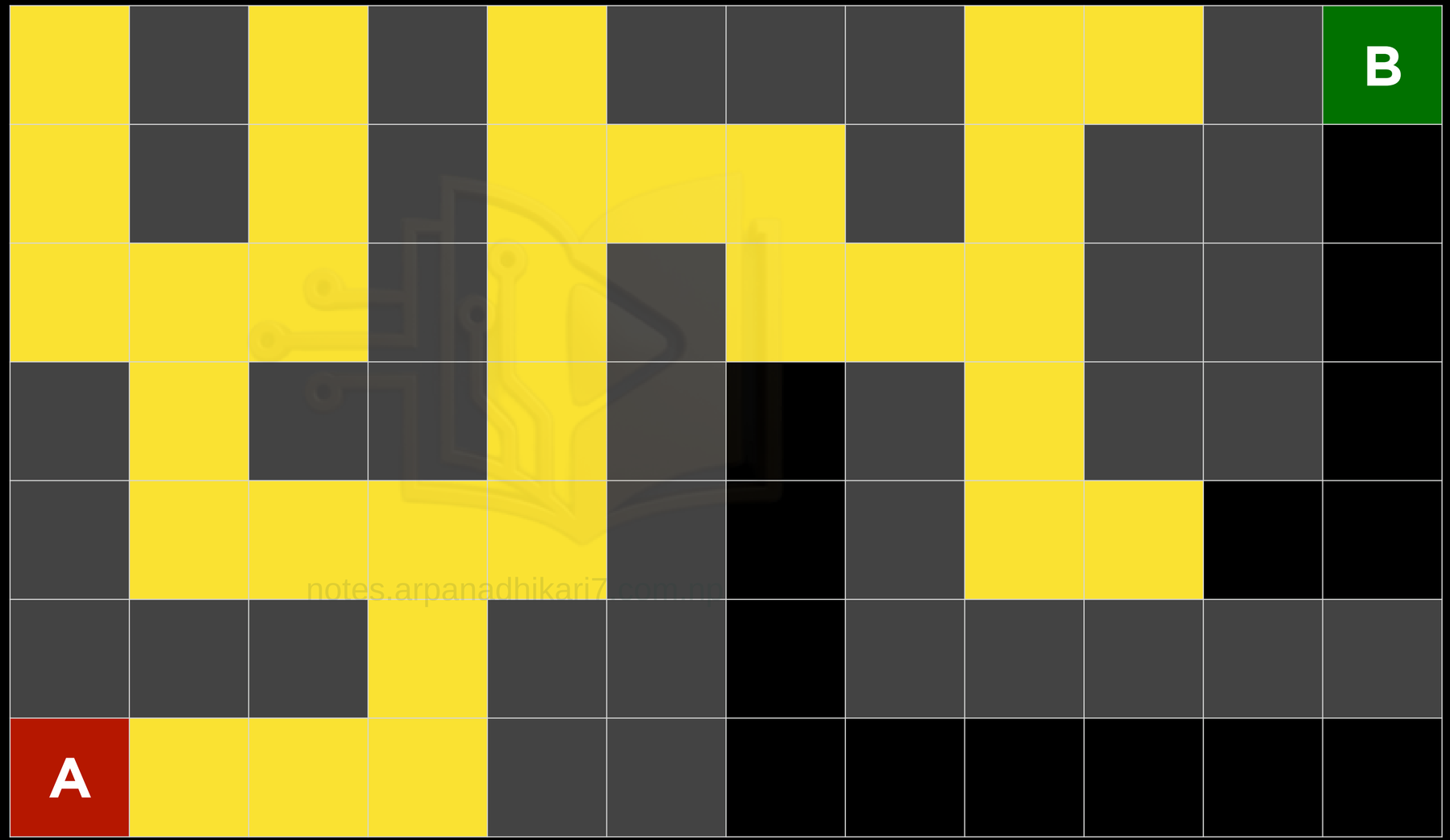
notes.arpanadhikari7.com.np

Depth-First Search



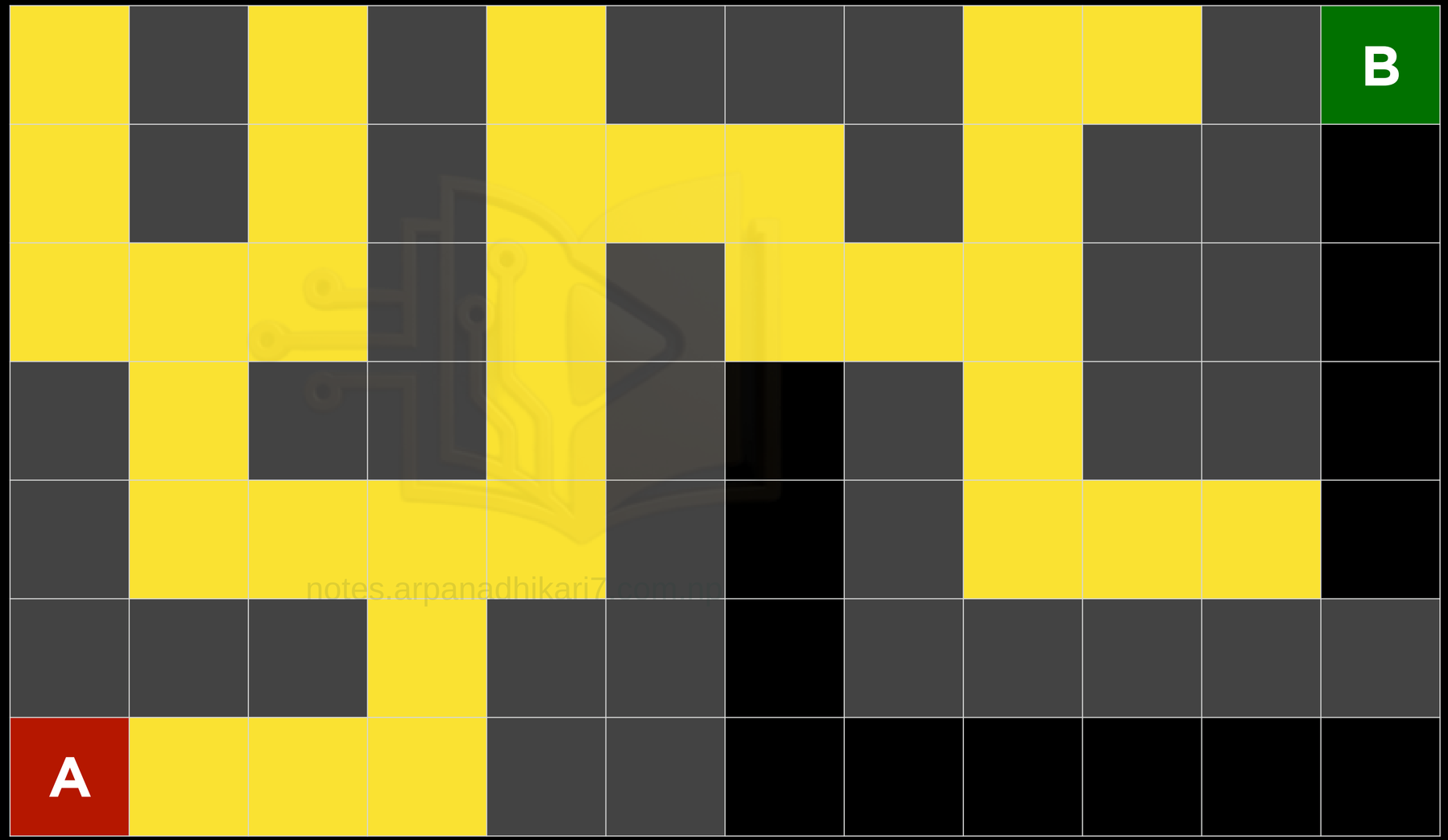
notes.arpanadhikari7.com.np

Depth-First Search



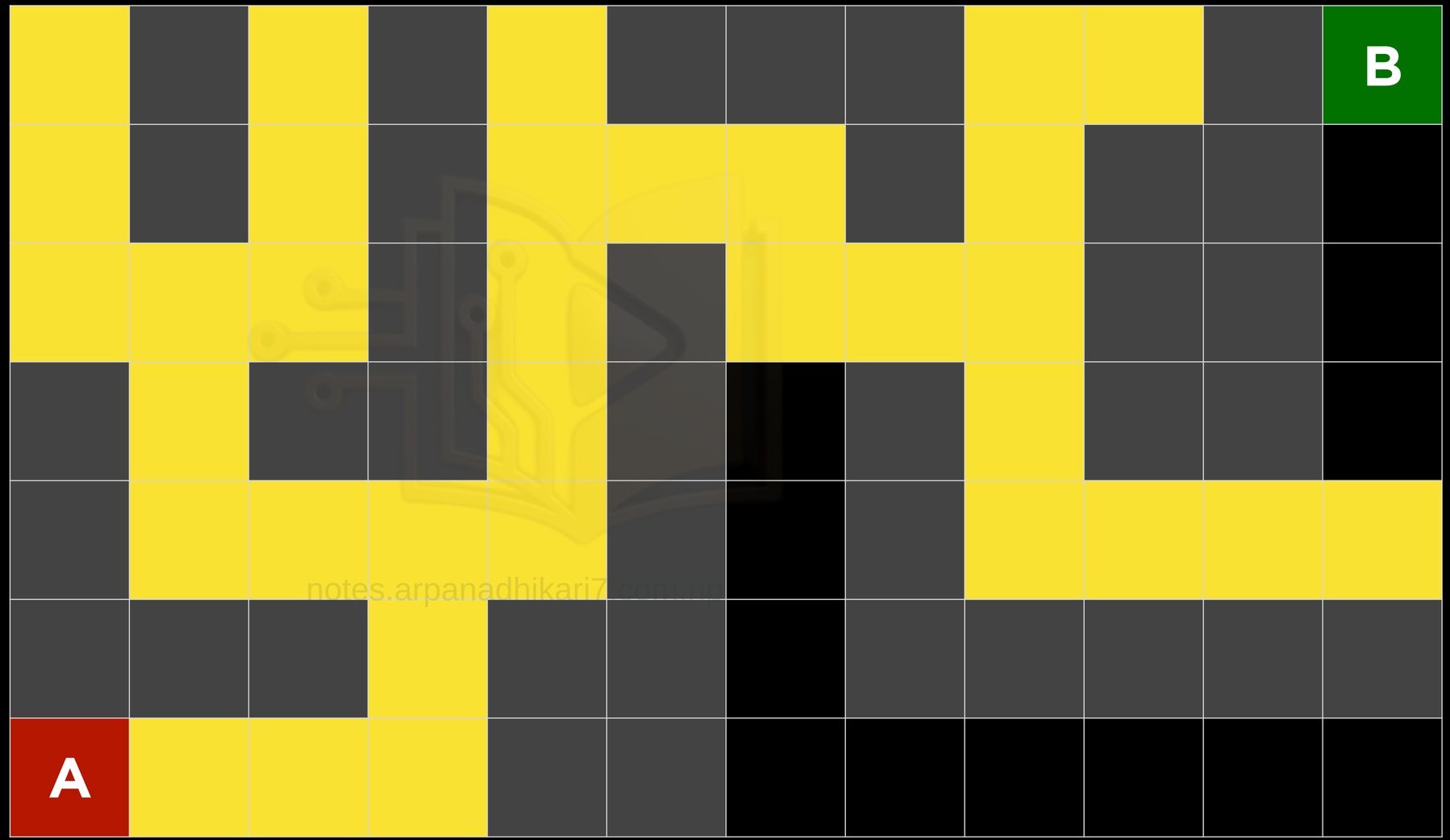
notes.arpanadhikari7.com.np

Depth-First Search



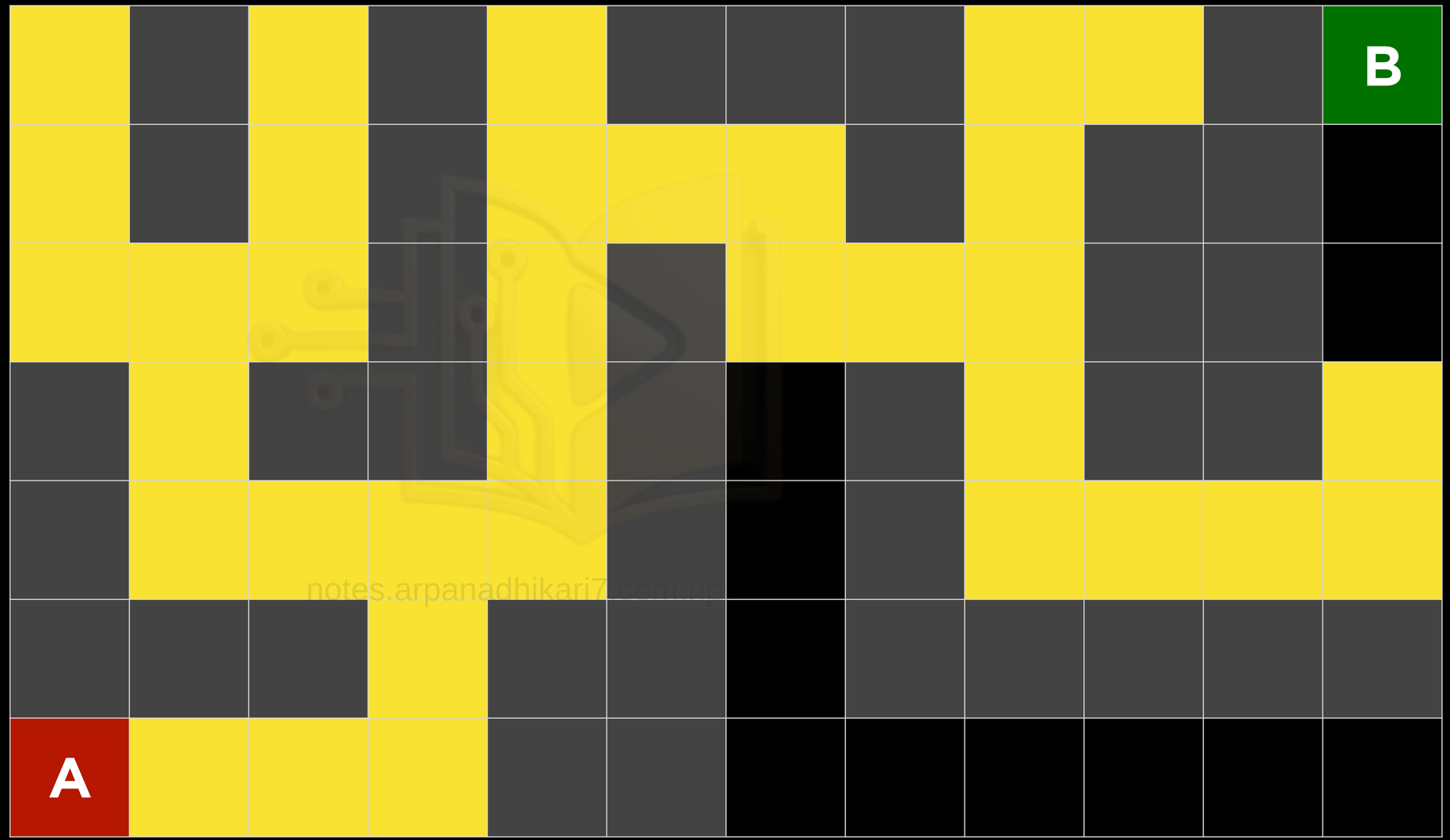
notes.arpanadhikari7.com.np

Depth-First Search



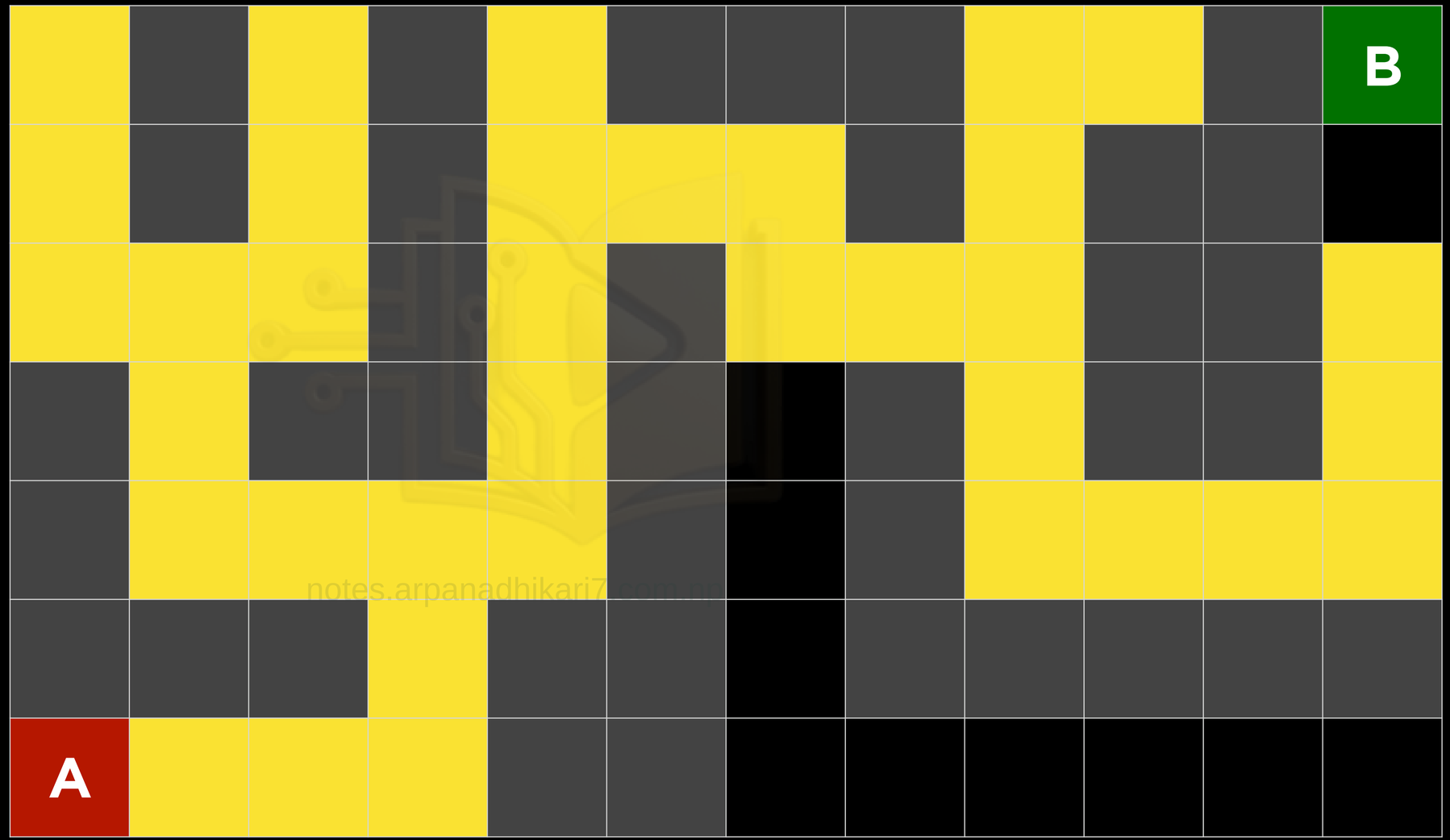
notes.arpanadhikari7.com.np

Depth-First Search



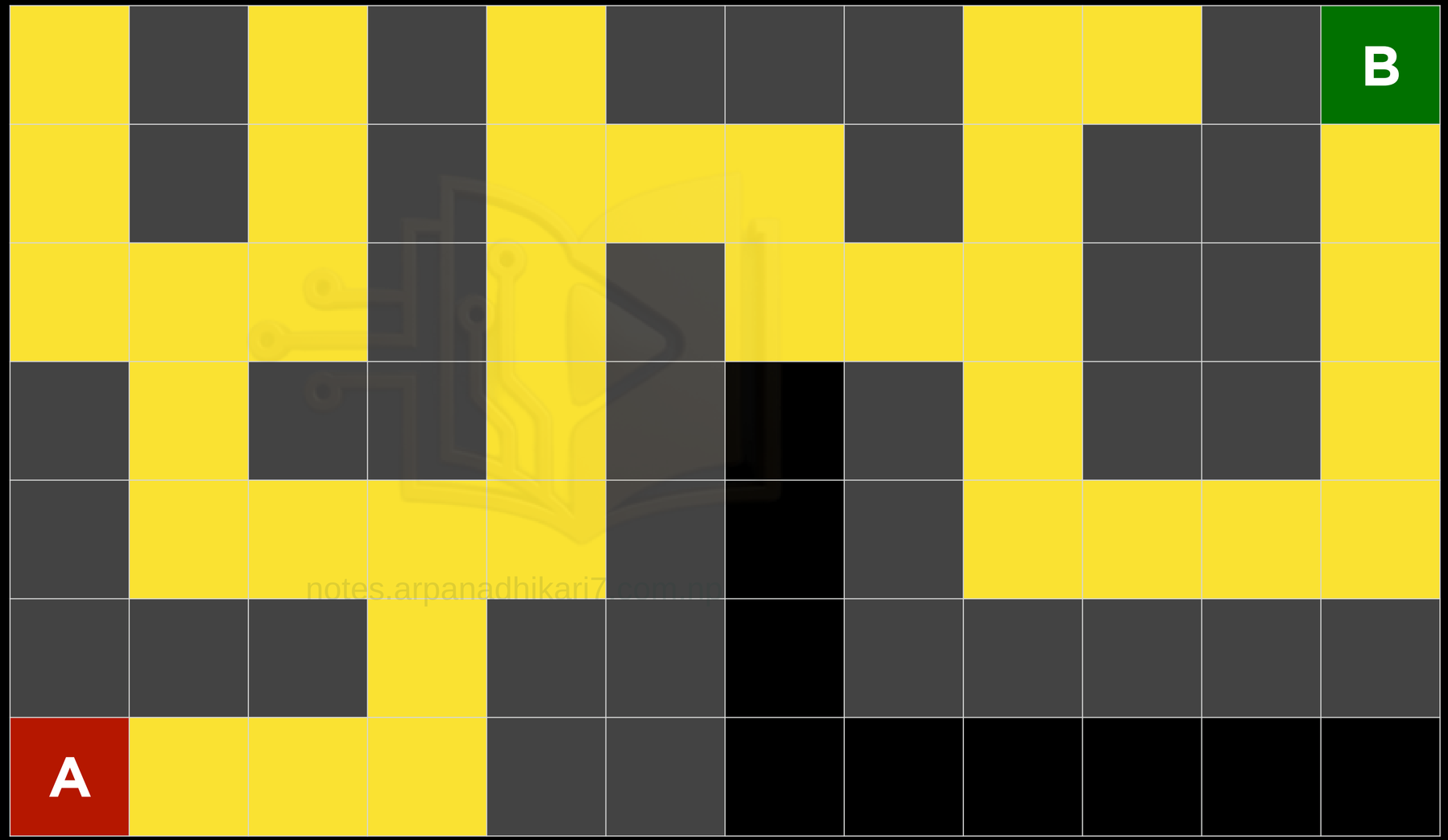
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Depth-First Search



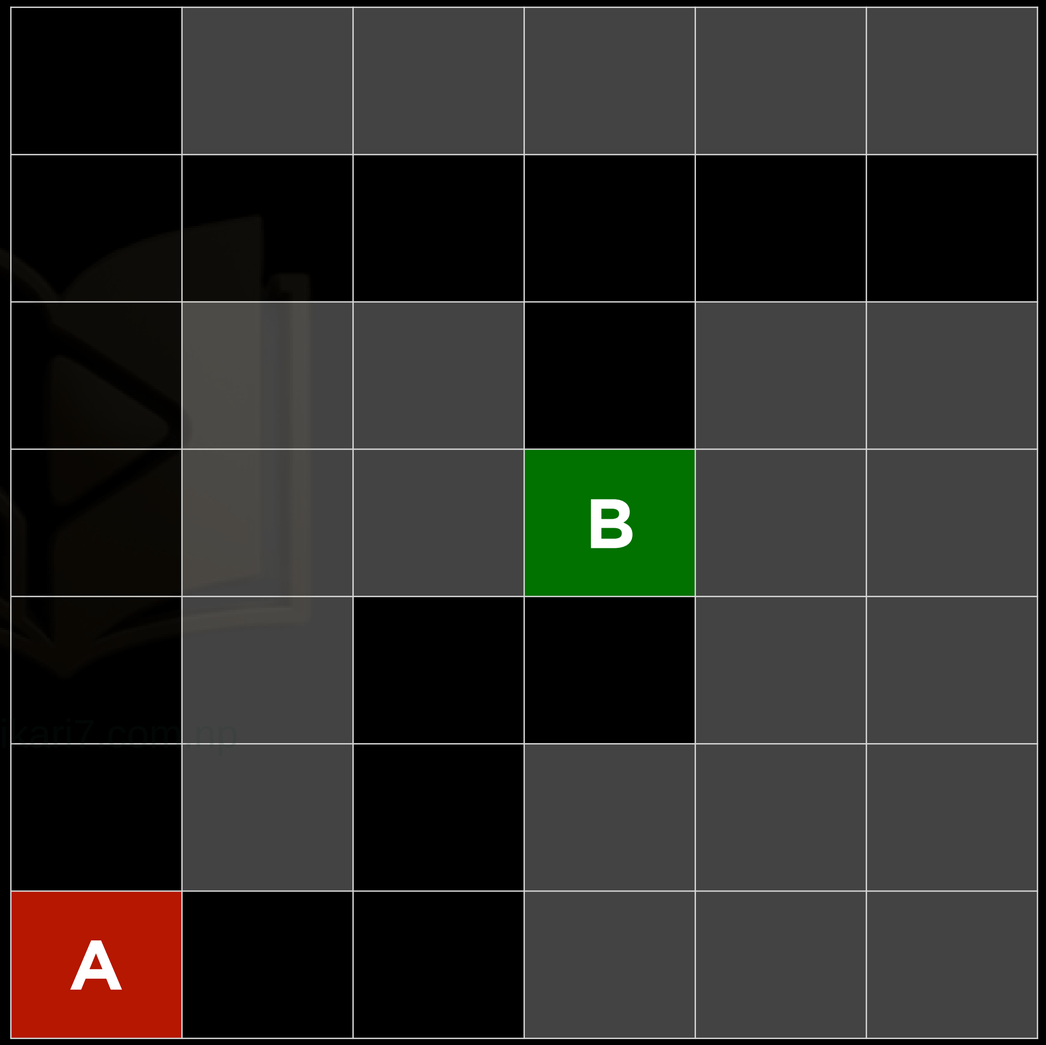
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Depth-First Search



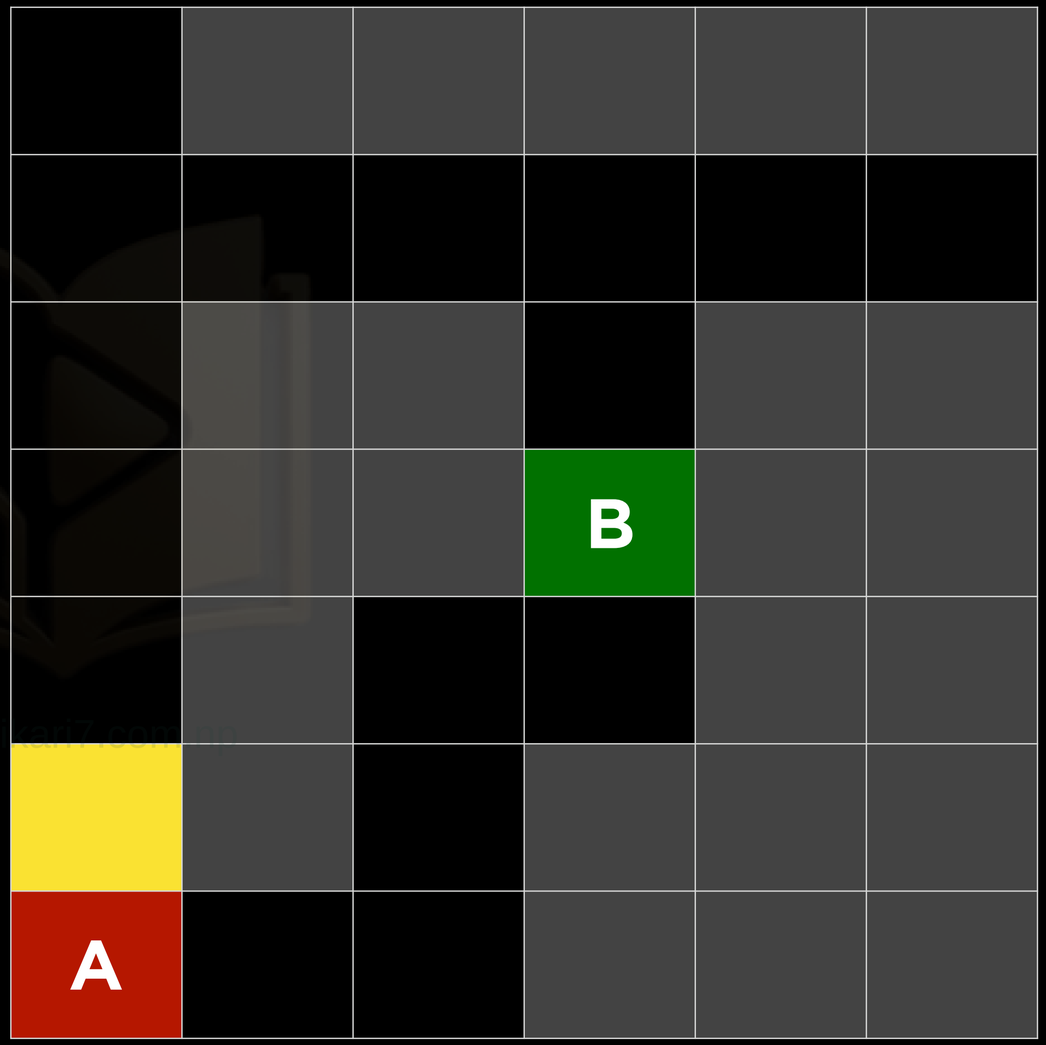
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Depth-First Search



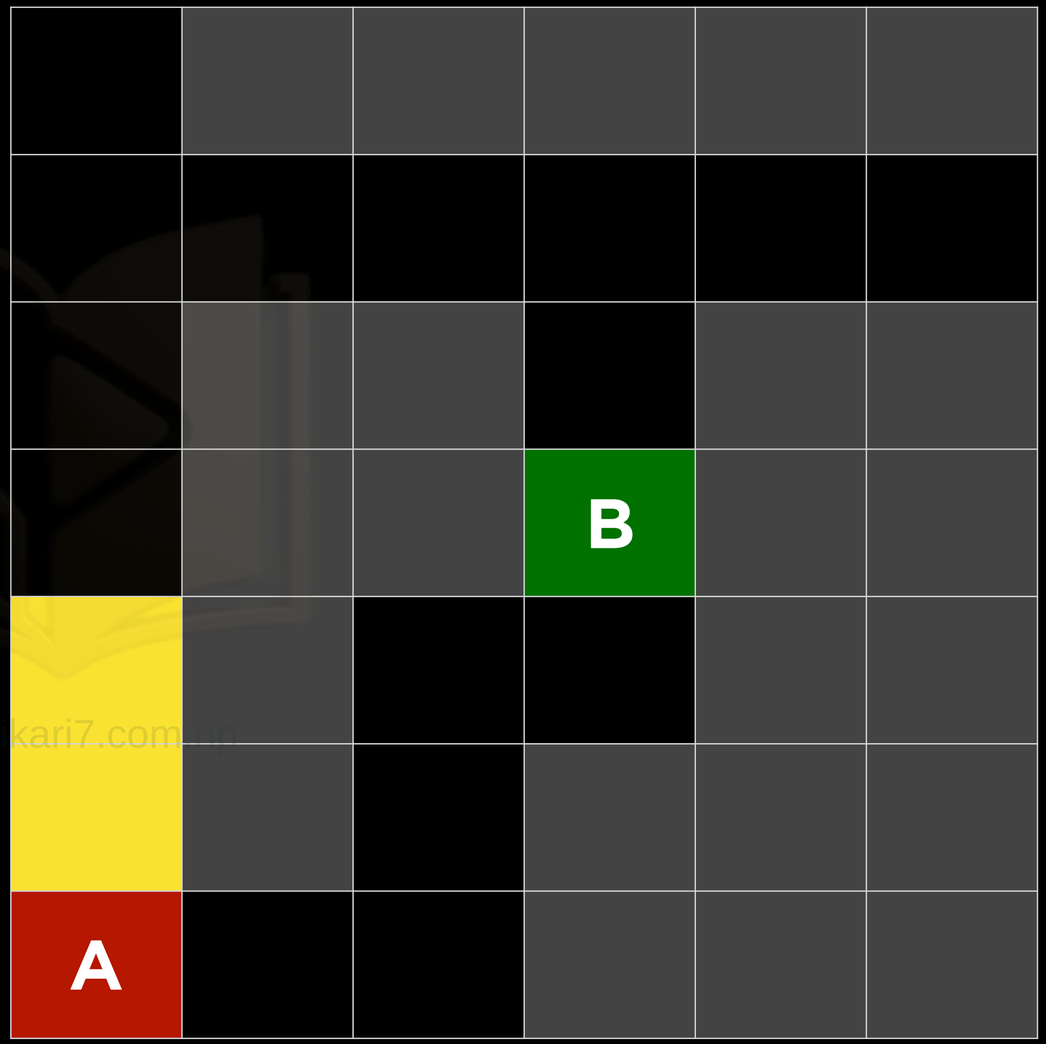
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Depth-First Search



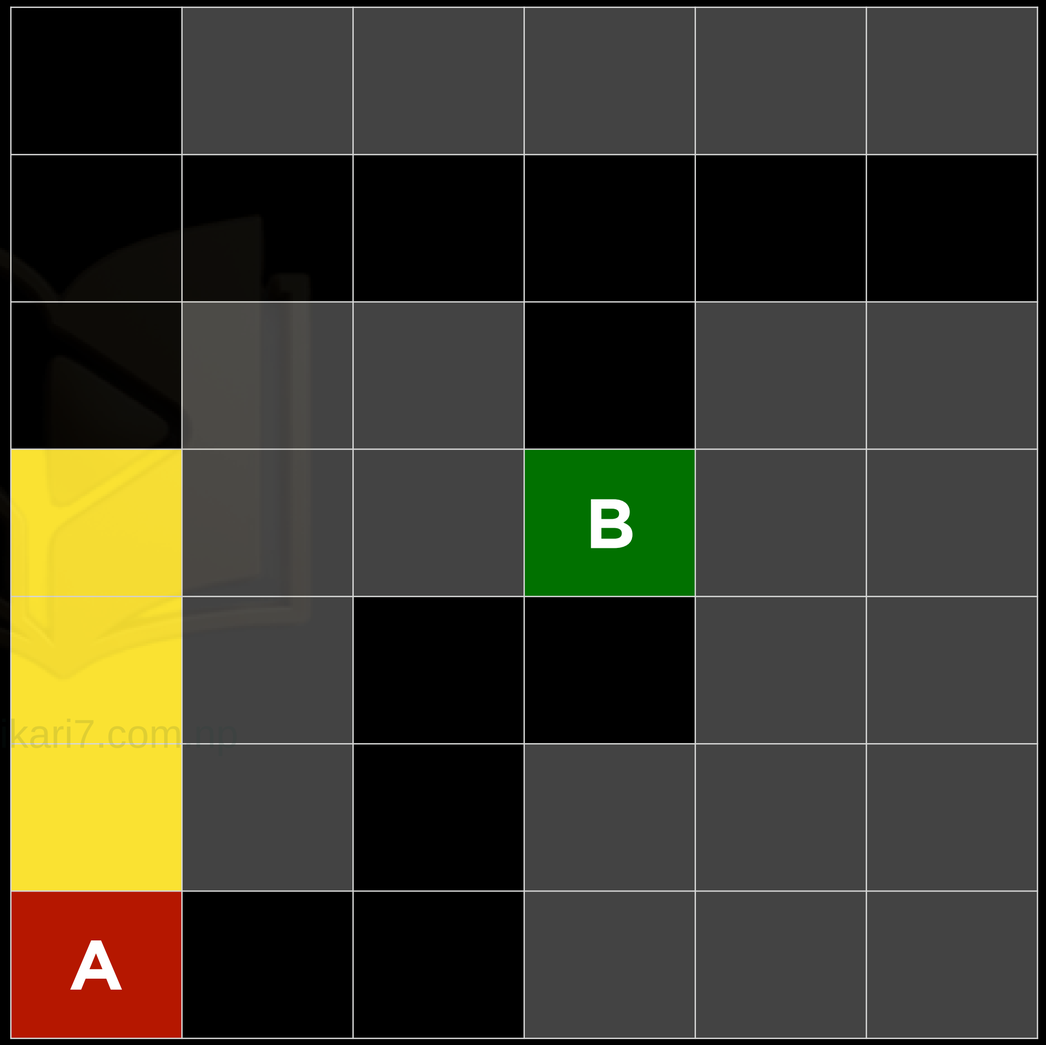
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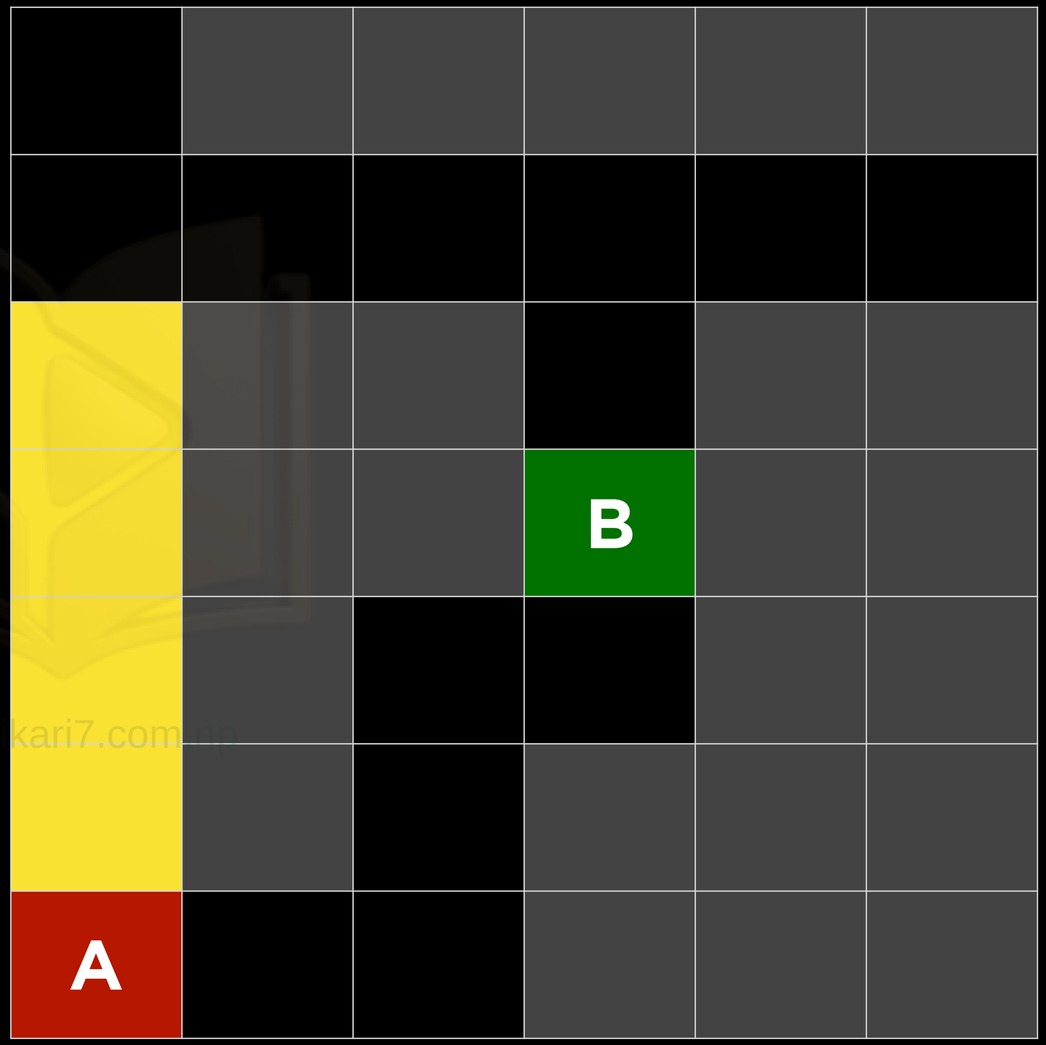
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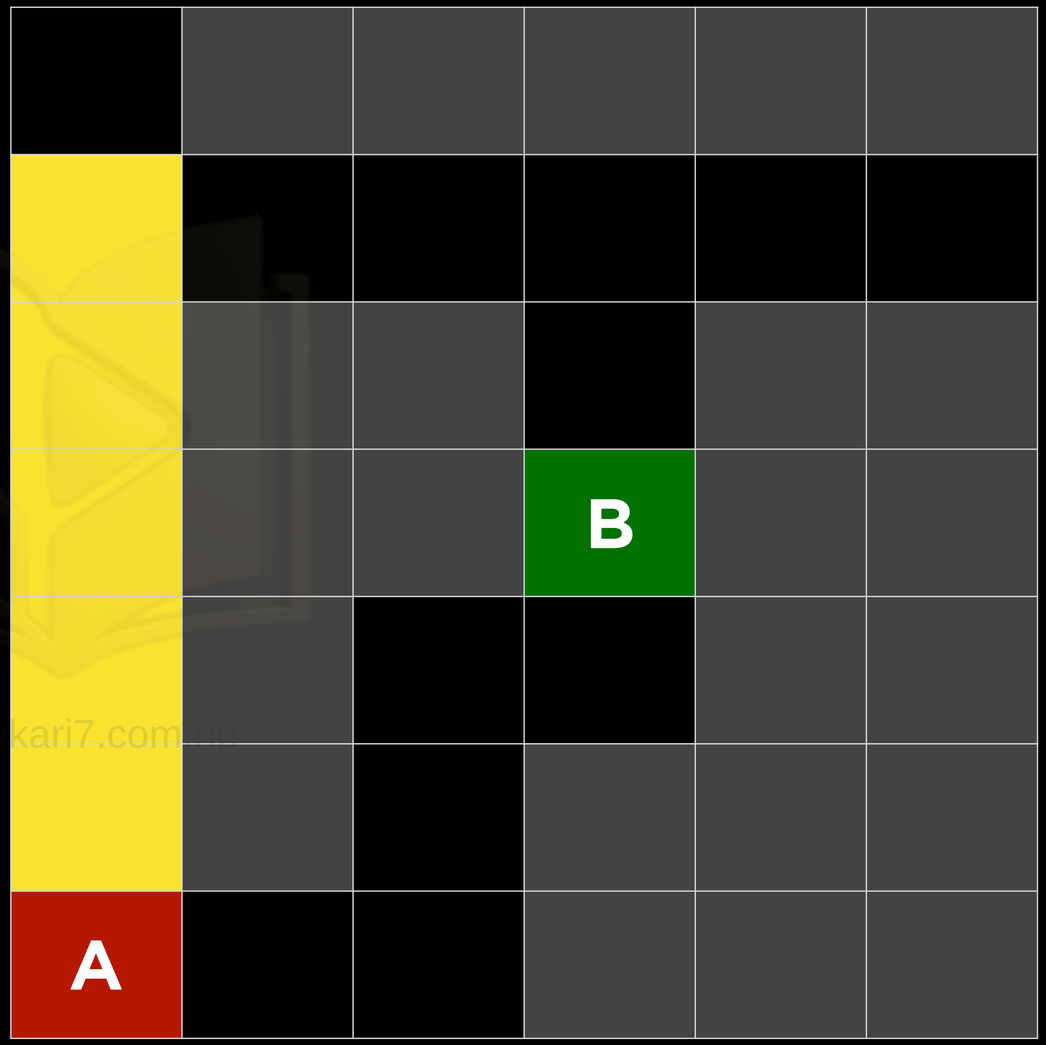
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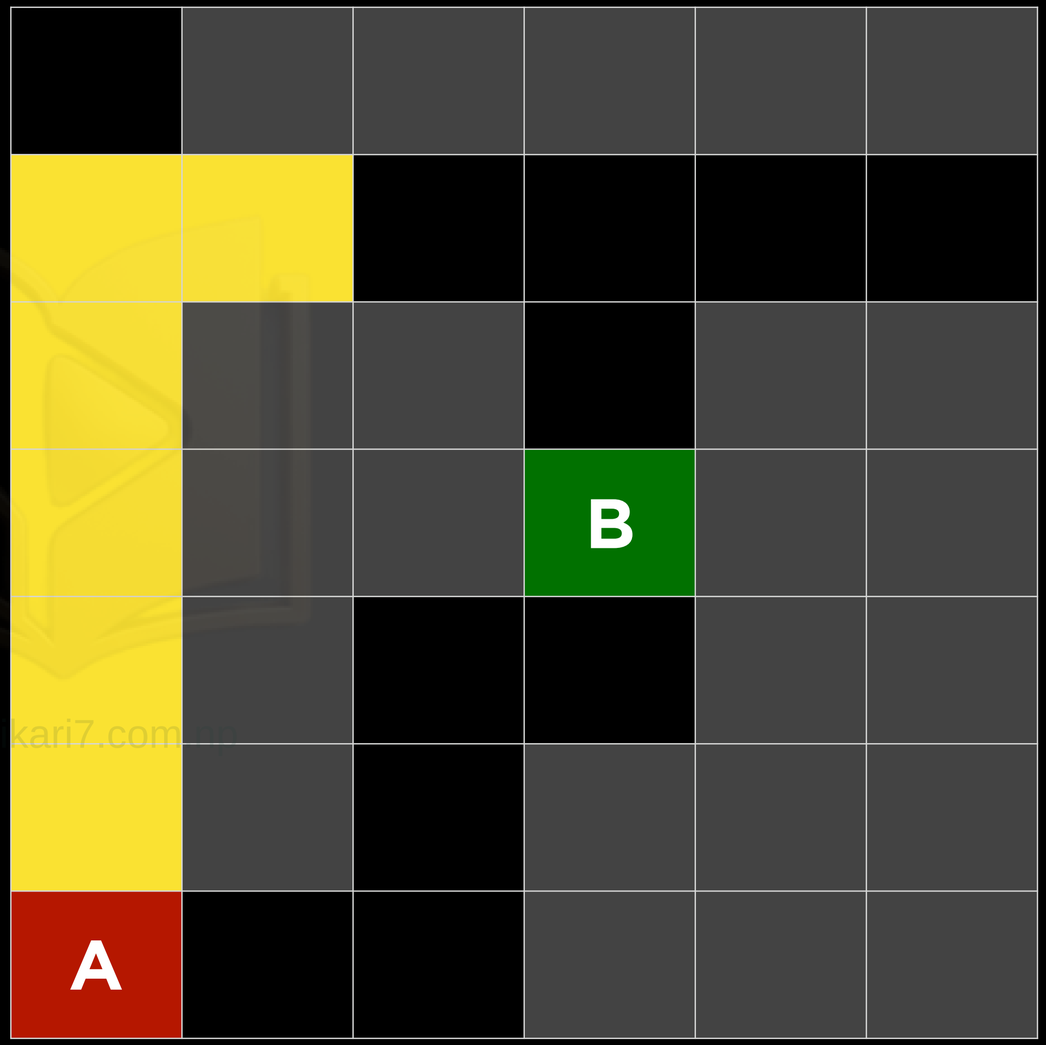
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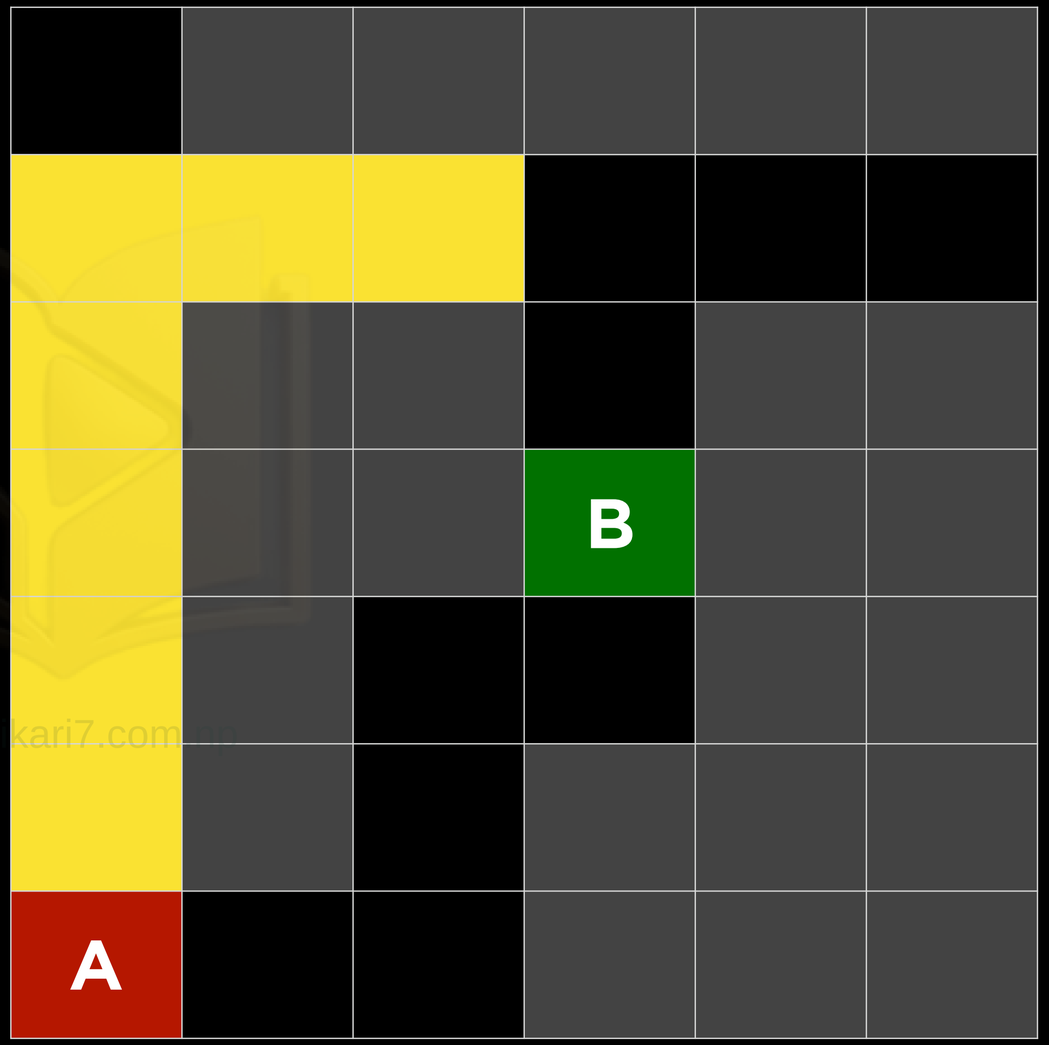
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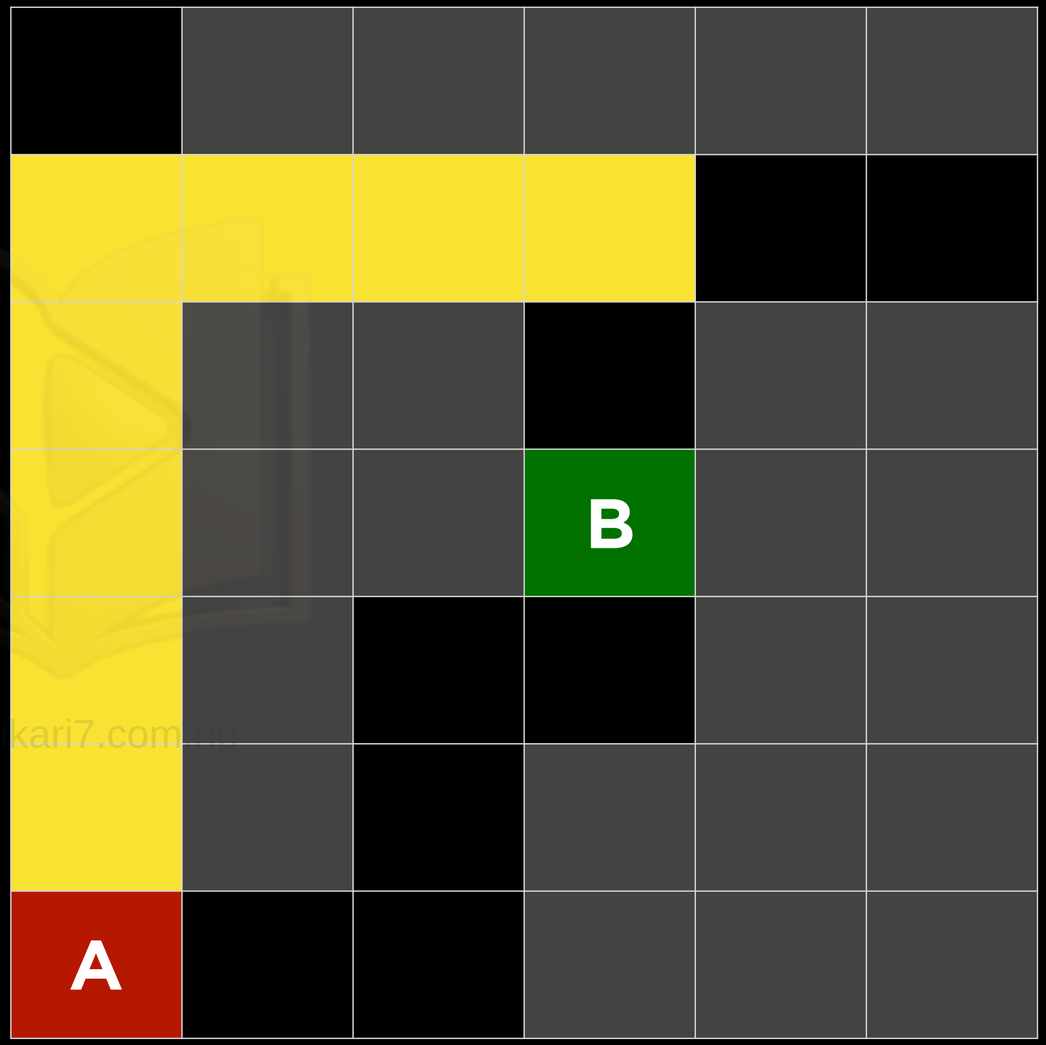
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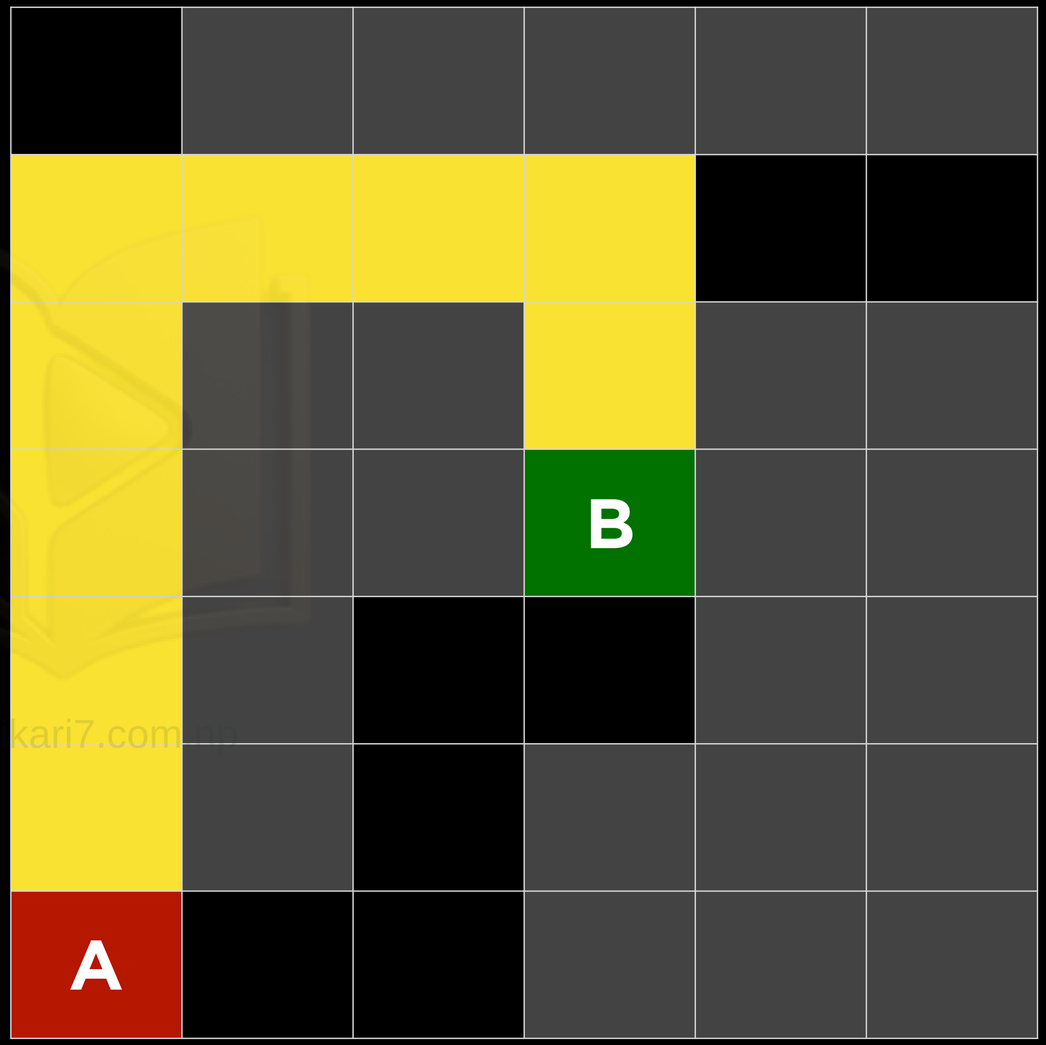
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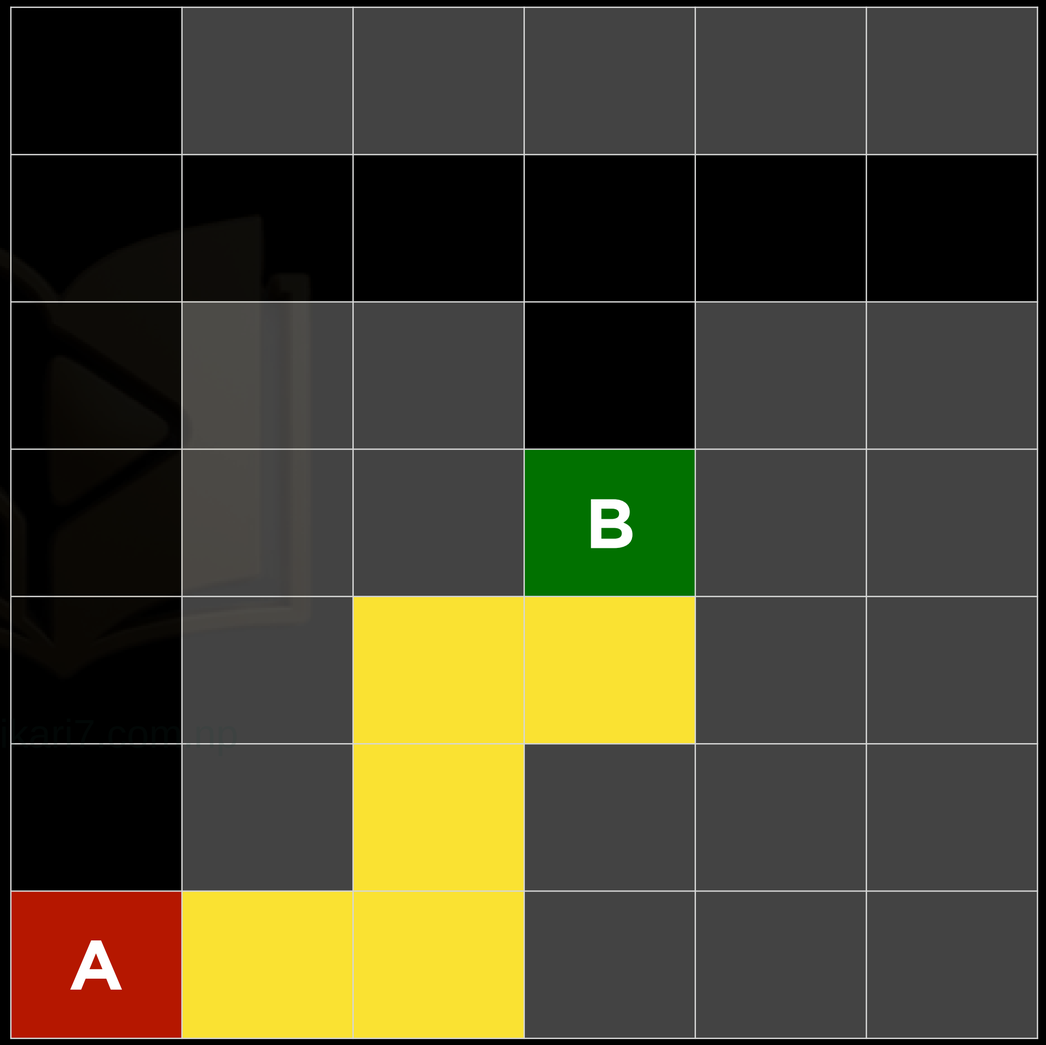
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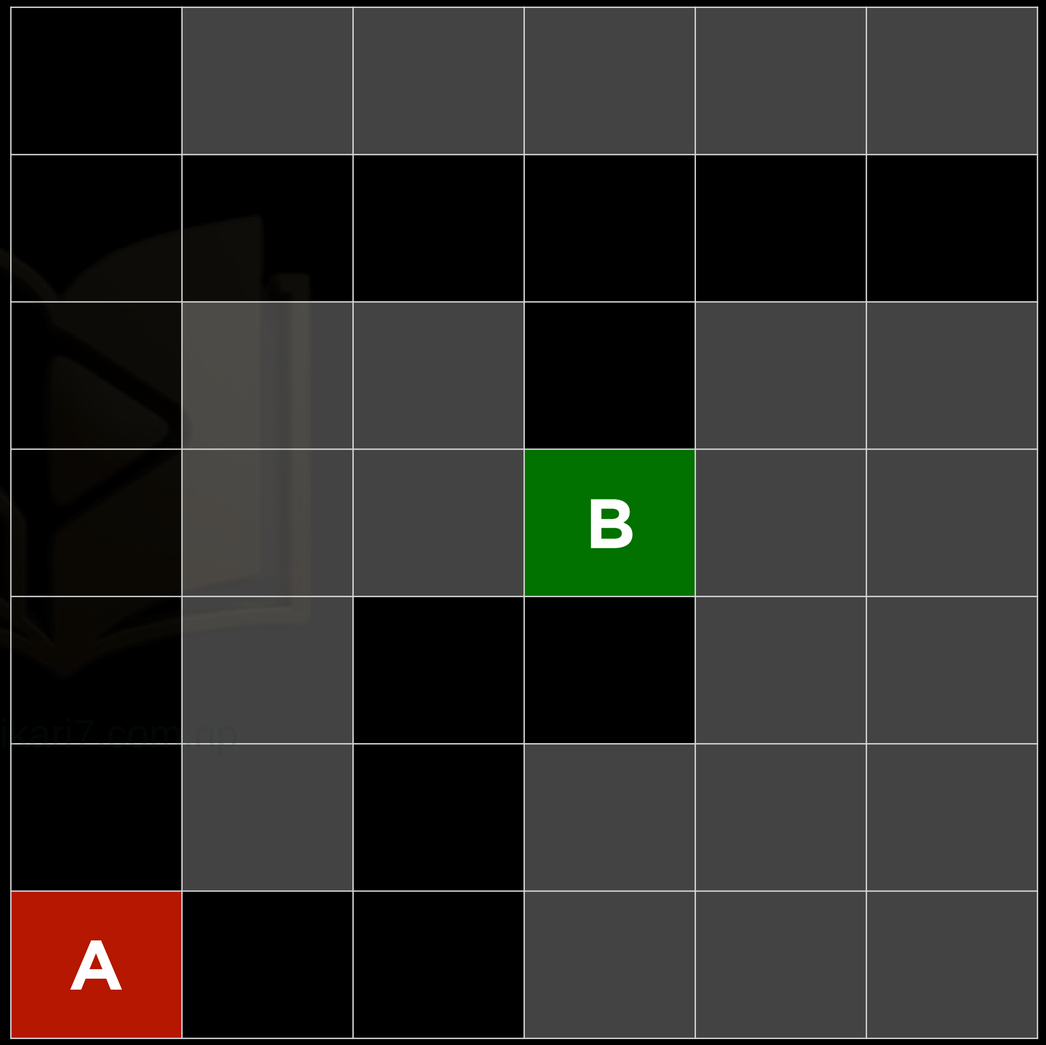
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Depth-First Search



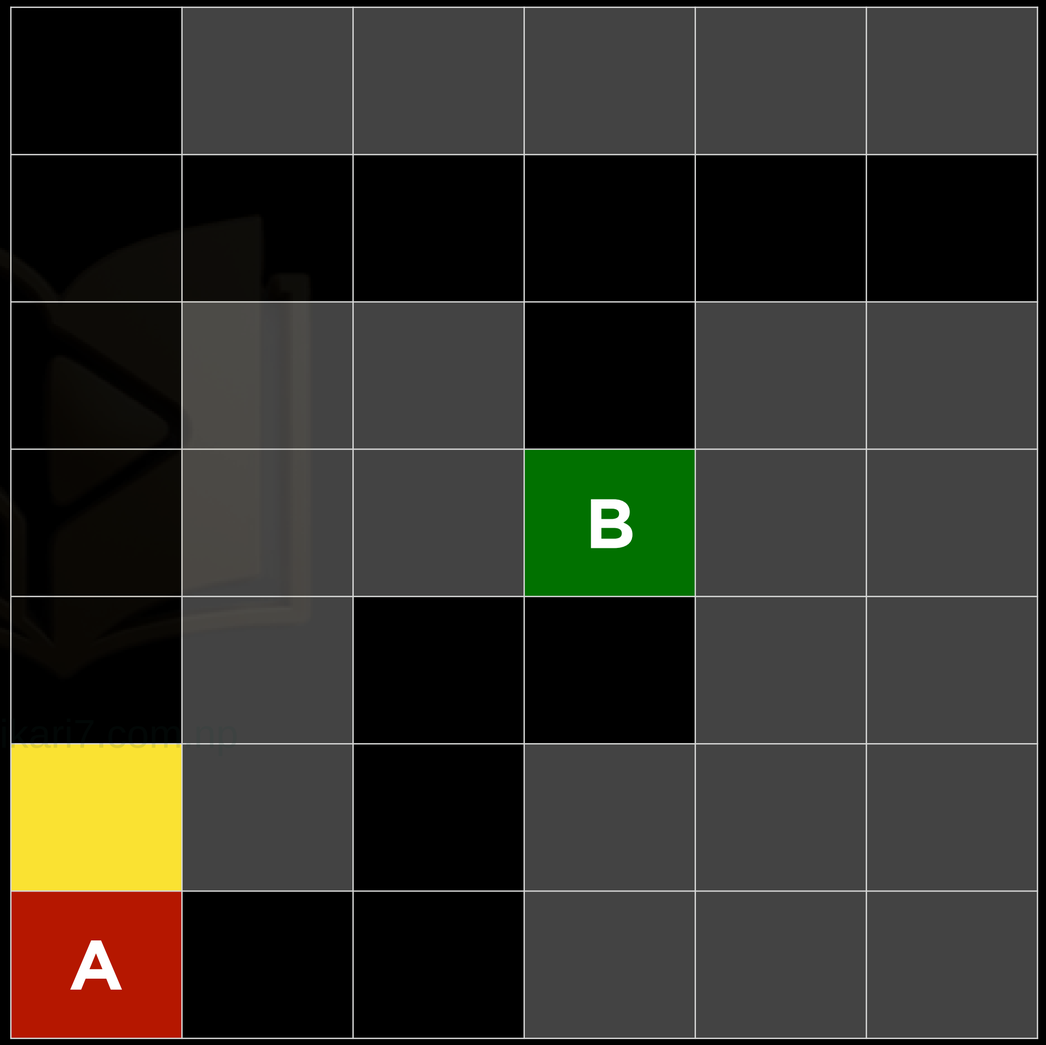
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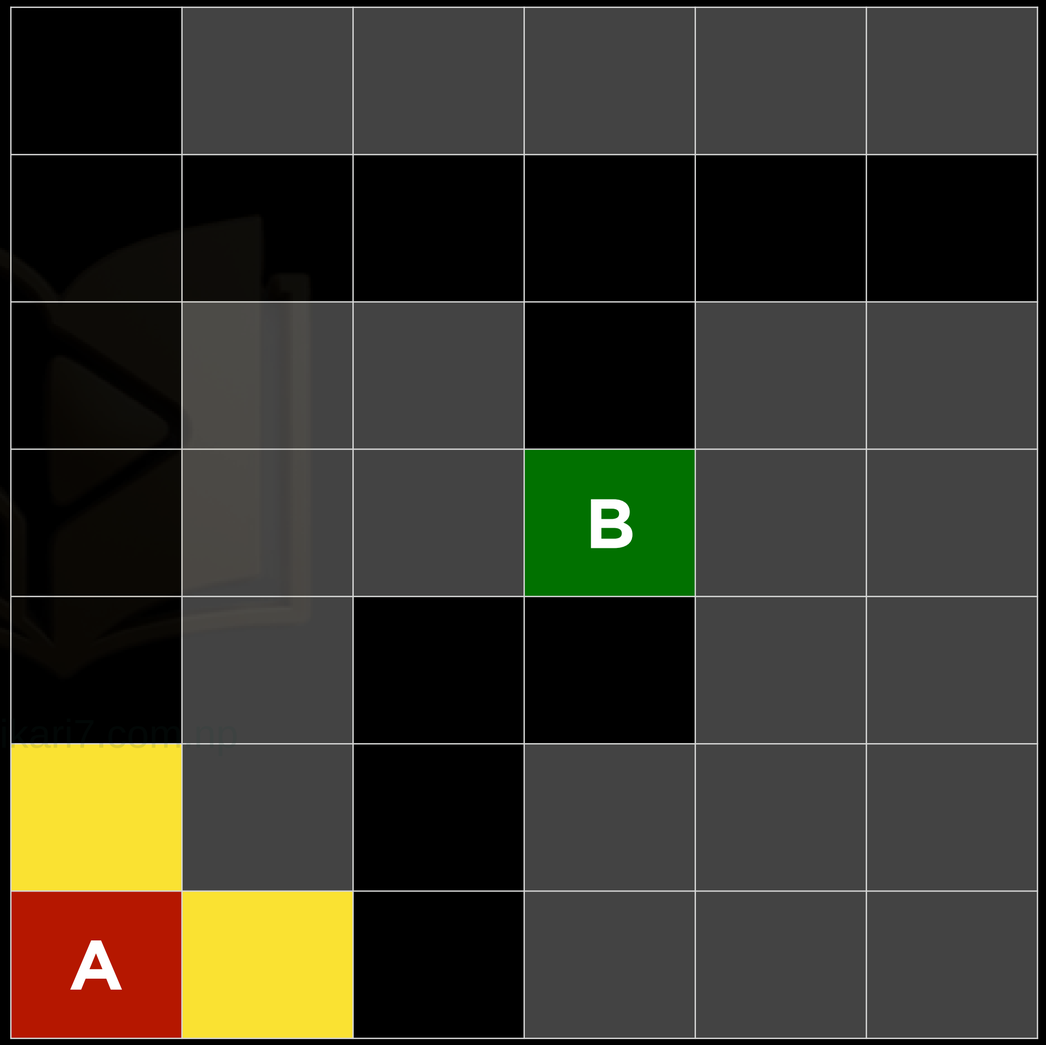
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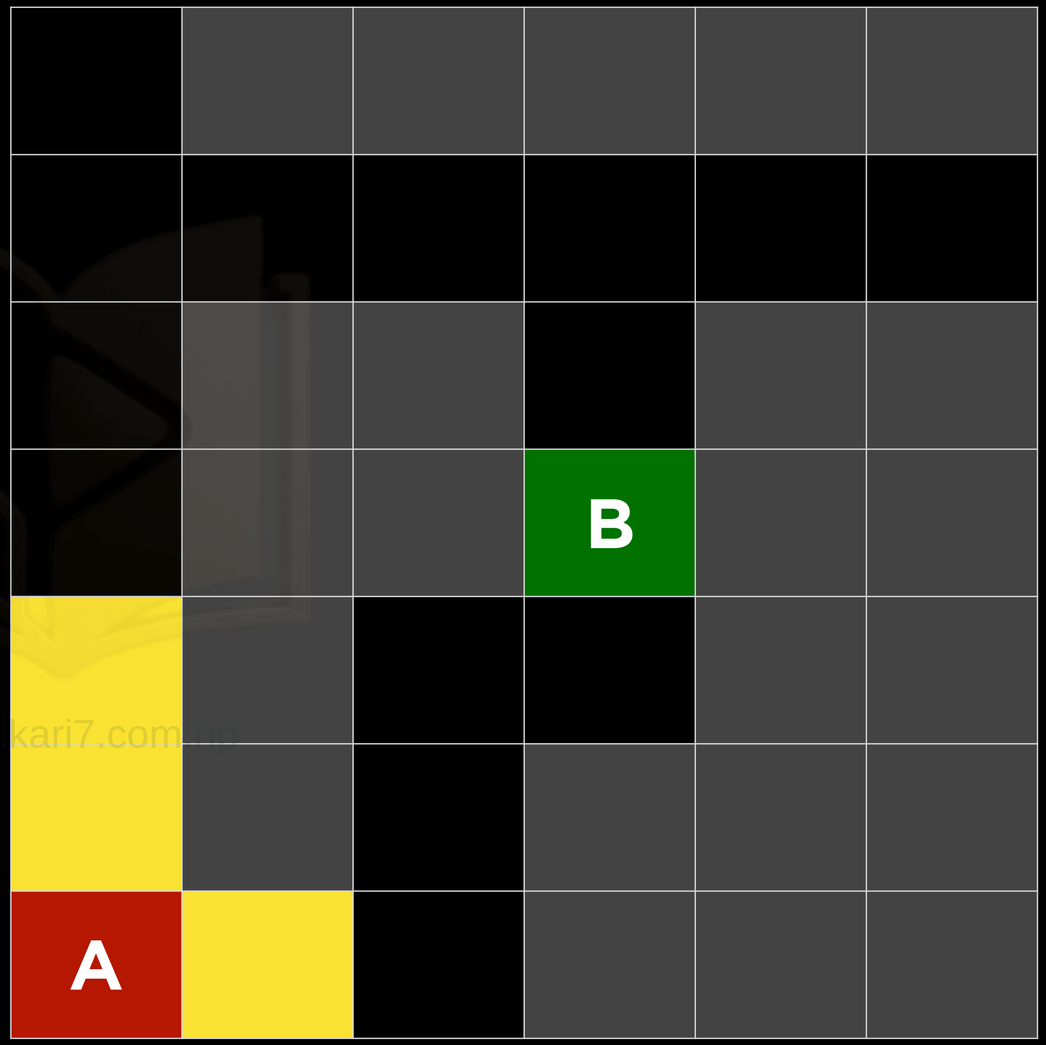
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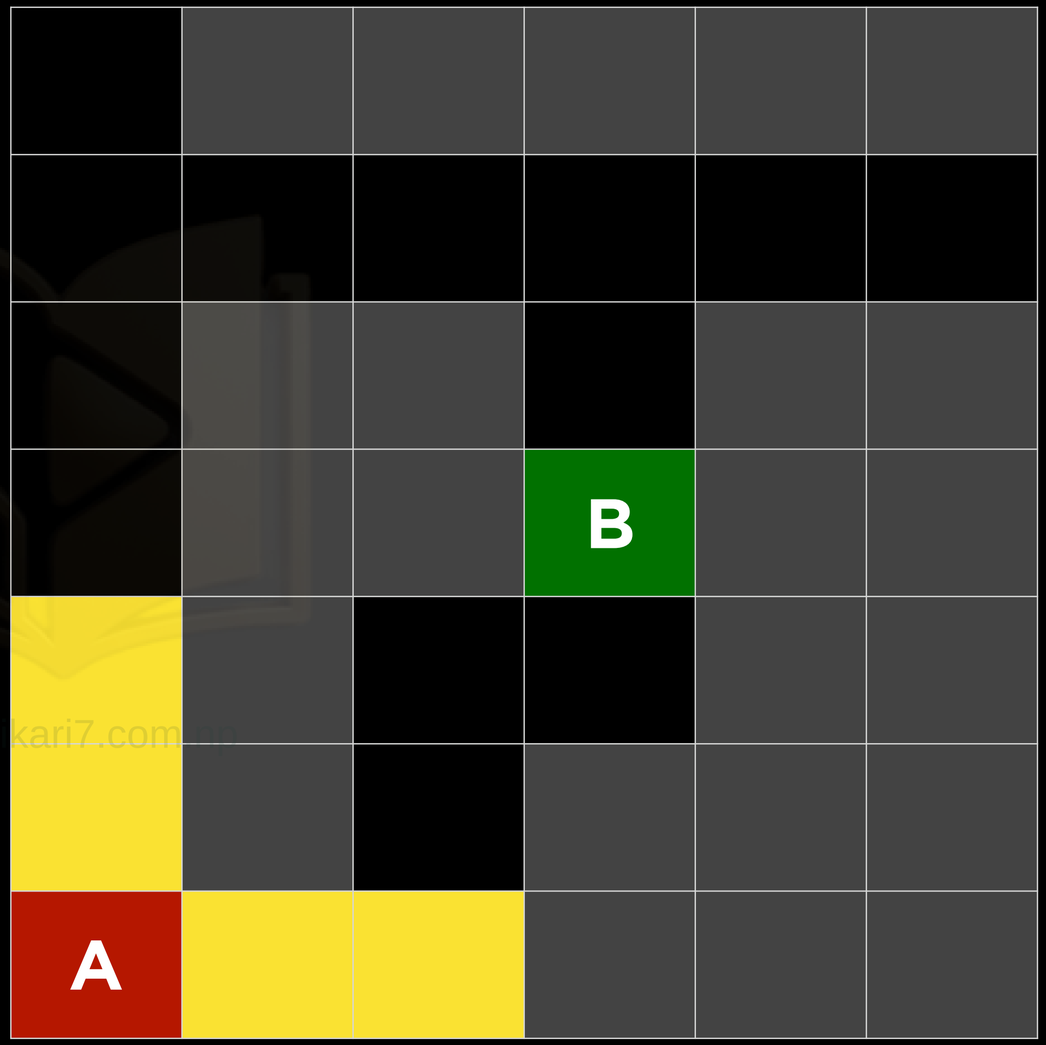
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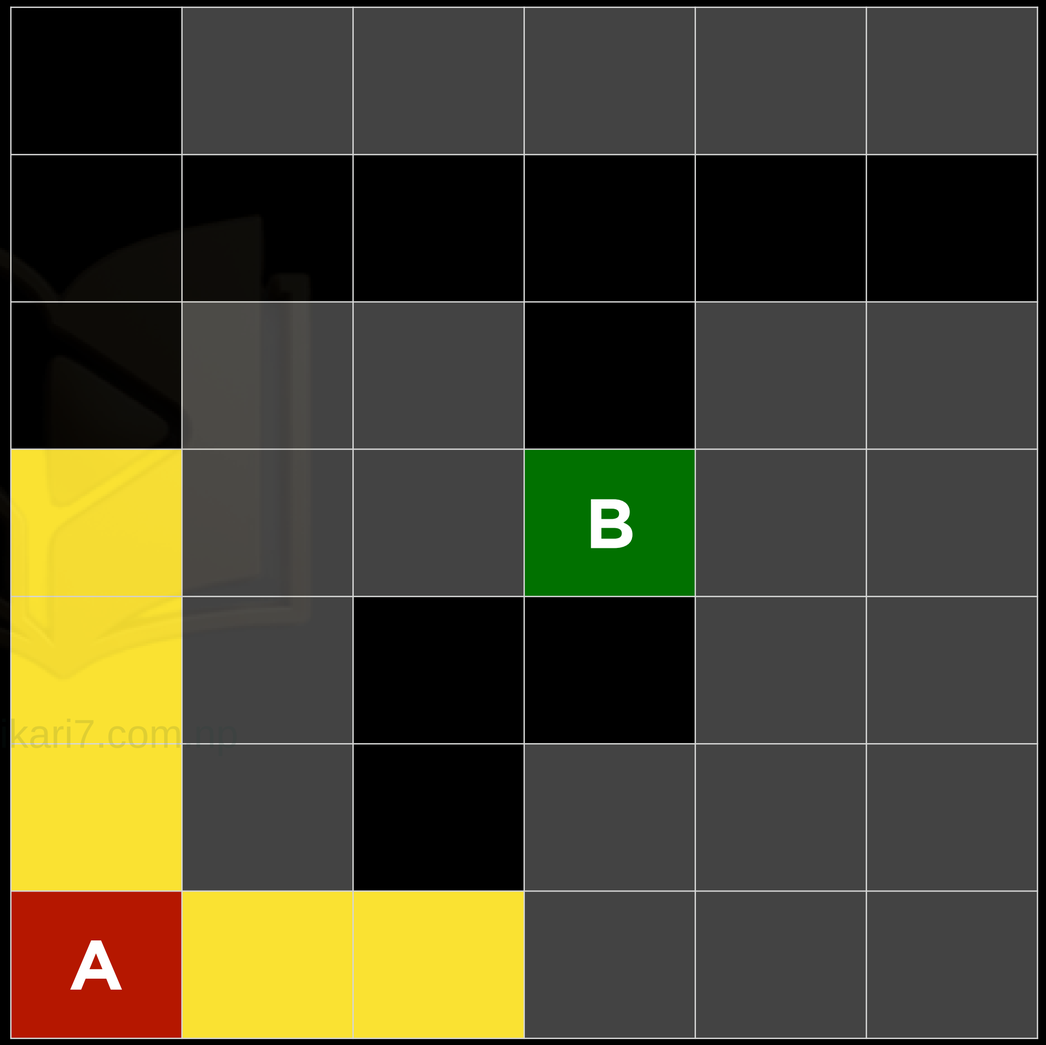
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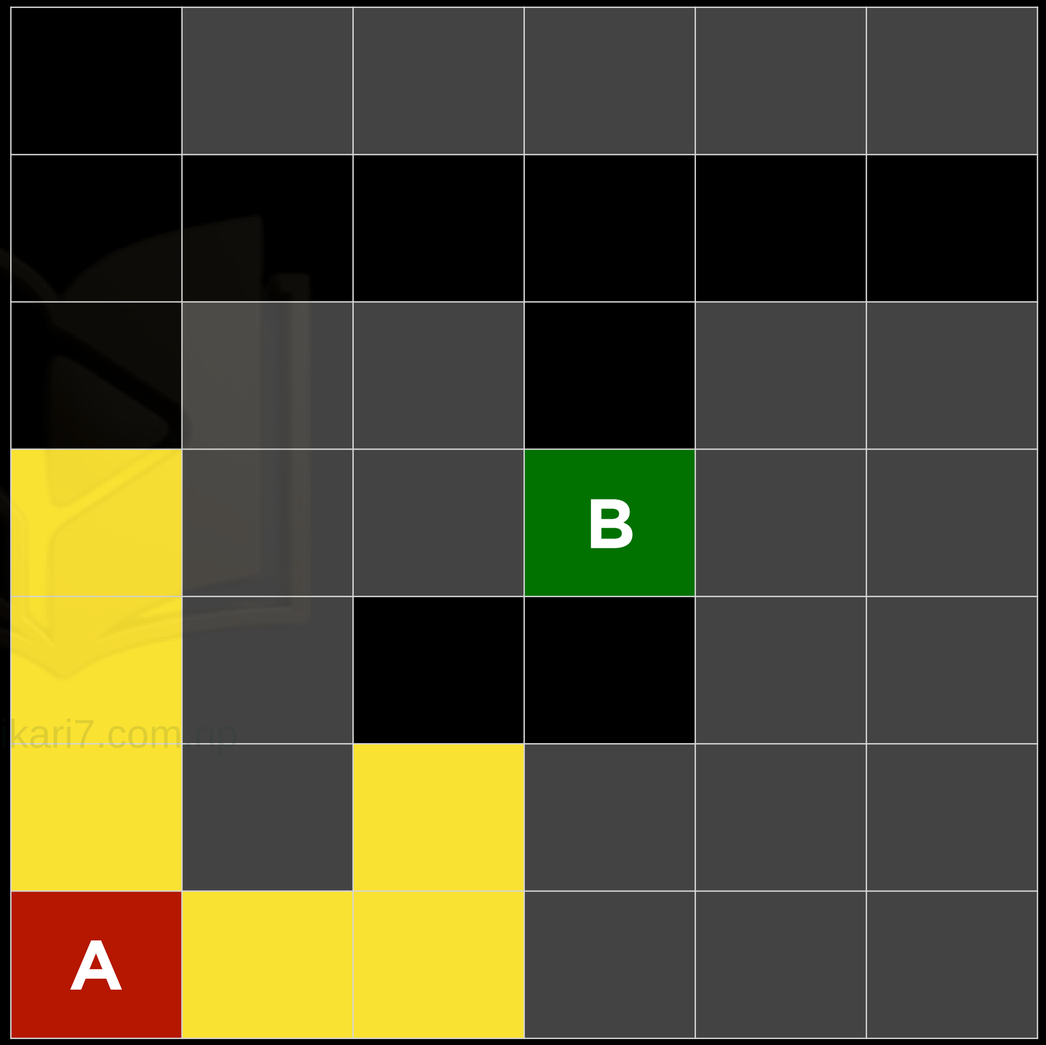
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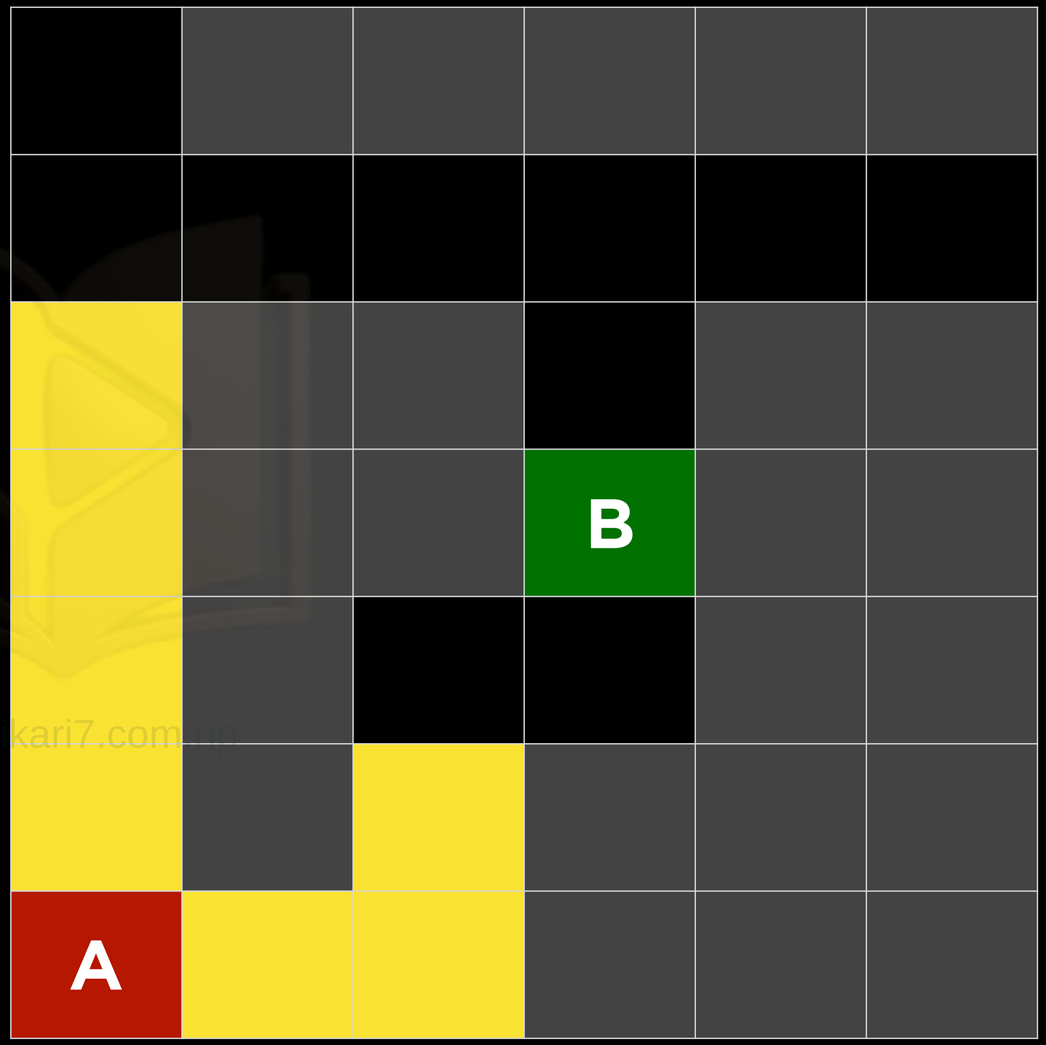
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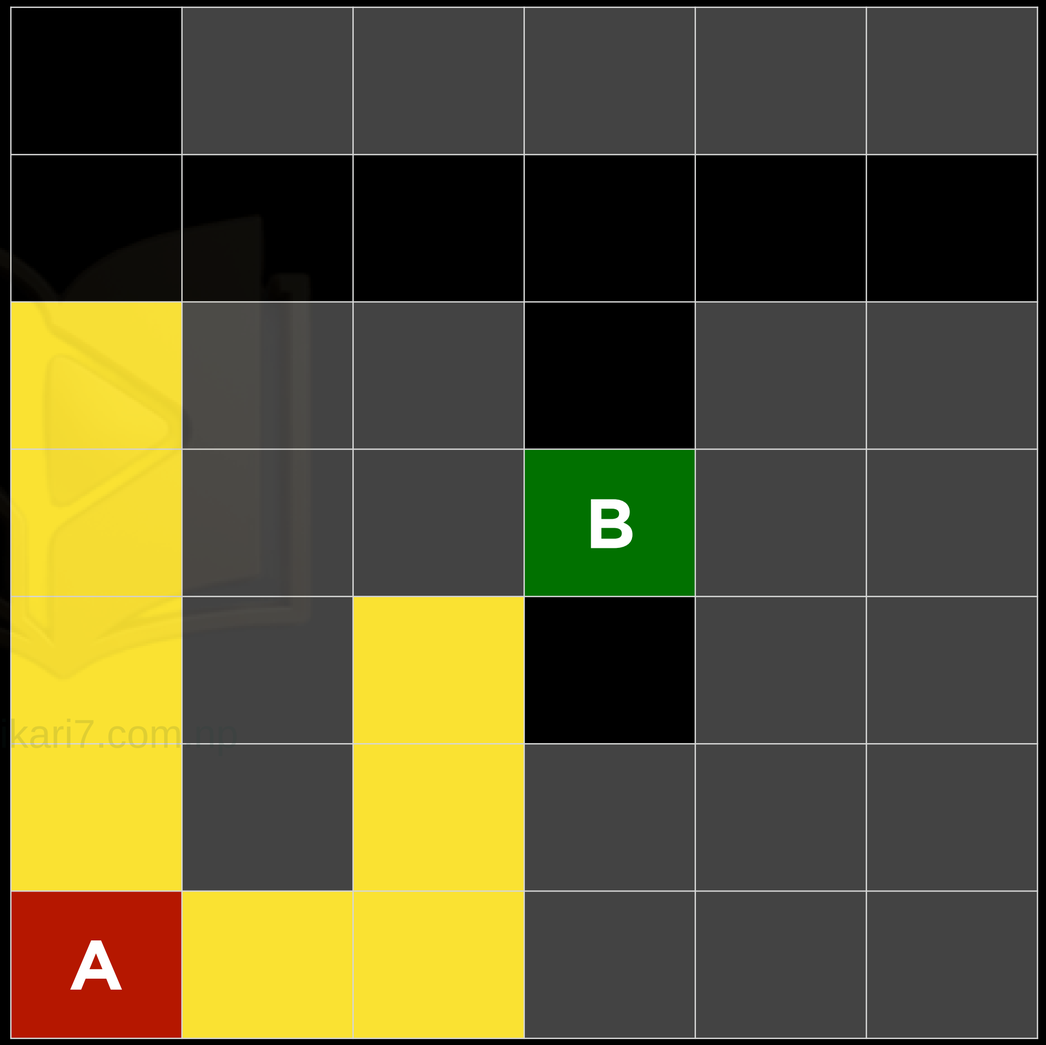
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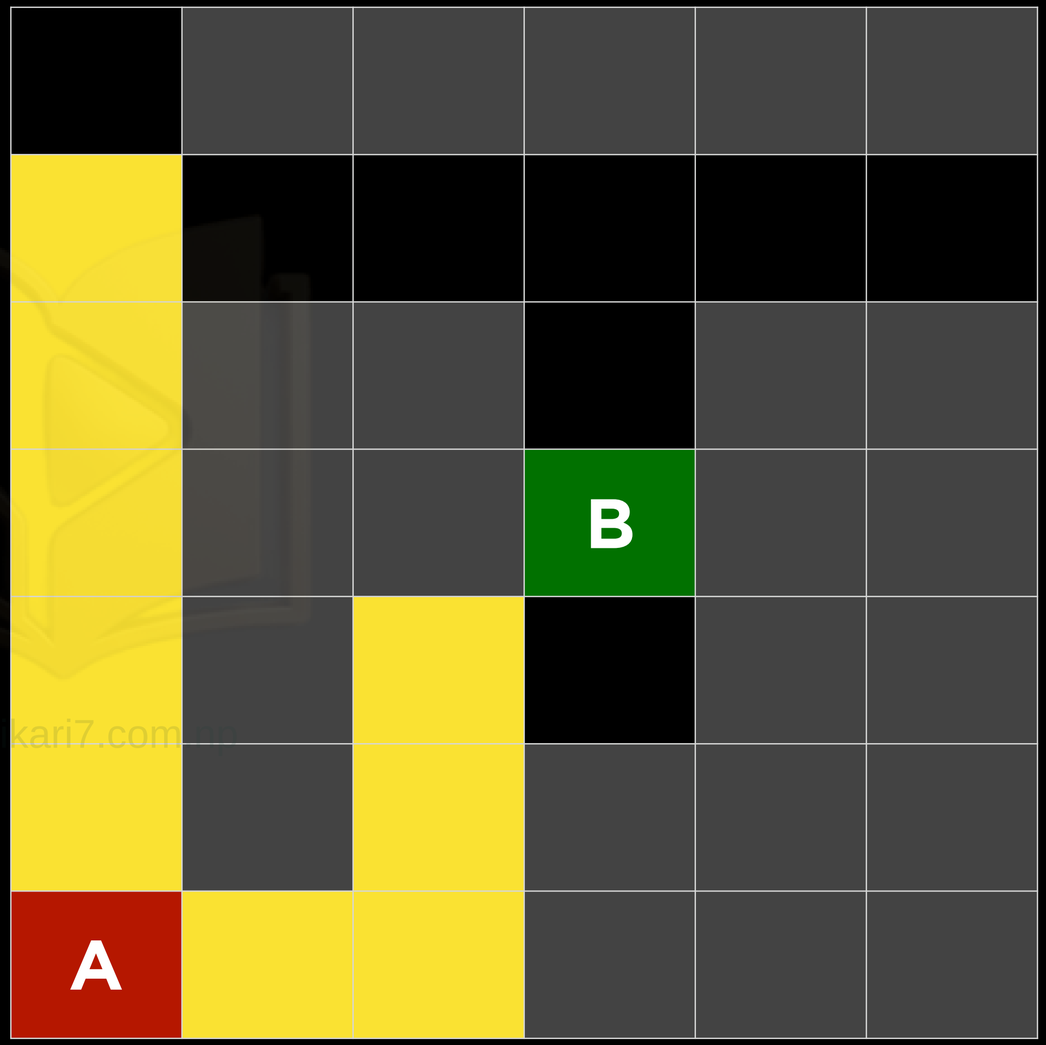
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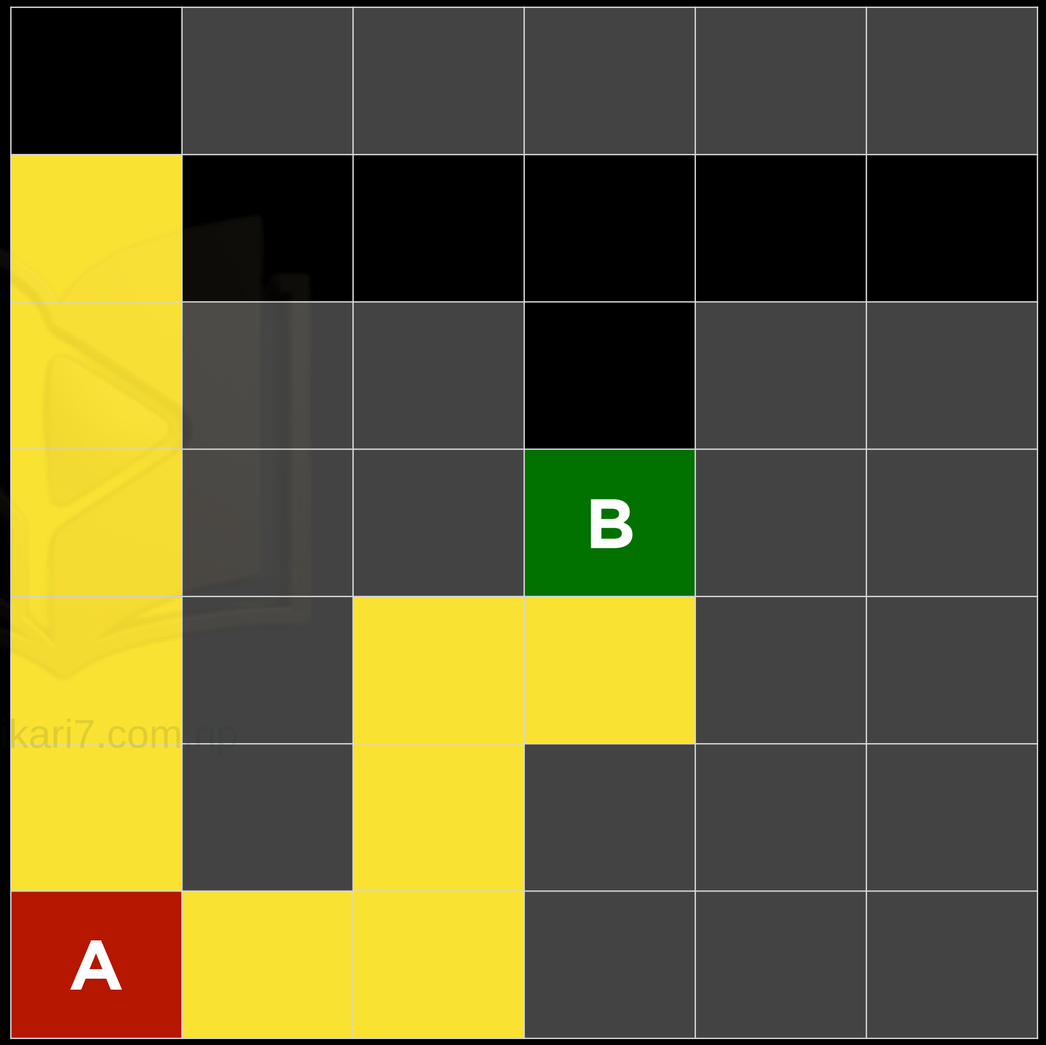
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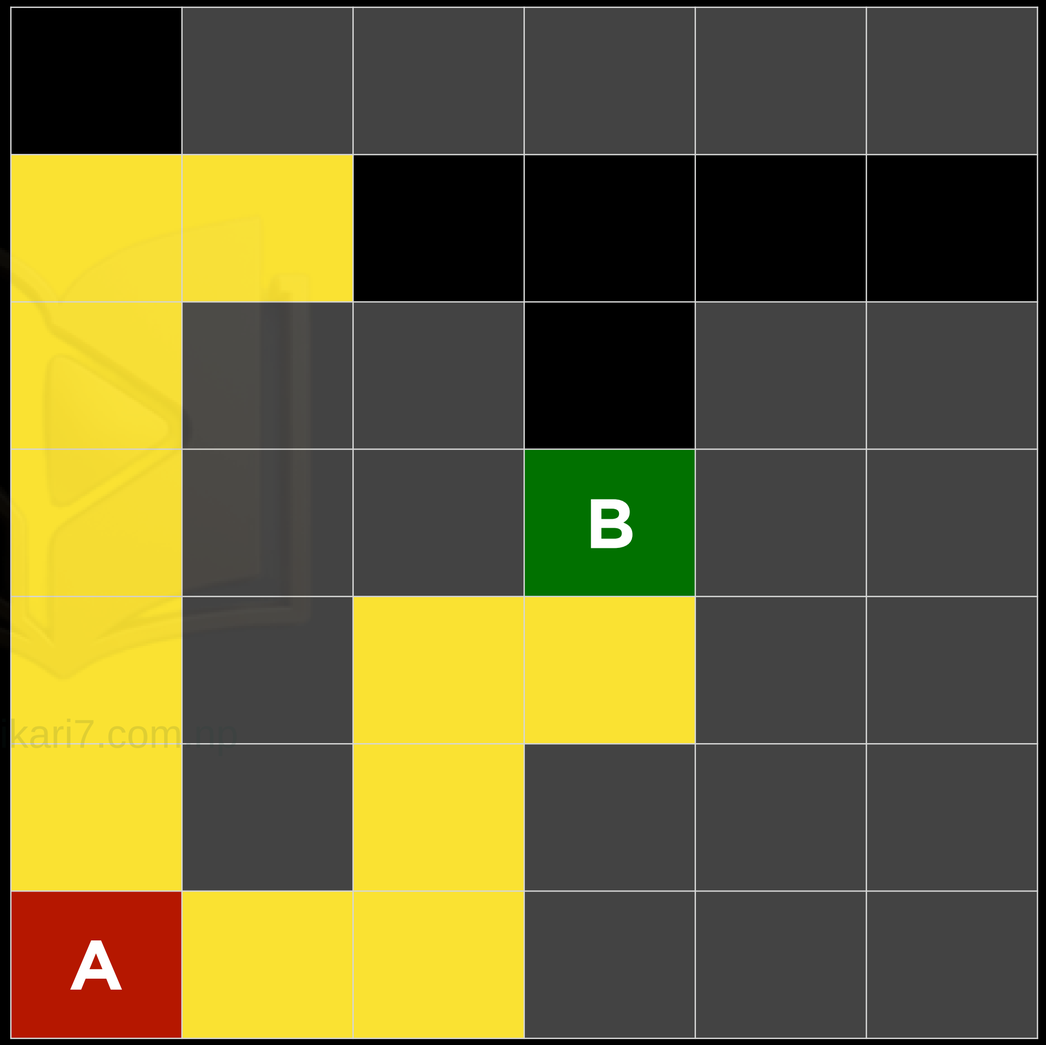
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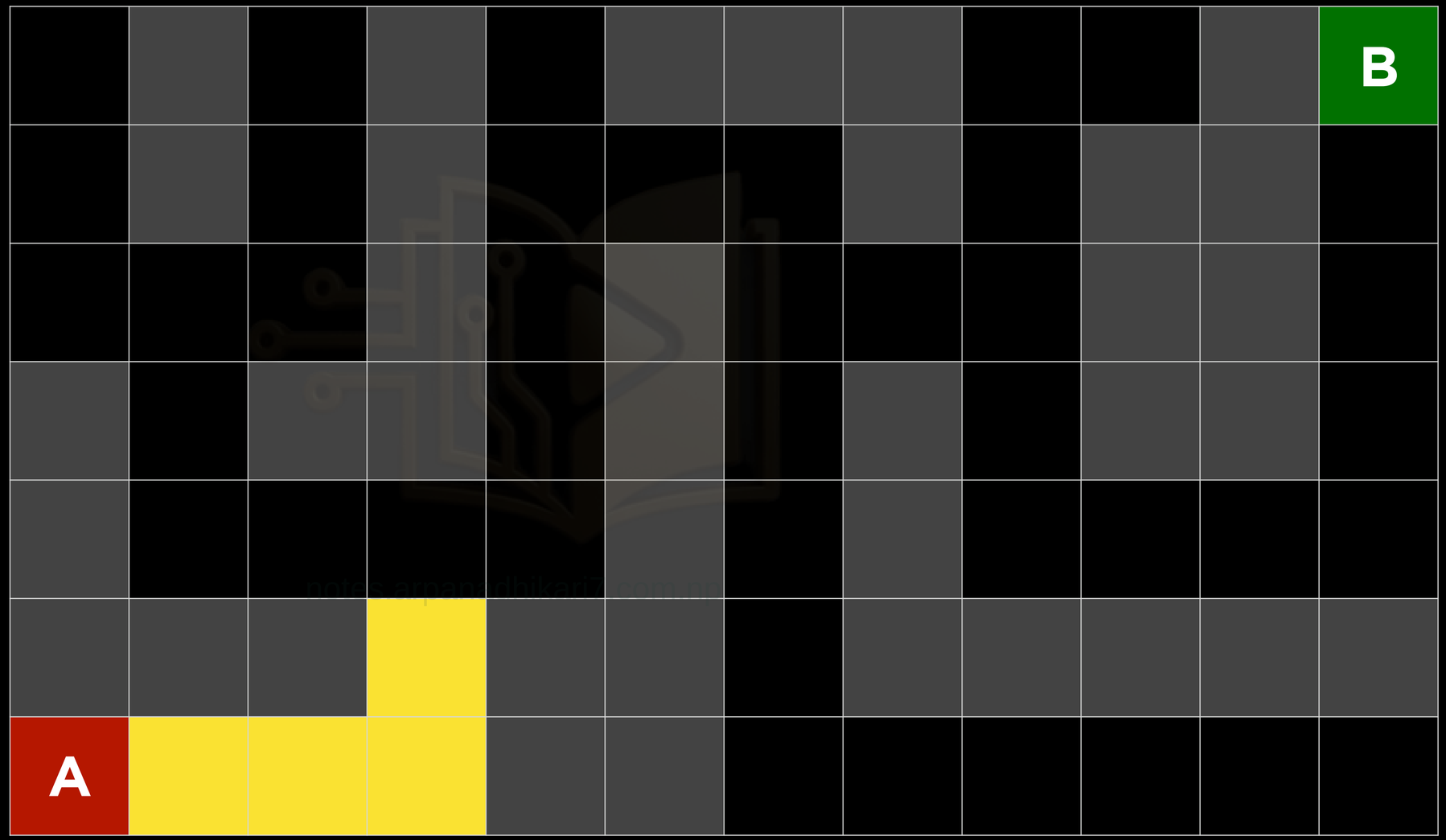
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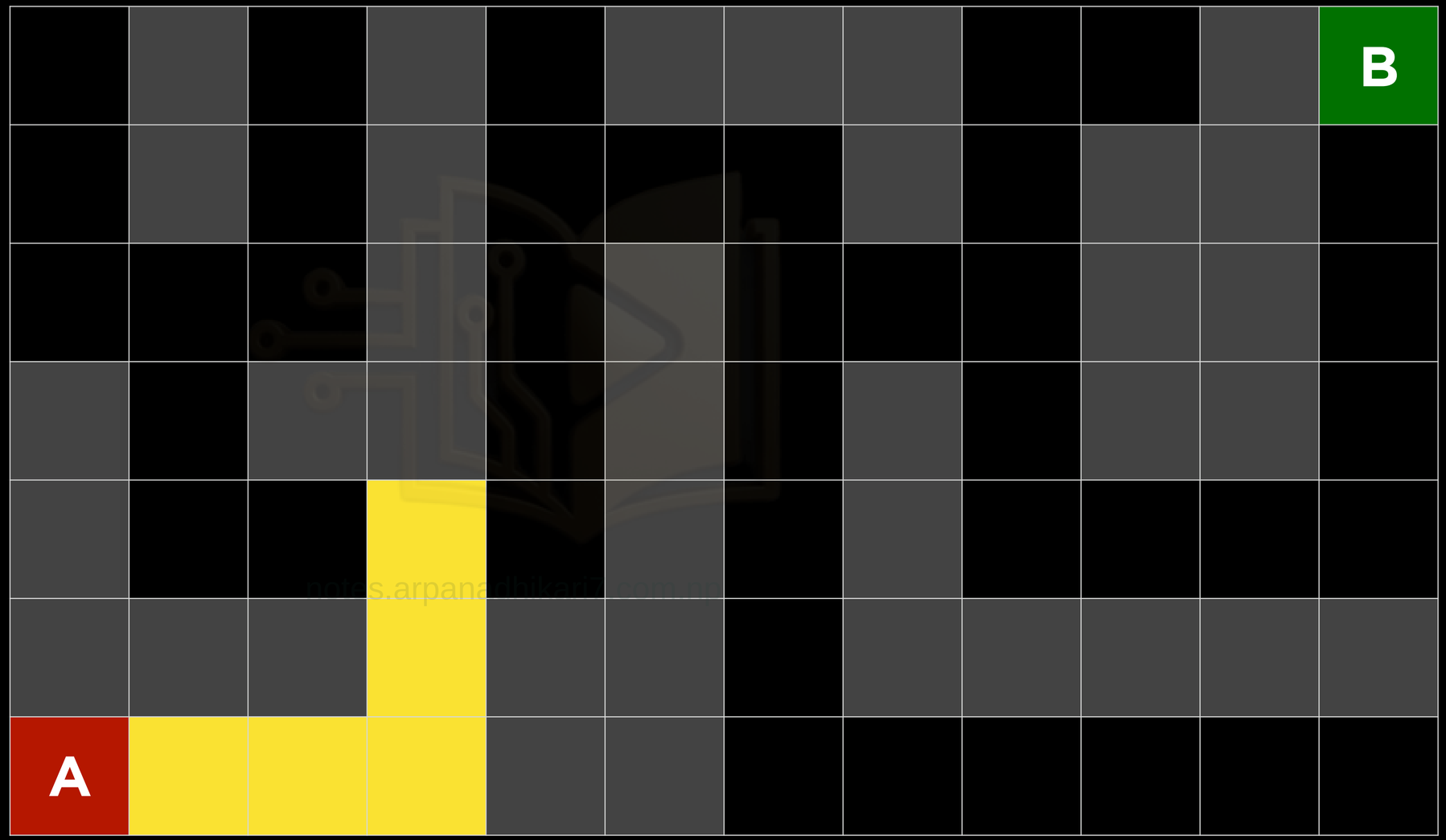
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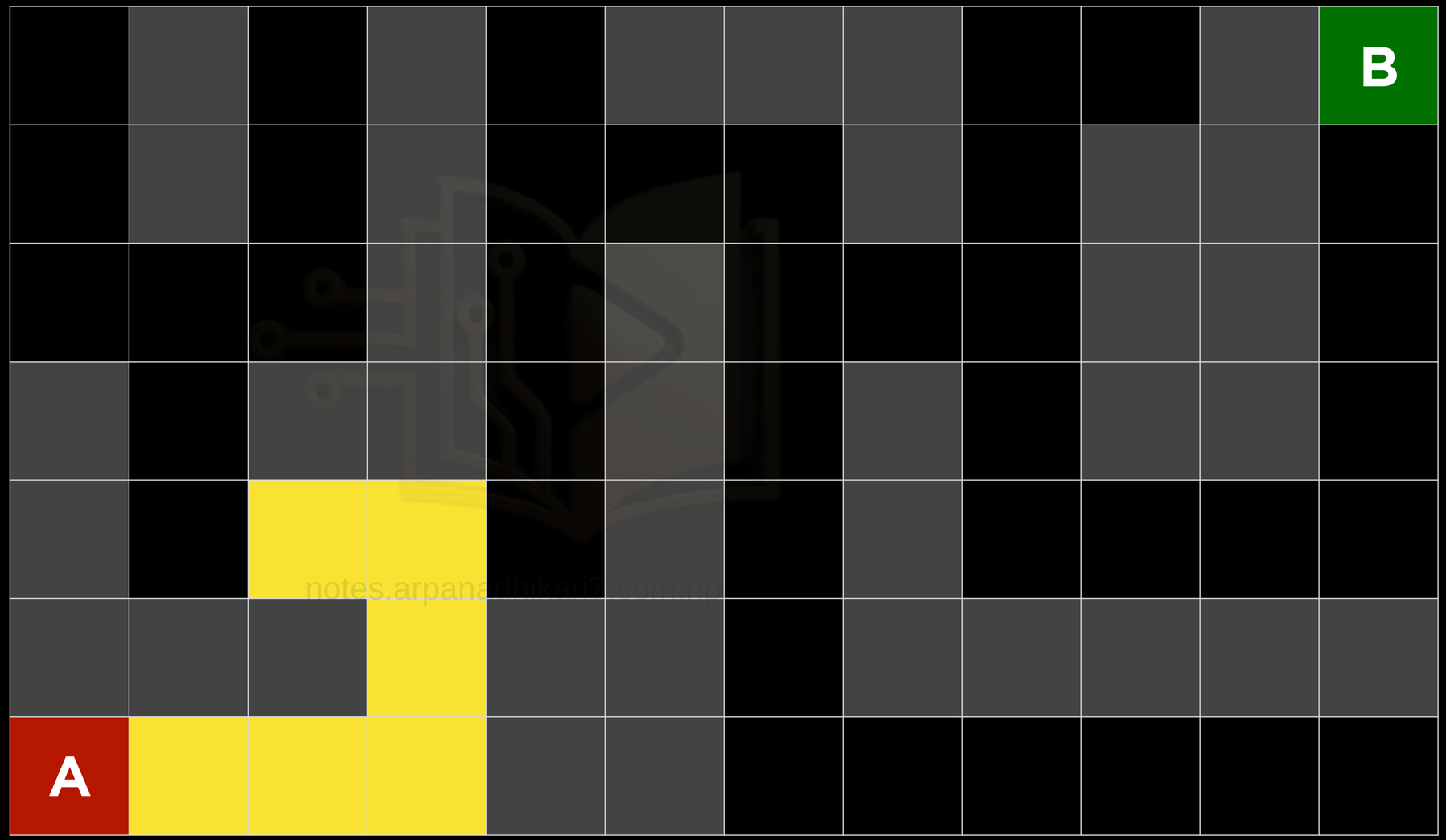
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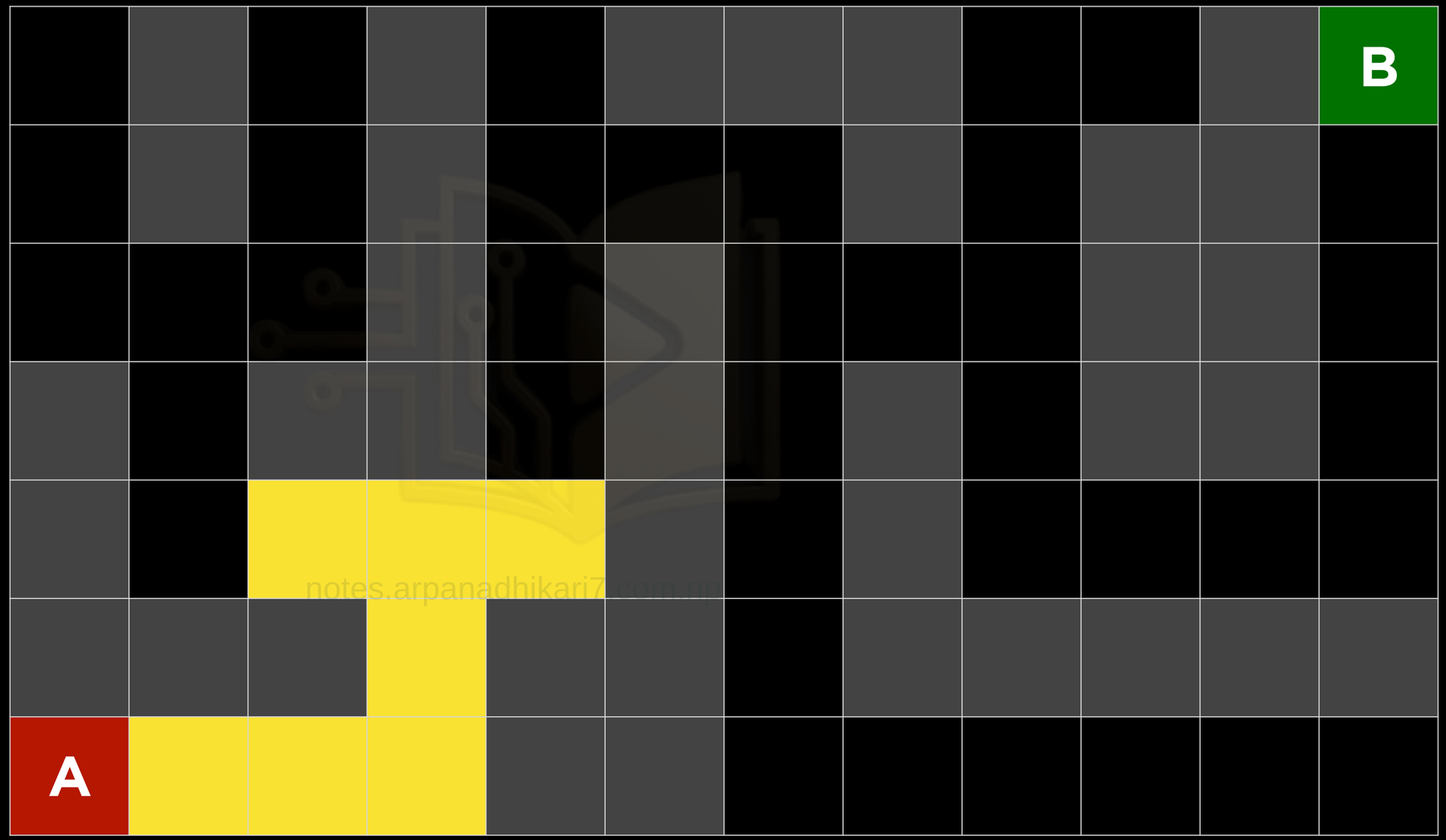
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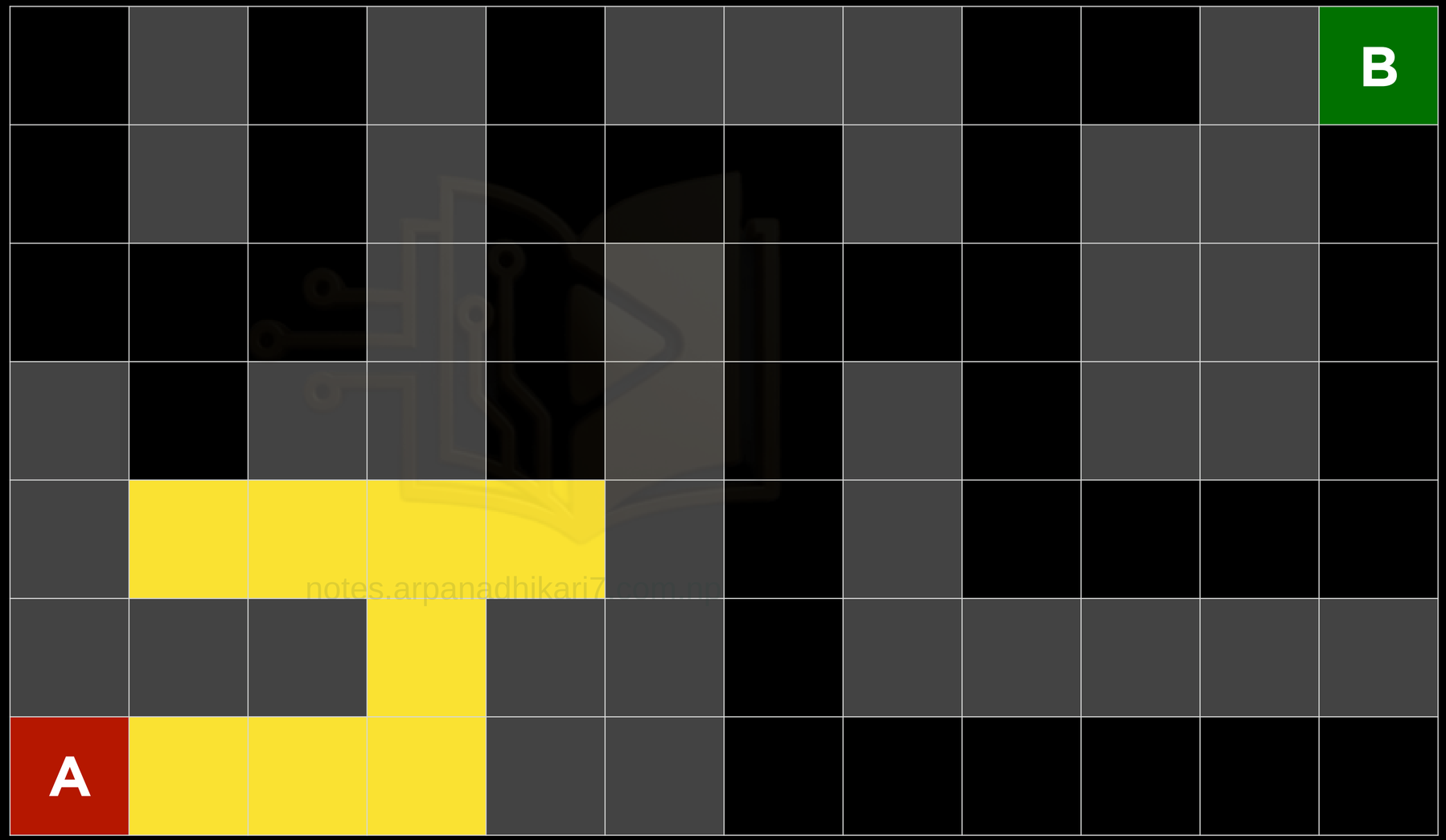
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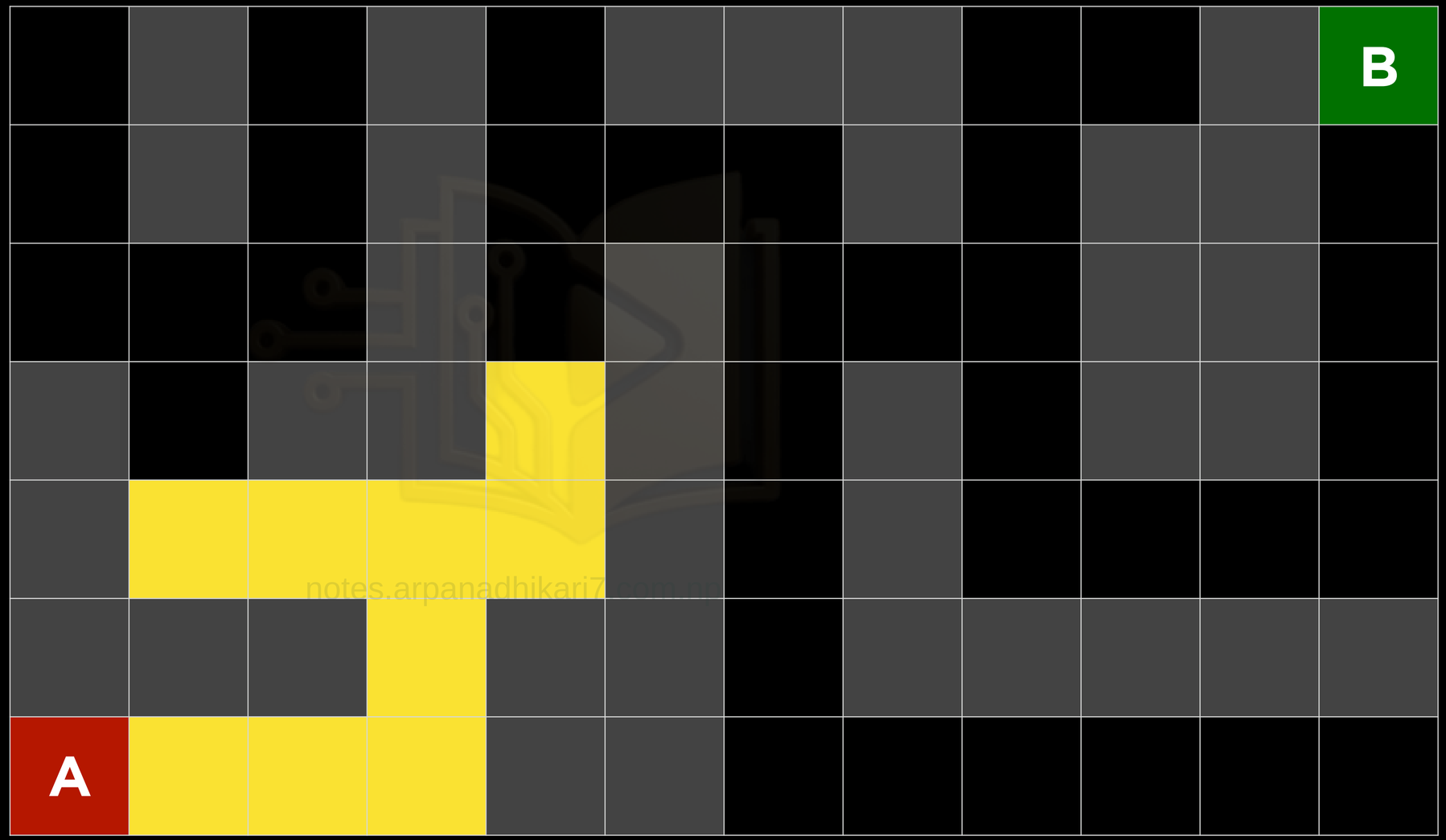
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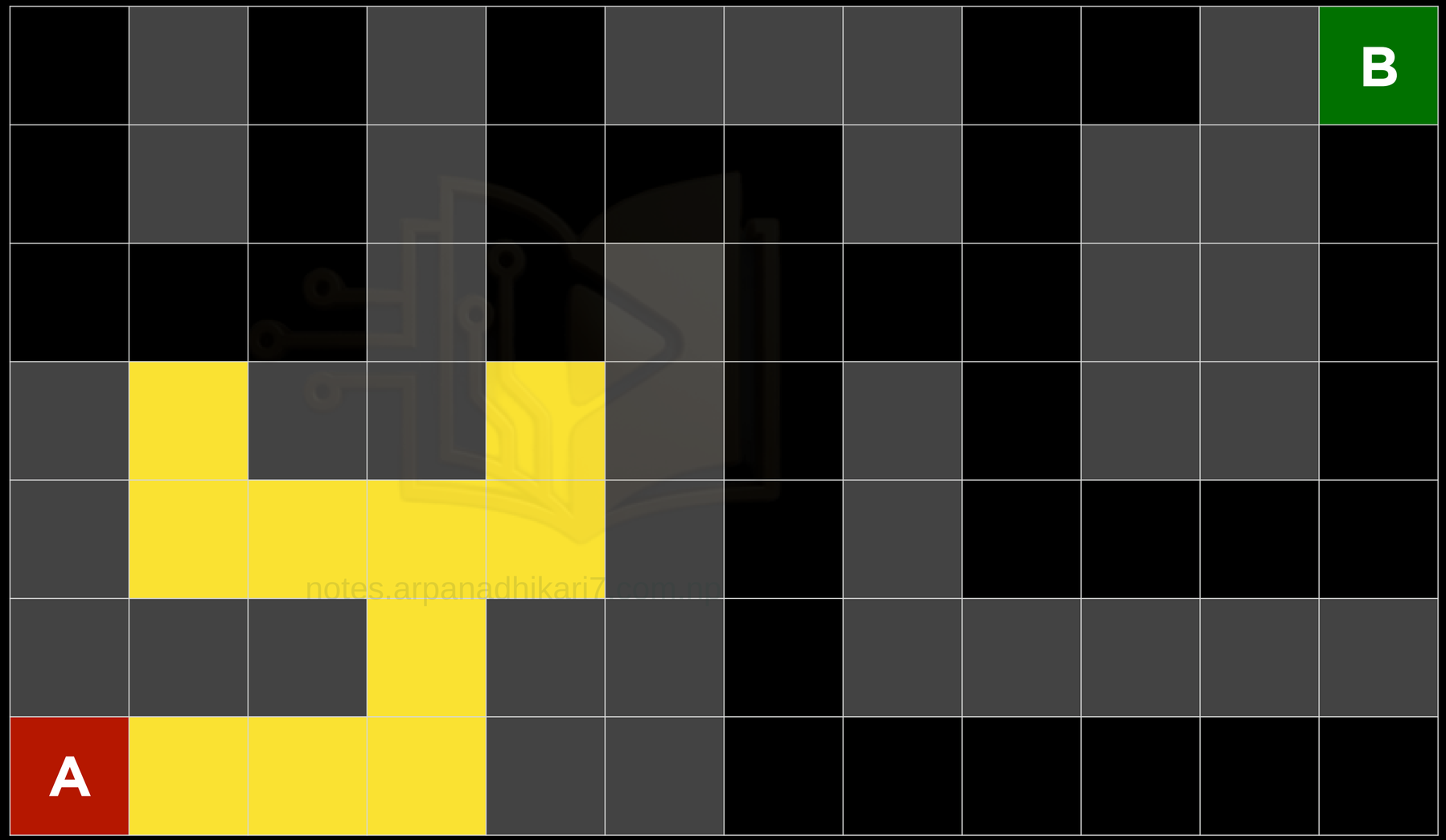
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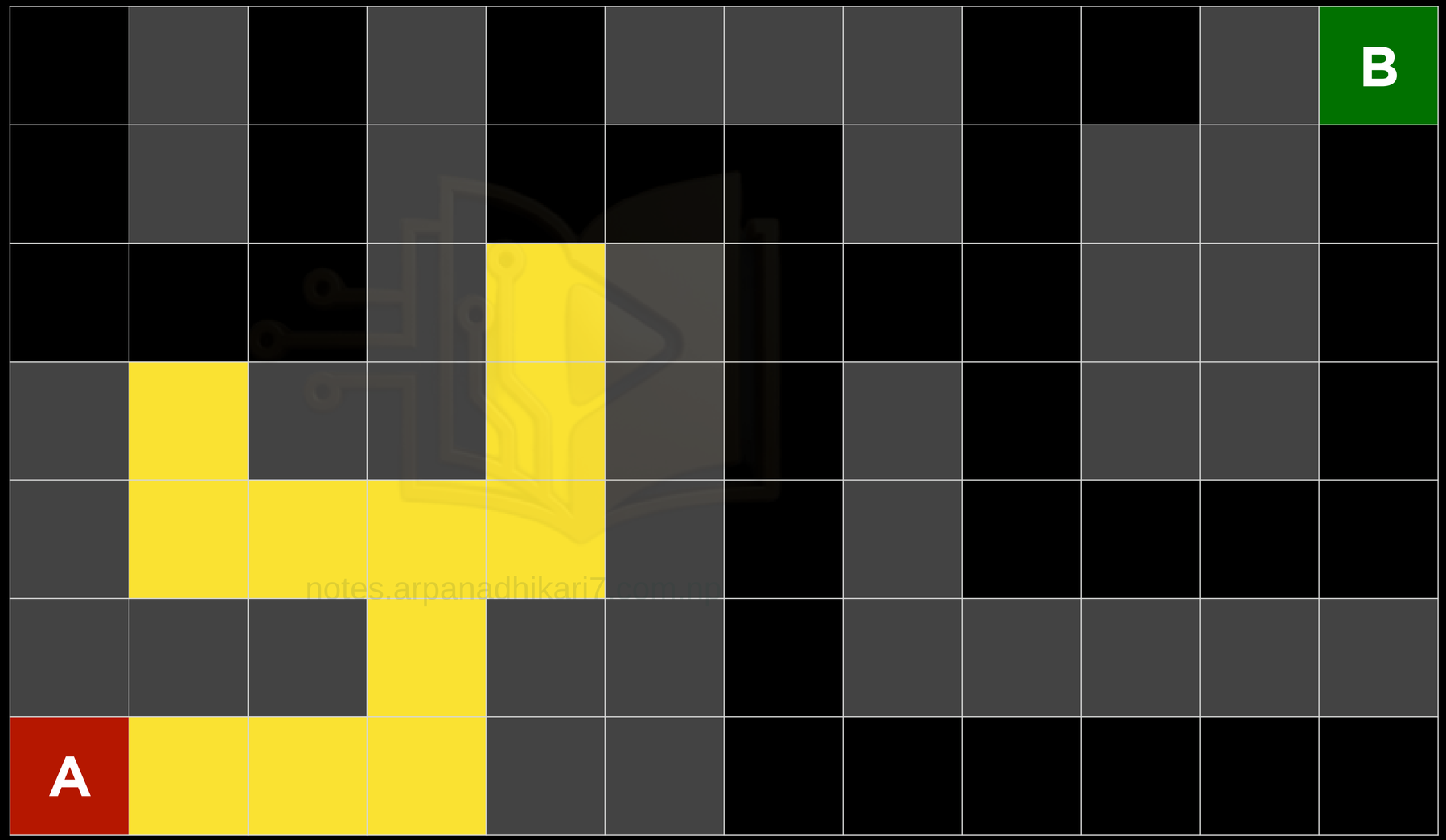
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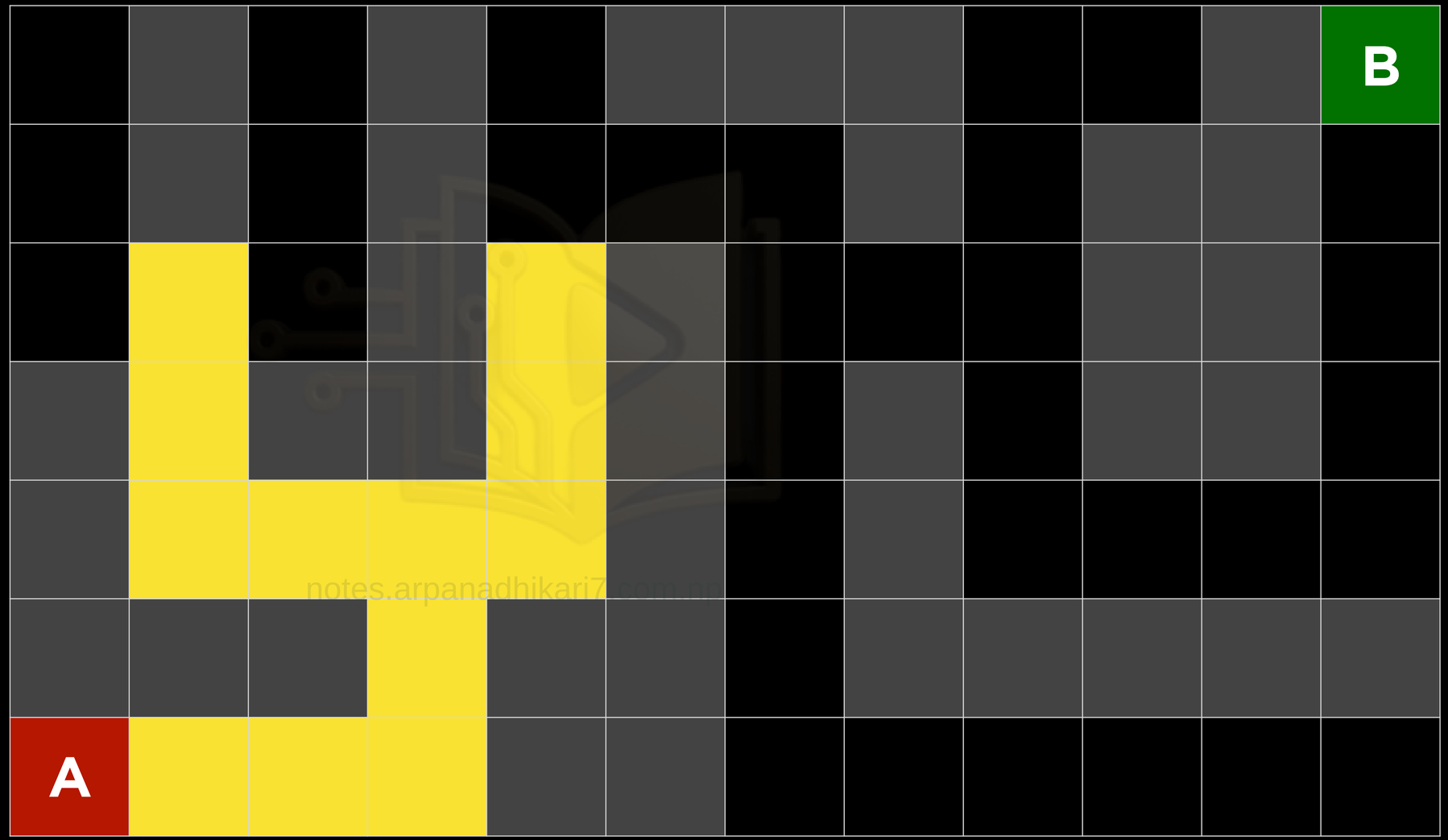
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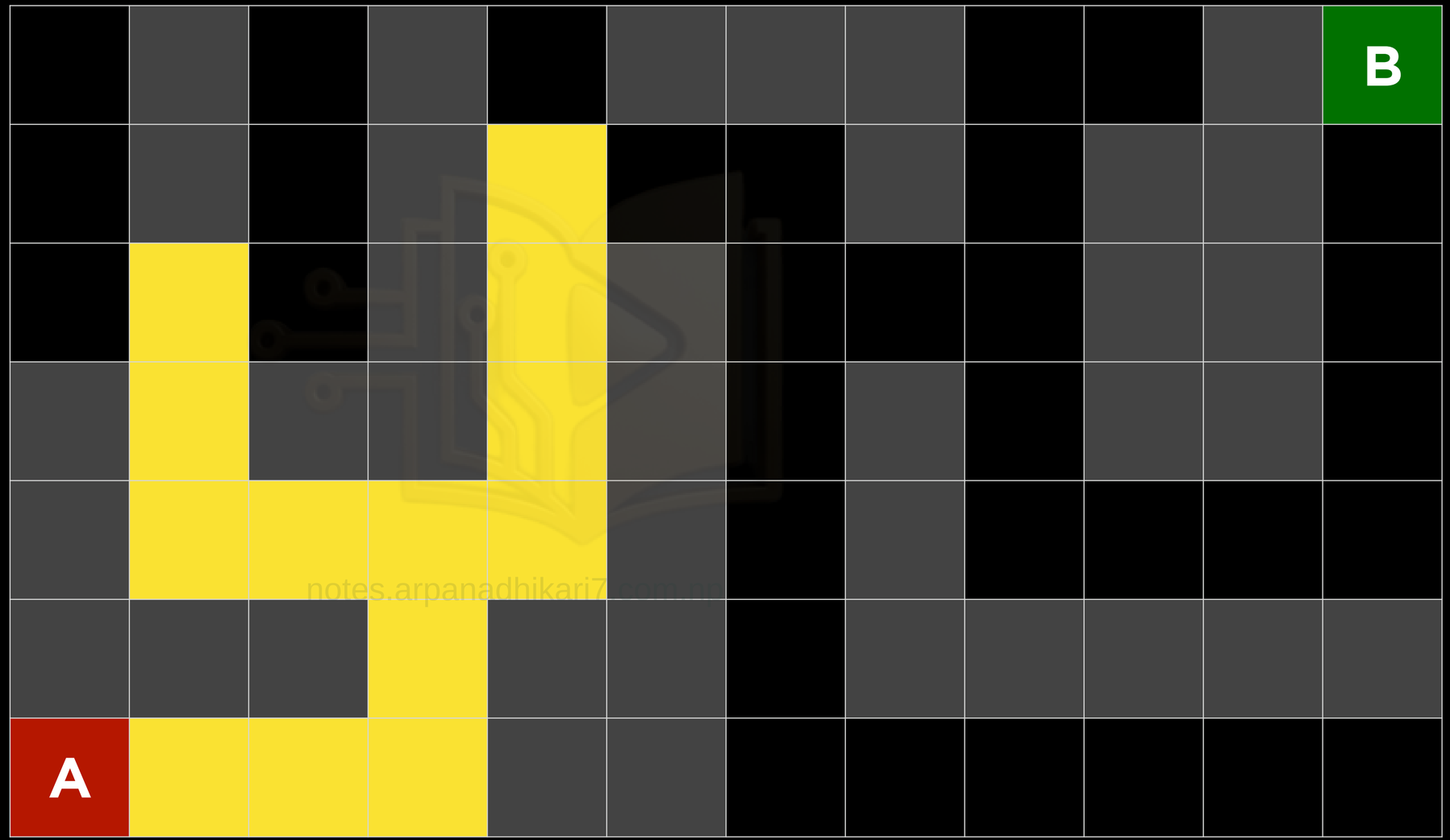
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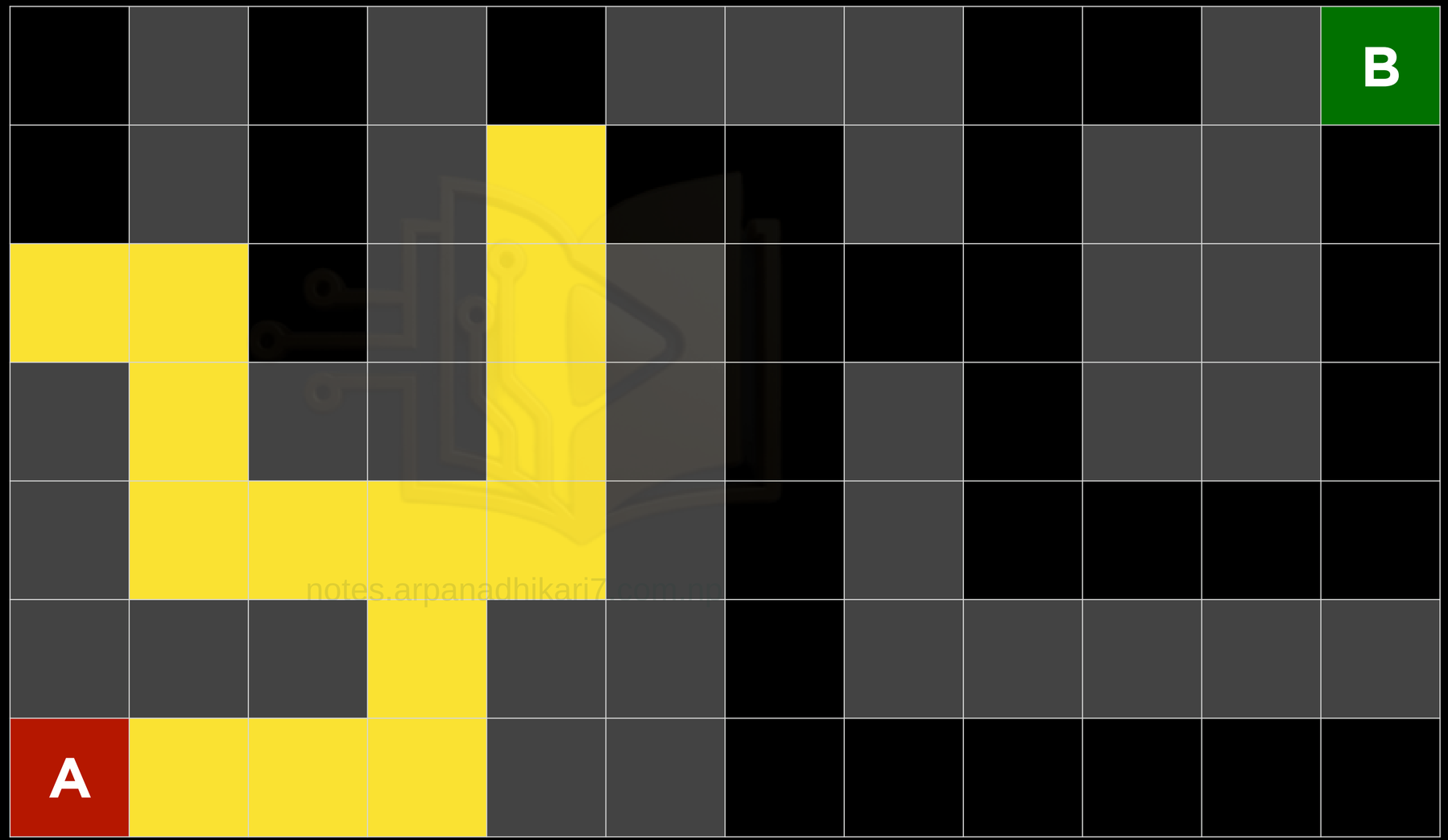
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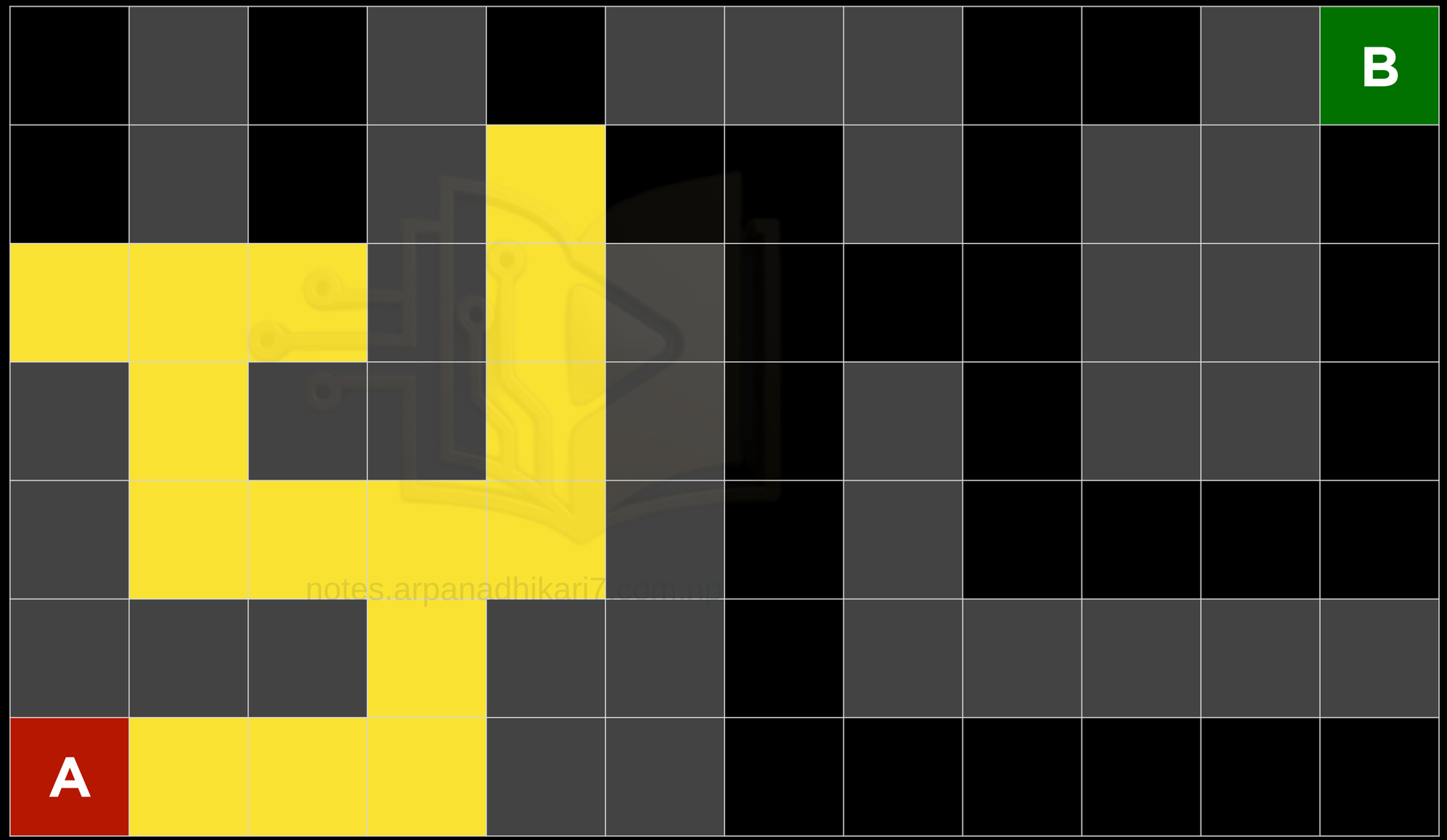
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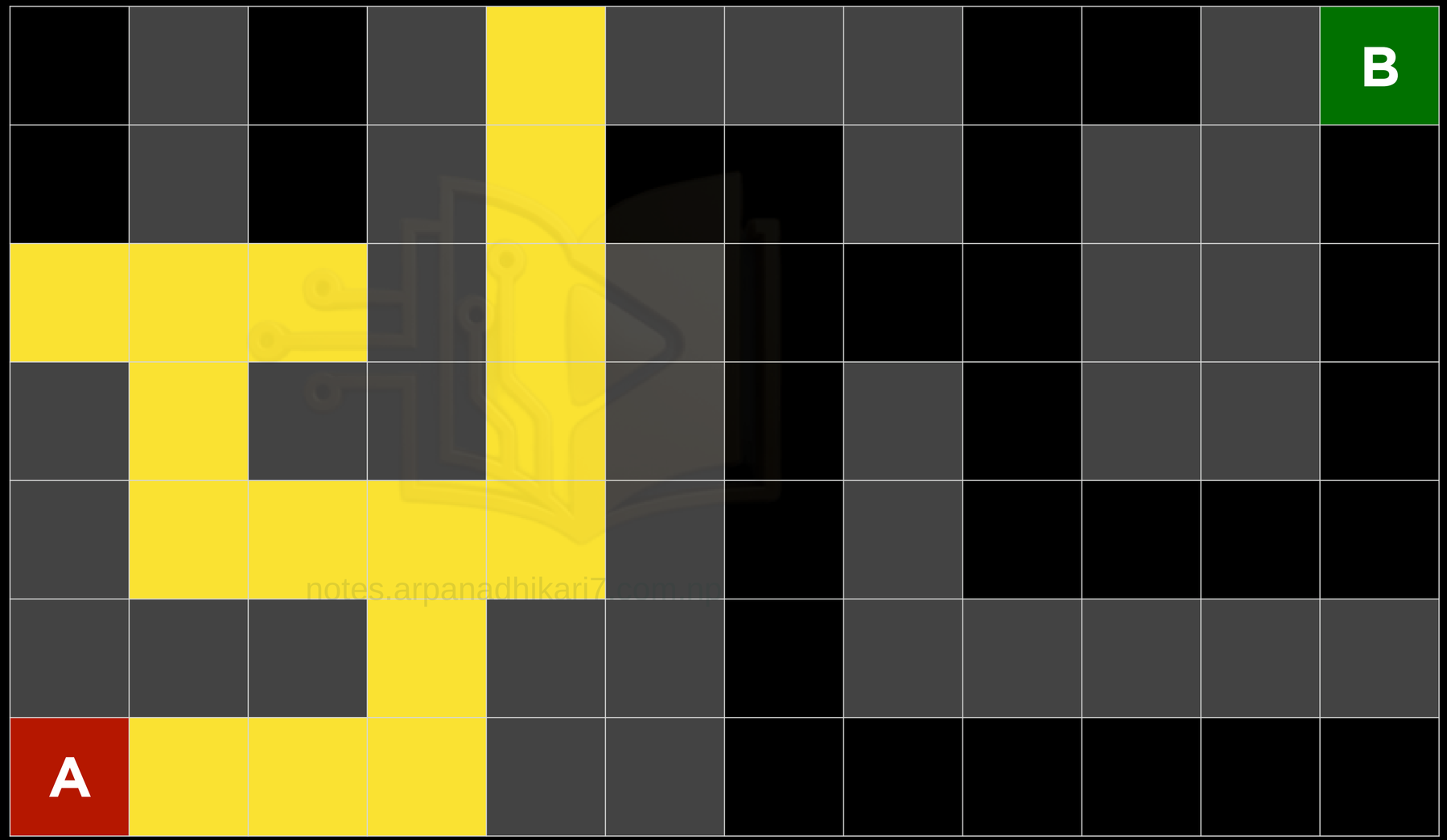
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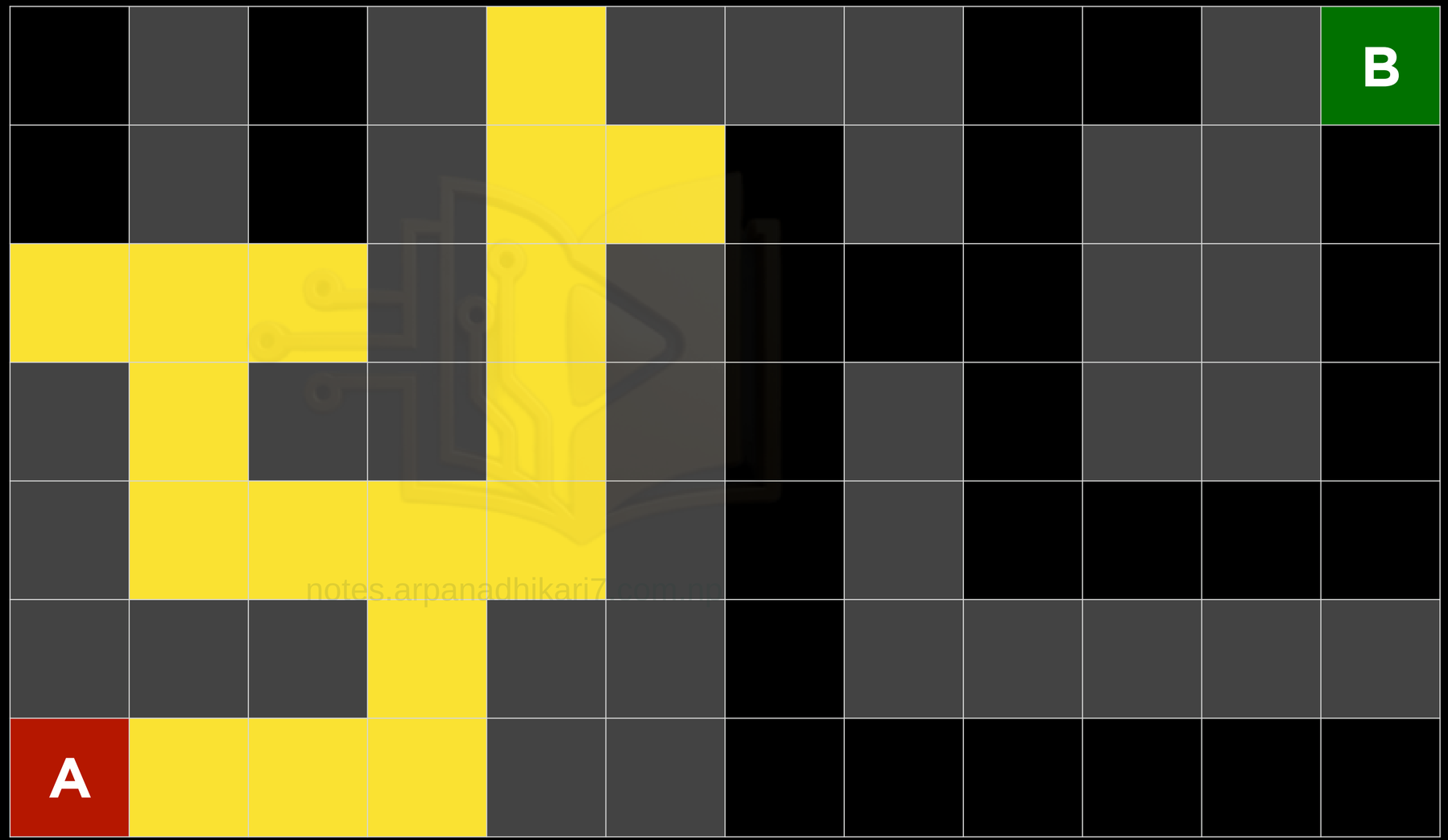
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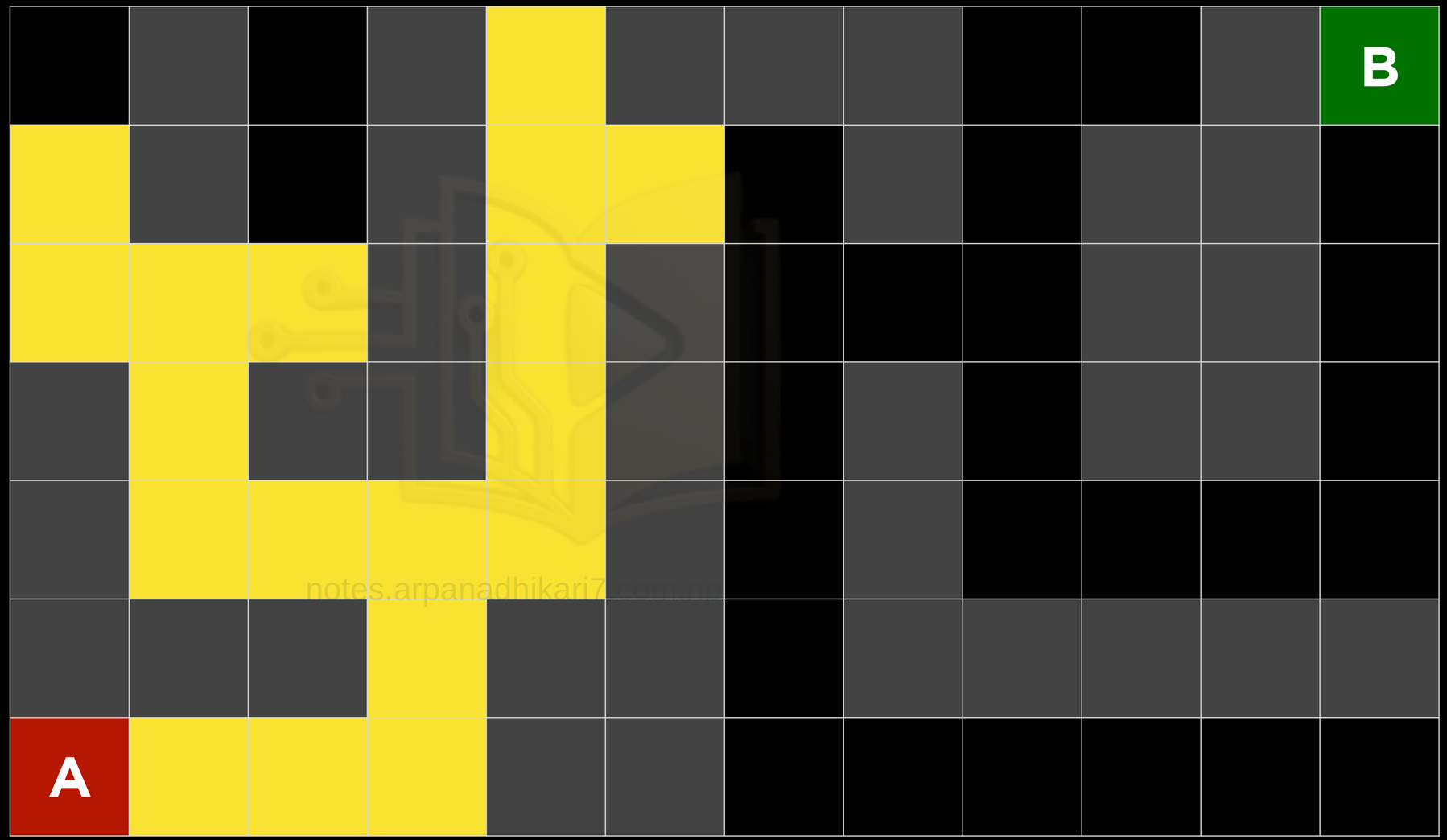
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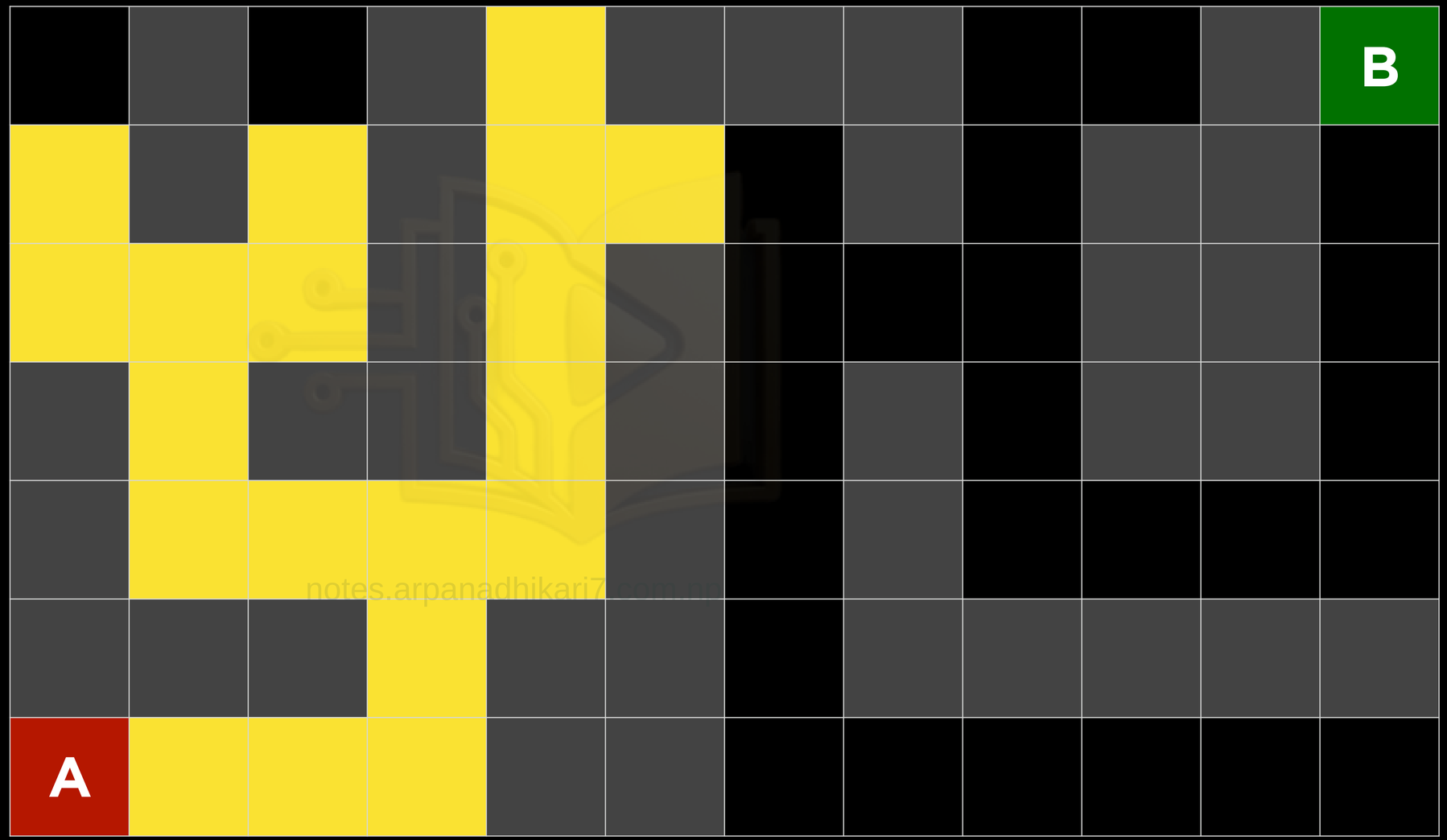
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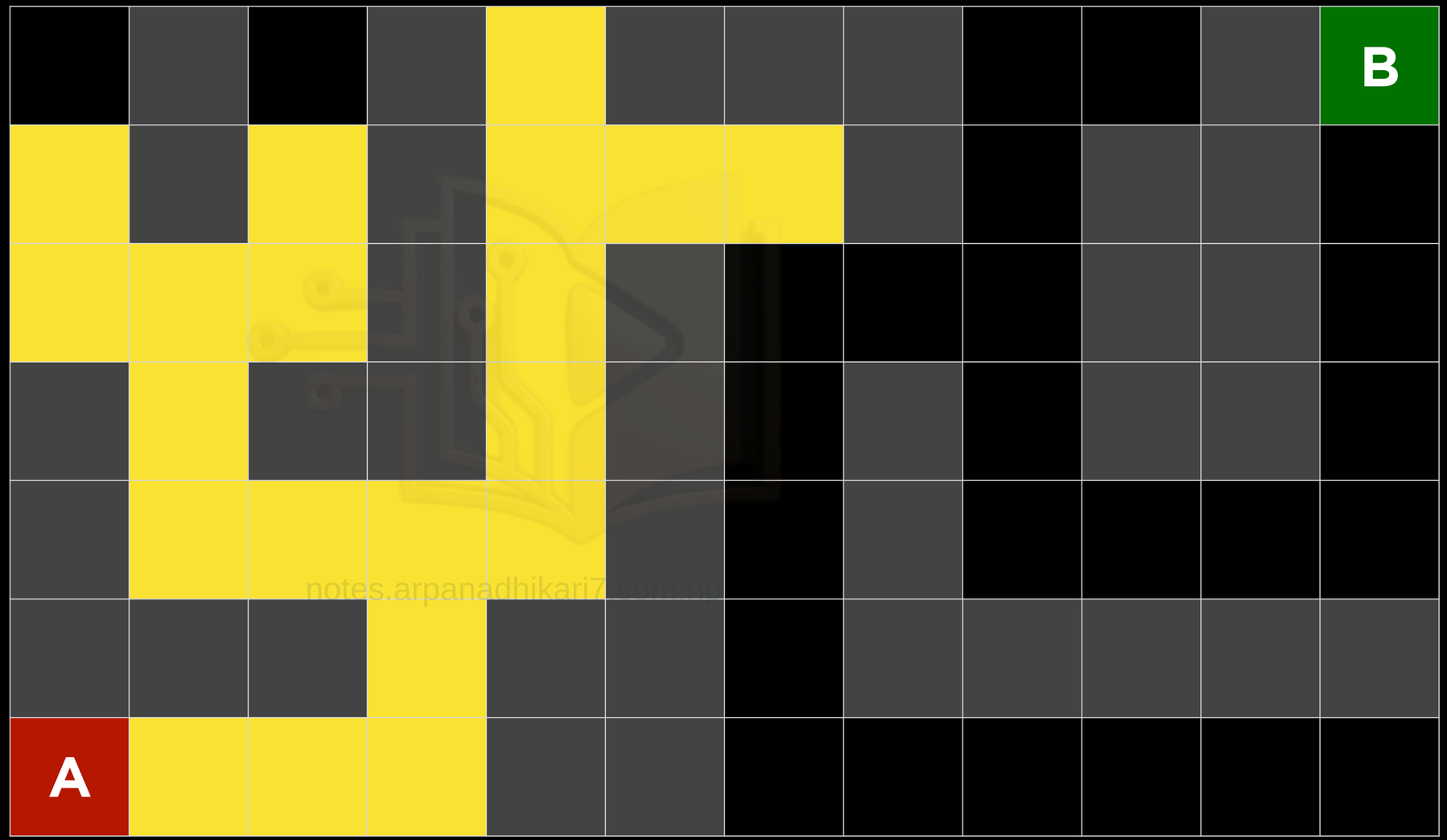
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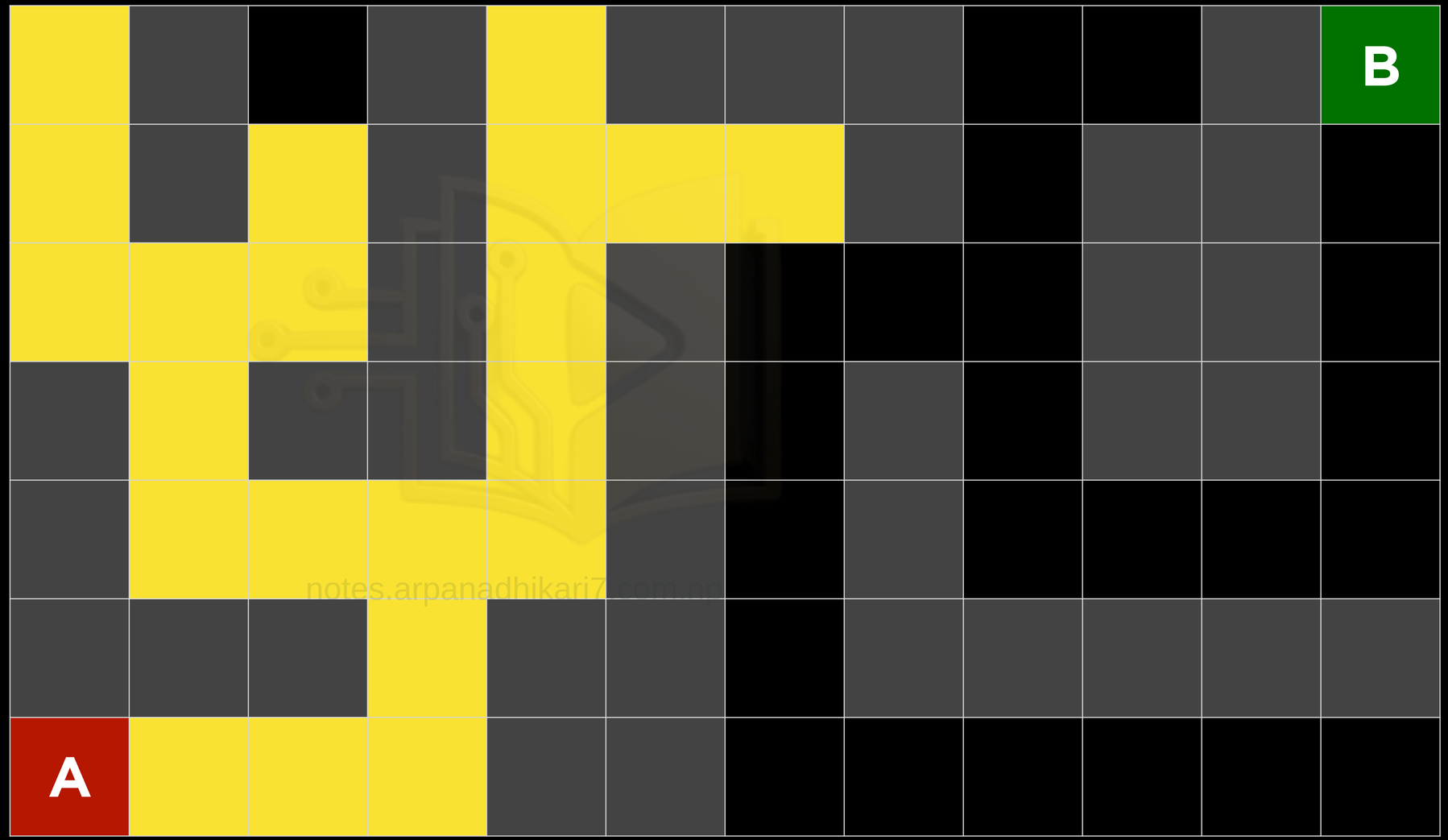
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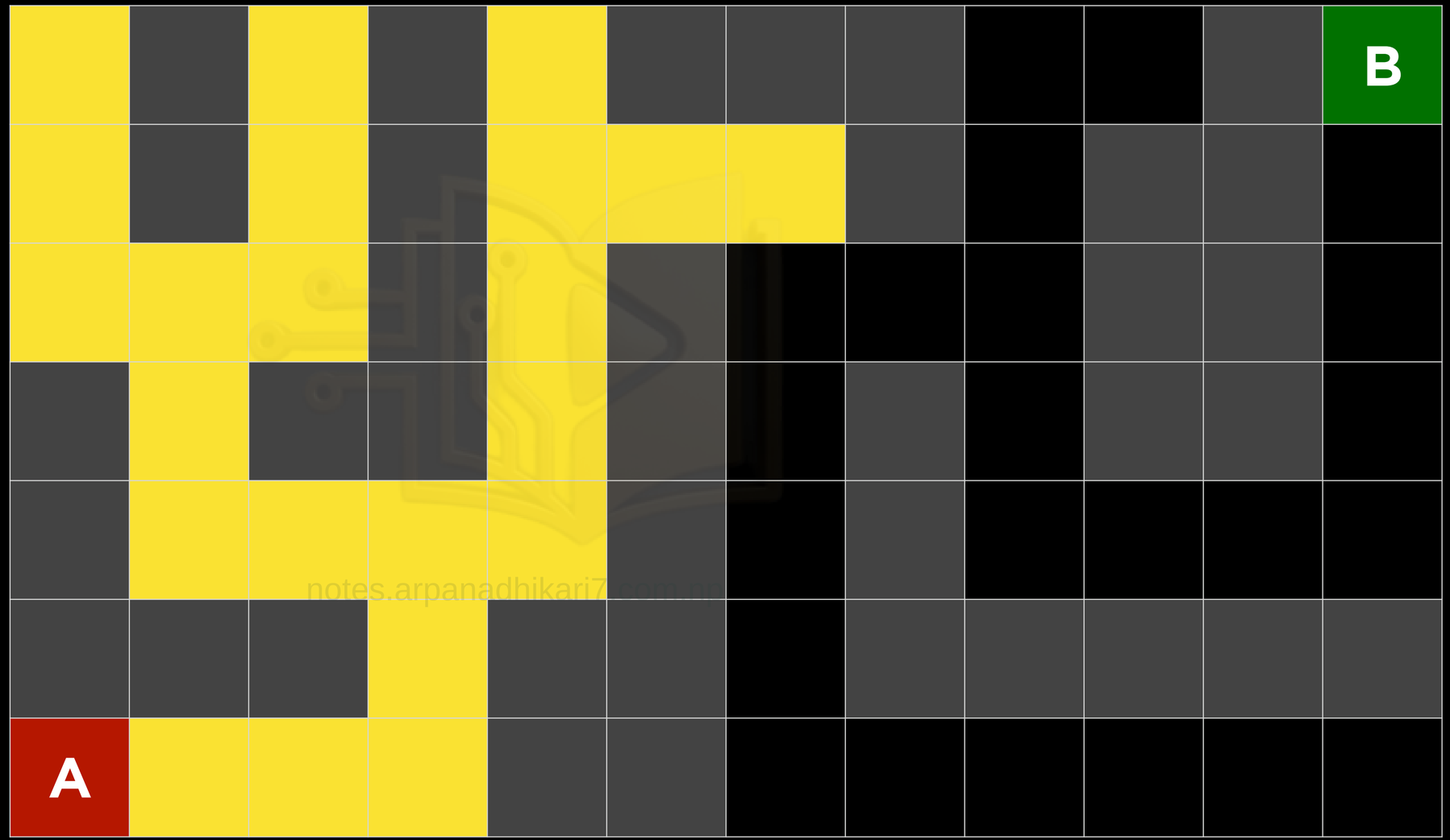
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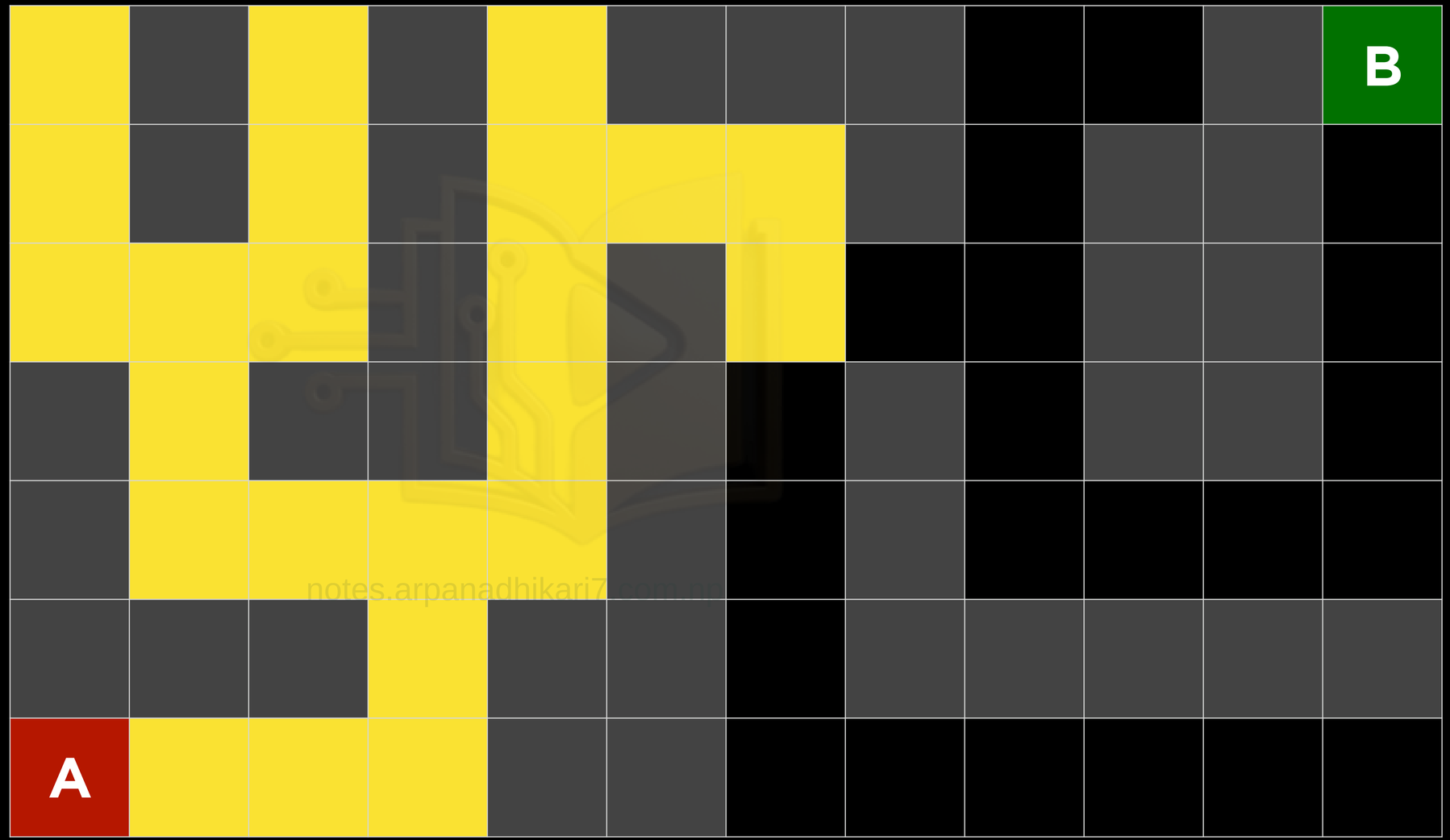
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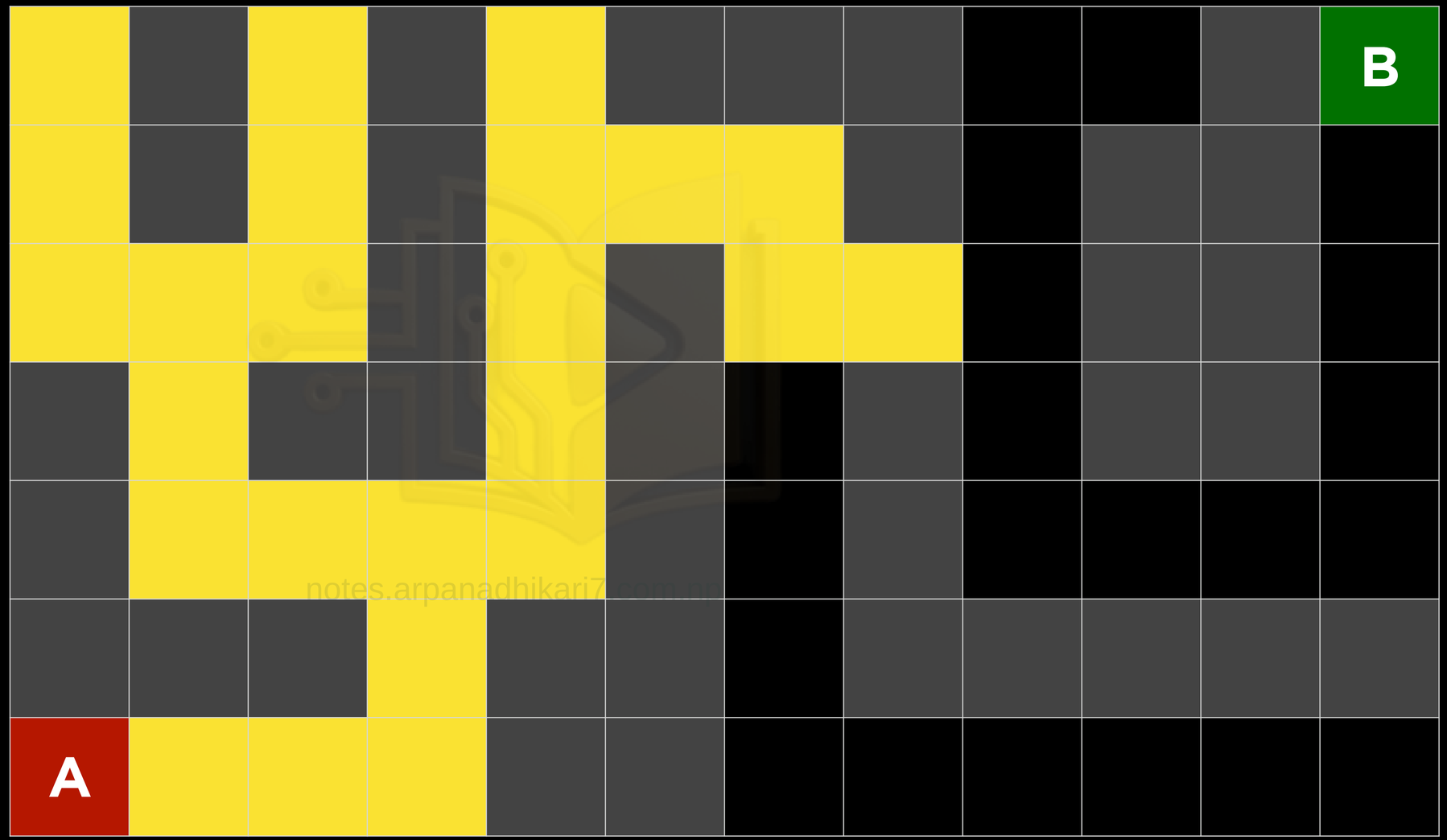
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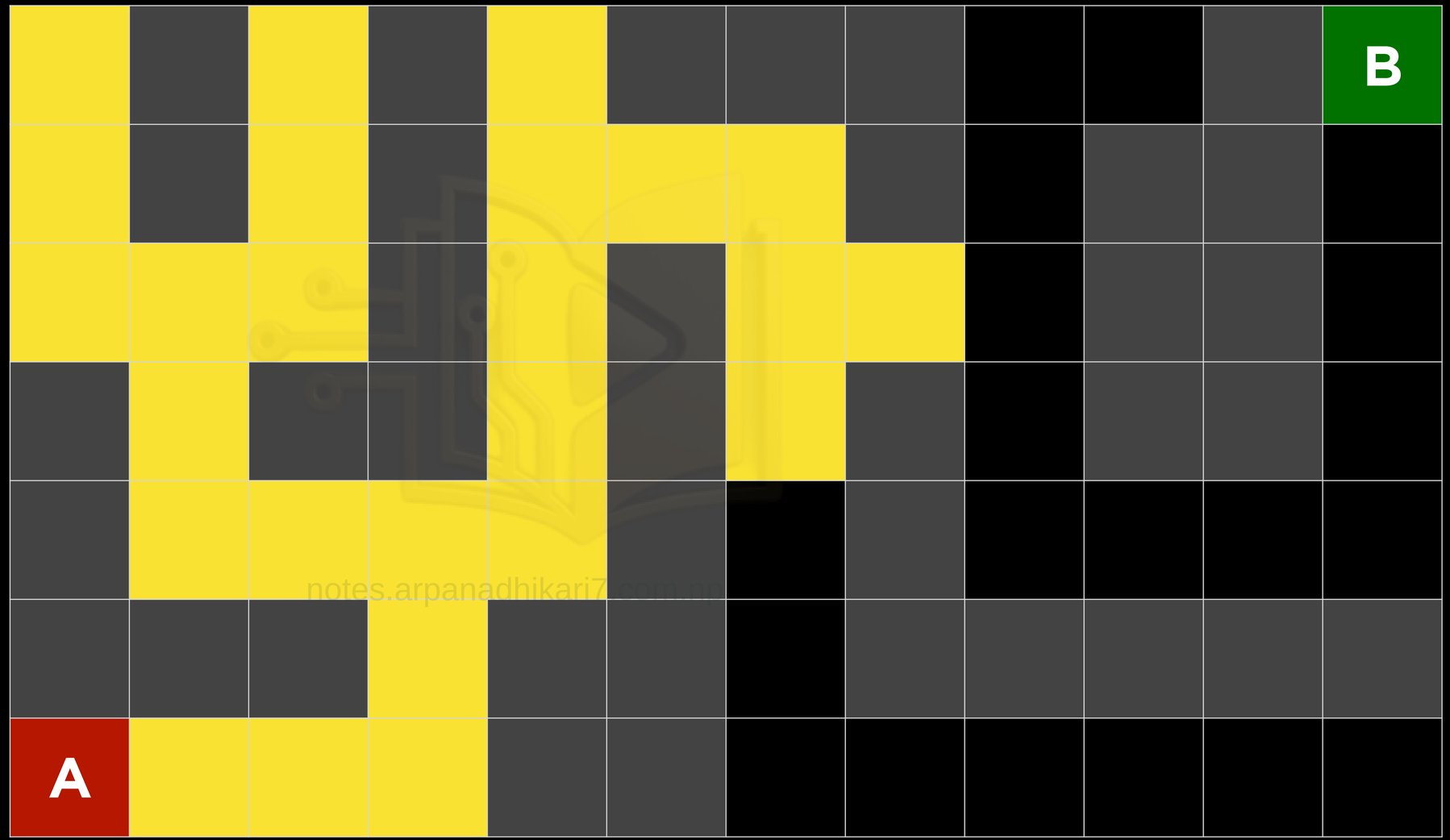
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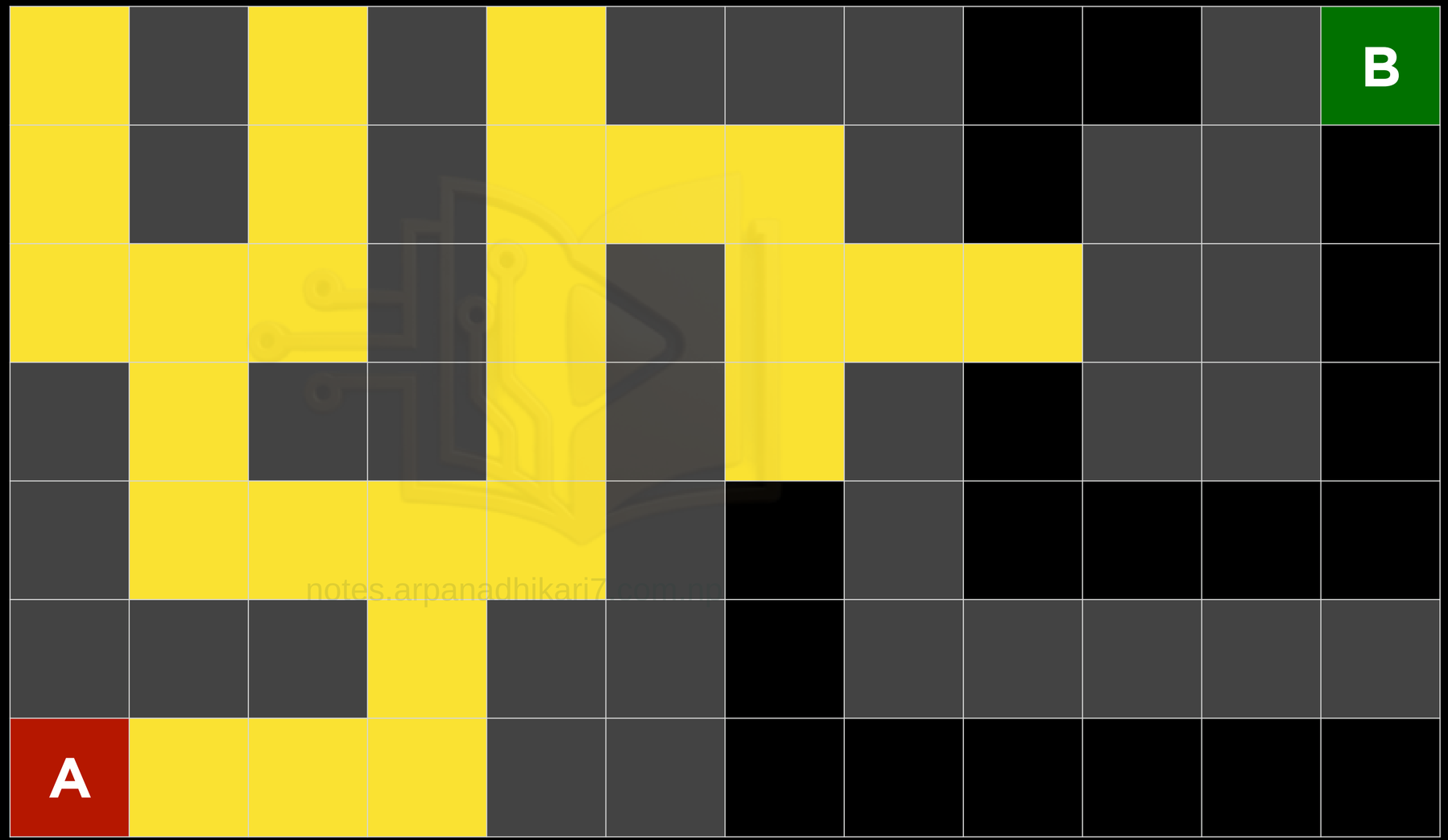
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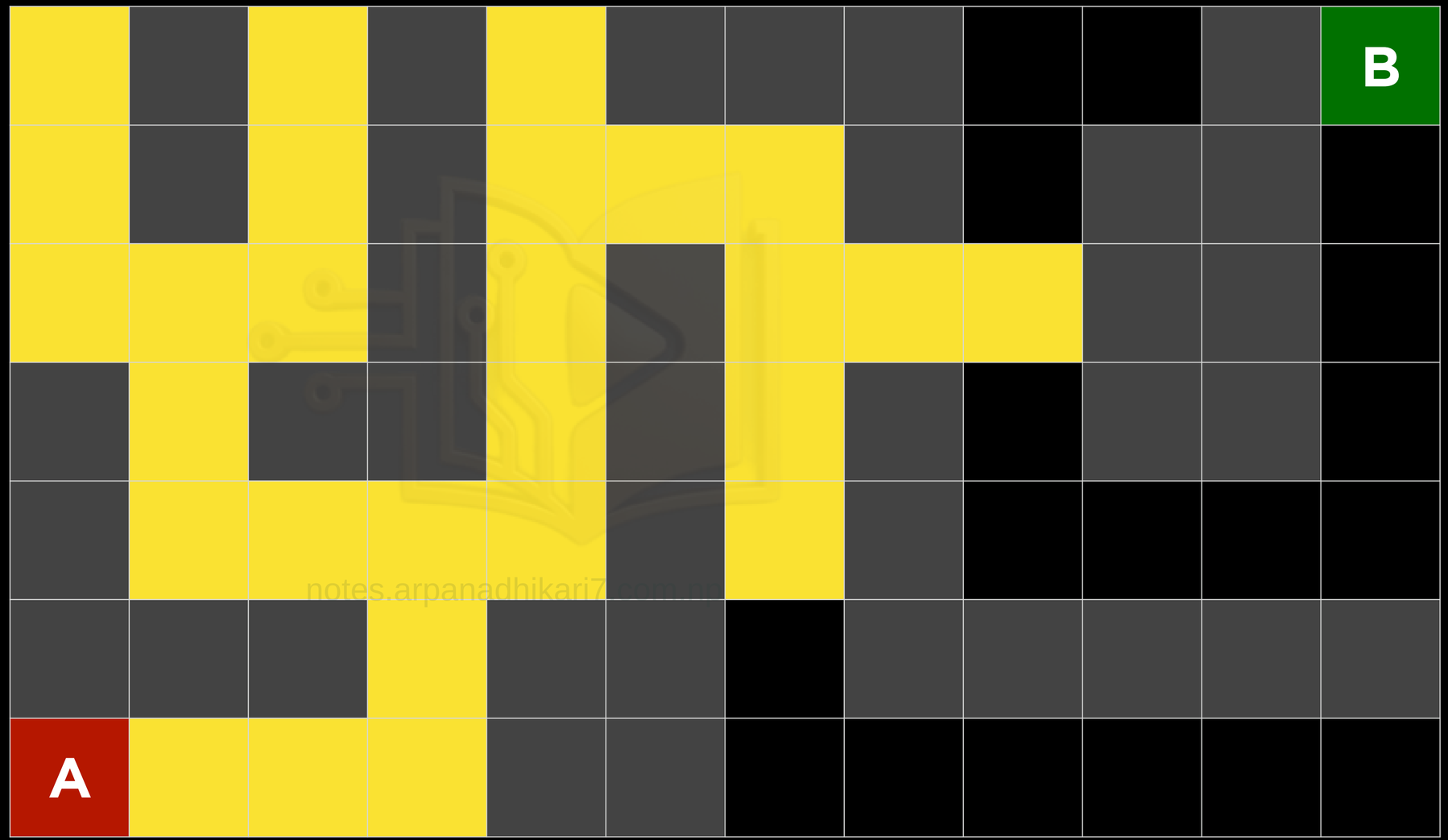
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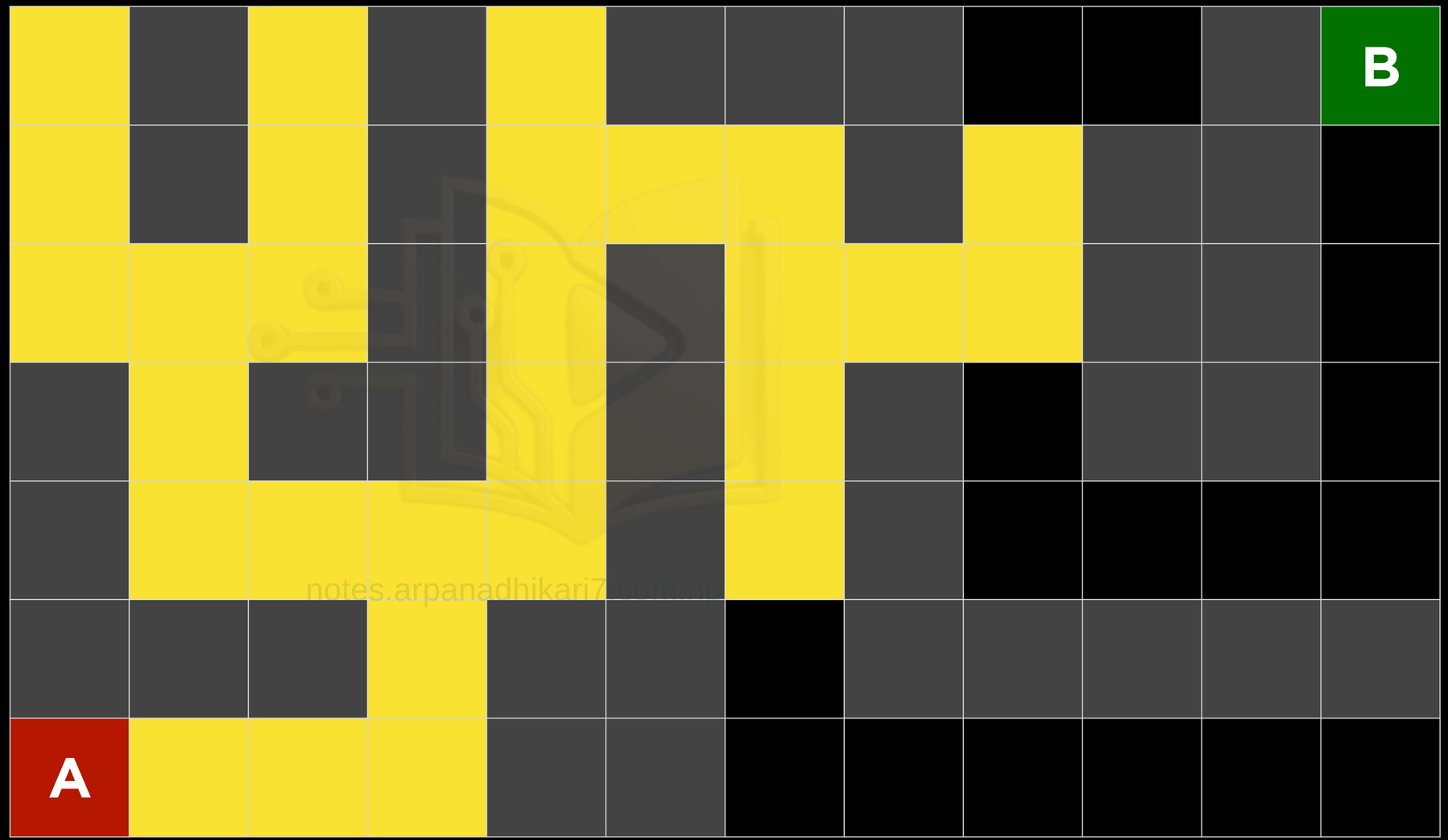
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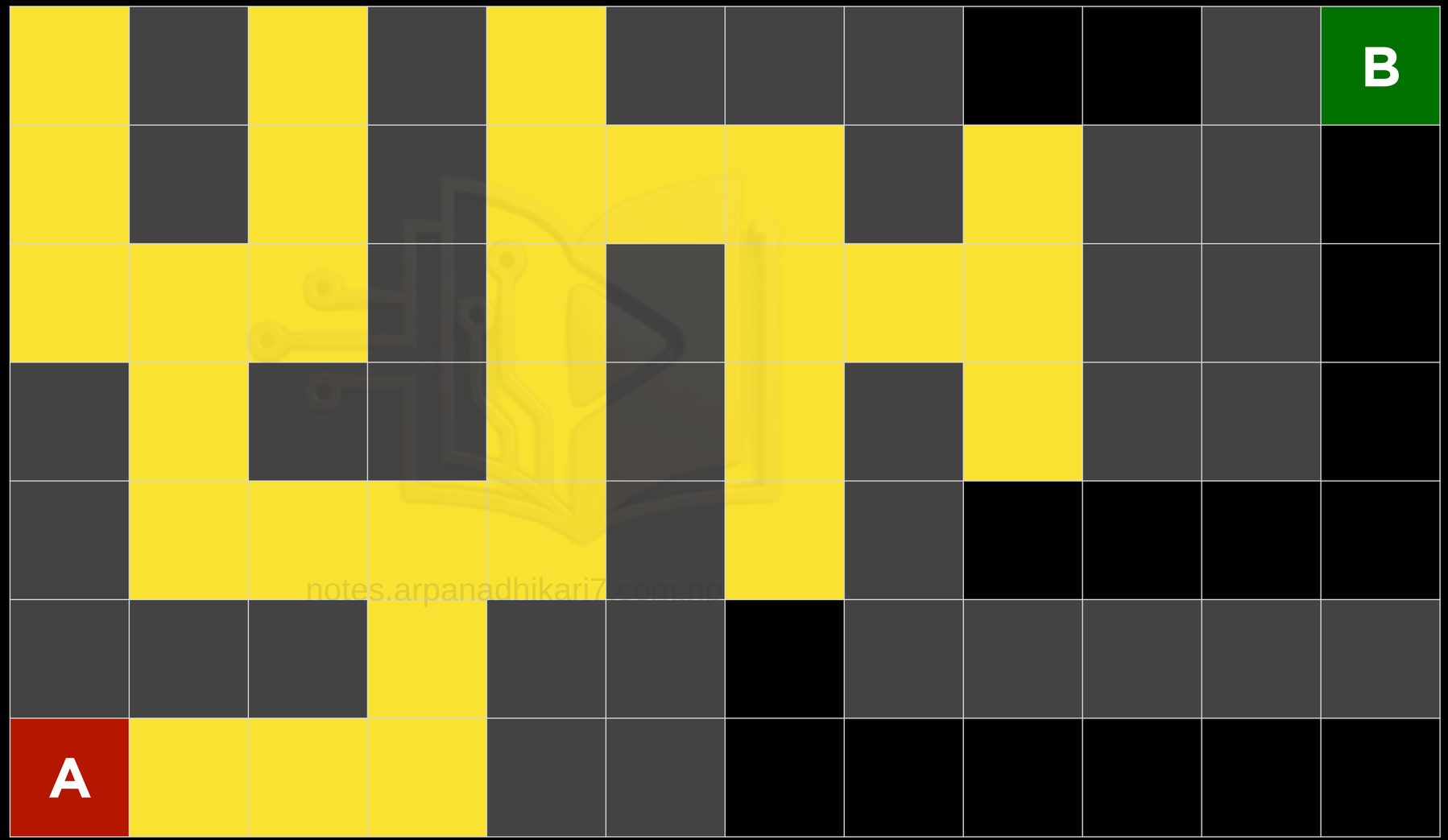
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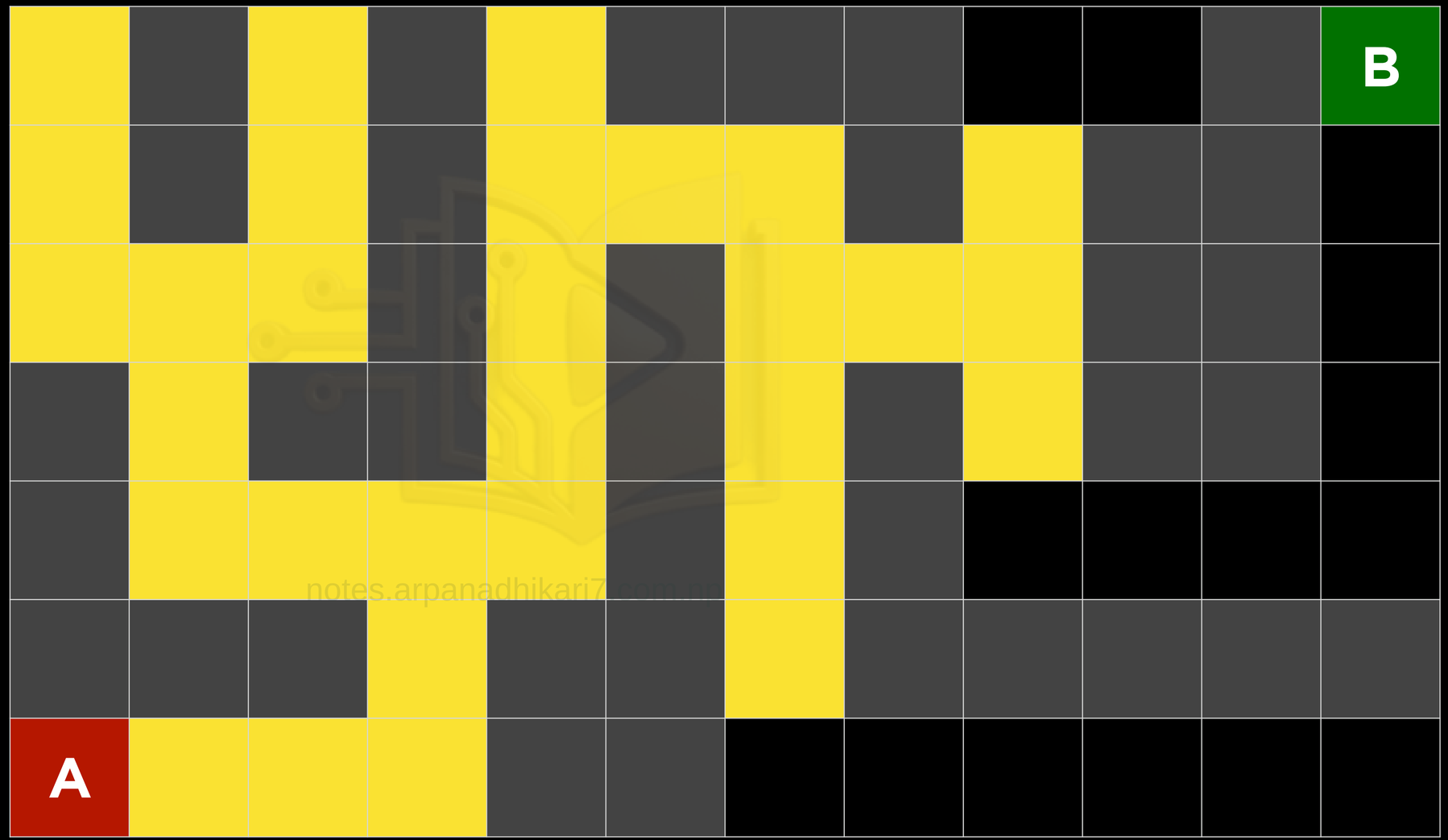
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Breadth-First Search



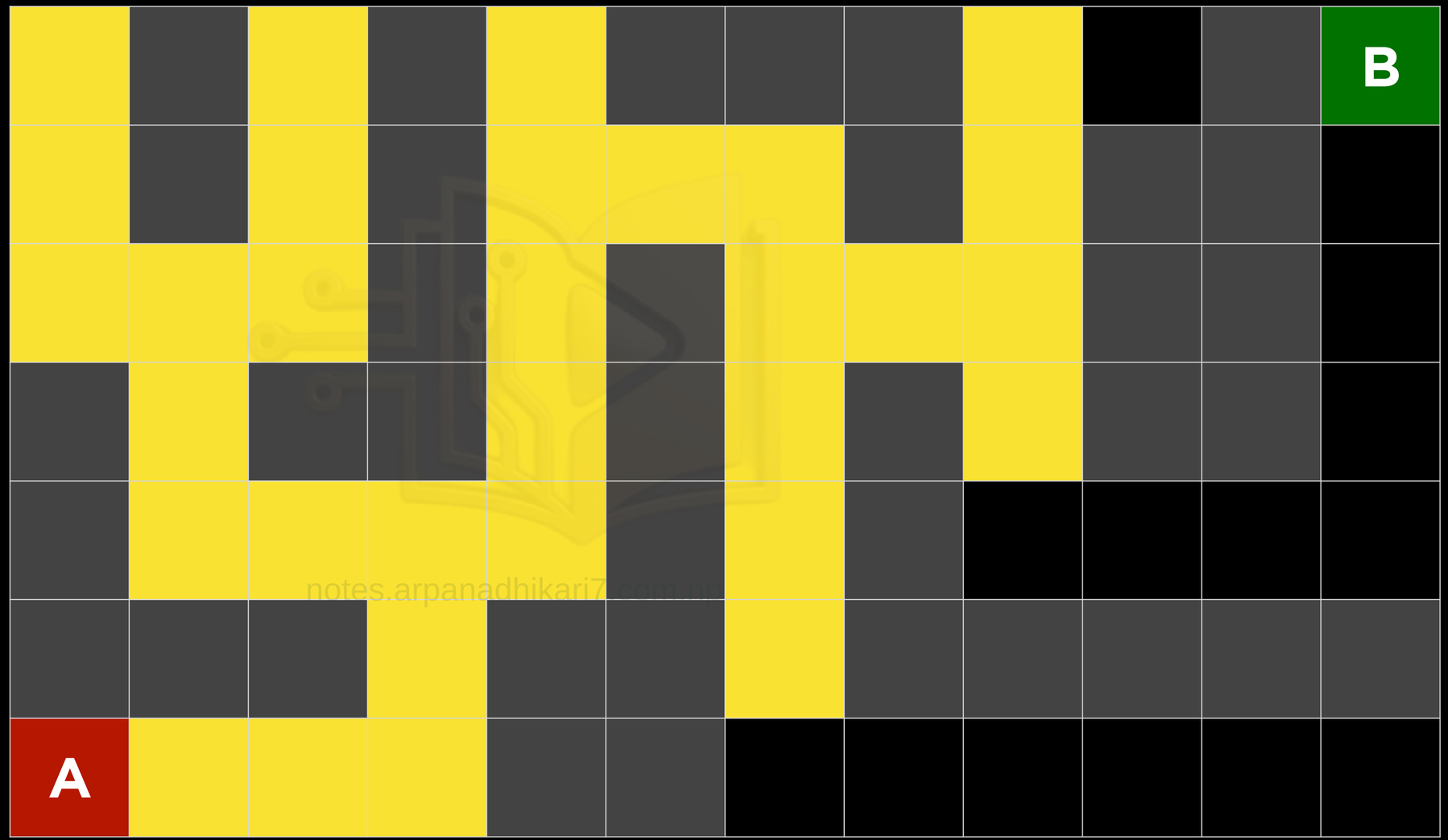
notes.arpanadhikari7.com.np

Breadth-First Search



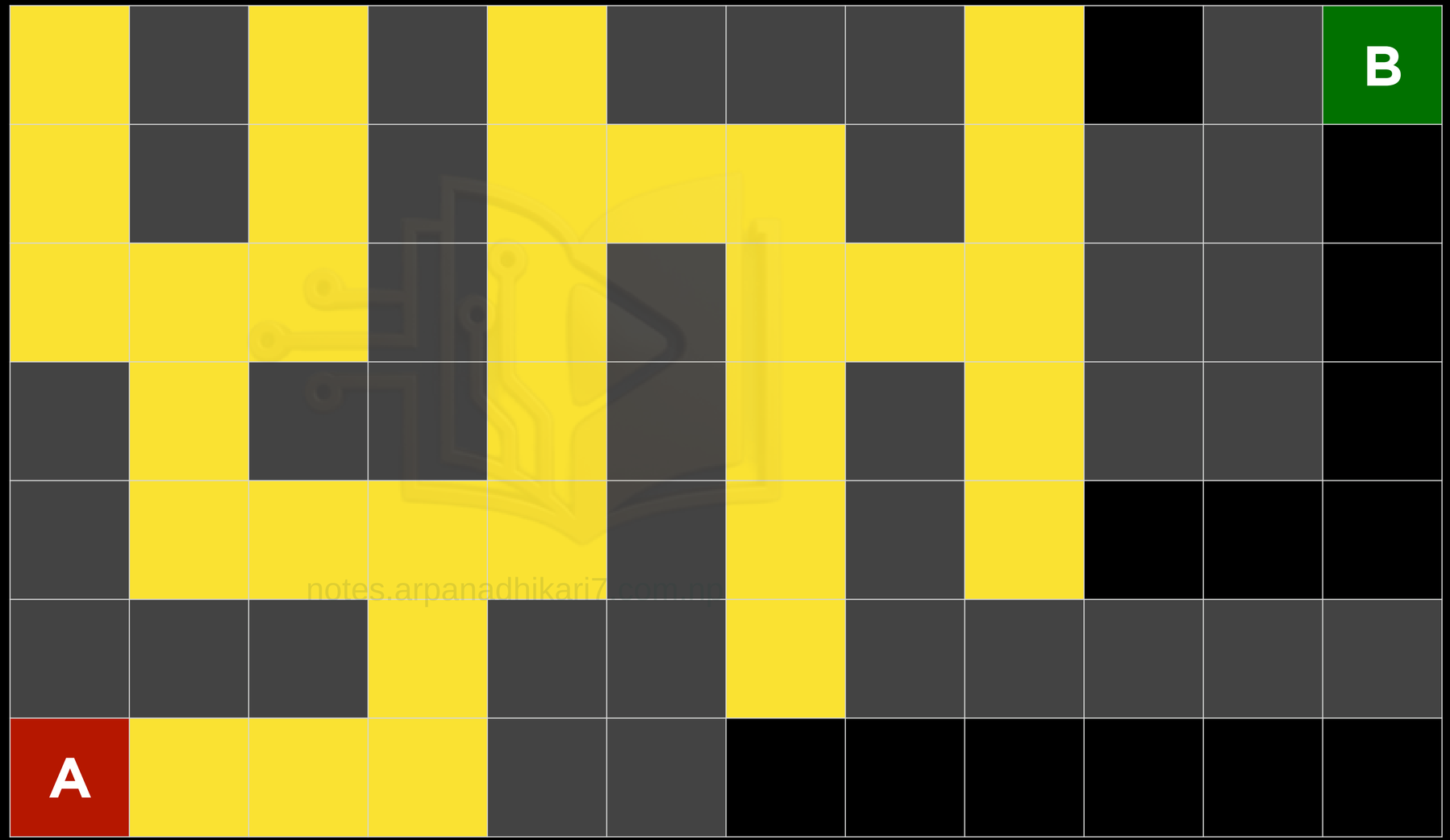
notes.arpanadhikari7.com.np

Breadth-First Search



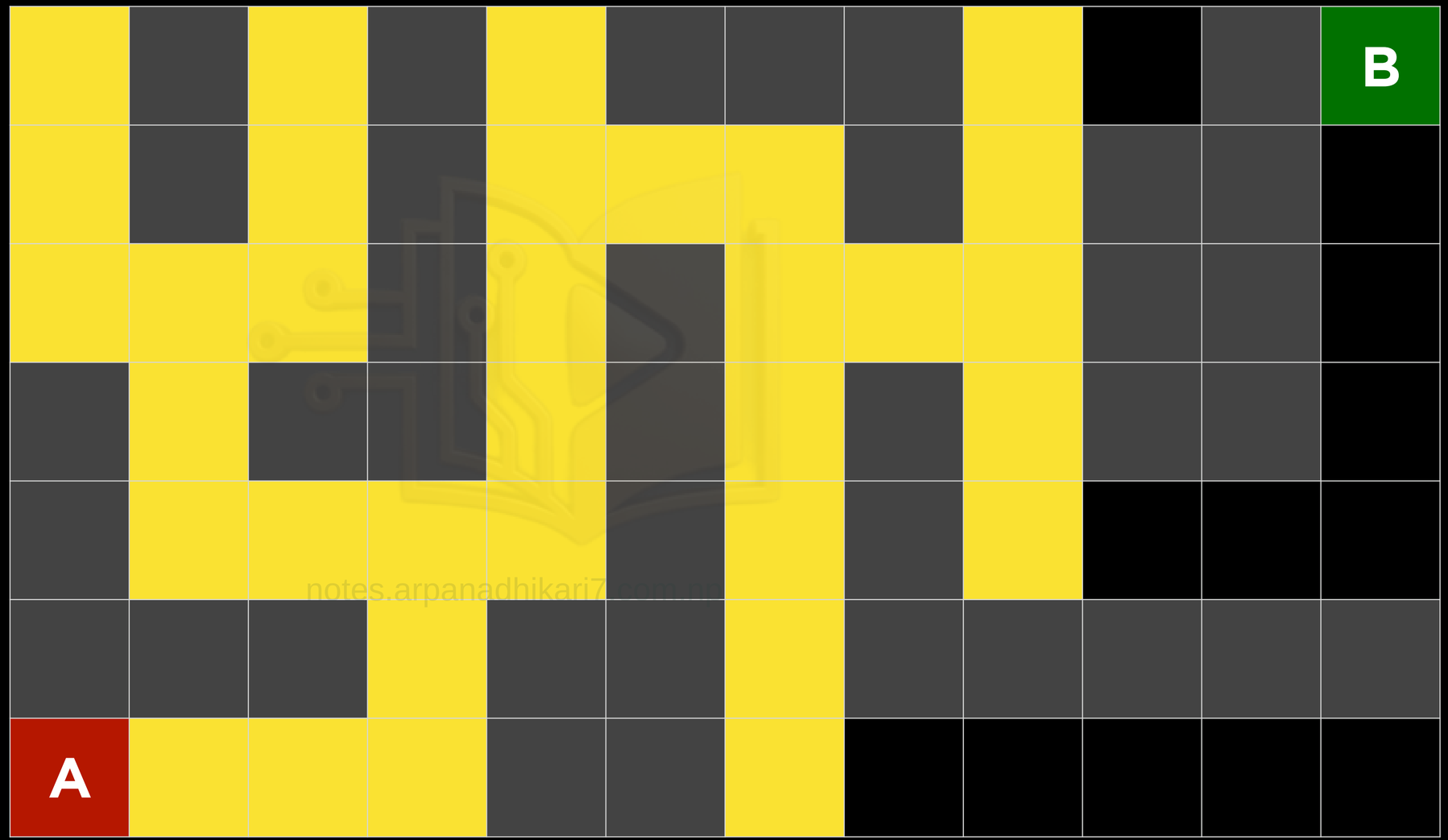
notes.arpanadhikari7.com.np

Breadth-First Search



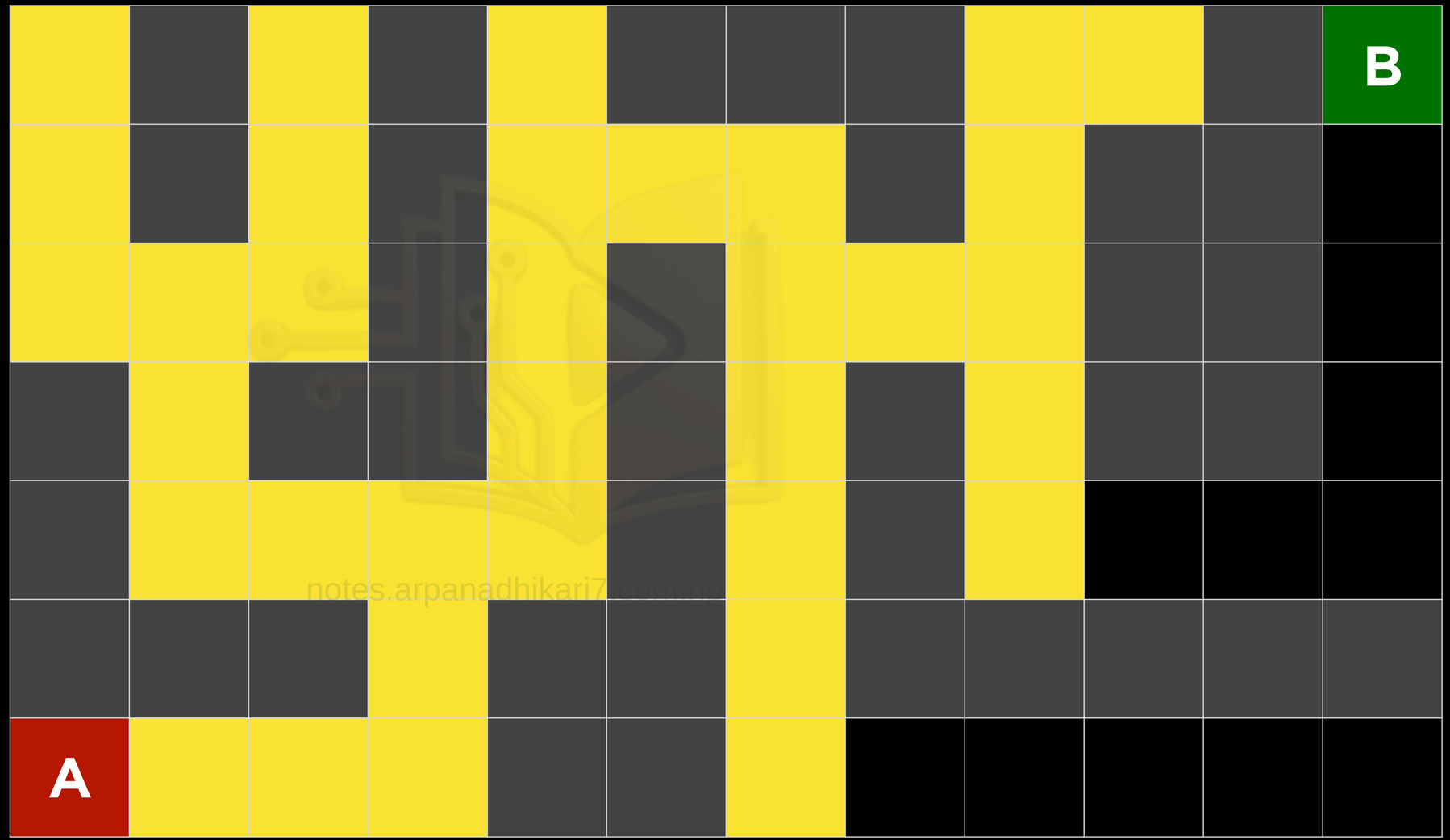
notes.arpanadhikari7.com.np

Breadth-First Search



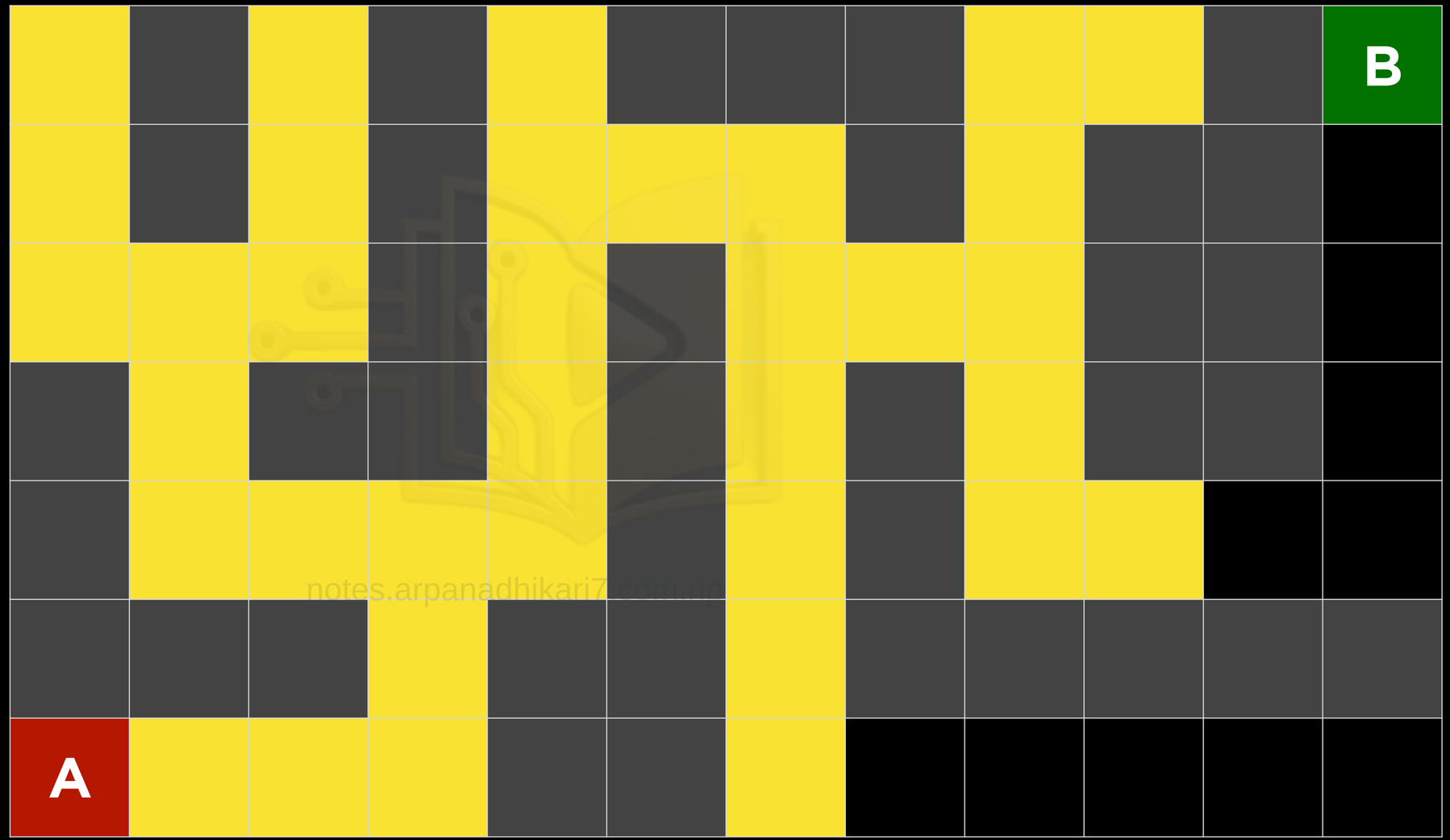
notes.arpanadhikari7.com.np

Breadth-First Search



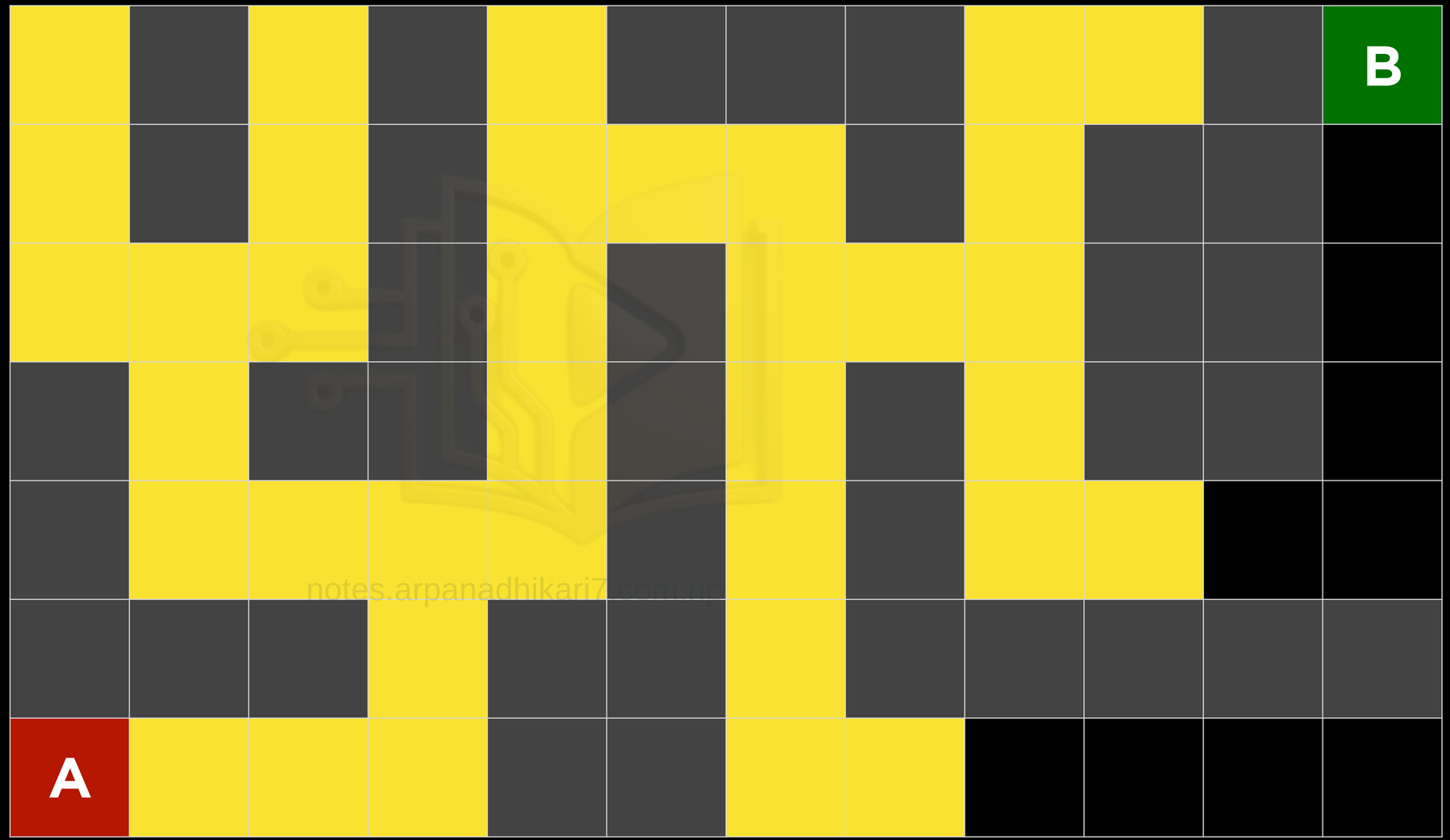
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Breadth-First Search



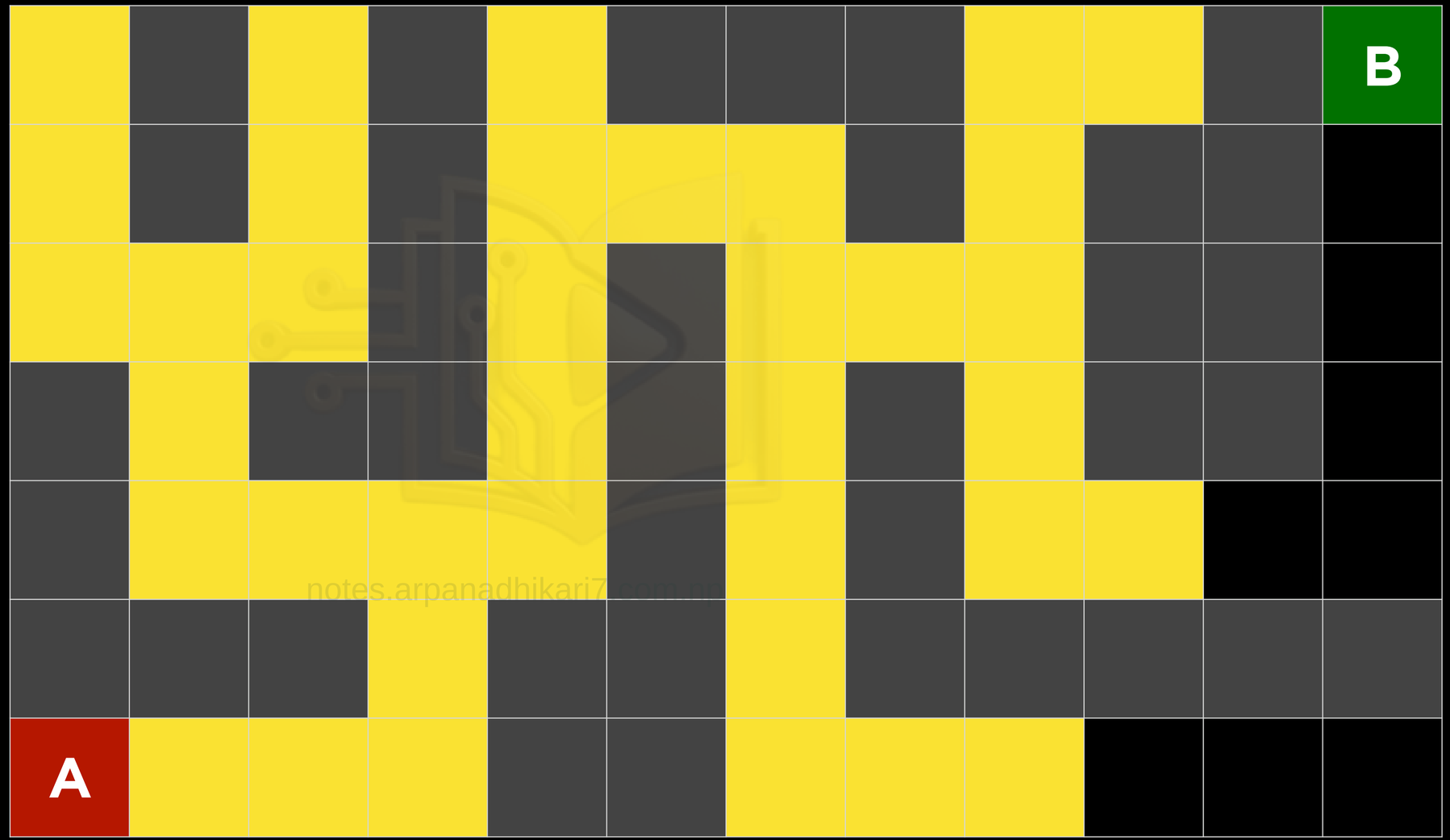
notes.arpanadhikari7.com.np

Breadth-First Search



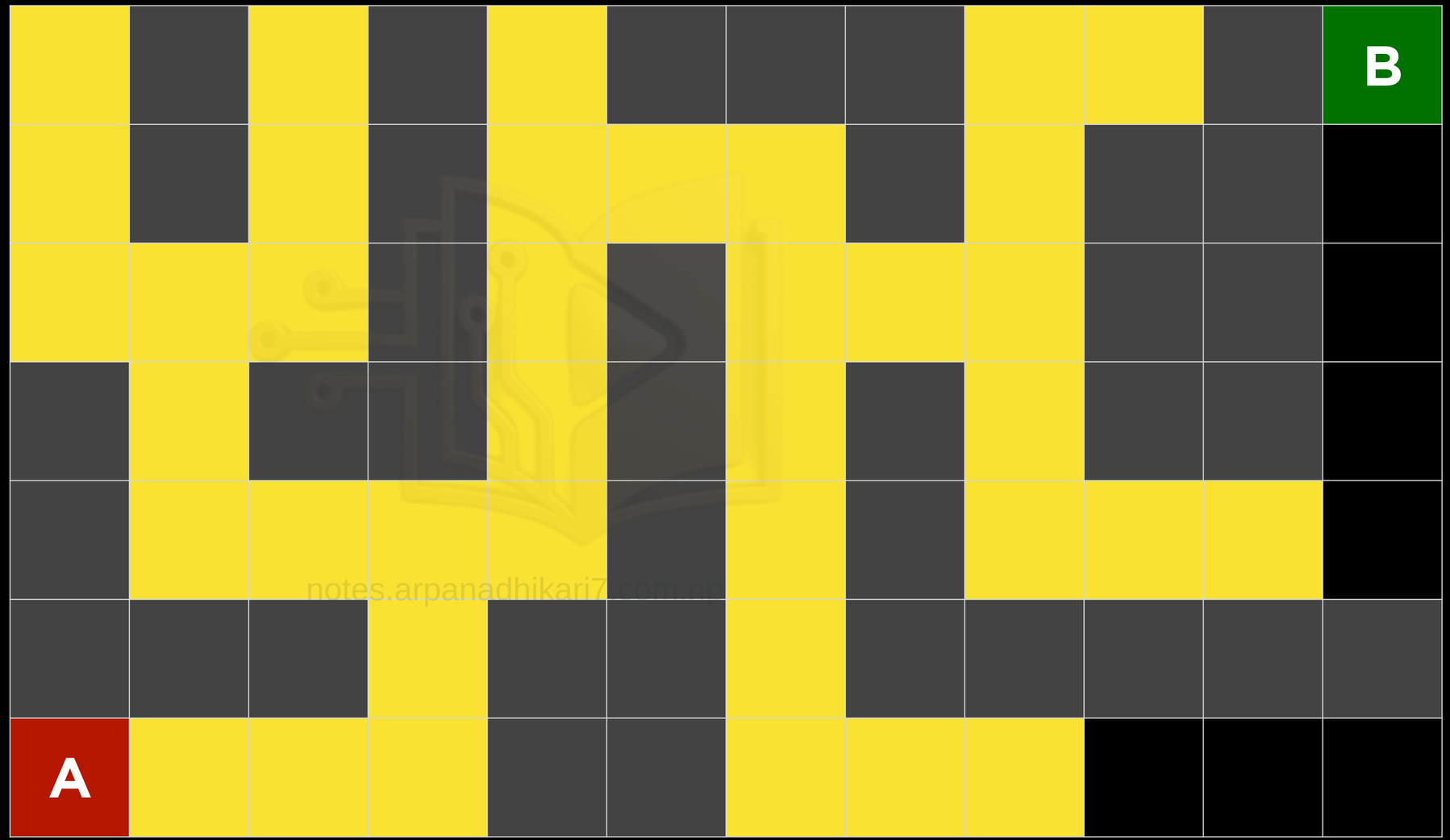
notes.arpanadhikari7.com.np

Breadth-First Search



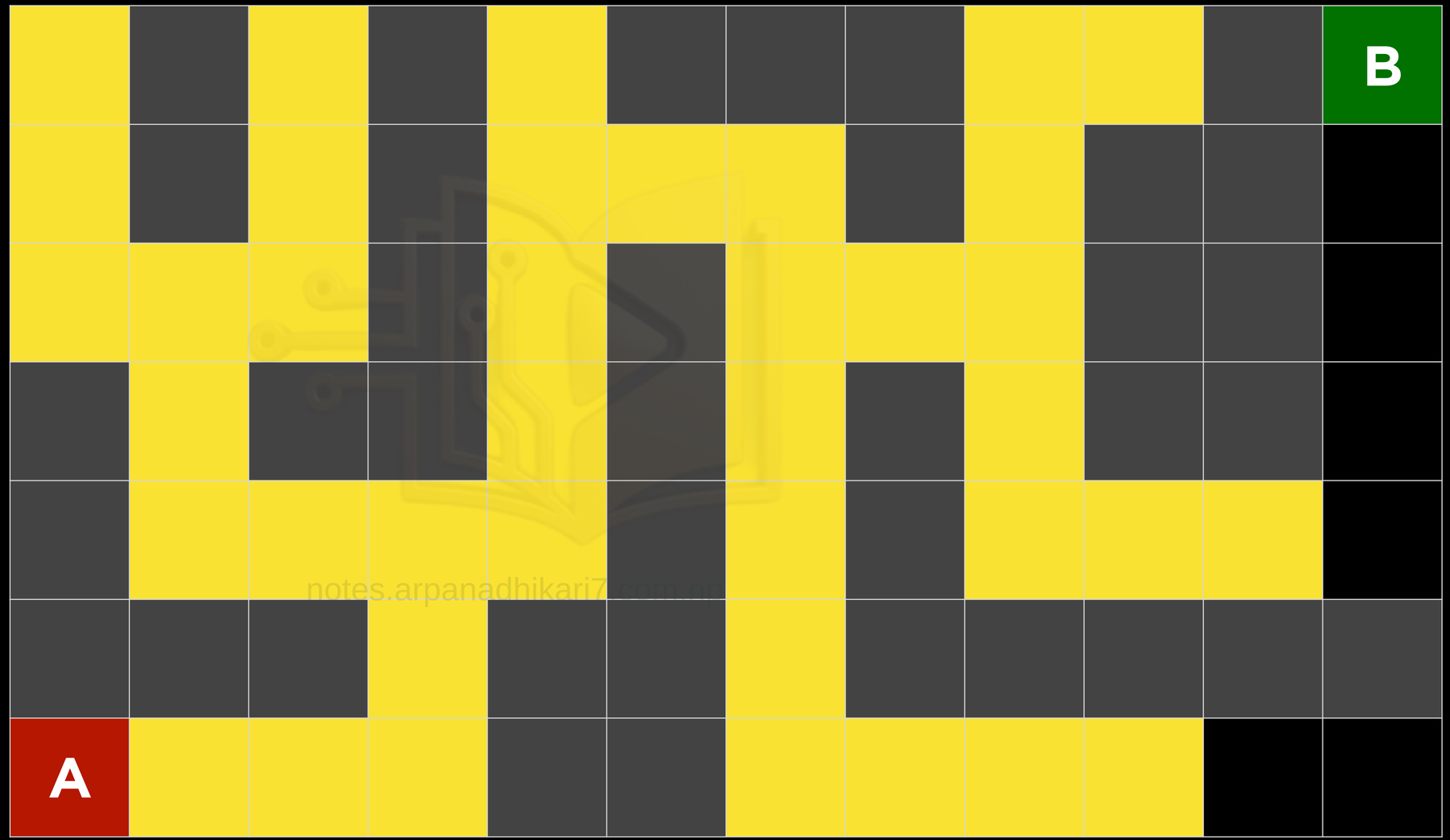
notes.arpanadhikari7.com.np

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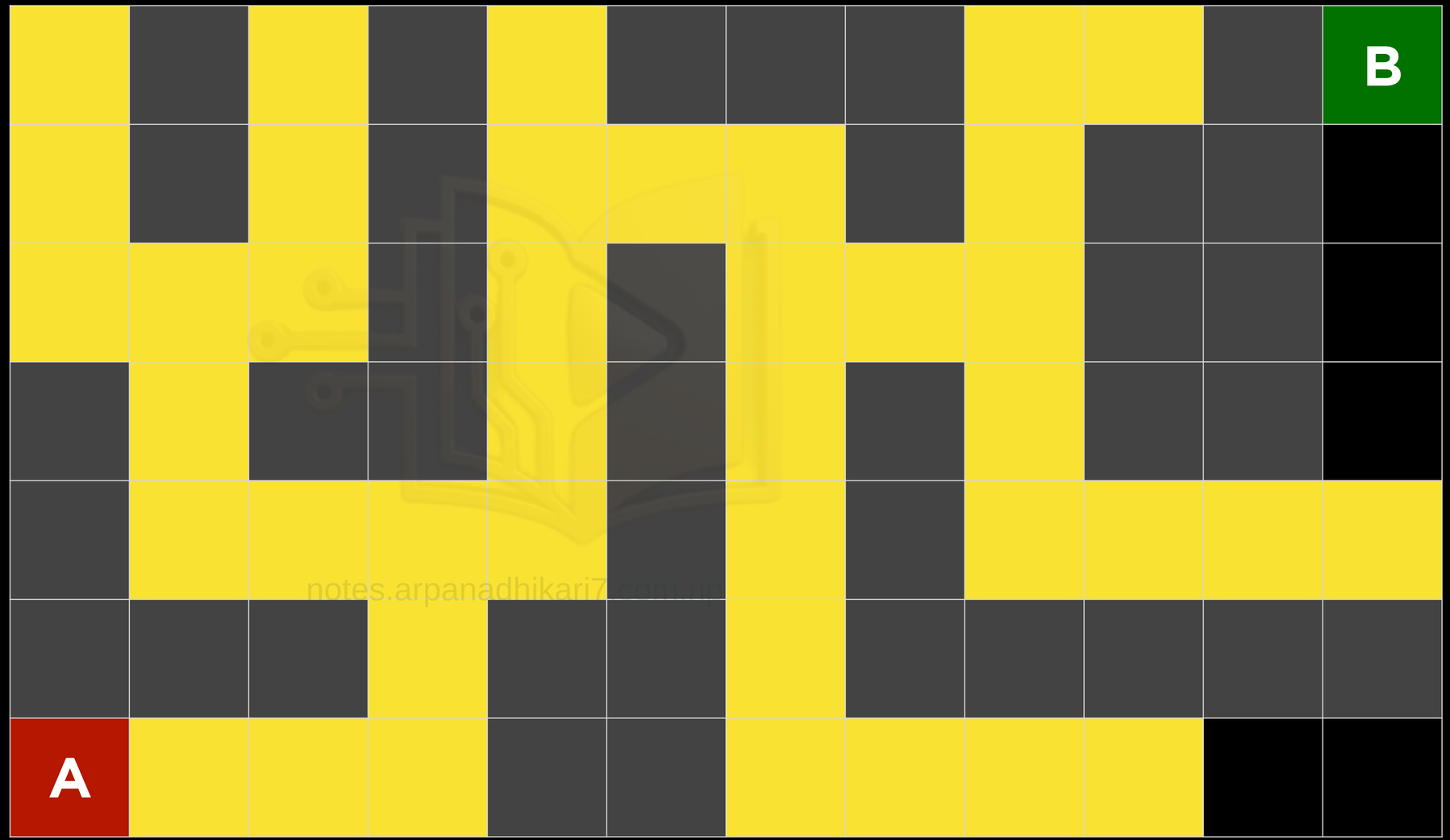
notes.arpanadhikari7.com.np

Breadth-First Search



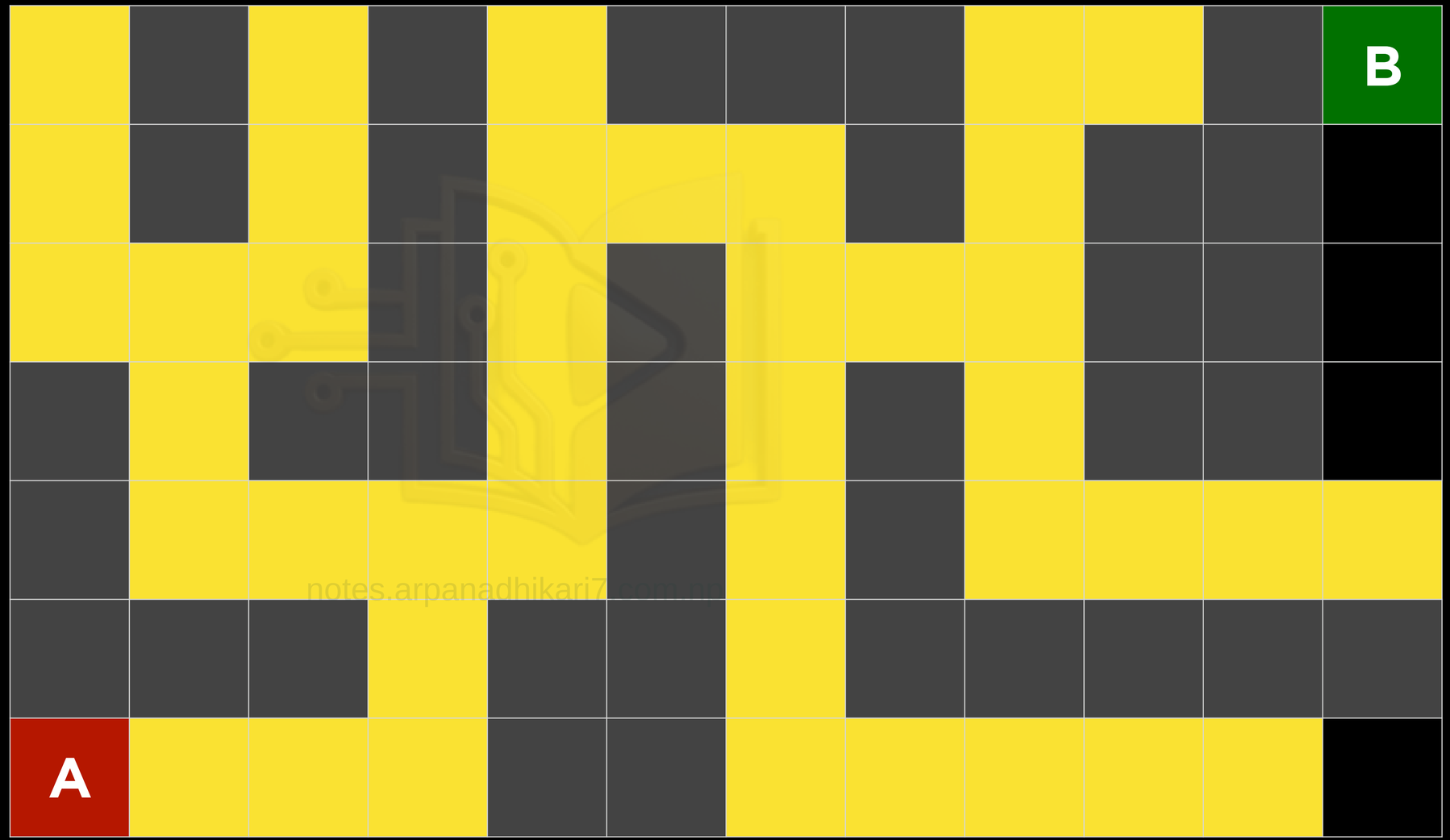
notes.arpanadhikari7.com.np

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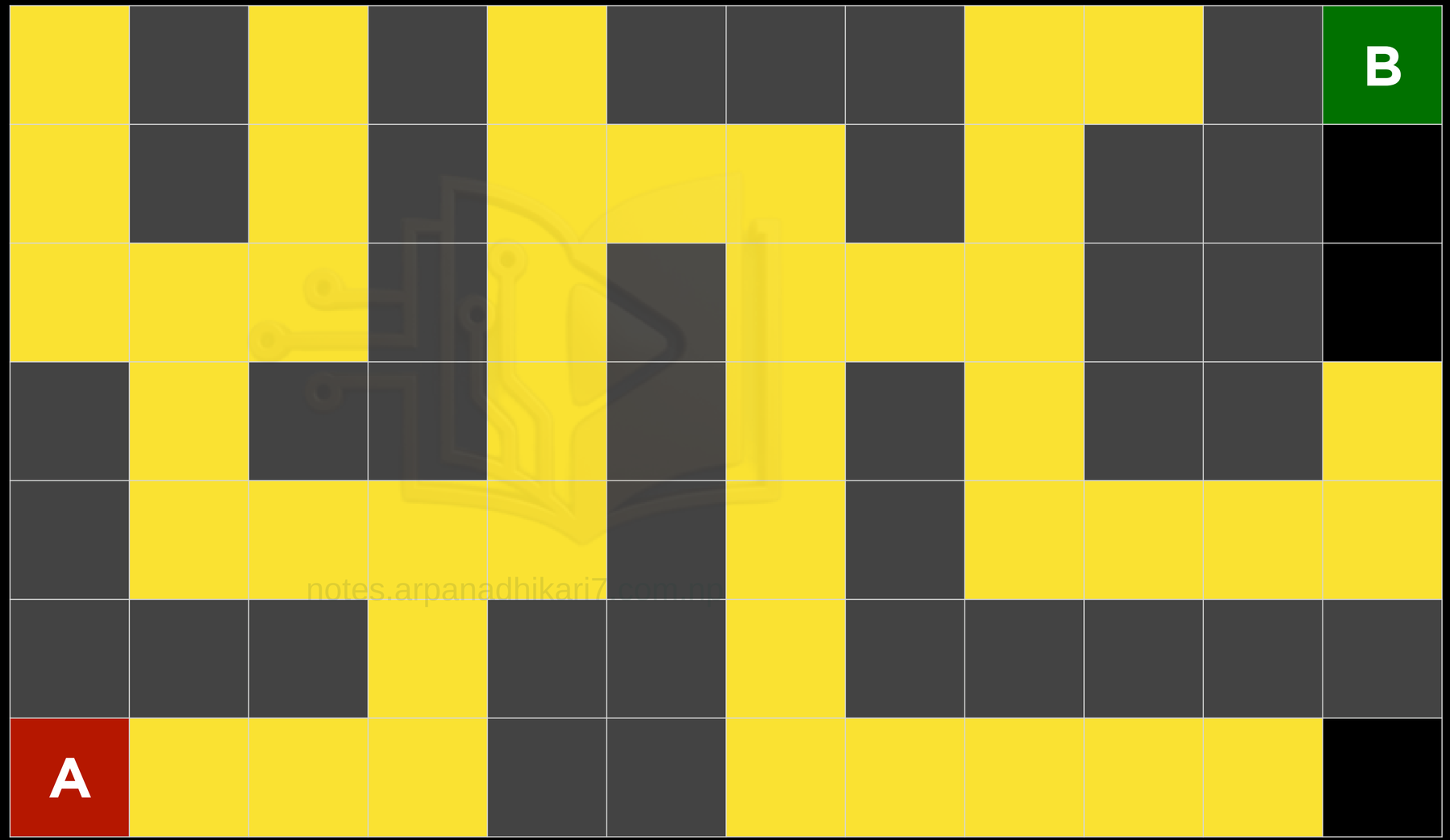
notes.arpanadhikari7.com.np

Breadth-First Search



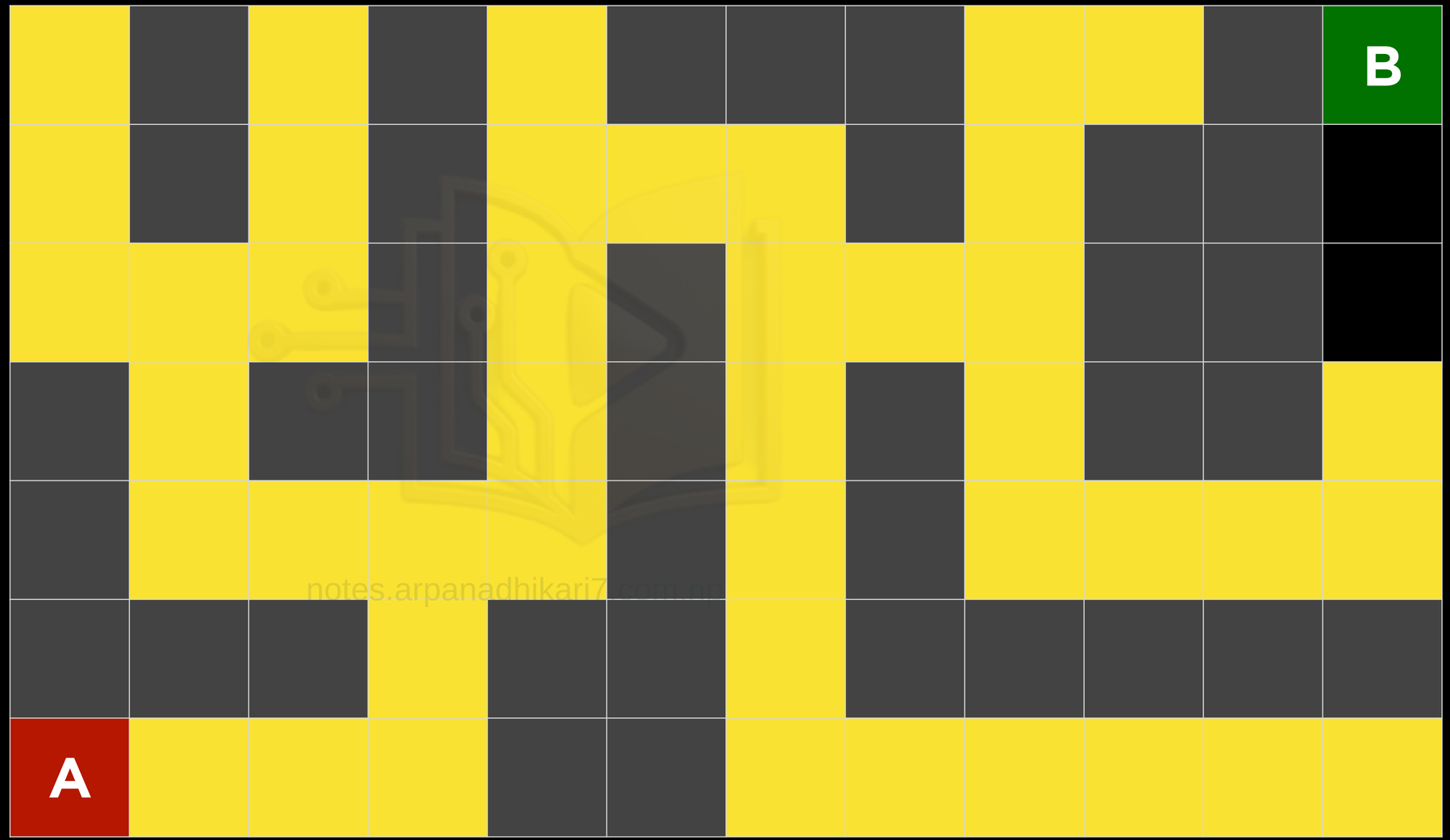
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Breadth-First Search



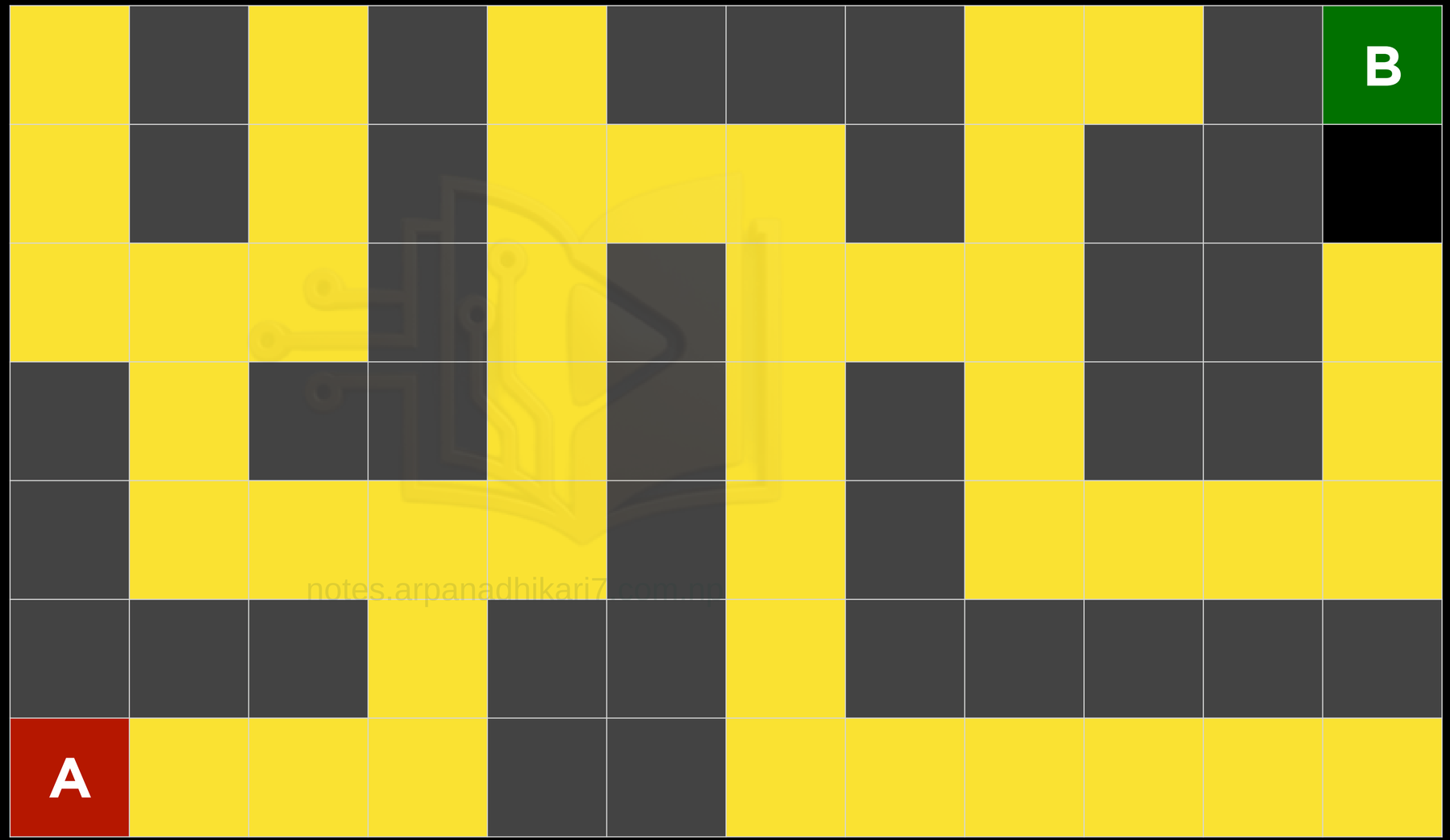
notes.arpanadhikari7.com.np

Breadth-First Search



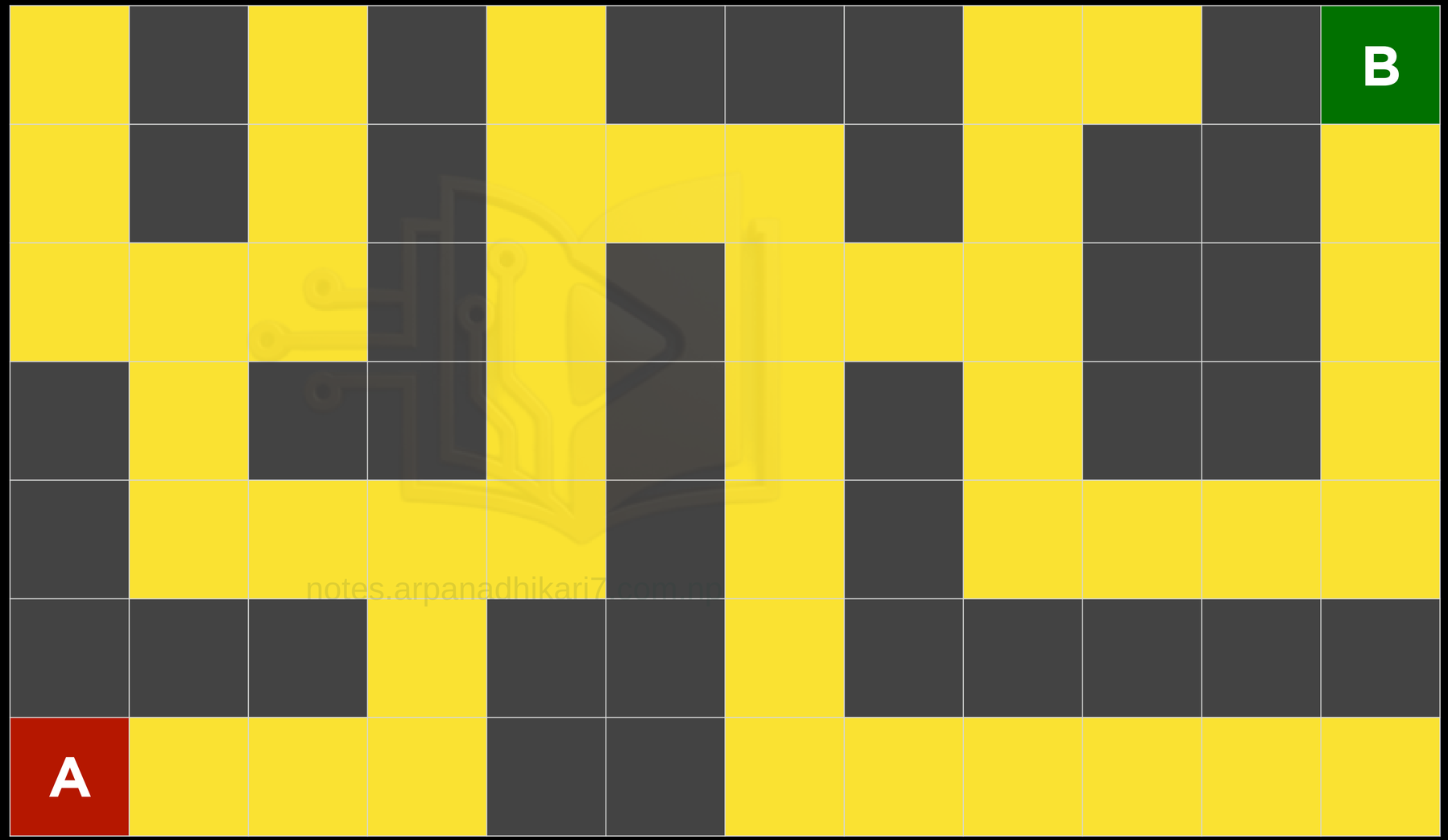
notes.arpanadhikari7.com.np

Breadth-First Search



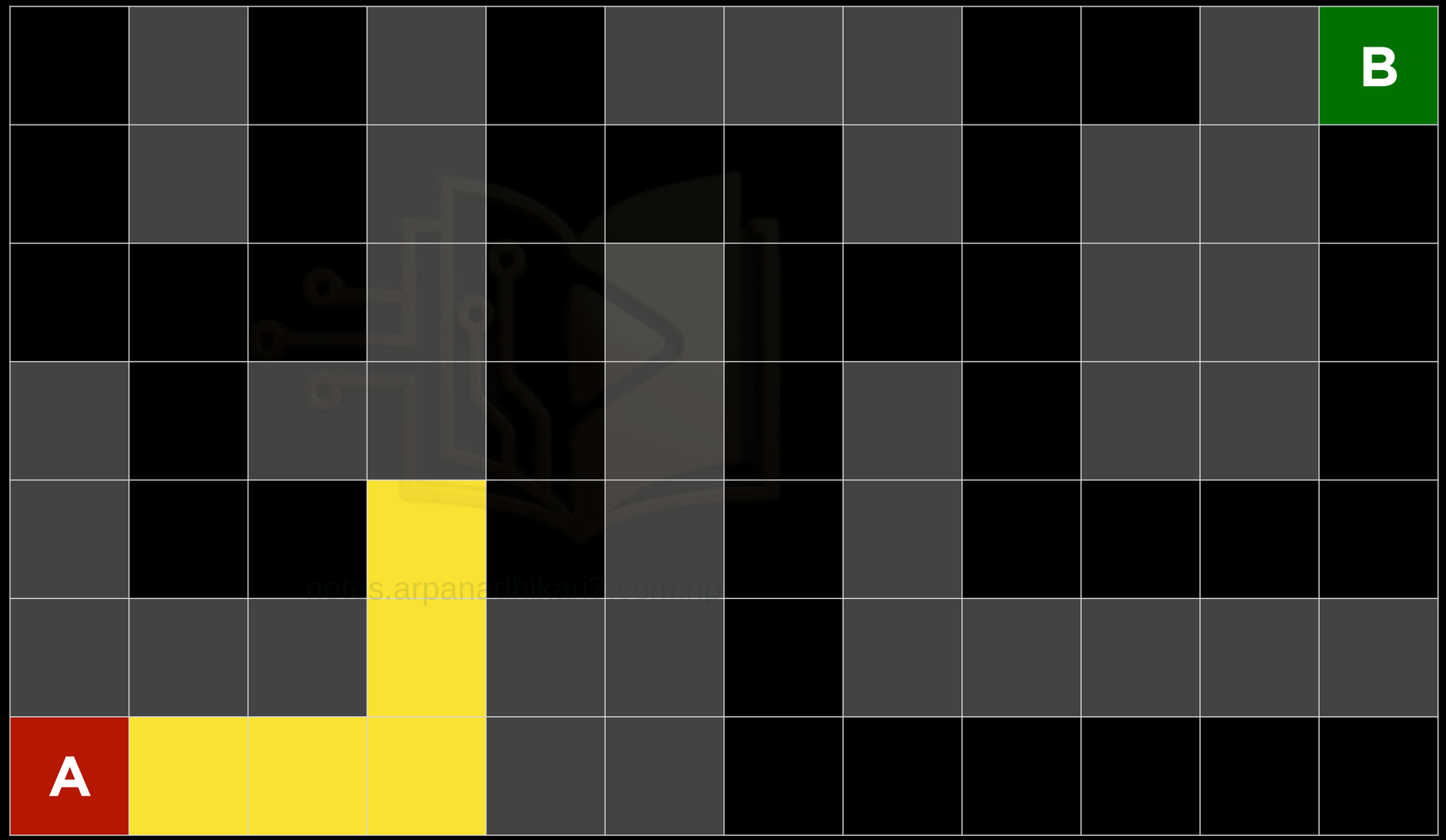
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Breadth-First Search



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Breadth-First Search



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uninformed search

search strategy that uses no problem-specific knowledge

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informed search

search strategy that uses problem-specific knowledge to find solutions more efficiently

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greedy best-first search

search algorithm that expands the node that is closest to the goal, as estimated by a heuristic function $h(n)$

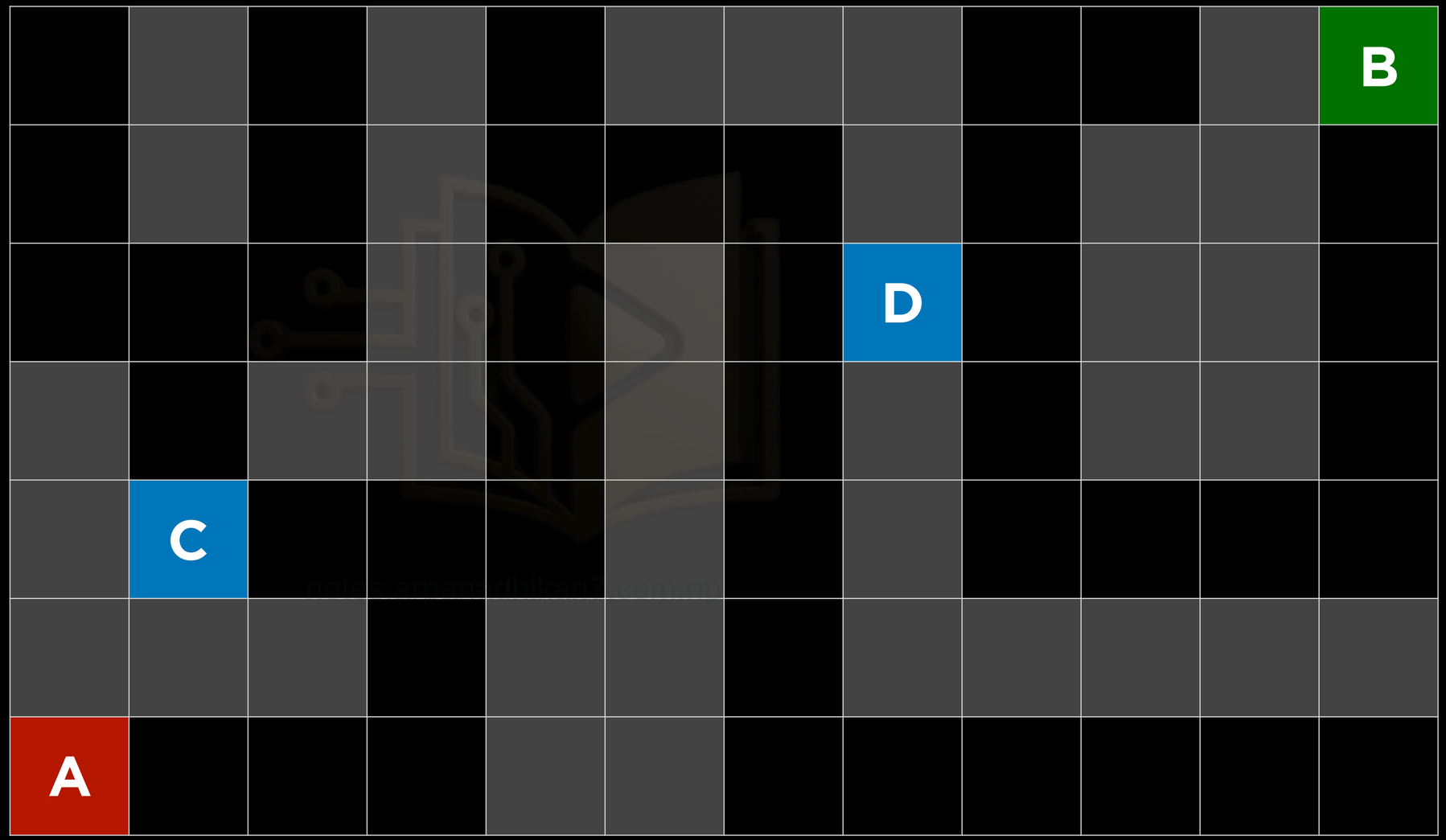
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Heuristic function?



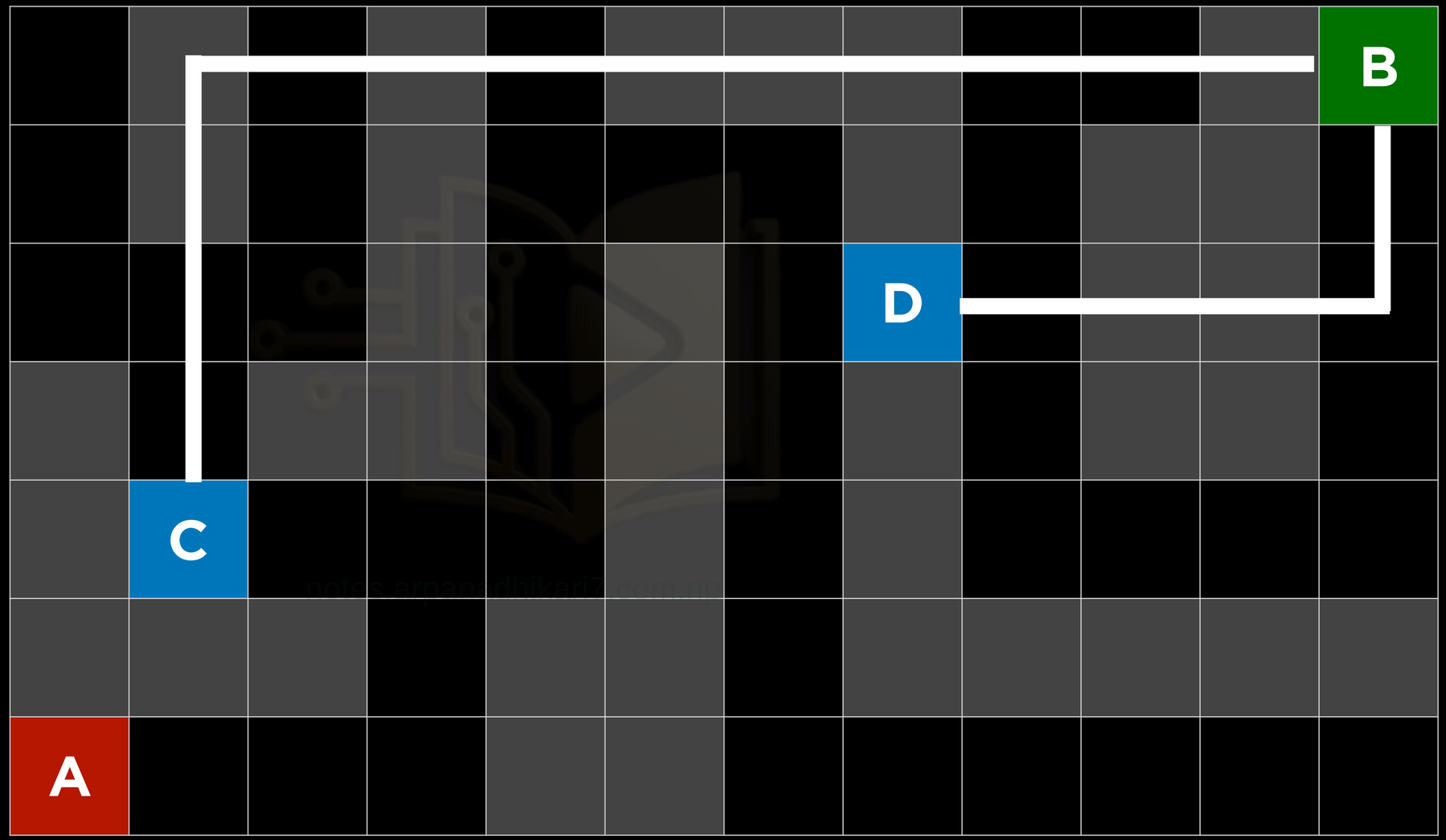
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Heuristic function?



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Heuristic function? Manhattan distance.



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Greedy Best-First Search

11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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Greedy Best-First Search

11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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11		9		7				3	2		**B**
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13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		B
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13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
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	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		**B**
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

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11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

notes.arpanadhikari7.com.np

Greedy Best-First Search

11		9		7				3	2		B
12		10		8	7	6		4			1
13	12	11		9		7	6	5			2
	13			10		8		6			3
	14	13	12	11		9		7	6	5	4
			13			10					
A	16	15	14			11	10	9	8	7	6

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Greedy Best-First Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	16	15	14		12	11	10	9	8	7	6

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Greedy Best-First Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	16	15	14		12	11	10	9	8	7	6

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Greedy Best-First Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	16	15	14		12	11	10	9	8	7	6

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A* search

search algorithm that expands node with lowest value of $g(n) + h(n)$

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$g(n)$ = cost to reach node
$h(n)$ = estimated cost to goal

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	16	15	14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	1+16	15	14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	1+16	2+15	14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

Available on notes.arpanadhikari7.com.np

A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

notes.arpanadhikari7.com.np

A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		10	9	8	7	6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	9	8	7	6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	8	7	6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	7	6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

notes.arpanadhikari7.com.np

A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	13		6+11						5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	B
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	13		6+11						14+5		3
	14	13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

notes.arpanadhikari7.com.np

A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	13		6+11						14+5		3
	14	6+13	5+12		10	9	8	7	6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	B
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	13		6+11						14+5		3
	14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	**B**
	11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	10	9	8	7	6	5	4	3	2	1	B
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	9	8	7	6	5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	8	7	6	5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	7	6	5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	6	5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	16+5	4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	16+5	17+4	3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	16+5	17+4	18+3	2	1	**B**
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	16+5	17+4	18+3	19+2	1	B
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* Search

	11+10	12+9	13+8	14+7	15+6	16+5	17+4	18+3	19+2	20+1	B
	10+11										1
	9+12		7+10	8+9	9+8	10+7	11+6	12+5	13+4		2
	8+13		6+11						14+5		3
	7+14	6+13	5+12		10	9	8	7	15+6		4
			4+13		11						5
A	1+16	2+15	3+14		12	11	10	9	8	7	6

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A* search

optimal if

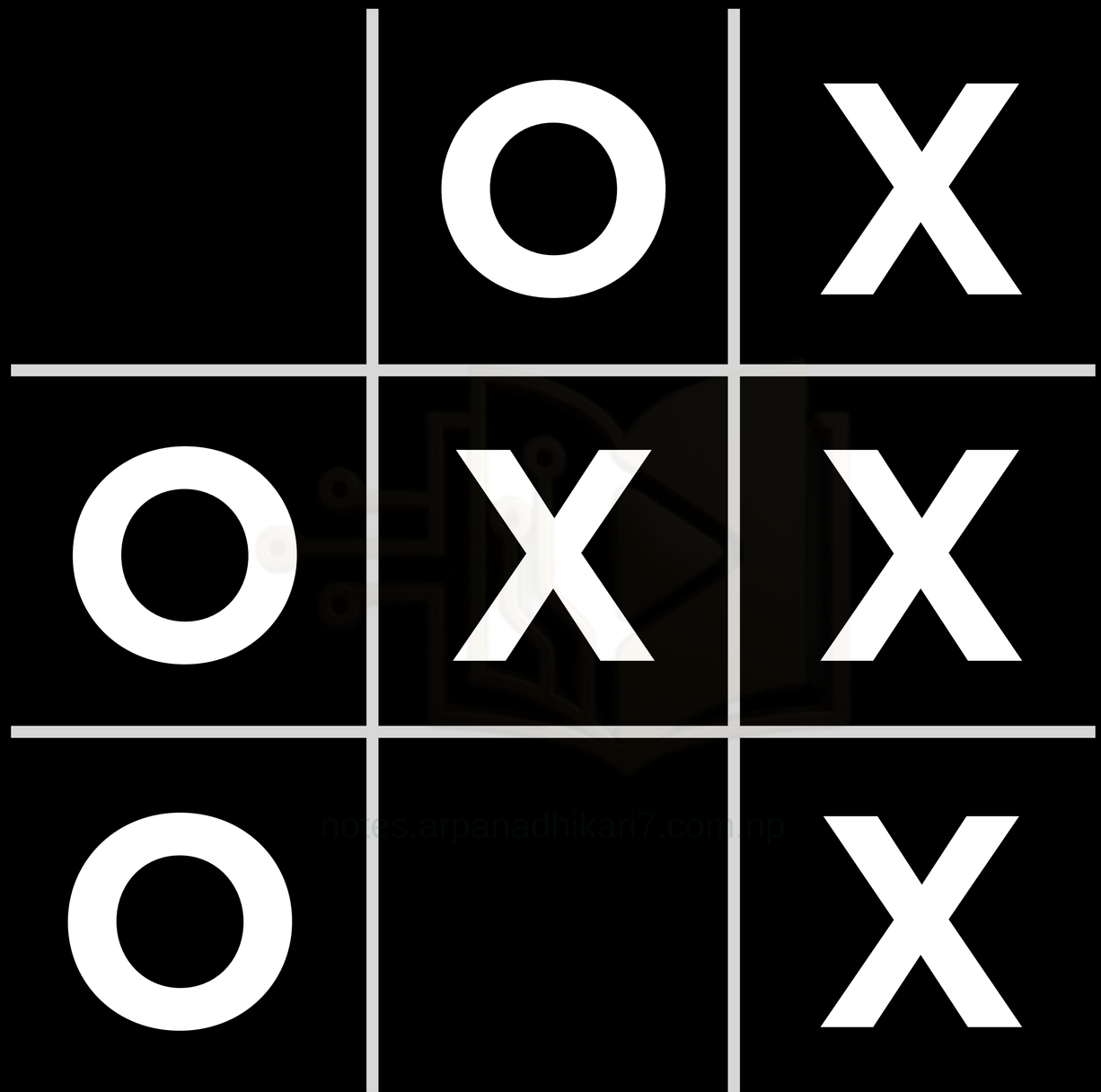
- $h(n)$ is admissible (never overestimates the true cost), and
- $h(n)$ is consistent (for every node n and successor n' with step cost c, $h(n) \leq h(n') + c$)

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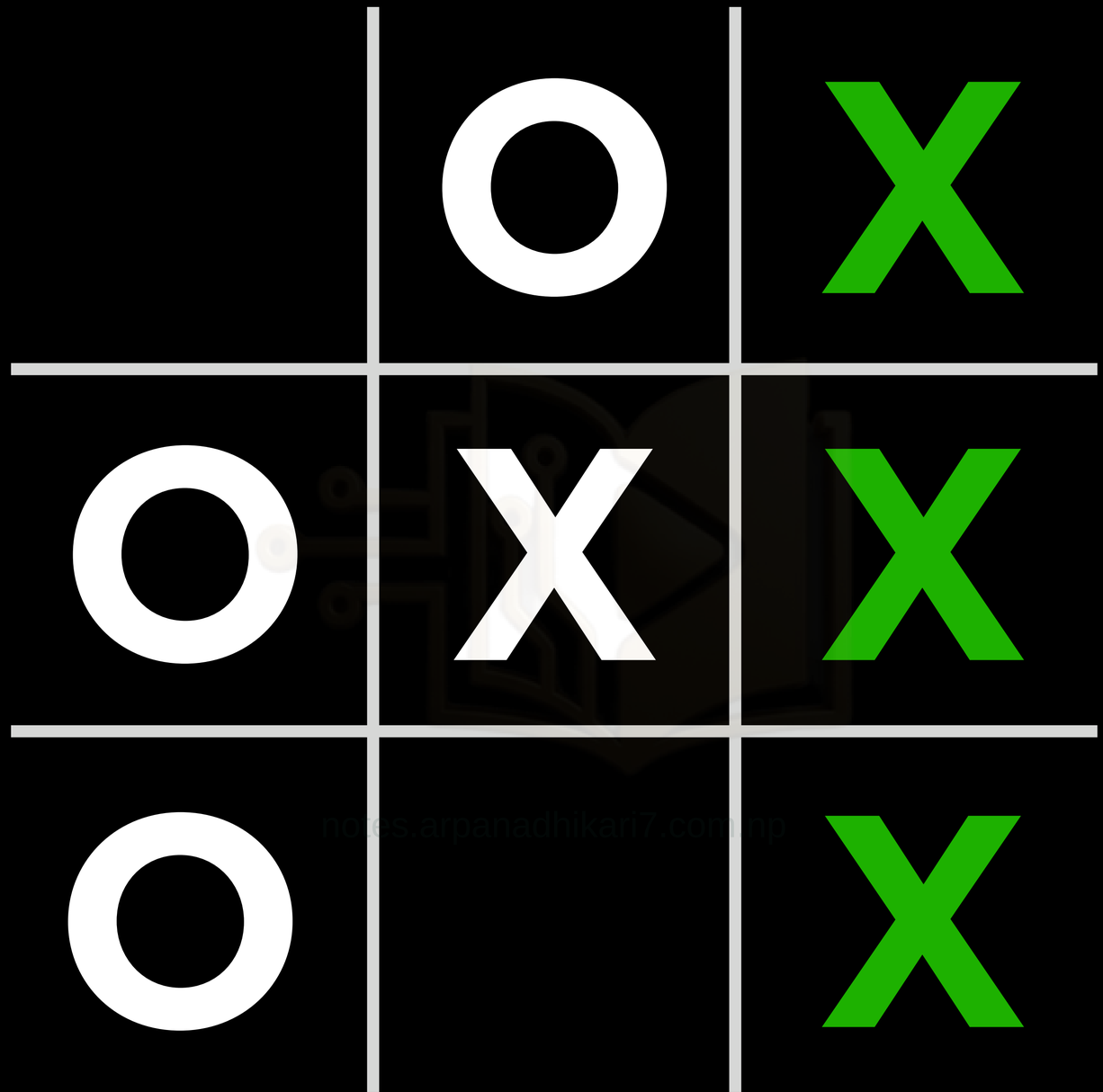
Available on notes.arpanadhikari7.com.np

Adversarial Search

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Minimax

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O	X	X
O	O	
O	X	X

-1

X	O	X
O	O	X
X	X	O

0

O		X
	X	O
X	O	X

1

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Minimax

- **MAX** (X) aims to maximize score.
- **MIN** (O) aims to minimize score.

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Game

- S_0 : initial state
- $\textsc{Player}(s)$: returns which player to move in state s
- $\textsc{Actions}(s)$: returns legal moves in state s
- $\textsc{Result}(s, a)$: returns state after action a taken in state s
- $\textsc{Terminal}(s)$: checks if state s is a terminal state
- $\textsc{Utility}(s)$: final numerical value for terminal state s

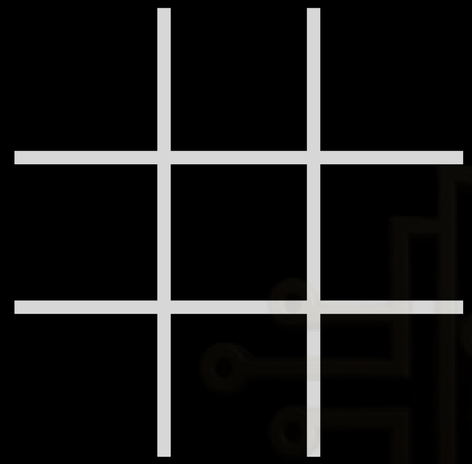
Available on notes.arpanadhikari7.com.np

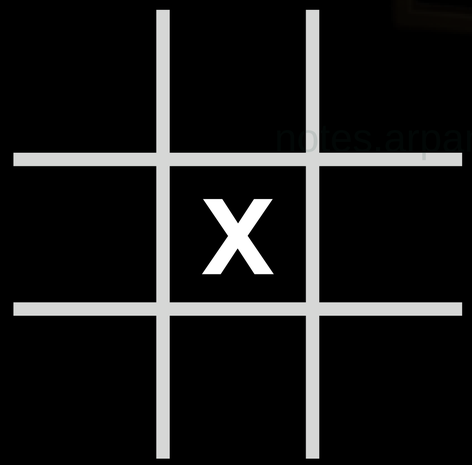
Initial State



notes.arpanadhikari7.com.np

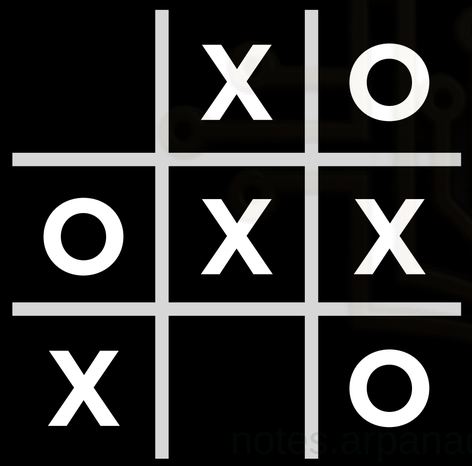
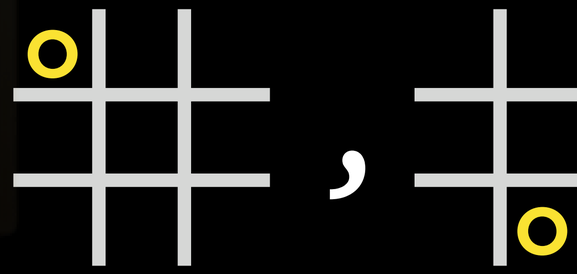
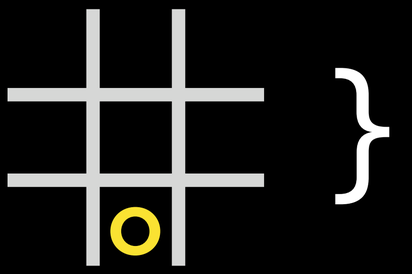
PLAYER(*s*)

PLAYER() = **X**

PLAYER() = **O**

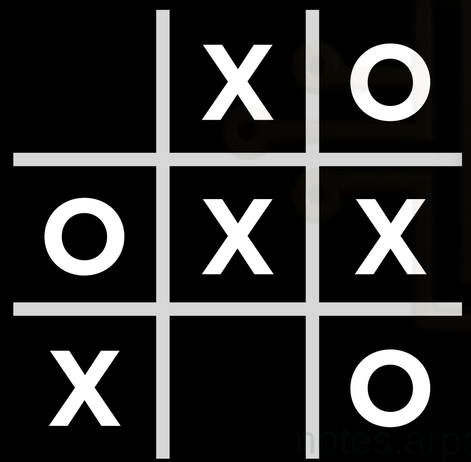
notes.arpanadhikari7.com.np

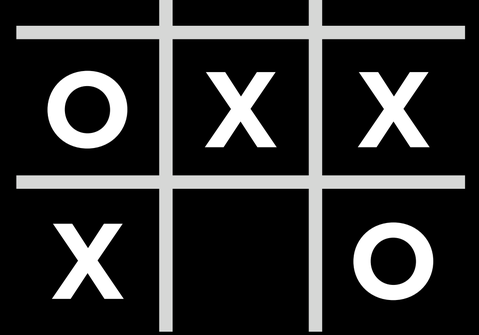
ACTIONS(s)

ACTIONS() = {  ,  }

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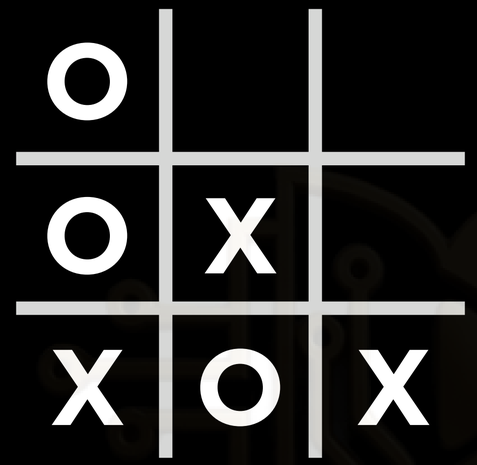
RESULT(s, a)

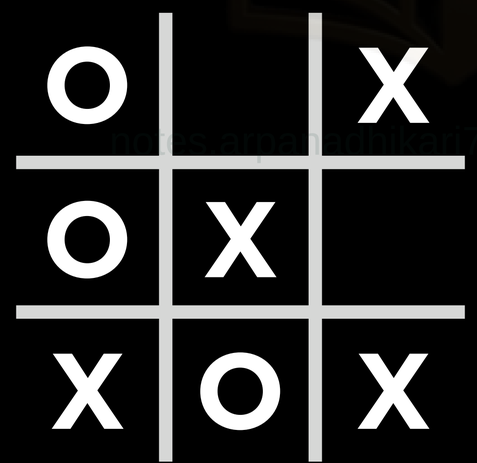
RESULT() =



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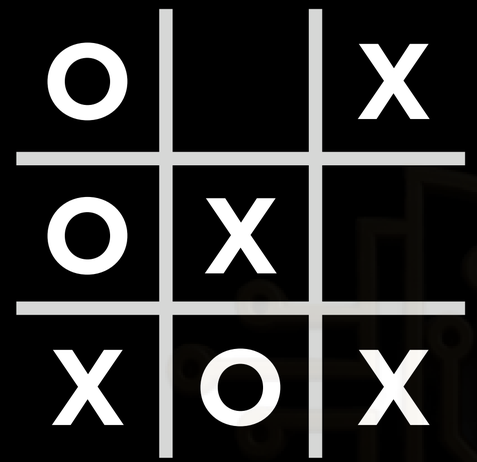
TERMINAL(s)

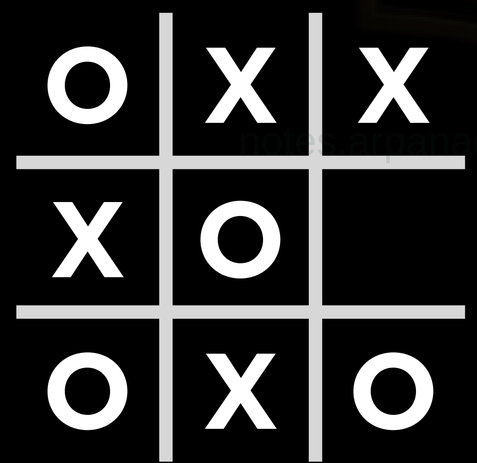
TERMINAL() = false

TERMINAL() = true

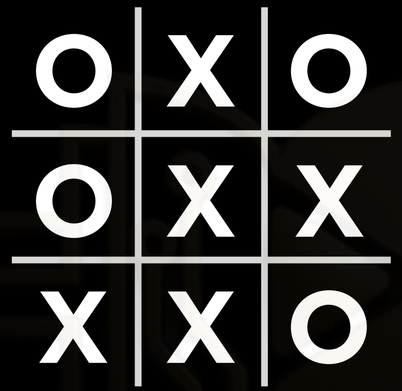
notes.arpanadhikari7.com.np

$\textsc{Utility}(s)$

$\textsc{Utility}($  $) = 1$

$\textsc{Utility}($  $) = -1$

notes.arpanadhikari7.com.np

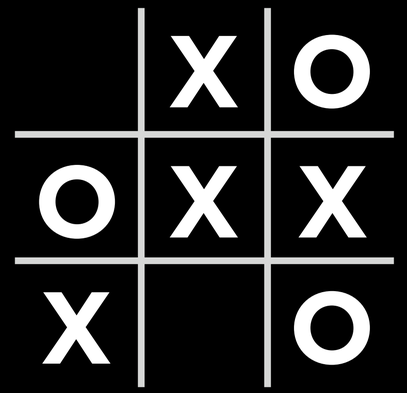


Value: 1

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PLAYER(s) = ○

MIN-VALUE: 0



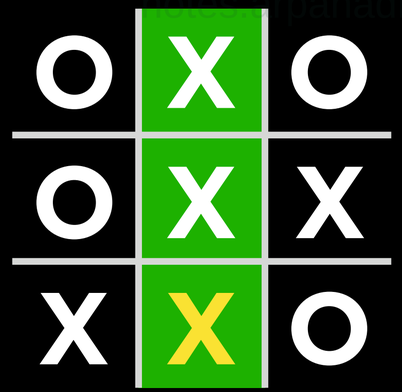
MAX-VALUE: 1



MAX-VALUE: 0



VALUE: 1



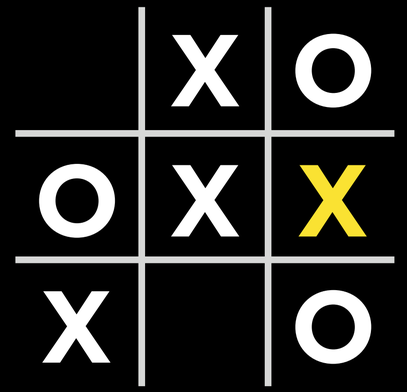
VALUE: 0



notes.arpanadhikari7.com.np

PLAYER(s) = ○

MIN-VALUE: 0





MAX-VALUE: 1



VALUE: 1

MAX-VALUE: 0



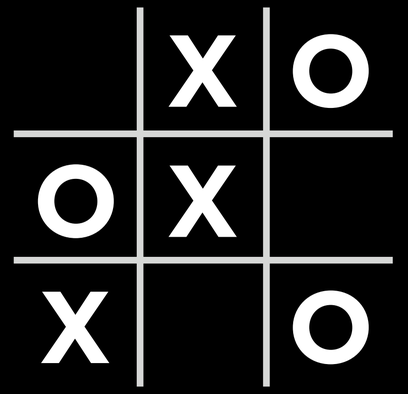
VALUE: 0



notes.arpanadhikari7.com.np

PLAYER(s) = X

MAX-VALUE:
1



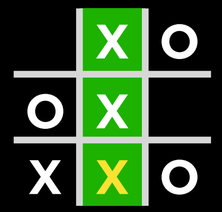
MIN-VALUE:
0



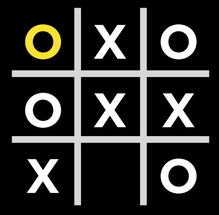
MIN-VALUE:
-1



VALUE:
1



MAX-VALUE:
1



MAX-VALUE:
0



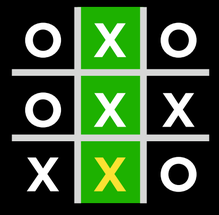
VALUE:
-1



MAX-VALUE:
0



VALUE:
1



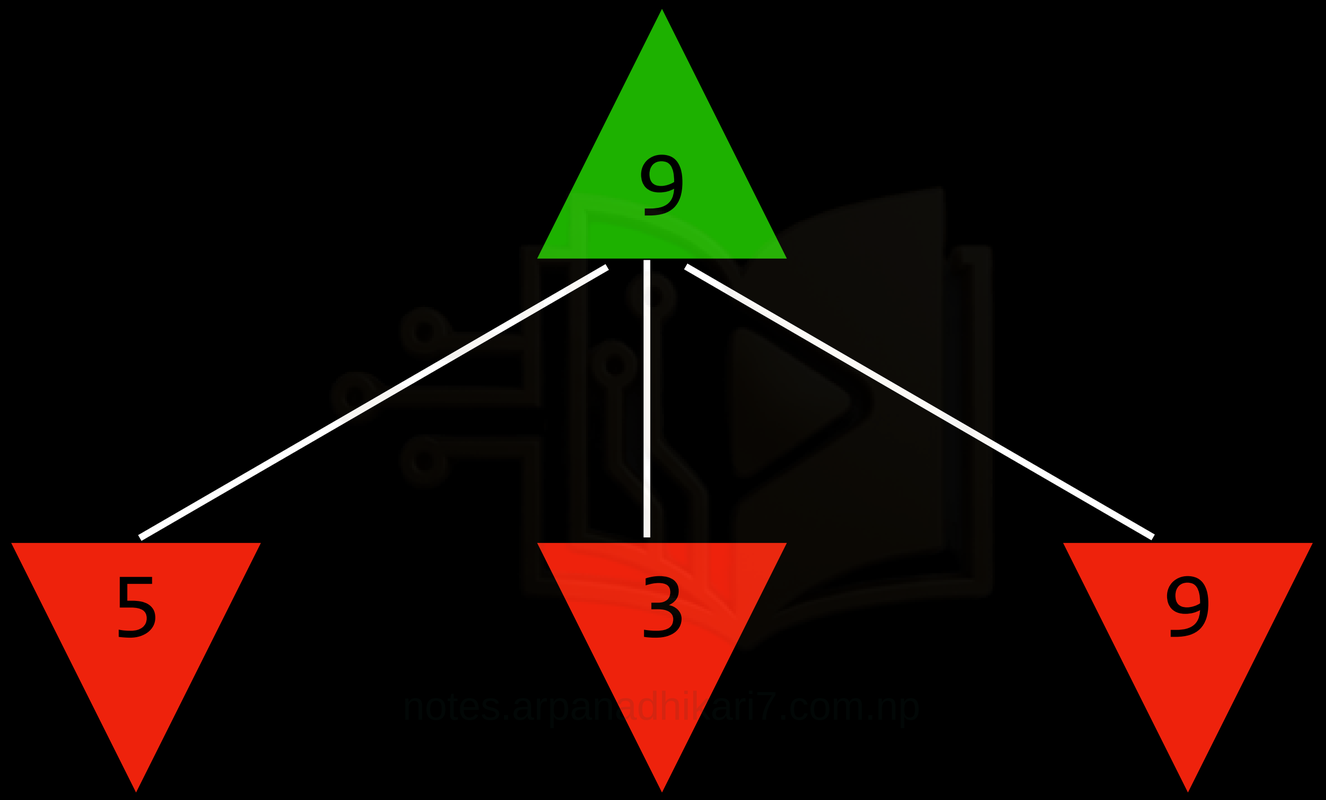
VALUE:
0



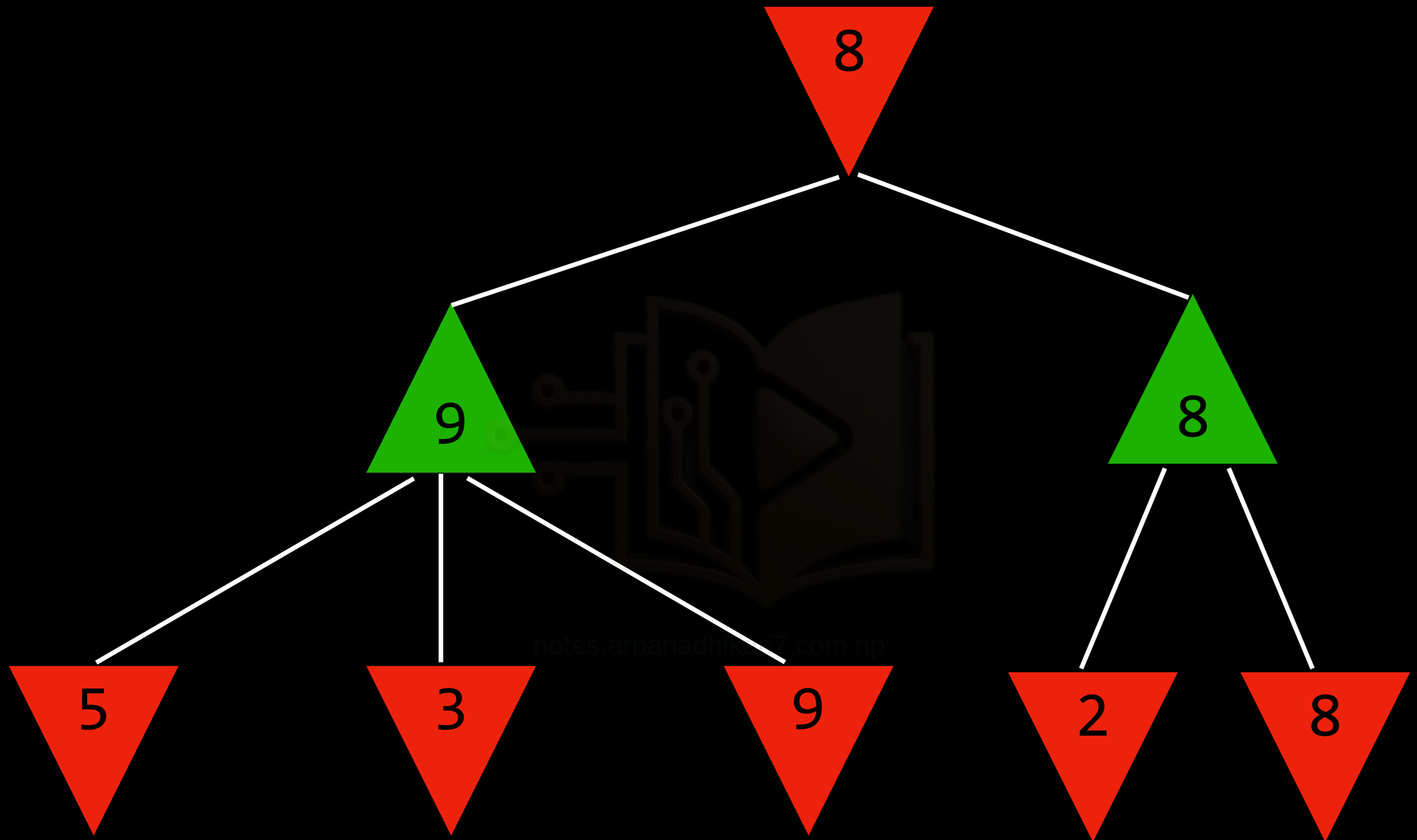
VALUE:
0



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Minimax

- Given a state s:
  - **MAX** picks action a in $\text{ACTIONS}(s)$ that produces highest value of $\text{MIN-VALUE}(\text{RESULT}(s, a))$
  - **MIN** picks action a in $\text{ACTIONS}(s)$ that produces smallest value of $\text{MAX-VALUE}(\text{RESULT}(s, a))$

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Minimax

function MAX-VALUE(*state*):
    if TERMINAL(*state*):
        return UTILITY(*state*)
    v = -∞
    for *action* in ACTIONS(*state*):
        v = MAX(*v*, MIN-VALUE(RESULT(*state*, *action*)))
    return *v*

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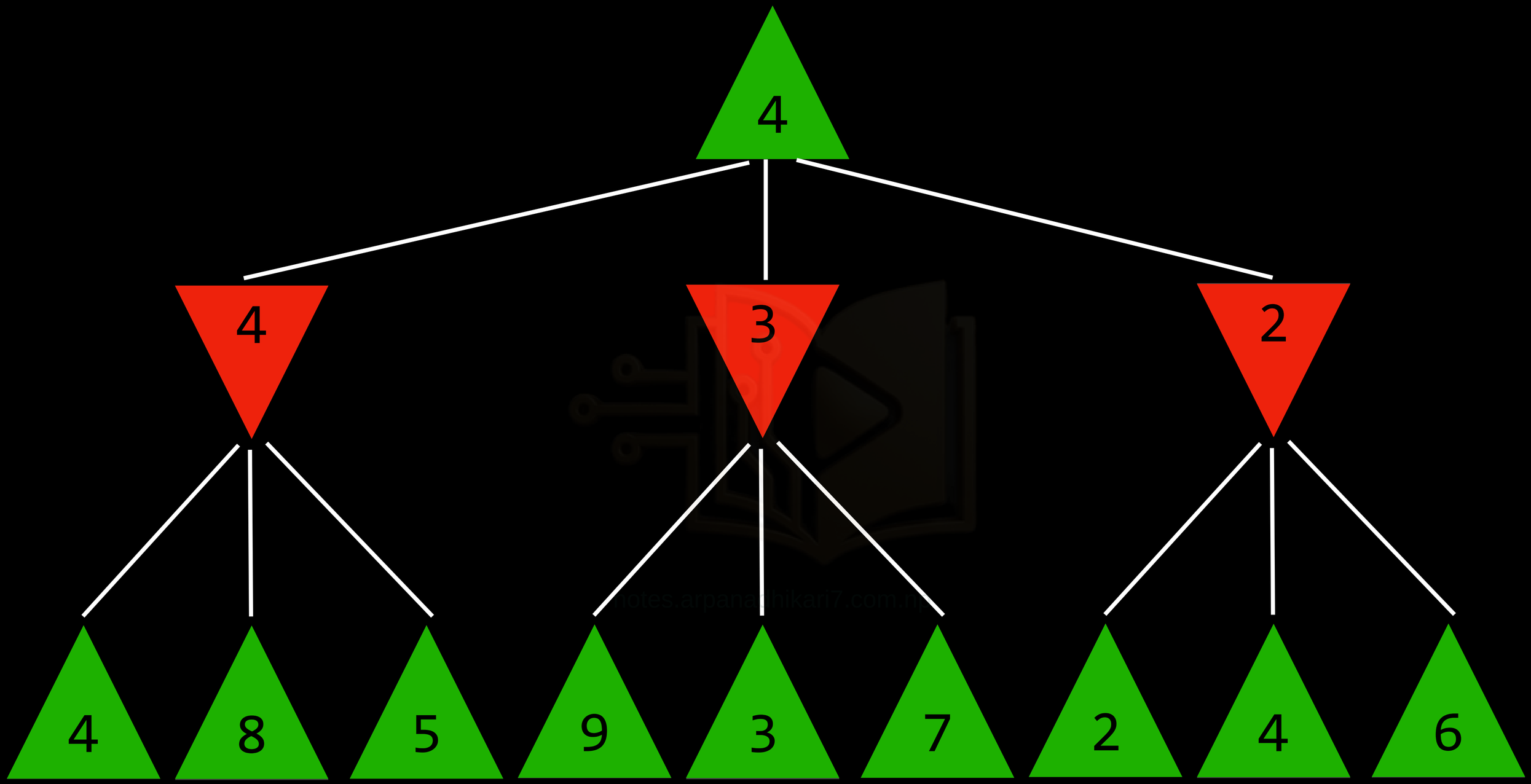
Minimax

function MIN-VALUE(*state*):
    if TERMINAL(*state*):
        return UTILITY(*state*)
    $v = \infty$
    for *action* in ACTIONS(*state*):
        v = MIN(v, MAX-VALUE(RESULT(*state*, *action*)))
    return v

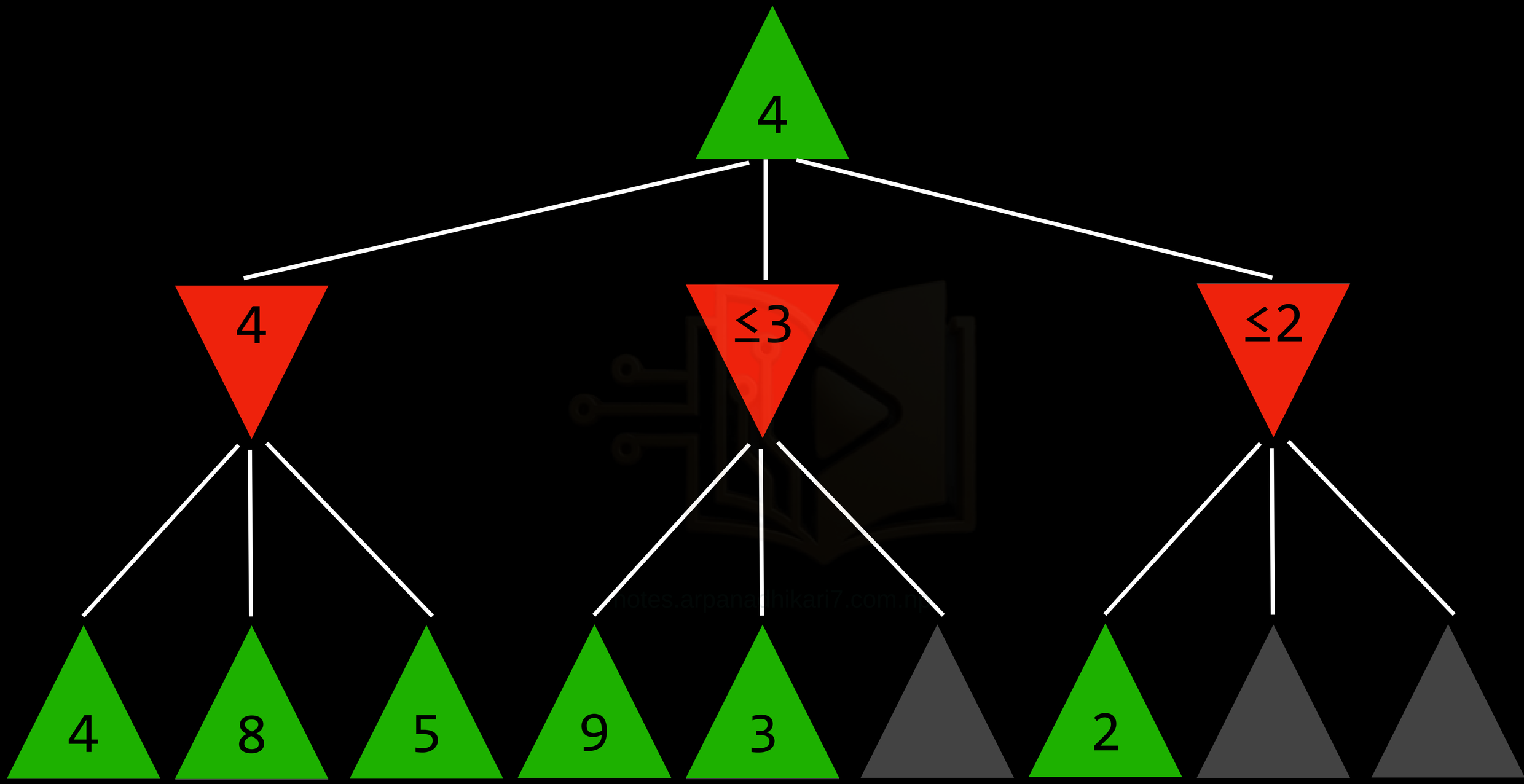
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Optimizations

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Alpha-Beta Pruning

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255,168

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total possible Tic-Tac-Toe games

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288,000,000,000

total possible chess games
after four moves each

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10^{29000}

total possible chess games
(lower bound)

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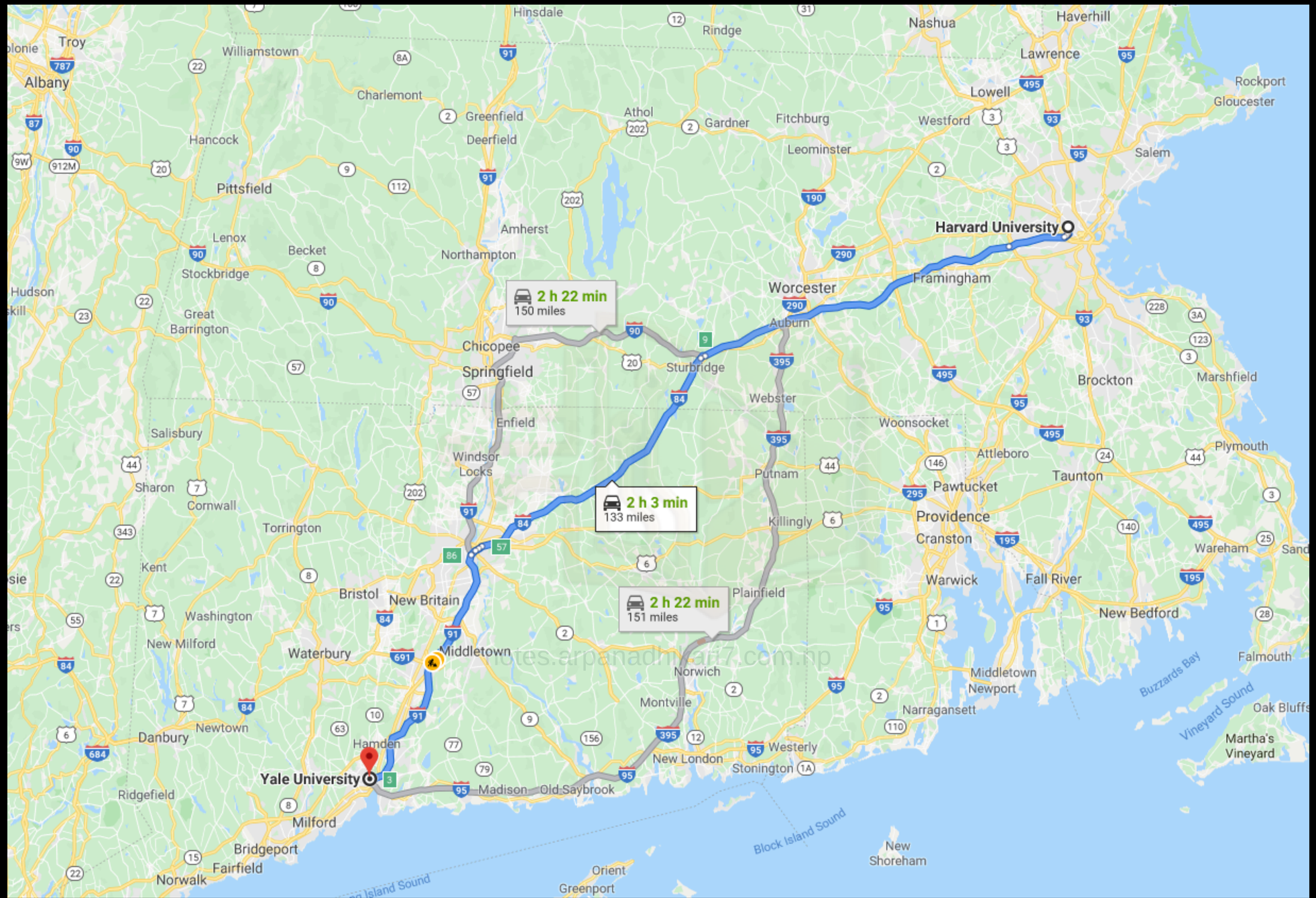
Depth-Limited Minimax

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evaluation function

function that estimates the expected utility of the game from a given state

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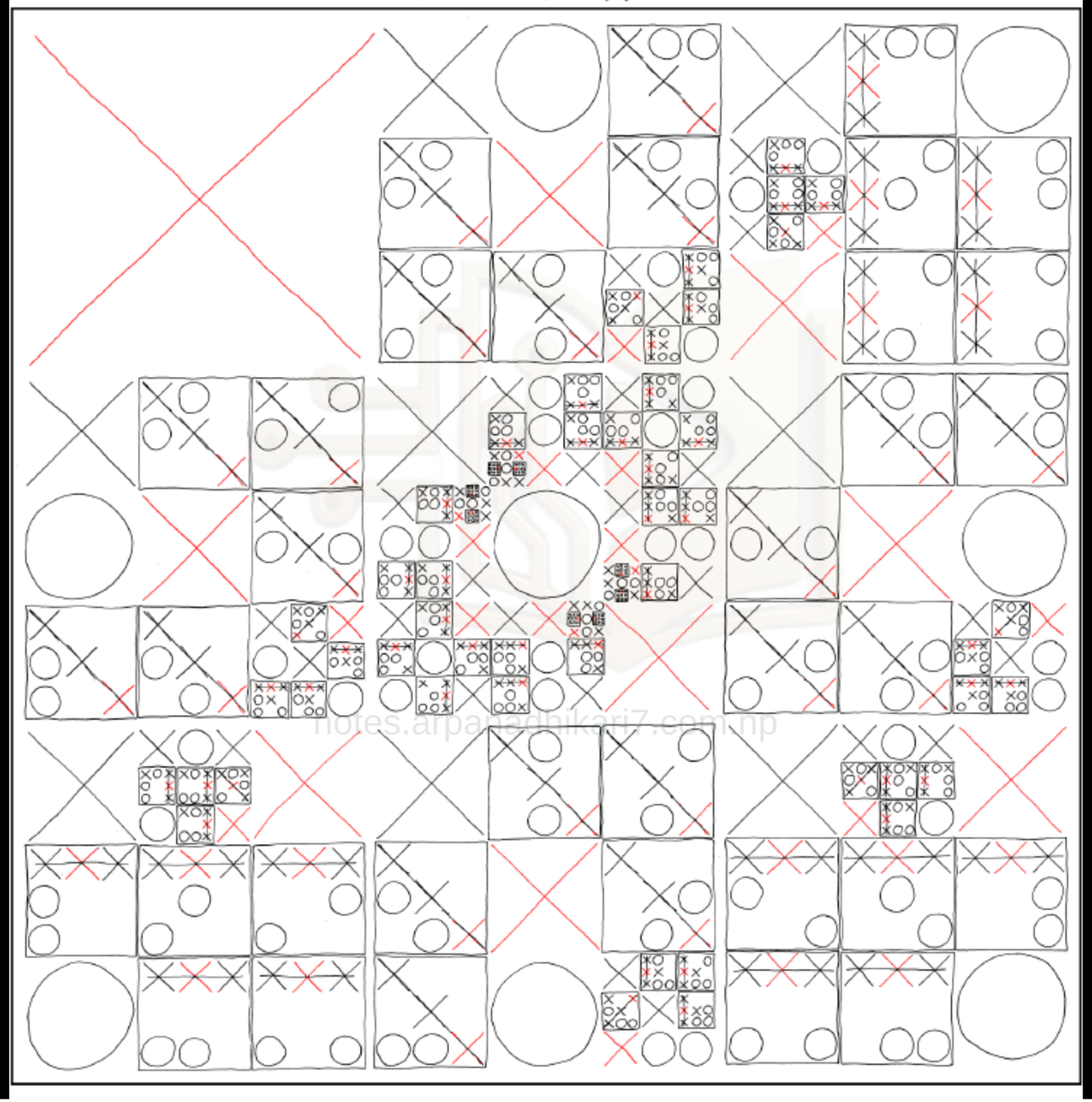


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COMPLETE MAP OF OPTIMAL TIC-TAC-TOE MOVES

YOUR MOVE IS GIVEN BY THE POSITION OF THE LARGEST RED SYMBOL ON THE GRID. WHEN YOUR OPPONENT PICKS A MOVE, ZOOM IN ON THE REGION OF THE GRID WHERE THEY WENT. REPEAT.

MAP FOR X:



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https://xkcd.com/832/

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